# Agent = Harness(Model):
## Five Mechanistic Primitives, and What Is Not One

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**Abstract**

An agent is a model wrapped in a harness, and the harness has five parts. INPUT captures what comes in. CONTEXT assembles what the model is shown. MODEL INTERFACE sends the request and returns the response. OUTPUT carries the response out. CONTROL FLOW decides whether to go again, hand back or stop. We propose this as the architecture of every harness, and show each primitive as the smallest program that adds it. Everything else said about a harness is a design decision within one primitive or a production concern, persistence, safety and security, observability, resilience, performance and evaluation, asked of every primitive. The vocabulary is needed because harnesses change results as much as models do, so a result belongs to the pair and comparison needs the harness stated in parts; it lets research compare primitives by their implementations, and lets debugging isolate a failure to the primitives an implementation touches. We deconstruct the popular harnesses against the five: six from their sources at pinned versions, Claude Code, Codex CLI, opencode, pi, Hermes Agent and a LangGraph workflow, and a wider set of thirteen from their sources or documentation. Every component found has a home. What the deconstruction cannot show is which implementation of a primitive is better; the vocabulary makes that comparison well-posed, by holding the model and four primitives fixed and varying one, and we set out the comparisons that would both use the framework and test it. The minimal harnesses and the evidence for every assignment are published with the paper.

## 1 The Architecture

A language model is a function from a sequence of tokens to a distribution over the next token, applied until a stop token is produced:

$$\mathit{TokensOut} = \mathit{Model}(\mathit{TokensIn}).$$

Viewed from outside, the model keeps no state between calls, cannot read from or act on its environment, and cannot call itself again. Chat templates, tool-call formats and reasoning tokens are conventions for laying out *TokensIn* and reading *TokensOut*; models learn them in training and can learn to emit calls to tools [29], but the calls are carried out, and the conventions written and parsed, by something else.

That something else is the harness: everything in an agent that is not the model. It turns the world into tokens for the model and the model's tokens into effects on the world.

$$\mathit{Agent} = \mathit{Harness}(\mathit{Model}).$$

The boundary is at the model's weights and its sampling. Memory across sessions, tool use and knowledge of the date or the working directory are all on the harness side. The software has

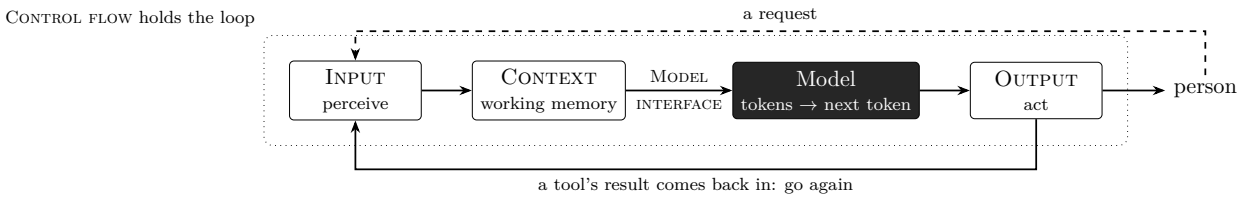



**Figure 1:** The agent loop. The five primitives surround the model; Control flow decides whether the loop runs again, hands back, or stops.

**Algorithm 1** A harness. Lines are labelled with the primitive they implement.

```
1:  procedure ControlFlow( )
2:      conversation ← [ ]
3:      loop                                          ▷ outer loop: one request at a time
4:          conversation.append(User(Input( )))        ▷ Input: from a person
5:          loop                                      ▷ inner loop: one request, carried through
6:              ctx ← Context(conversation)           ▷ Context
7:              response ← ModelInterface(ctx)        ▷ Model interface
8:              conversation.append(Assistant(response))
9:              results ← Output(response)            ▷ Output: show words, run actions
10:             if results = ∅ then break             ▷ nothing ran: hand back
11:             conversation.append(User(results))    ▷ Input: from the world
```

many names: evaluations call it a *scaffold* [3, 20], libraries call themselves *frameworks*, *SDKs* or an *orchestration framework and runtime* [19], and *harness* is common among people who build coding agents [37]. None of the names says what the thing contains.

We propose that it contains five parts, and that the five are the same in every harness.

Input
: captures what comes in.

Context
: assembles what the model is shown.

Model interface
: sends the request and returns the response.

Output
: carries the response out: words to a person, actions on the world.

Control flow
: decides whether to go again, hand back, or stop.

Figure 1 shows how they sit around the model, and Algorithm 1 gives the loop they form, each line labelled with its primitive. Something comes in: a request from a person, an event, or the result of an earlier action. The harness assembles a context around it and sends the context to the model. The harness handles the response: it shows words to a person, or carries out the actions the response requests. If an action produced a result, the result comes back in and the loop runs again; if not, control returns to the person, or the agent stops. Every harness in §5 is an elaboration of this loop.

Everything else said about a harness is at one of two other levels. A *design decision* is one way of implementing a primitive: one tool or forty, a loop or a graph, a repository map or a summarizer. A *production concern* is a question asked of every primitive: what should outlast the session, what could do harm, what did it do, what if something breaks, what is slow, how do we know it still works. Neither adds a primitive (§3).

**Scope.** The primitives describe what happens on the path of a request at run time. Software that ships with a harness but never acts on that path is out of scope: hosting and transport, configuration loading, packaging, the user-interface framework, and the infrastructure that carries telemetry. The model is held fixed. A program that makes one model call and prints the reply has a harness, a trivial one; what makes a harness an agent's is a Control flow that feeds the results of actions back to the model.

**What this paper does.** §2 shows each primitive as the smallest program that adds it. §3 separates the primitives from design decisions and production concerns. §4 says why a shared vocabulary for the harness is needed and what it is for: comparing primitives by their implementations, and isolating a failure to the primitives an implementation touches. §5 deconstructs the popular harnesses against the five: six from their sources at pinned versions, and a wider set from their sources or documentation at summary level. §6 sets out the comparisons that would both use the framework and test it. The minimal harnesses are in Appendix A, the rules for the cases at the boundaries in Appendix B, and the evidence for every assignment in Appendix C.

## 2 The Primitives

Each primitive is defined by what it does and by what it never does. The negative definitions make the five mutually exclusive: every piece of a harness has one home, assigned by its function rather than by its name or its location in the code. Table 1 collects the definitions.

To show each primitive rather than describe it, we use five programs, each the previous one plus one primitive, all run on the same task: a two-file Python project whose `total()` subtracts a discount as a flat amount where the test expects a percentage, and the request *"The tests are failing. Find out why and fix it."* The model is Claude Sonnet through Anthropic's Messages API. The complete programs are in Appendix A; the excerpts below are the lines each primitive adds. In the transcripts, `>` is what the person typed, `<` what the model said, and `$` a command it ran.

### 2.1 Model interface: reach the model

**Does.** Send the assembled context to the model and bring back its response in a form the rest of the harness can read: words, and structured requests to act. The client, the wire format, the translation to the model's chat template, streaming, retries, and switching to a backup model when a request could not be served. **Never.** Decide what the request contains (Context). Act on the response (Output).

```
from anthropic import Anthropic

model = Anthropic()


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


response = request_response({"messages": [{"role": "user", "content": "The tests are failing. Find out why and
     fix it."}]})
print(response.model_dump_json(indent=2))
```

The model answers, but it can only answer: *"I don't have access to your code, test files, or test output. Nothing came through in your message besides the request."* The request is written into the program, so nobody can ask it anything else.

### 2.2 Input: capture what comes in

**Does.** Capture what enters the agent and hand it to Context. Two sources: a person (a message, a file, an interrupt, an answer to a question the model asked) and the world (the result of an action, an event, a scheduled trigger). **Never.** Decide what the model is shown (Context). Run anything (Output).

```
def capture_input():
    return input("\n> ")
```

Anyone can now type the request, and the answer is the same: *"I don't have the code or the test output yet, so I can't tell why the tests are failing. Could you share…"* It can only talk.

### 2.3 Output: carry the response out

**Does.** Handle what the model returned: show its words to a person, and carry out each requested action, collecting its result. The tool set, the executor that runs an action, and the environment it runs in. **Never.** Decide whether to go again (Control flow). Assemble the next request (Context). The model only ever *requests* an action; Output decides whether and how it happens, which is where sandboxes and never-lists live.

```
model = Anthropic()

bash = {"name": "bash", "description": "Run a shell command",
        "input_schema": {"type": "object", "properties": {"command": {"type": "string"}}, "required": ["
     command"]}}


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def capture_input():
    return input("\n> ")


def handle_output(response):
    tool_results = []
    for block in response.content:
        if block.type == "text":
            print("< " + block.text.replace("\n", "\n  "))
        if block.type == "tool_use":
```

A tool is described to the model, and `handle_output` runs each command the model asks for and collects the result:

```
> The tests are failing. Find out why and fix it.
< I'll start by exploring the project and running the tests.
$ pwd; ls -la; git status 2>/dev/null | head
```

Then it stops. The command ran and printed the files, and the result went nowhere.

### 2.4 Control flow: decide what happens next

**Does.** Define the shape of the loop: send results back and go again, hand back to the person, or stop. Step and token budgets, approval gates the harness consumes, delegation to sub-agents, and routing to a model keyed to the work. **Never.** Capture input, assemble context, call the model or run a tool. It sequences the other four.

```
def control_flow():
    conversation = []
    while True:
        conversation.append({"role": "user", "content": capture_input()})
        while True:
            response = request_response({"tools": [bash], "messages": conversation})
            conversation.append({"role": "assistant", "content": response.content})
            tool_results = handle_output(response)
            if not tool_results:
```

With the results sent back and the model called again until it stops requesting actions, the same model, tool and request finish the task:

```
> The tests are failing. Find out why and fix it.
< Let me start by looking at the project and running the tests.
$ pwd; ls -la; git status 2>/dev/null | head
$ cat prices.py test_prices.py; python -m pytest -q 2>&1 | tail -30
$ python -m unittest -v 2>&1 | tail -20
< The docstring says the discount is a percentage, but the code subtracts it
  as a flat amount. So 25 - 10 = 15 instead of 25 x 0.9 = 22.5. I'll fix the
  code to match the documented behavior.
$ sed -i 's|return round(subtotal - discount, 2)|return round(subtotal * (1 - discount / 100), 2)|' prices.py
   && cat prices.py && python -m unittest 2>&1 | tail -5
< The tests pass now. The bug was in prices.py, not in the tests.
```

This is an agent. Asked a question about itself, it has to go looking:

```
> Who are you, and where are you working?
< I'll check the environment first so I can answer accurately.
$ echo "user: $(whoami)"; echo "host: $(hostname)"; echo "cwd: $(pwd)"; uname -a; ls -la | head -20
< I'm an AI assistant. I can run shell commands, and I use that to read files,
  run code and make changes in the environment I'm given.
```

It does not know its name, its body, its working directory or the date.

### 2.5 Context: assemble what the model is shown

**Does.** Assemble, for one call, everything the model is shown: instructions, the tools' descriptions, the conversation so far, and anything retrieved from memory or the environment; and fit it to the window by truncation, compaction or selection. Context is the agent's working memory, rebuilt on every call. **Never.** Call the model (Model interface). Act on the world (Output). Reading a memory store is Context; writing one, if the model requests it, is an action and so Output.

```
import subprocess, os, datetime
from anthropic import Anthropic

model = Anthropic()
bash = {"name": "bash", "description": "Run a shell command",
        "input_schema": {"type": "object", "properties": {"command": {"type": "string"}}, "required": ["
     command"]}}


def capture_input():
    return input("\n> ")


def assemble_context(conversation):
    instructions = f"You are quark, an agent. You act through bash, in {os.getcwd()}. Today is {datetime.date.
     today()}."
    return {"system": instructions, "tools": [bash], "messages": conversation}


def request_response(context):
```

**Table 1:** The five primitives. Each is defined by what it does and what it never does; the last column is what it hands to the next.

| Primitive | Does | Never | Hands on |
|---|---|---|---|
| Input | captures what comes in, from a person or from the world | decides what the model is shown; runs anything | the captured item, to Context |
| Context | assembles what the model is shown on one call, and fits it to the window | calls the model; acts on the world | one request, to Model interface |
| Model interface | sends the request and returns the response as words and requests to act | decides what the request contains; acts on the response | the response, to Output |
| Output | shows words to a person; carries out each requested action and collects its result | decides whether to go again; assembles the next request | the results, to Control flow |
| Control flow | decides: go again, hand back, or stop | captures, assembles, calls, or runs | results back to Input, or control to the person |

```
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def handle_output(response):
    tool_results = []
    for block in response.content:
        if block.type == "text":
            print("< " + block.text.replace("\n", "\n  "))
        if block.type == "tool_use":
            print(f"$ {block.input['command']}")
            ran = subprocess.run(block.input["command"], shell=True, capture_output=True, text=True)
            tool_results.append({"type": "tool_result", "tool_use_id": block.id, "content": ran.stdout + ran.
     stderr or "(no output)"})
    return tool_results


def control_flow():
    conversation = []
    while True:
        conversation.append({"role": "user", "content": capture_input()})
        while True:
            response = request_response(assemble_context(conversation))
            conversation.append({"role": "assistant", "content": response.content})
            tool_results = handle_output(response)
            if not tool_results:
                break                                   # done: hand back to the person
            conversation.append({"role": "user", "content": tool_results})   # a tool ran: go again


control_flow()
```

One sentence of context, assembled on every call, answers the question without any action:

```
> Who are you, and where are you working?
< I'm quark, an agent that works by running shell commands through bash. My
  working directory is /tmp/shop. Today's date is 2026-10-07.
```

This is the whole harness: fifty lines, five functions, one per primitive. It runs commands on the host without asking and forgets everything when it exits; the production concerns of §3 are what it lacks.

**Each primitive removes a failure the others do not.** With only Model interface, the model answers but cannot act and nobody can ask it anything else; with Input, anyone can ask and it still cannot act; with Output, it acts once and the result goes nowhere; with Control flow, it finishes the task but spends actions finding out where it is; with Context, it knows. The same five programs, run with a local model in place of the hosted one, needed one function changed: `request_response`, which is Model interface. That is the porting property of §4 in practice.

# 3 Design Decisions and Production Concerns

Descriptions of harnesses mix three kinds of statement. "It has a loop" names a primitive. "It gives the model forty tools" names a decision within one. "It runs commands in a sandbox" names a measure taken for a reason that applies to every primitive. Keeping the three apart is the main use of the vocabulary. A primitive is a mechanism every harness has, defined by its function on the request path; a harness has exactly five. The other two levels are where harnesses differ and where they are made fit to run.

## 3.1 Design decisions

A design decision is one way of implementing a primitive. It is the level at which harnesses differ, and the level at which they are usually described. A design decision never adds a primitive; it fills one. The test for whether something is a design decision is whether a different harness could make a different choice and still be a harness. Table 2 lists the decisions found in the deconstruction of §5, by primitive, with the choices the harnesses made.

## 3.2 Production concerns

A production concern is a question asked of every primitive, whose answer changes how some primitives are implemented. We use six (Table 3): what should outlast the session (persistence); what could do harm, and who could make it (safety and security); what did it do (observability); what if something breaks (resilience); what is slow or costly (performance); and how do we know it still works (evaluation). A concern is not a part of the harness. A sandbox is not a sixth primitive; it is the safety answer in Output. A memory store is not a sixth primitive; it is a store in the world that Context reads and Output writes, kept for the persistence concern. A trace is not a sixth primitive; it is the observability answer, recorded wherever the primitive it watches runs. The test for whether something is a concern is whether it is motivated by a question that could equally be asked of a different primitive.

The concerns interact. Persistence creates a security surface: a poisoned memory steers every later session. Streaming changes safety: an interrupt can stop a reply mid-sentence, and the harness must record what was said rather than a finished answer. Summarizing changes persistence: the summary must say where the originals are kept. Treating concerns as questions asked of each primitive makes these interactions visible; treating them as separate components hides them.

## 3.3 Bundles and the cases at the boundaries

Much of what a harness's documentation names is a bundle. A mode, a skill, a hook system or an agent profile packages decisions across several primitives: a planning mode is a context template, a restricted tool set and an approval gate; a hook system is a dispatcher in Control flow whose events each affect one primitive. Decomposed, each element has one home. The negative definitions

decide most placements. Thirteen rules, each needed for a component found in §5, decide the rest, and are given in Appendix B. Three are used often enough to state here. A component is assigned by its function, not its location: an approval decision made inside a tool's runtime is CONTROL FLOW. A sub-agent is a harness of its own, and delegation to it is CONTROL FLOW of the outer harness. And one primitive may use another's mechanism to do its job: CONTEXT may ask the model to summarize old turns before compacting them, and the component is assigned by what it decides, with the call it makes a use of MODEL INTERFACE.

# 4 Why a Vocabulary

## 4.1 Harnesses change results as much as models do

If $Agent = Harness(Model)$, a benchmark score belongs to the pair. The evidence that the harness term is large is now direct. Holding GPT-4 Turbo fixed, Yang et al. [35] report 1.31% of SWE-bench [16] resolved with retrieval-augmented generation and 12.47% with their agent, and on SWE-bench Lite 11.00% for a shell-only agent against 18.00% for SWE-agent, with only the harness changing. Anthropic notes that performance "can vary significantly based on this scaffolding, even when using the same underlying AI model" [3]. METR found that tailoring a scaffold to one model more than doubled the length of tasks it could complete [20], and later that two mature harnesses did not outperform their default scaffolds [21], which suggests the effect is largest where the design decisions differ most. Zhang et al. [39] argue that harness configuration can account for more variance in performance than the choice of model, and ask that the harness be disclosed when agents are compared; Kapoor et al. [18] evaluate models, scaffolds and benchmarks as separate dimensions.

A disclosure needs a schema, and a comparison needs the harness stated in parts. Without one, two agents built on the same model are compared as wholes, and a difference between them cannot be attributed to anything smaller than "the scaffold". The five primitives are the schema: the input sources; what the context holds and how it is fitted to the window; the provider, decoding settings and retry policy; the tools, their output formats and the execution environment; and the loop's shape and stopping rules. Table 5 and Table 6 are such disclosures for nineteen harnesses.

## 4.2 Comparing primitives by their implementations

With the harness stated in five parts, a question that could not be asked becomes well-posed: which implementation of a primitive is better, for which model, on which task? The form of the study is to hold the model and four primitives fixed, vary the fifth, and measure. Fan et al. [10] do this for three decisions in a coding harness, planning, the action space and context management, across four models, and find that each changes behaviour. Their three dimensions were chosen independently of this paper and each falls within one primitive: planning in CONTROL FLOW, the action space in OUTPUT, context management in CONTEXT. The vocabulary says what the other dimensions are, and says which parts must be held fixed for a result to be attributed to the one varied. §6 lists the comparisons the framework makes well-posed.

The same property makes a harness portable. Because only MODEL INTERFACE knows which model is inside, a harness can be moved to a new model or provider by changing that primitive alone. A harness loses this property when model-specific details leak into other primitives: a provider's tool-call format assumed in OUTPUT, or a provider's error strings matched in CONTEXT management. The deconstruction records such leaks where it found them. It also says what a model trained for one harness must learn: the context layout it will be shown (CONTEXT), the format in

which it must request actions and the shape of the results it will read (Output), and the loop it runs in (Control flow).[1]

### 4.3 Isolating a failure

When an agent fails, the primitives give a short list of places to look, in order. The order matters: each question rules out a place the next question assumes.

1. Context**: was the model shown what it needed?** Read the request that was sent. If the file, the result or the instruction is not in it, the model could not have used it, and nothing downstream is at fault.
2. Output**: did what it asked for happen?** Compare the action the model requested with what ran. A parser that dropped a call, a sandbox that refused it, a tool that returned a truncated result, and an error message the model could not read are all Output.
3. Control flow**: did the loop stop early or run too long?** A step limit reached before the task was done, a hand-back on a response that still had work in it, and a loop that repeated the same action are Control flow.
4. Model interface**: was the response read correctly?** A tool call emitted in the right format but not recognized, a stream cut off and treated as complete, and a backup model substituted without notice are Model interface.
5. Input**: did the right thing come in?** A result delivered to the wrong turn, an interrupt lost, a person's answer consumed by the harness when the model needed it.
6. **Only then, the model.** If the model was shown what it needed, asked for the wrong thing, and its request was carried out as asked, the failure is the model's: its knowledge, its habits, or its size.

An implementation touches some primitives and not others, and a failure is isolated to the ones it touches. Two examples from our own transcripts. A small model, after instruction-tuning, sometimes emitted a second tool call as bare JSON without its `<tool_call>` tags. The harness read the JSON as words, found no action, and handed back. Question 4 locates this in Model interface, whose parser defines what counts as a call; the fix was on the model side, but the diagnosis was the harness's. In the same experiments, four kinds of task could never pass during reinforcement learning because the check compared the folder before and after, and the harness's own memory directory changed the folder on every run. No question in the list locates this, because the fault was in the evaluation, a concern, not in a primitive. Building the harness around a capable model first, until it passed every task in the evaluation, is what let every later failure be attributed to the model swapped in.

### 4.4 Existing vocabularies

The loop that harnesses implement goes back to ReAct [36], which interleaves a model's reasoning with actions whose results return to it as observations; in artificial intelligence more broadly, an agent perceives its environment and acts on it [28]. Later patterns live in particular primitives: Reflexion [31] writes reflections on failed attempts and carries them into the next, a decision in Control flow and Context; MemGPT [27] pages information in and out of a limited window, a decision in Context; Toolformer [29] trains the model to emit tool calls, moving policy into the

[1] In the companion book, a 0.6-billion-parameter model instruction-tuned on sessions recorded in its own harness rose from 2 to 18 of 32 held-out tasks in that harness, where the same base model post-trained by its developers for general use scored 14. The experiment, its data and its checks are described there [9]; this paper makes no claim that rests on it.

model while the calls are still carried out by OUTPUT; CodeAct [34] has the model write code as its action format, a decision in OUTPUT.

Sumers et al. [32] organize language agents by memory (working, episodic, semantic, procedural), an action space of external and internal actions, and a decision cycle. CoALA describes what an agent *is*, as a cognitive architecture; this paper describes what software an agent is *made of*. The correspondence is in Table 4. Two things here have no counterpart there: MODEL INTERFACE as a component, and the production concerns as a level of description distinct from the architecture. CoALA's memory modules, with the retrieval and learning actions that read and write them, are design decisions in CONTEXT and OUTPUT here, not a component.

Yang et al. [35] name the agent-computer interface, the commands a model is given, their output format and guardrails such as linting edits, and show it changes performance on software tasks; in our terms the interface is a set of design decisions in OUTPUT and CONTEXT, and this paper asks what the complete set of components is, of which the interface is part. Schluntz and Zhang [30] distinguish workflows, in which code fixes the sequence of model calls, from agents, in which the model directs its own process, and describe patterns built on an "augmented LLM"; we treat the distinction as a property of CONTROL FLOW, and decompose the augmented LLM into CONTEXT (retrieval, memory) and OUTPUT (tools) around a MODEL INTERFACE. Jin et al. [17] use "primitives" for recurring patterns of model calls in multi-agent systems (review, voting, planning and execution); in our terms these are shapes of CONTROL FLOW, and the two uses of the word are complementary. Tool protocols such as MCP [23] provide pluggable OUTPUT, with their other features assigned by function (Appendix B).

## 5 Deconstruction

If the architecture holds, a harness built by someone else, for a different purpose, at a different scale, should decompose into the five primitives with nothing left over, and its differences from other harnesses should be locatable as design decisions within a primitive. We test this on the popular harnesses, in two sets.

**Selection.** The first set is six harnesses in wide use whose primary sources were available, read in depth: two command-line coding agents from model developers (Claude Code [4] and Codex CLI [25]), two open-source coding agents built outside the model developers (opencode [2] and pi [38]), one general-purpose agent built for messaging platforms as well as the terminal (Hermes Agent [24]), and one workflow framework (LangGraph [19]), for which we deconstruct the canonical agent it ships, a ReAct loop, together with the runtime that executes it. The workflow framework tests the claim that a workflow and an agent differ only in CONTROL FLOW. The second set is thirteen more, read at summary level: the coding agents SWE-agent [33], OpenHands [1], Aider [11], Cline [7], Gemini CLI [13] and Goose [6]; the agent frameworks OpenAI Agents SDK [26], Google ADK [14], AutoGen [22], CrewAI [8] and smolagents [15]; and two closed harnesses read from their documentation alone, Cursor [5] and GitHub Copilot's coding agent [12]. Claude Code is likewise deconstructed from its documentation, since its source is not published. The nineteen are the harnesses we could read, not a sample of all harnesses; §5.8 says what that leaves untested.

**Method.** For each harness we (i) read the source at a pinned commit, or the documentation on a stated date, and mapped the path of a request from the moment a message arrives to the next model call; (ii) assigned every component on that path to exactly one primitive, or to one production concern together with the primitive it changes, by the definitions of §2 and the rules

of Appendix B, citing a file and line range or a page and section for each; and (iii) recorded every component that was hard to place and how it was resolved, or that it was not. For the first set every component on the path was assigned; for the second, two to five components per primitive, chosen as the ones that define the harness's design decision there. Components outside the request path (hosting, packaging, UI frameworks) were noted and left unassigned. The evidence is in Appendix C and in the repository as a file per harness; the subsections below summarize the first set.

### 5.1 Claude Code

Anthropic's coding harness, deconstructed from its documentation as read on 15 October 2026 [4]; its source is not published. The loop is the open one of Algorithm 1: the model is called until it returns a response with no tool calls, and a turn's result carries why it ended (done, a turn limit, a cost budget, an error). Input has one entry point for everything that starts a turn: typed prompts, piped input, messages queued mid-turn and delivered after the current tool batch, scheduled prompts, and reports from background sub-agents. Context is built as a stable cached prefix (the system prompt, the tools, then project files such as `CLAUDE.md`) with later updates appended as messages or system reminders; tool schemas for MCP servers are deferred behind a search tool; and compaction, on reaching the window, is a separate request that re-injects the project files, memory, recently read files and invoked skills. Model interface is the Messages API with streaming, up to ten retries that never re-issue after a completed tool block, idle watchdogs, and a fallback chain the user configures. Output is a fixed built-in tool set plus MCP tools; read-only calls run concurrently and mutating calls in sequence; shell commands run under an OS sandbox with a domain-allowlist proxy; large results spill to files; file edits are checkpointed for rewind. Control flow adds a six-step permission pipeline (hooks, deny rules, ask rules, the permission mode, allow rules, then a prompt), delegation through an agent tool to a depth of three, and, at scale, a model-written JavaScript workflow the runtime executes, which sequences model calls and is therefore Control flow by the rule on model-written programs.

Three components were hard to place. The web-fetch tool calls the model itself to condense a page before the main model sees it: the call is Output, the condensing is a Context decision, and the extra request is Model interface. The default permission gate in "auto" mode is a second model asked to classify each action: a Control flow gate implemented as a model call with its own stripped context. And a server-side variant of that classifier runs inside the API during the session's own requests, so the harness only consumes its verdict; we return to this in §5.7.

### 5.2 Codex CLI

OpenAI's terminal and IDE coding agent, at commit `f0c5077` [25], written in Rust. A session is one thread that receives typed operations over a channel; a dispatcher matches each one (a turn's input, an interrupt, an approval, the answer to a question, a compaction request) and runs each user turn as a task. Input is that channel: text typed during a running turn is queued as steering rather than starting a new turn, an inter-agent mailbox and asynchronous hook results merge into the same pending-input queue, and the model's questions to the person come back as a typed answer. Context is the Responses API's own item list, with the environment and instructions injected as typed "world-state" sections sent as diffs when they change, project files walked from the root to the working directory, and tool outputs truncated in the live history only; compaction has three implementations (a local summary, a remote one performed by the server, and a token-budget reset), chosen by explicit threshold arithmetic, and can run speculatively in the background once usage passes a percentage of the window. Model interface is the Responses

API only, WebSocket first with a session-wide fallback to HTTP, five retries with jittered backoff that honour the server's retry timing, an idle timeout, and requests sent without server-side storage, with encrypted reasoning replayed. OUTPUT routes every risky action through one orchestrator (approval, then the platform sandbox, then the run, then a retry if the sandbox denied it), with Seatbelt on macOS, seccomp with bubblewrap and Landlock on Linux and a restricted token on Windows; tools run concurrently under a read–write gate that lets parallel-safe tools overlap; and a "code mode" runs model-written JavaScript in V8, each nested tool call re-entering the harness's approval and sandbox path. CONTROL FLOW is one task per turn, looping until the model emits no tool call, with stop hooks, roll-over into compaction mid-turn, an approval precedence of hooks, then an optional model reviewer ("Guardian"), then the person, and sub-agents that are full sessions of their own with depth and thread caps.

Hard cases: the Guardian is a CONTROL FLOW gate whose verdict comes from a nested harness; the remote compaction is a CONTEXT decision whose summarizer is the provider; over WebSocket the request omits the tools and instructions the server already holds, so what is sent depends on transport state while what the model is shown does not; and the plan-mode parser, which reads the streamed text into a proposed-plan item and can redirect it, is OUTPUT rendering that also feeds the stop decision. We record the last as straddling OUTPUT and CONTROL FLOW by design.

### 5.3 opencode

An open-source coding harness by Anomaly, at commit `654b79b` [2]; it was rewritten since earlier versions, and the deconstruction describes the current one. Each session is an event-sourced record in SQLite, and one fibre per session drains a durable inbox: INPUT is a row in that inbox, from a person, a shell command, a skill, or the harness itself, marked for delivery either at the next step boundary ("steer") or after the current turn ("queue"). CONTEXT re-projects the event history into model messages on every call, with instructions placed in a cached prefix and changes to them frozen into the history as dated events; compaction is triggered at about ninety percent of the window, using provider-measured token counts plus a local estimate, keeps the newest fifteen thousand tokens verbatim, and can be delegated to the provider's native compaction. MODEL INTERFACE is an in-house request type compiled per provider protocol (Anthropic Messages, OpenAI Responses, Bedrock Converse, Gemini), streamed over HTTP or a stateful WebSocket, with a retry policy split by whether output had already started. OUTPUT runs tool calls concurrently as they stream, and most tools are reachable only through one `execute` tool that runs model-written JavaScript in an interpreter that cannot call the model, which the rule on model-written programs places in OUTPUT. CONTROL FLOW continues while a step made tool calls, stops on a text-only finish, enforces an optional per-agent step budget, and delegates to child sessions (depth one by default), whose results return as tool text or as a synthetic message.

Hard cases: every tool asserts its own permission, so an approval gate the harness consumes lives inside the tool's code (CONTROL FLOW by function); the step budget is enforced by appending an assistant message and disabling tools, a CONTROL FLOW decision carried out by editing what the model sees; and the compaction service both decides what to summarize and issues the model call itself. Native provider-side compaction leaves the harness's CONTEXT holding an item it cannot inspect.

### 5.4 pi

A terminal coding agent by Mario Zechner, at commit `5ab7fe7` [38], built in layers: a minimal agent loop, a provider layer, and a coding-agent session on top. INPUT comes from a terminal

editor, a print mode, or a JSON-RPC mode, all into one `prompt()` call, with mid-run messages split into steering (after this turn) and follow-ups (after the agent would have stopped). CONTEXT is a projection of a persisted session tree: a compaction entry, the kept tail, and context edits, with the system prompt kept as named sections and diffed into patches, so that replaying the file reproduces the exact request; compaction triggers when the context exceeds the window less a reserve, and keeps the most recent twenty thousand tokens. MODEL INTERFACE is one streaming function behind about fifteen provider adapters, with the SDKs' own retries disabled so that the harness owns a two-tier retry policy, and an overflow classifier that routes a too-long request to compaction rather than to a retry. OUTPUT is a set of plain tool objects run in parallel by default, with mutations to one file serialized; there is no sandbox in the core (OS sandboxing and permission prompts are extensions), and a "code mode" tool runs model-written scripts in a QuickJS sandbox. CONTROL FLOW is a two-level loop (tool calls and steering inside, follow-ups outside) plus a session-level loop for retries, overflow recovery and extension continuations; sub-agents are not in the core, though an example spawns child processes; and per-request routing between "virtual models" is keyed to the reason for the request.

Hard cases: a wire-level hook lets an extension rewrite the request body, which is CONTEXT by function though it sits in the adapter; the virtual-model router has a "retry" branch that overlaps the backup-model rule (we place it in CONTROL FLOW, since the router decides by the state of the work); and the code-mode script can call auxiliary models from inside the sandbox, a model call issued from inside a tool, which we discuss in §5.7. A telemetry interface exists, but no span is emitted at this commit.

### 5.5 Hermes Agent

A personal agent by Nous Research, at commit `7bf3766` [24], with one agent core behind a terminal interface, about twenty-five messaging-platform adapters, an HTTP server, an ACP adapter and a cron scheduler. INPUT is one normalized message event shared by all of these; mid-turn messages become steering rather than new turns. CONTEXT is a three-tier system prompt, kept byte-stable so that caching holds: identity, then project files (capped by size), then volatile memory, skills and the time; compaction at half the window prunes, then summarizes with an auxiliary model; oversized tool output is spilled to files rather than dropped; and a tool-search bridge lets the model fetch tool schemas on demand. MODEL INTERFACE is an OpenAI-style chat-completions client by default with pluggable transports for Anthropic, Responses and Bedrock, streaming on, three retries with backoff, then a configurable fallback chain, then a bounded auto-recovery wait. OUTPUT sends a fixed core tool set on every call, splits each batch into parallel-safe segments, runs terminal commands behind pluggable sandboxes (local, Docker, Modal, and others) and a regex-plus-human approval gate, and persists each tool-call row before executing it. CONTROL FLOW is a loop of named phases, each returning a verdict, with an unlimited iteration budget by default, "stop-gate" nudges that append a synthetic message and go again, flat delegation to threaded children, and task-keyed routing to auxiliary models.

Two components do not fit cleanly. In one mode the whole turn is handed to an external Codex process, and Hermes keeps only input, persistence and a fallback: the five primitives collapse into one opaque external harness that is neither a sub-agent (there is no outer loop) nor the model. And "providers" reached over ACP make the thing behind the MODEL INTERFACE another agent. Both are nested harnesses by the rule of Appendix B, but of an unusual shape: the nesting is at the MODEL INTERFACE, not in CONTROL FLOW.

### 5.6 A LangGraph workflow

LangGraph [19], at commit `5eb60ae`, is a framework for writing control flow as a graph over a checkpointed state, executed by a runtime in bulk-synchronous supersteps. We deconstruct the agent it ships, `create_react_agent`, together with that runtime. Input is a write to a named channel of a typed state, merged by a user-chosen reducer, whether it is a person's message, a tool result, or a resume value after an interrupt. Context is exactly the state's message list, optionally prefixed by one system message; fitting it to the window is left entirely to a user-written hook before the model call, and the framework provides only a primitive for removing messages. Model interface is one call to a LangChain chat model per step, and the framework contains no client of its own, no streaming of its own, and no backup model; retries and timeouts are per-node policies, off by default. Output is a tool node that runs every tool call of a response concurrently in-process, turns results and (configurably) errors into messages, and lets a tool return a routing command. Control flow is the graph: "go again" is a conditional edge from the agent node to the tool node and back, "stop" is an empty task set or a recursion limit, and a pause is an exception the runtime persists as a checkpoint write and resumes by re-running the node.

This is the clearest case of the three-level distinction. The framework fixes the five primitives' mechanisms and leaves most of the design decisions to user code: the edges, the stopping rule, the window-fitting hook, the approval gate around an interrupt, dynamic model selection, retry policies. A workflow and an agent built with it differ in the graph, which is Control flow, and nowhere else. The hard cases are the step counter injected into the state (it looks like Context but no model sees it; it is a Control flow budget), and a scheduling packet that carries a state snapshot to the tool task (Control flow carrying Context).

### 5.7 Summary

Table 5 gives the six side by side, one row per primitive, and Table 6 the thirteen of the second set. Every component we found has a home. The differences between the harnesses are in the rows: what comes in and from where, what the context holds and how it is fitted, how the model is reached, which tools exist and where they run, and what shape the loop has. None of the nineteen required a sixth row.

**Table 6:** Thirteen more harnesses by primitive, read from their sources or documentation at summary level. Each cell is the design decision that harness made within the primitive; the evidence for each is in Appendix C, which also gives the pinned commit or documentation date.

| | Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|---|
| **SWE-agent** | single problem statement (text, file, GitHub issue, images) plus templated bash observations; no mid-run messages | Jinja system/instance templates, demos, history processors eliding old observations, 100k-char clipping, cache marks; no compaction | synchronous litellm.completion, tenacity 20 random-exponential retries, litellm fallbacks, no streaming | exactly one shell command per step in a SWE-ReX Docker bash session; blocklist; patch from /root/model.patch | step until submit or exit; cost, call and timeout budgets; requery on format errors; RetryAgent attempts with reviewer |
| **OpenHands** | Canvas posts user events to an Agent Server; SDK wraps them as MessageEvents; tool results are ObservationEvents | section-registry system prompt with AGENTS.md skills; event view condensed by LLM summary on token or size limit | litellm chat or Responses API; 5 exponential retries; profile fallbacks; optional token streaming | typed ToolDefinitions with executors in a local or Docker workspace; batch truncated at Finish; optional parallelism | status-machine loop with iteration and budget caps, stuck detector, risk-based confirmation, hooks, threaded delegation |

| | Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|---|
| **Aider** | terminal REPL, scripted message, or AI! file comments; results re-enter as plain text message pairs | ordered chunks: system, examples, PageRank repo map, files, weak-model summarized history; optional cache headers | LiteLLM completion with streaming, exponential backoff to 60 s, assistant-prefill continuation on length cutoff | text edit formats parsed and fuzzily applied to files, auto-commit, lint/test, approved unsandboxed shell commands | reflection loop up to 3 re-sends on edit/lint/test failure; confirm gates; architect-to-editor two-coder handoff |
| **Cline** | host runTurn with text, images, files; steer queue consumed per request; tool results as tool messages | templated system prompt plus rules, cloned transcript, hook context, estimator-driven compaction at 90% via summarizer | gateway over Vercel AI SDK streamText, native tools, 5 SDK retries plus 3 turn retries with backoff | native tool calls in parallel groups with timeouts; shell spawn, editor, MCP; git-ref checkpoints | loop until no tool calls or completion tool; maxIterations, mistake and loop trackers, approval gates, sub-agents |
| **Gemini CLI** | CLI/ACP parts with @file expansion; tool results and AskUser answers re-enter as functionResponse parts | snippet-composed system prompt, hierarchical GEMINI.md memory, curated history, 50% threshold snapshot compression, tool masking | @google/genai or Code Assist streaming, 10-attempt backoff, mid-stream retries, policy-chain model fallback | registry of core and MCP tools, parallel batches, node-pty shell, seatbelt/b-wrap/container sandboxes, 40k truncation | caller-driven tool loop, 100-turn cap, next-speaker and loop checks, policy/confirmation gate, hooks, classifier routing, sub-agents |
| **Goose** | user message persisted first; steer queue and elicitation answers; tool results batched into one message | templated system prompt plus hints, per-turn context event, 0.8 threshold summarization, tool-pair summaries, prompt caching | provider trait with required stream; reqwest SSE; 3 retries exponential; empty-response retries; no backup model | all tools are MCP extensions; concurrent select_all execution; optional Docker container; 50k/200k output spill | ordered operation pipeline with one inference step; 1000-turn cap; approval wait and rerun; stop hook; foreground sub-agents |
| **OpenAI Agents SDK** | string or Responses items plus Session history; tool outputs re-enter as run items; RunState resumes | instructions callable, tools and handoffs as functions, full item list; server compaction and trimmer | OpenAI Responses API via SDK; prefix-routed providers; configurable retry, timeout, streaming | function tools as concurrent asyncio tasks; hosted tools; Docker or local sandbox for shell and patches | NextStep loop with max_turns=10; guardrail tripwires; approval interruptions; handoffs swap agent in place |
| **Google Agent Development Kit** | types.Content user message appended as an Event; tool results and HITL answers re-enter as function-response events | ordered request processors rebuild LlmRequest from session events each step; instructions, memory tool, compaction, caching | BaseLlm adapters resolved by model name; Gemini via google-genai; SSE streaming; SDK retries; FallbackModel | BaseTool subclasses run concurrently as asyncio tasks with callbacks; code executors from in-process to Docker | Runner picks agent, flow loops until final response; max_llm_calls; confirmation pauses; transfers run sub-agents in place |
| **AutoGen** | chat messages from user or peers; tool results batched as one function-result message | one system message plus a pluggable ChatCompletionContext (unbounded, buffered, token-limited); memory hooks | ChatCompletionClient protocol; OpenAI SDK client with usage logging and streaming; SDK retries | workbench tools run concurrently via asyncio.gather; Docker or local code executors | bounded max_tool_iterations loop, then reflect or summarize; teams via actor runtime, termination conditions |
| **CrewAI** | Crew inputs interpolated into templates; tool results and human feedback appended as messages | role/goal/backstory system prompt, task plus memory and knowledge; LLM summarization near window limit | LLM class over native provider clients or litellm; rate-limit retries; streaming; native or ReAct text | in-process BaseTool execution with cache and repeat check; parallel native calls; no sandbox | Flow state machine per task to max_iter, task sequencing by crew, guardrail retries, human gate |

| | Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|---|
| **smolagents** | task string plus optional images and named args; tool results and errors re-enter as text observations | Jinja system prompt plus full step memory replayed each call; truncation only, no compaction | pluggable Model adapters over provider SDKs; rate-limit retries; native tool calls or code blobs | JSON tool calls run in threads, or Python snippets in an AST interpreter or remote sandbox | ReAct while-loop to max_steps, stops on final_answer; optional periodic planning; managed agents as tools |
| **Cursor Agent** | chat or CLI prompt, queued or steered messages delivered at next tool call, Ask-questions tool answers mid-work | rules at prompt start, skills on demand, local grep search, compaction near capacity, no native memory | Cursor backend to many providers, Auto routing by classifier, 256k to 1M windows, guardrail reroute | nine first-party tools plus MCP, Seatbelt or Landlock sandbox, network blocked by default, local checkpoints | unbounded tool loop, Run Mode gate with LLM classifier, two-level subagents, stop hook loops, coordinator Projects |
| **GitHub Copilot cloud agent and agent mode** | issues, @copilot mentions, chat handoff, API and automations; steering applied after the current tool call; write-access only | layered instruction files, custom agents, skills and validated 28-day memory; summarization not documented | GitHub-hosted Copilot API, Auto routing on health and task complexity, LTS fallback, AI-credit metering | ephemeral Actions runner with firewall, Bash plus read-only GitHub MCP, pushes to copilot/ branch, draft PR | background session with 59-minute cap, no in-session approvals, edge gates on triggers and workflows, automations |

**What was hard to place.** The hard cases are collected in Appendix C, and four recur across the first set.

- **Primitives that call the model.** In Claude Code, opencode and pi, compaction decides what to summarize and then calls the model to summarize it; Claude Code's web-fetch tool and pi's code mode call models from inside a tool. Read as a call graph, these break the rule that Context and Output never call the model. Read as responsibility, they do not: the component still decides one thing, and uses Model interface to do it. We added the rule on primitives as services (Appendix B) for these cases.
- **Approval gates inside tools.** In opencode and Hermes the code that asks the person sits inside the tool that would act, and in Codex the approval requirement depends on what the sandbox allows. By function the gate is Control flow and the enforcement is Output; by location they are one file. The two also read each other, which is the interaction of concerns noted in §3.
- **Decisions carried out by editing the context.** opencode enforces a step budget by appending an assistant message and disabling tools; Hermes's stop gates append a synthetic message and go again; LangGraph replaces the model's answer with a canned message when the step budget is reached. Each is a Control flow decision whose mechanism is a change to what the model is shown. The decision and the mechanism have different homes, and we record both.
- **Nesting at the model interface.** Hermes can hand a whole turn to an external Codex process, or reach an agent over ACP as if it were a model. The thing behind Model interface is then another harness. The nested-harness rule covers it, but the nesting is at a different primitive from the sub-agent case, and the outer harness keeps only Input, persistence and a fallback.

In the wider set the same four recurred, and two more appeared. *Model calls whose only effect is a decision.* Gemini CLI's next-speaker and loop checks, Goose's permission judge, Cursor's auto-review classifier and AutoGen's reflection step each call the model and consume the answer as a Control flow decision. The nested-harness rule places them, and Cursor's classifier makes read-only tool calls of its own, so it is a small harness in fact. *The loop outside the client.* Gemini CLI's core makes one model call per invocation and its caller re-sends the tool results; OpenHands's loop, tools and client are spread over three repositories; Copilot's coding agent runs inside a GitHub

Actions job that is at once its executor and its time budget. The primitives are present in each case; they are not in one place. Two placements went by function against appearance: Aider's edit formats bundle the prompt that asks for a format (CONTEXT) with the parser and applier (OUTPUT) in one class, and smolagents' `final_answer` is a tool whose stop signal travels as an exception raised inside the sandbox, an OUTPUT mechanism carrying a CONTROL FLOW decision.

One finding is not a hard case but a boundary question. Claude Code's server-side classifier runs inside the API during the session's own requests, and opencode can delegate compaction to the provider. In the wider set the split is more common: the OpenAI Agents SDK can keep the conversation and compact it on the server, and hosted tools in the OpenAI Agents SDK, Google ADK and Cline (web search, a code interpreter) run inside the model call with no local OUTPUT step. In each, a function the vocabulary places in the harness runs on the provider's side of the API, and the local harness holds only a verdict, an opaque item or a result it did not produce. The primitives still apply: the harness is split across two machines, not reduced to four parts. But it means the boundary of §1 is at the model's weights, not at the API, and a deconstruction from a client's source alone will miss the part that runs behind the API. We found no component in the nineteen that satisfied none of the five definitions.

### 5.8 What the deconstruction does not show

**Selection.** Nineteen harnesses, most of them coding agents or frameworks used for coding examples. Harnesses for browsing, voice, robotics and customer service were not examined, and the design decisions in Table 2 are drawn from what these nineteen do. The selection criterion was wide use and the availability of sources; it is not a random sample.

**Documentation-only analysis.** Claude Code, Cursor and Copilot are deconstructed from documentation, which states what the harness does but not always how. Negative definitions are harder to confirm from documentation than from source, and a documented feature may differ from the shipped one.

**The harness is not always in one place.** Two of the six, and at least four of the thirteen, run part of their harness behind the provider's API. A deconstruction from client source or documentation describes the local part; the part on the provider's side is visible only through its effects. As providers absorb more harness functions, this gap will grow.

**Products change.** Every assignment is pinned to a commit or a documentation date. Several of these harnesses change weekly. The deconstruction describes them on those dates, and a reader checking it later should use the pinned versions.

**Who made the assignments.** The assignments were made by the author with an AI coding agent as a research instrument (see the disclosure), and each cites its evidence. No second, independent decomposition was made; §6 returns to this. The rules in Appendix B are where disagreement would most likely arise: they follow from the definitions, but another reader could draw some boundaries elsewhere.

**Coverage is not comparison.** That every component has a home shows the five cover what exists. It does not show that any implementation is better than another, which is the subject of §6.

# 6 Further Work: Comparing Implementations of a Primitive

The deconstruction shows that the five primitives cover what exists: every component in nineteen harnesses had a home in one primitive or one concern. It does not show which implementation of a primitive is better, for which model, on which task. That is the comparison the vocabulary makes well-posed, and it has not been done.

**The form of the study.** Hold the model fixed. Hold four primitives fixed. Vary the fifth, one design decision at a time, and measure on a task set with a check that does not depend on the primitive varied. The harness is stated in the five parts of Table 1, so that the varied part is the only difference between arms, and the statement is published with the result. Fan et al. [10] is the nearest existing study, with three decisions varied across four models; the vocabulary names the remaining decisions and says what must be held fixed.

**One comparison per primitive.** The decisions in Table 2 are the candidates. Within CONTEXT: window fitting, where the harnesses in §5 drop old turns, summarize, compact at a threshold with a verbatim tail, or leave it to the provider, and no two agree on when; and the placement of project instructions, preloaded or searched. Within OUTPUT: the action format, structured calls against model-written code run in an interpreter, which some of the nineteen offer and the rest do not; and the execution environment, host, OS sandbox or container, which changes what the model is allowed to do and what it is told it did. Within CONTROL FLOW: the stopping rule, a step budget against a cost budget against none; and the approval gate, a rule pipeline against a classifier model against per-tool assertion, measured on both task completion and the rate of harmful actions. Within MODEL INTERFACE: the retry and backup policy, measured under induced provider failure. Within INPUT: the delivery of mid-turn messages, at the next step boundary or after the turn, measured on tasks where the person changes the request part-way.

**What such a study needs.** A task set with a harness-independent check. A model held at a fixed version, since the harness that suits one model may not suit another [20]. The four fixed primitives reported in the same five-part schema, so that a later study can hold the same things fixed. And the evidence files of Appendix C are a starting point for the arms: each cell of Table 5 is an implementation that exists and can be ported into a common harness, because the primitives give the seam along which to port it.

**These comparisons are also the test of the framework.** The deconstruction tests coverage: can every component be placed? A comparison tests something the deconstruction cannot: can a result be attributed to the primitive varied? If an implementation cannot be placed in one primitive, or if varying one primitive changes behaviour only through another, the framework is wrong in a way that reading source could not show. The five questions of §4.3 make the same prediction in the small: a failure isolated to one primitive should be fixed by a change to that primitive alone. Both predictions are checkable, and neither has been checked at scale.

**A second coding of the deconstruction.** The assignments in Appendix C were made by the author with an AI coding agent as a research instrument, against the definitions of §2 and the rules of Appendix B. No independent second coding was made, so the agreement between two people applying the definitions is unmeasured. The brief, the definitions and the component lists are published for that purpose.

## 7 Conclusion

An agent is a model wrapped in a harness, and the harness has five parts: Input, Context, Model interface, Output and Control flow. Each is the smallest program that adds it, and each removes a failure the others do not. What distinguishes one harness from another is the design decisions made within those primitives, and what makes a harness fit to run is the answers it gives to six questions asked of each. Nineteen harnesses in wide use, six read from source at pinned versions and thirteen from their sources or documentation at summary level, fit this description with nothing left over. The vocabulary lets a result be reported with the harness that produced it, lets research compare primitives by their implementations, and lets an engineer isolate a failure to the primitives an implementation touches. The comparison of implementations is the work that remains, and it is also the test of the framework.

## Disclosure of AI use

AI tools were used in two roles.

**As a research instrument.** The deconstruction of §5 was carried out with an AI coding agent (Claude, Anthropic) under the author's direction. The agent read each harness's source at the pinned commit, or its documentation on the stated date, applied the definitions of §2 and the rules of Appendix B, and recorded every assignment with a file-and-line or URL citation. The author reviewed the assignments and the hard cases. Every assignment in Appendix C can be checked against its cited source without the agent.

**As a writing assistant.** The author set the position, the structure, the definitions and the three-level vocabulary, and reviewed every section. An AI assistant drafted prose from the author's outline and the companion repository, and edited for consistency. The minimal harnesses of §2 and Appendix A, and the transcripts they cite, are the author's, recorded in the companion repository.

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# A The Minimal Harnesses

The five programs of §2, complete, without their dependency headers. Each is the previous one plus one primitive. They are `harness/1_model_interface.py` to `harness/quark.py` in the companion repository [9], and the transcripts quoted in §2 are from the recorded runs there. The companion repository adds the production concerns of §3 to the last program one question at a time, and later replaces `request_response` with a function that runs a local model and translates between the Messages format and the model's chat template, leaving the other four primitives unchanged.

## A.1 Model interface alone

```
from anthropic import Anthropic

model = Anthropic()


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


response = request_response({"messages": [{"role": "user", "content": "The tests are failing. Find out why and
     fix it."}]})
print(response.model_dump_json(indent=2))
```

## A.2 Plus Input

```
from anthropic import Anthropic

model = Anthropic()


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def capture_input():
    return input("\n> ")


response = request_response({"messages": [{"role": "user", "content": capture_input()}]})
print(response.model_dump_json(indent=2))
```

## A.3 Plus Output

```
import subprocess
from anthropic import Anthropic

model = Anthropic()

bash = {"name": "bash", "description": "Run a shell command",
```

```
        "input_schema": {"type": "object", "properties": {"command": {"type": "string"}}, "required": ["
     command"]}}


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def capture_input():
    return input("\n> ")


def handle_output(response):
    tool_results = []
    for block in response.content:
        if block.type == "text":
            print("< " + block.text.replace("\n", "\n  "))
        if block.type == "tool_use":
            print(f"$ {block.input['command']}")
            ran = subprocess.run(block.input["command"], shell=True, capture_output=True, text=True)
            tool_results.append({"type": "tool_result", "tool_use_id": block.id, "content": ran.stdout + ran.
     stderr or "(no output)"})
    return tool_results


response = request_response({"tools": [bash], "messages": [{"role": "user", "content": capture_input()}]})
handle_output(response)
```

## A.4 Plus Control flow

```
import subprocess
from anthropic import Anthropic

model = Anthropic()

bash = {"name": "bash", "description": "Run a shell command",
        "input_schema": {"type": "object", "properties": {"command": {"type": "string"}}, "required": ["
     command"]}}


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def capture_input():
    return input("\n> ")


def handle_output(response):
    tool_results = []
    for block in response.content:
        if block.type == "text":
            print("< " + block.text.replace("\n", "\n  "))
        if block.type == "tool_use":
            print(f"$ {block.input['command']}")
            ran = subprocess.run(block.input["command"], shell=True, capture_output=True, text=True)
            tool_results.append({"type": "tool_result", "tool_use_id": block.id, "content": ran.stdout + ran.
     stderr or "(no output)"})
    return tool_results


def control_flow():
    conversation = []
    while True:
        conversation.append({"role": "user", "content": capture_input()})
        while True:
```

```
            response = request_response({"tools": [bash], "messages": conversation})
            conversation.append({"role": "assistant", "content": response.content})
            tool_results = handle_output(response)
            if not tool_results:
                break                                   # done: hand back to the person
            conversation.append({"role": "user", "content": tool_results})   # a tool ran: go again


control_flow()
```

## A.5 Plus Context: the complete harness

```
import subprocess, os, datetime
from anthropic import Anthropic

model = Anthropic()
bash = {"name": "bash", "description": "Run a shell command",
        "input_schema": {"type": "object", "properties": {"command": {"type": "string"}}, "required": ["
     command"]}}


def capture_input():
    return input("\n> ")


def assemble_context(conversation):
    instructions = f"You are quark, an agent. You act through bash, in {os.getcwd()}. Today is {datetime.date.
     today()}."
    return {"system": instructions, "tools": [bash], "messages": conversation}


def request_response(context):
    return model.messages.create(model="claude-sonnet-5-5", max_tokens=16384, **context)


def handle_output(response):
    tool_results = []
    for block in response.content:
        if block.type == "text":
            print("< " + block.text.replace("\n", "\n  "))
        if block.type == "tool_use":
            print(f"$ {block.input['command']}")
            ran = subprocess.run(block.input["command"], shell=True, capture_output=True, text=True)
            tool_results.append({"type": "tool_result", "tool_use_id": block.id, "content": ran.stdout + ran.
     stderr or "(no output)"})
    return tool_results


def control_flow():
    conversation = []
    while True:
        conversation.append({"role": "user", "content": capture_input()})
        while True:
            response = request_response(assemble_context(conversation))
            conversation.append({"role": "assistant", "content": response.content})
            tool_results = handle_output(response)
            if not tool_results:
                break                                   # done: hand back to the person
            conversation.append({"role": "user", "content": tool_results})   # a tool ran: go again


control_flow()
```

## B Rules for the Cases at the Boundaries

The negative definitions of §2 decide most cases. The following rules decide the rest. Each was needed for a component found in §5.

**Function, not location.**
A component is assigned by what it does, not by the file or the name it has. Code inside a tool that decides what the model sees is Context. An approval decision made inside a tool's runtime is Control flow.

**The tool channel.**
The model drives other primitives through tool calls: spawning a sub-agent, entering a planning mode, writing a plan, searching for tools, asking the person a question. The call is Output; its effect is assigned to the primitive it changes.

**Backup versus routing.**
A model change triggered by a request that could not be served is Model interface. A model change keyed to the work is Control flow. A fixed model for one class of request, such as compaction summaries, is a request setting and Model interface.

**Receiving versus rendering.**
Receiving a response as a stream is Model interface; rendering each fragment is Output.

**Parallel actions versus parallel calls.**
Several tool calls from one response run concurrently: Output. Several model calls issued concurrently, to divide a task or to vote: Control flow.

**The reader of a record.**
A record kept for the model (a searchable transcript) is Context. A record kept for the operator (a trace) is observability. An artifact for the user (a diff view) is Output. One file may serve several readers, and each use is assigned separately.

**The consumer of an answer.**
An answer the harness consumes (an approval) is Control flow. An answer the model consumes (the result of an ask-the-user tool) is Input.

**Direct actions by a person.**
A command the person runs through the harness's own executor, whose output enters the conversation, is Input, although it reuses Output's executor.

**Model-written programs.**
A program the model writes at run time is Control flow if it sequences model calls and Output if it sequences only tools.

**Bundles.**
A mode, a skill, a hook system or an agent profile is packaging that bundles decisions across several primitives. Decomposed, each element has one home: a planning mode is a context template, a restricted tool set and an approval gate; a hook system is a dispatcher (Control flow) whose events each affect one primitive.

**Nested harnesses.**
A sub-agent is a harness of its own, and delegation to it is Control flow of the outer harness. The same holds for model-backed components inside a harness, such as an approval reviewer or a termination judge: each is Control flow with a Model interface call.

**Primitives as services.**
One primitive may use another's mechanism to do its job: Context asks the model to summarize old turns before compacting them; a web-fetch tool in Output asks the model to condense a page before returning it. The component is assigned by what it *decides* (what to compact, what to return), and the call it makes is a use of Model interface. The negative

definitions are about responsibility, not about which function calls which.

**Tool protocols.**
A protocol such as MCP [23] provides pluggable OUTPUT (third-party tools). Its other features are assigned by function: resources and prompts are CONTEXT; sampling borrows the client's MODEL INTERFACE; elicitation is INPUT.

# C Evidence for the Deconstruction

Every assignment in §5, with its evidence. For harnesses with published source, a citation is a path and a line range at the pinned commit; for a harness read from documentation it is a page and section heading on the date read. The same files are published in the repository under `paper/evidence/`. **Path conventions** are given at the start of each harness; **not found** means the mechanism was looked for and not found, not that it is absent. The first six harnesses are deconstructed in full; the thirteen that follow at summary level, two to five components per primitive.

## C.1 Claude Code

**Source:** https://code.claude.com/docs/en/overview (official docs; source not published) · **Version:** documentation as of 2026-10-15; the docs state no single CLI version (pages reference behaviours by version, the highest seen being v2.1.299; agent-loop page bundles "Claude Code v2.1.290 or later" in SDK v0.3.290). Engineering posts: claude.dev/blog "Prompt caching is everything" (2026-04-30), "A harness for every task" (2026-06-02), "Getting started with loops" (2026-06-30) · **Language:** not documented (docs describe a native binary and an npm package; troubleshooting refers to a "JavaScript heap") **Path conventions:** `D` = https://code.claude.com/docs/en ; `SDK` = `D/agent-sdk` ; `B` = https://claude.dev/blog ; every citation is `page § Section heading` and was read on 2026-10-15. Where a page did not say how something works I write "not documented". **What it is:** Claude Code is Anthropic's agentic coding harness: a terminal/IDE/desktop/web/CI program that wraps Claude models with file, search, shell and web tools, permission gating, memory loading and the loop that chains actions (`D/glossary § Agentic harness`). It reaches the model over the Anthropic Messages API (`/v1/messages`, SSE streaming) or via Amazon Bedrock, Google Cloud's Agent Platform, Microsoft Foundry or an LLM gateway (`D/llm-gateway-protocol § API formats`). The same loop is exposed as a library (Agent SDK, Python/TypeScript) and as `claude -p` for scripts (`SDK/overview § Compare the Agent SDK to other Claude tools`).

### Control flow

- **Core loop:** "Claude continues calling tools and processing results until it produces a response with no tool calls"; each cycle (model output with tool calls → tools run → results fed back) is one turn, with no control returned to the caller between turns. `SDK/agent-loop § The loop at a glance`, `§ Turns and messages`
- **Stop condition / result:** the loop ends on a text-only response; the `ResultMessage.subtype` is `success`, `error_max_turns`, `error_max_budget_usd`, `error_during_execution` or `error_max_structured_output_retries`; `stop_reason` carries `end_turn`, `max_tokens` or `refusal`. `SDK/agent-loop § Handle the result`
- **Step budget:** `max_turns`/`--max-turns` "counts tool-use turns only"; no limit by default (print mode only). A message still queued when the limit ends a turn starts a new turn with a fresh count. `SDK/agent-loop § Turns and budget`; `D/cli-reference § CLI flags` (`--max-turns`)
- **Token/cost budget:** `max_budget_usd` is compared "after model responses arrive", so spend can pass the cap by one response; subagent spend counts; at the cap, spawning fails with `Budget limit reached` and background subagents are stopped. `SDK/agent-loop § Budget headroom`
- **Permission gate order (harness consumes the answer):** hooks → deny rules → ask rules → permission mode → allow rules → `canUseTool`/prompt; a hook `allow` never skips deny/ask; `dontAsk` denies at the last step. `SDK/permissions § How permissions are evaluated`; rule order "deny, then ask, then allow … first match" `D/permissions § Manage permissions`

- **Permission modes:** `default` (Manual), `acceptEdits`, `plan`, `auto`, `dontAsk`, `bypassPermissions`; auto is the built-in starting mode for interactive terminal/VS Code on v2.1.283+. `D/permission-modes § Available modes`, `§ Which mode a session starts in`
- **Auto-mode classifier gate:** a second model decides most tool calls; "runs on Claude Sonnet 5 by default rather than on your /model selection"; reads/working-dir edits skip it; it sees user messages, non-read-only tool calls and CLAUDE.md but "Tool results are stripped"; falls back to prompting after 3 blocks in a row or 20 total. `D/permission-modes § How auto mode evaluates actions`, `§ When auto mode falls back`
- **Plan mode:** edits stay blocked until a plan is approved; the approval prompt (`Yes, and use auto mode` / `Yes, manually approve edits` / `No, keep planning`) switches the session's permission mode. `D/permission-modes § Analyze before you edit with plan mode`, `§ Review and approve a plan`. Entering/leaving is itself a tool call (`EnterPlanMode`/`ExitPlanMode`) so the tool set never changes. `B/lessons-from-building-claude-code-prompt-caching-is-everything § Using Plan Mode to design around the cache`
- **Stop hook as loop extender:** a `Stop` hook returning `decision: "block"` keeps Claude working; capped at 8 consecutive continuations (reset on tool use), `CLAUDE_CODE_STOP_HOOK_BLOCK_CAP`. `D/hooks § Stop`; `D/env-vars § Variables` (`CLAUDE_CODE_STOP_HOOK_BLOCK_CAP`)
- **Goal loop:** `/goal` "is a wrapper around a session-scoped prompt-based Stop hook"; after each turn the condition plus conversation go to the small fast model, which returns Not yet met / Met / Impossible; the loop halts if no tool use for several turns. `D/goal § How evaluation works`
- **PostToolBatch gate:** fires once after all tool calls in a batch resolve, "before the next model request"; `decision: "block"` or `continue: false` stops the loop before the next model call. `D/hooks § PostToolBatch`
- **Sub-agent delegation (Agent tool):** the model spawns a subagent in its own context; depth default 3 layers (`CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH`, `1` disables nesting); at most 20 concurrent (`CLAUDE_CODE_MAX_CONCURRENT_SUBAGENTS`); only the final message returns to the parent. `D/sub-agents § Let subagents spawn their own subagents`, `§ Concurrent subagent limit`; `SDK/ subagents § What subagents inherit`
- **Foreground vs background sequencing:** order of decision: teammates → `CLAUDE_CODE_DISABLE_ BACKGROUND_TASKS=1` → fork mode on (background, Claude can't request foreground) → fork mode off (background by default, foreground when Claude needs the result); background results arrive later as a completion notification. `D/sub-agents § Run subagents in foreground or background`
- **Model keyed to work:** `opusplan` "Uses opus in plan mode, then switches to sonnet for execution"; subagent model order: per-invocation `model` → frontmatter `model` → `CLAUDE_CODE_SUBAGENT_MODEL` → main model; background functionality (titles, summaries, goal evaluator) uses the Haiku-class model. `D/model-config § Model aliases`, `§ Environment variables`; `D/sub-agents § Choose a model`; `D/ sessions § Name your sessions`
- **Dynamic workflows (model-written program sequencing model calls):** "A dynamic workflow is a JavaScript script that orchestrates many subagents"; `agent()`, `parallel()`, `pipeline()`; up to 16 concurrent agents, 4,096 items per call, 1,000 agents per run; no mid-run user input; `Date.now()`/`Math. random()` throw so a relaunch replays deterministically. `D/workflows § What the saved script looks like`, `§ Behavior and limits`, `§ Edit a saved script`
- **Workflow resume:** completed agents return saved results; the first agent whose prompt differs, and every agent after it, runs again. `D/workflows § Which agents run again`
- **Agent teams (outer harness of peer sessions):** lead + teammates with a shared task list and a per-agent mailbox JSON file at `~/.claude/teams/{team-name}/inboxes/{agent-name}.json`; experimental, `CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1`; no nested teams. `D/agent-teams § Architecture`, `§ Limitations`
- **Prompt- and agent-based hooks (harness-sequenced model calls):** `type: "prompt"` sends hook input to a model (default: the background-functionality model, 30 s timeout) to allow/block; `type: "agent"` spawns a subagent with tools (60 s). `D/hooks § Prompt-based hooks`, `§ Agent-based hooks`
- **Scheduled turns:** `/loop` creates cron tasks; "The scheduler checks every second for due tasks and enqueues them at low priority"; fires between turns, never mid-response; 7-day expiry; self-paced loops end via `ScheduleWakeup` with `stop: true`. `D/scheduled-tasks § How scheduled tasks run`, `§ Stop a`

loop
- **Non-interactive exit wait:** after the turn ends, a `claude -p` run stays open for background work; "stops whatever is still running after 10 minutes of continuous idle waiting" (`CLAUDE_CODE_PRINT_BG_WAIT_CEILING_MS`). `D/headless § Background tasks at exit`

### Input

- **Prompt capture:** interactive terminal prompt, `claude "query"`, `claude -p "query"`, and piped stdin ("Piped stdin is capped at 10MB"). `D/cli-reference § CLI commands`; `D/headless § Pipe data through Claude`
- **Mid-turn message queue:** a message typed while Claude works is queued; "Claude Code passes it to Claude as soon as those tool calls finish, within the same turn"; commands are held until the turn ends. `D/interactive-mode § When Claude Code sends what you queued`
- **Interrupt:** `Esc` cancels the running tool call and Claude waits; queued messages are sent next. `D/how-claude-code-works § Interrupt and steer`; `D/interactive-mode § General controls`
- **Model-consumed answers:** `AskUserQuestion` returns the user's choice to the model; questions stay open until answered unless `askUserQuestionTimeout` (`60s`/`5m`/`10m`) is set. `D/tools-reference § AskUserQuestion tool behavior`
- **Tool results as input:** each tool execution yields a `UserMessage` carrying the tool result back to the model. `SDK/agent-loop § Message types`
- **Event/stream inputs:** `--input-format stream-json` accepts user messages over stdin; `Monitor` feeds "each output line back to Claude" and can treat WebSocket messages as events; channels push external events into the session. `D/cli-reference § CLI flags`; `D/tools-reference § Tools reference` (`Monitor` row); `D/glossary § Channel`
- **Non-human turns captured through the same entry:** `UserPromptSubmit` "also fires for scheduled tasks, background subagent reports, and messages from other sessions". `D/hooks § UserPromptSubmit`
- **Rich input:** `Ctrl+V` pastes an image as an `[Image #N]` chip; `@path` mentions expand file content; `@server:protocol://resource` fetches MCP resources as attachments. `D/interactive-mode § General controls`; `SDK/modifying-system-prompts § Turn off the context your agent replaces` (`CLAUDE_CODE_DISABLE_FILE_MENTIONS`); `D/mcp § Use MCP resources`
- **MCP elicitation:** a server's mid-task form/URL request is shown as a dialog and the answer goes back to the server (not the model); an `Elicitation` hook can answer it. `D/mcp § Respond to MCP elicitation requests`

### Context

- **Request layering (what one call contains):** system prompt (core instructions + tool definitions) → project context (CLAUDE.md, auto memory, unscoped rules) → conversation (messages, tool results); ordered most- to least-stable for prefix caching. `D/prompt-caching § How the cache is organized`
- **Two system prompts:** full vs "a shorter one with abbreviated tool descriptions"; newer models get the short one unless `CLAUDE_CODE_SIMPLE_SYSTEM_PROMPT=0`. `D/env-vars § Variables` (`CLAUDE_CODE_SIMPLE_SYSTEM_PROMPT`)
- **System prompt attribution block:** "a short attribution block to the system prompt containing the client version and a fingerprint"; stripped by `api.anthropic.com`; `CLAUDE_CODE_ATTRIBUTION_HEADER=0` omits it. `D/llm-gateway-protocol § System prompt attribution block`
- **CLAUDE.md delivery:** loaded "as a user message after the system prompt"; all discovered files are concatenated root→working directory, `CLAUDE.local.md` after `CLAUDE.md`; subdirectory files load "once Claude reads, writes, or edits another file in that subdirectory". `D/glossary § CLAUDE.md`; `D/memory § How CLAUDE.md files load`
- **Imports and rules:** `@path` imports expand at launch; `.claude/rules/*.md` load every session, or on demand when `paths:` frontmatter matches a file Claude touches; a CLAUDE.md over 4 MiB is skipped. `D/memory § Import additional files`, `§ Path-specific rules`, `§ How it works`
- **Auto memory (model-kept record):** `~/.claude/projects/<project>/memory/MEMORY.md`; "The first 200 lines of MEMORY.md, or the first 25KB, whichever comes first, are loaded"; topic files read on

demand; a write past the limit returns an error telling Claude to rewrite the index. D/memory § How it works, § Storage location

- **System reminders:** harness-added context delivered "wrapped in <system-reminder> tags inside a user message, or ... a separate message with the system role": CLAUDE.md, output style, hook additionalContext, skill list, file-changed notes, attribution lines. D/glossary § System reminder
- **Output style injection:** "Claude Code sends the active style's instructions with every request" as a conversation message, so the cached prefix survives a switch. D/output-styles § How output styles work; D/prompt-caching § Changing output style
- **Skills in context:** descriptions (truncated at 1,536 chars) are always in context; the body loads on invocation as a user message and stays for the session; disable-model-invocation: true keeps the description out. D/skills § Frontmatter fields, § Controlling visibility and context cost; D/prompt-caching § Invoking skills and commands
- **Skill dynamic context:** !command runs a shell command before the model sees the skill and splices the output in. D/skills § Dynamic context injection
- **MCP tool search (deferred definitions):** only names and server instructions load; schemas arrive via the ToolSearch tool; ENABLE_TOOL_SEARCH auto loads upfront while definitions are under 10% of context; description caps 4,096 (upfront) / 16,384 (deferred) chars. D/mcp § Scale with MCP tool search, § Configure tool search
- **Environment and git snapshot:** working directory, platform, shell, OS and a git branch/status/recent-commits block are added at session start; includeGitInstructions: false drops the git snapshot. D/context-window (Environment info event); SDK/modifying-system-prompts § Turn off the context your agent replaces; D/prompt-caching § Cache scope
- **Fitting tool results:** Bash output inline "up to roughly 30,000 characters"; past that a file path plus 2,000-char preview; MCP results capped at 25,000 tokens and text over 50,000 chars spilled to a file; Read returns at most 25,000 tokens per call and refuses whole files over 256 KB. D/tools-reference § Output limits, § Large files; D/mcp § Raise the limit for a specific tool; D/env-vars § Variables (MAX_MCP_OUTPUT_TOKENS)
- **Auto-compaction trigger:** by default "compacts when the conversation reaches the model's context limit"; native-1M models compact "at about 967K tokens"; window settable 100K–1M via /autocompact, autoCompactWindow, --autocompact, CLAUDE_CODE_AUTO_COMPACT_WINDOW. D/model-config § Default auto-compact thresholds, § Set the auto-compact window
- **Compaction order:** "It clears older tool outputs first, then summarizes the conversation if needed." D/how-claude-code-works § When context fills up
- **Compaction as a forked request:** the summarization call reuses the same system prompt, tools and history "plus a summarization instruction appended as a final user message"; inherits the session's thinking config. D/prompt-caching § Compacting the conversation; D/context-window § What survives compaction; B/lessons-from-building-claude-code-prompt-caching-is-everything § COMPACTING WITHOUT BREAKING THE CACHE
- **What survives compaction:** system prompt/output style still apply; project-root CLAUDE.md, auto memory, plan file re-injected from disk; up to five recently modified files re-read (over 5,000 tokens → path reference); invoked skill bodies re-injected "capped at 5,000 tokens per skill and 25,000 tokens total"; hook context is summarized away. D/context-window § What survives compaction
- **Hook-supplied context:** additionalContext is wrapped in a system reminder; additionalContext, systemMessage, initialUserMessage capped at 10,000 chars (longer saved to file with a 2,000-char preview); PostToolUse.updatedToolOutput replaces what Claude sees. D/hooks § JSON output, § Add context for Claude, § PostToolUse
- **Sub-agent context:** a non-fork subagent gets its own system prompt plus environment details, the task message, CLAUDE.md hierarchy (not for Explore/Plan), git snapshot, preloaded skills; not the parent history, output style or auto memory. A fork gets the full conversation, system prompt and tools. D/sub-agents § What loads at startup, § How forks differ from other subagents
- **Sub-agent output scanning:** before the parent reads a report, control-tag imitations get a backslash and a [harness: ...] marker line is prepended; nothing is removed. D/sub-agents § Subagent output scanning; SDK/subagents § What subagents inherit

### Model interface

- **Wire format:** Anthropic Messages format, "Inference requests post to `/v1/messages?beta=true`"; optional `/v1/messages/count_tokens` (falls back to a character estimate); headers `anthropic-version: 2023-06-01`, `anthropic-beta`, `x-claude-code-session-id`, `x-claude-code-agent-id`. `D/llm-gateway-protocol § API formats`, `§ Optional endpoints and startup traffic`, `§ Request headers`
- **Providers/backends:** `CLAUDE_CODE_USE_BEDROCK`, `CLAUDE_CODE_USE_VERTEX`, `CLAUDE_CODE_USE_FOUNDRY`, Claude Platform on AWS, or any gateway via `ANTHROPIC_BASE_URL`; Bedrock uses `invoke-with-response-stream` binary eventstream. `D/env-vars § Variables`; `D/llm-gateway-protocol § API formats`
- **Streaming receipt:** "Claude Code reads each streaming inference response event by event as it arrives"; expects the full sequence through `message_delta`/`message_stop`, else treats it as a dropped connection. `D/llm-gateway-protocol § Streaming`
- **Retry policy:** "retries transient failures up to 10 times with exponential backoff" (`CLAUDE_CODE_MAX_RETRIES`, capped at 15); retries server errors/529/timeouts before output starts, dropped connections (up to 2 after thinking), stalled streams (once), no-headers (once per request), temporary 429s; never re-runs after a completed text or tool block "because that could execute the same tool calls twice". `D/errors § Automatic retries`
- **529 backoff:** `CLAUDE_CODE_OVERLOADED_RETRY_BASE_DELAY_MS` 500 → `CLAUDE_CODE_OVERLOADED_RETRY_MAX_DELAY_MS` 32000; `retry-after` honoured, and >60 s stops retries outside watchdog sessions; `x-should-retry` read. `D/errors § Automatic retries`; `D/llm-gateway-protocol § Response headers`
- **Indefinite retry for unattended runs:** `CLAUDE_CODE_RETRY_WATCHDOG=1` "Retries 429 and 529 capacity errors indefinitely", backing off up to 5 minutes. `D/env-vars § Variables`
- **Timeouts and watchdogs:** `API_TIMEOUT_MS` default 600000; first-byte deadline 180 s (direct API) / 300 s (elsewhere) + 1 s per 32 KB of body; event-level watchdog 300 s; byte-level watchdog 180 s/300 s; body idle timeout 5 min. `D/env-vars § Variables` (`API_TIMEOUT_MS`); `D/network-config § Streaming idle watchdogs`
- **Backup model when the request can't be served:** `--fallback-model sonnet,haiku` / `fallbackModel` runs "when the primary model is overloaded, unavailable, or returns another non-retryable server error"; chain of ≤3; applies to the current turn only; subagents use it too. `D/model-config § Fallback model chains`
- **Safety-flag model fallback:** when a server safety classifier flags a request, Claude Code "can re-run it on a fallback model" (e.g. Opus 4.8 for cybersecurity); `switchModelsOnFlag`. `D/model-config § Automatic model fallback`
- **Capability-rejection recovery:** on a 400 rejecting `thinking`, a mid-conversation system message, `cache_control`, the advisor tool or `output_config.effort`, Claude Code retries without that capability for the rest of the conversation; a rejected thinking signature strips earlier thinking blocks. `D/llm-gateway-protocol § Automatic retry and error forwarding`
- **Context-limit rejection:** a request rejected because input + `max_tokens` exceeds the limit is retried "with a reduced max_tokens", then compacted when no reduction fits. `D/errors § Automatic retries`
- **Prompt-cache markers and TTL:** `cache_control` on system blocks and messages; main conversation gets 1 h TTL on subscription usage, 5 min otherwise; subagents/compaction/titles get 5 min; `promptCacheTtl`, `FORCE_PROMPT_CACHING_5M`, `DISABLE_PROMPT_CACHING`. `D/prompt-caching § Which TTL each request gets`, `§ Choose the TTL yourself`, `§ Disable prompt caching`
- **Reasoning parameters on the wire:** `thinking: {"type": "adaptive"}` for 4.6+ models; effort and structured output in `output_config` with paired beta headers; `MAX_THINKING_TOKENS` fixed budget (≥1,024, ≤ max output - 1); `CLAUDE_CODE_MAX_OUTPUT_TOKENS` (default 32000 / cap 128000 for unknown IDs). `D/llm-gateway-protocol § Feature pass-through`; `D/env-vars § Variables`
- **Retry visibility:** a `system/api_retry` event (attempt, max_retries, retry_delay_ms, error_status, error category) is emitted in stream-json before each retry. `D/headless § Handle API retries`

**Output**

- **Built-in tool set:** Agent, AskUserQuestion, Bash, Edit, EndConversation, Glob, Grep, LSP, Monitor, NotebookEdit, PowerShell, Read, SendMessage, Skill, TaskCreate/TaskUpdate/TodoWrite, ToolSearch, WebFetch, WebSearch, Workflow, Write (+ MCP tools `mcp__<server>__<tool>`). `D/tools-reference § Tools reference` (table); `SDK/agent-loop § Built-in tools`
- **Concurrent execution of one response's calls:** "Read-only tools (like Read, Glob, Grep, and MCP tools marked as read-only) can run concurrently. Tools that modify state (like Edit, Write, and Bash) run sequentially"; custom tools opt in via `readOnlyHint`. `SDK/agent-loop § Parallel tool execution`
- **Bash executor:** default timeout 2 min (`BASH_DEFAULT_TIMEOUT_MS`), ceiling 10 min (`BASH_MAX_TIMEOUT_MS`); a foreground command that hits its timeout "moves it to the background instead of stopping it"; background commands get 30 min (max 2 h); env vars don't persist between commands; shell aliases/options captured in `~/.claude/shell-snapshots/`. `D/tools-reference § Timeout and output limits`, `§ Foreground commands that move to the background`, `§ Time limit for background commands`, `§ What persists between commands`; `D/claude-directory § Cleaned up automatically`
- **Sandbox technology:** macOS Seatbelt; Linux/WSL2 bubblewrap + socat; no sandbox on native Windows; covers Bash, PowerShell and Monitor commands only, not file tools, MCP servers or hooks. `D/sandboxing § OS-level enforcement`, `§ What runs outside the sandbox`
- **Sandbox network path:** "Claude Code runs the sandbox proxy on your machine, outside the sandbox, and directs commands to it with HTTP_PROXY, HTTPS_PROXY, ALL_PROXY"; the proxy checks hostnames against `allowedDomains`/`deniedDomains`, refuses names resolving only to local addresses, and can mask credentials with a sentinel (requires `tlsTerminate`). `D/sandboxing § Network isolation`, `§ Hostnames that resolve to local addresses`, `§ Mask credentials`
- **Sandbox filesystem path:** writable: working dir + per-user `$TMPDIR`; "Protected paths" (settings, hooks, MCP config) are denied even inside writable dirs; `excludedCommands` run unsandboxed; strict mode `allowUnsandboxedCommands: false`. `D/sandboxing § Filesystem isolation`, `§ Protected paths`, `§ Turn off the retry with strict sandbox mode`
- **WebFetch executor:** converts HTML to Markdown, "runs the prompt against the content in a separate model call, and Claude receives the result of that call rather than the raw page"; 15-min response cache; hostname first checked against an Anthropic blocklist at `api.anthropic.com` (`skipWebFetchPreflight`). `D/tools-reference § WebFetch tool behavior`; `D/data-usage § WebFetch domain safety check`
- **WebSearch:** server-side tool; "An interactive terminal session has a limit of 200 WebSearch calls" shared with subagents. `D/tools-reference § WebSearch tool behavior`, `§ Session search limit`
- **Read/Edit/Write executors:** Read returns numbered lines, resizes images (>500 KB re-encoded as JPEG), reads PDFs in ≤20-page ranges via `pdftoppm`; Edit/Write enforce read-before-edit on older models. `D/tools-reference § Read tool behavior`, `§ Edit tool behavior`, `§ Write tool behavior`
- **MCP executor:** stdio, HTTP (falls back to SSE), WebSocket transports; `MCP_TIMEOUT` 30 s startup, `MCP_TOOL_TIMEOUT` ~28 h, 60 s per HTTP request, idle timeout 5 min remote/30 min stdio, main-conversation calls auto-background after 2 min (`CLAUDE_CODE_MCP_AUTO_BACKGROUND_MS`); invalid schemas and names >128 chars are excluded per tool. `D/mcp § Transports`, `§ Timeouts`, `§ Tools with invalid input schemas`, `§ Tools with names longer than 128 characters`
- **Monitor executor:** Claude writes a script run in the background, or opens a WebSocket; deadline 5 min by default, max 30 min (10 min under `-p`). `D/tools-reference § Monitor tool`
- **Rendering the stream:** an `AssistantMessage` is emitted per content block; `stream_event` deltas (`text_delta`, `input_json_delta`) with `--include-partial-messages`; the spinner shows a live output-token count. `SDK/streaming-output § Message flow`; `D/costs § Read the token count beside the spinner`
- **Structured output:** `--output-format json|stream-json`, `--json-schema` validated output in `structured_output`; up to 5 attempts (`MAX_STRUCTURED_OUTPUT_RETRIES`). `D/headless § Get structured output`; `D/env-vars § Variables`
- **Hook executors:** handler types `command`, `http`, `mcp_tool`, `prompt`, `agent`; "All matching hooks run in parallel. Identical handlers from multiple settings files run once"; defaults 600 s command/http/mcp_tool; exit 2 blocks (PreToolUse) with stderr fed to Claude; `async: true` detaches; `PreToolUse.updatedInput` rewrites tool arguments before execution. `D/hooks § Configuration`, `§ Exit code output`, `§ Run`

hooks in the background, § PreToolUse

- **User-facing artifacts:** file checkpoints snapshot files "before each prompt you send that starts a turn", 100 most recent kept, `/rewind` restores code and/or conversation; Bash-made and subagent edits are not tracked. `D/checkpointing § Automatic tracking, § Bash command changes not tracked, § Subagent edits not restored`
- **EndConversation:** never prompts, PreToolUse hooks don't run for it, and deny rules/`--tools` can't remove it while any other tool remains. `D/tools-reference § EndConversation tool behavior`
- **Permission prompt surface:** Manual mode asks for Bash (except a built-in read-only set: `ls`, `cat`, `git status`…), file modification, WebFetch (except preapproved doc domains), WebSearch; "Yes, don't ask again" persists per repository. `D/permissions § Permission system, § Read-only commands`

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Input/Context (transcript, memory); Output (checkpoints); Control flow (resume) | Every message/tool use is appended to `~/.claude/projects/<project>/<session-id>.jsonl`; swept after `cleanupPeriodDays` (30 d default); auto memory excluded from the sweep; `file-history/<session>/` snapshots, `plans/`, `tool-results/`; `--resume`/`--continue` restore history, model, agent, permission mode, goal and cron tasks; `--fork-session`/`/branch` copy to a new ID; SDK `session_store` mirrors transcripts; `CLAUDE_CODE_TRANSCRIPT_LOCAL_GC` trims history before the last compaction once the file passes 5 MB | `D/sessions § Where transcripts are stored, § What a resumed session restores; D/claude-directory § Cleaned up automatically; D/memory § Storage location; SDK/agent-loop § Sessions and continuity; D/env-vars § Variables` |
| Safety & security | Control flow (gates); Output (sandbox, executors); Context (scanning) | deny→ask→allow rules plus modes; auto-mode classifier with tool results stripped and a "server-side probe" over incoming tool results; critical paths (`rm -rf /`, `~`, cwd) never approvable by allow rule or hook; protected paths; OS sandbox + domain allowlist + credential masking; subagent output scanning; teammate messages can't approve prompts; WebFetch summarised by a separate model call; `-p` runs project hooks/MCP without a trust dialog (`--bare` to skip); credentials in Keychain or a 0600 file; prompt-injection list | `D/permission-modes § What the classifier blocks by default, § Critical paths, § Protected paths; D/sandboxing § What the sandbox restricts; D/sub-agents § Subagent output scanning; D/agent-teams § Messages between agents; D/security § Protect against prompt injection, § Additional safeguards; D/headless § Start faster with bare mode` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Observability | Model interface (api events); Output (tool events); operator records | OpenTelemetry metrics (`claude_code.cost.usage`, `token.usage`, `session.count`, `lines_of_code.count`…) and events (`user_prompt`, `tool_result`, `api_request`, `api_error`, `tool_decision`) via `CLAUDE_CODE_ENABLE_TELEMETRY=1`; content opt-ins incl. `OTEL_LOG_RAW_API_BODIES=file:<dir>`; `/usage` session cost + `Prompt cache (main)` line; status-line `current_usage`/`prompt_cache`; debug log `~/.claude/debug/<session-id>.txt` with hook durations; stream-json `system/init`, `api_retry`, `compact_boundary`, `permission_denials`; gateway hint headers | `D/monitoring-usage § Metrics, § Events, § Content opt-in; D/costs § Prompt cache statistics; D/hooks § Debug hooks; D/headless § Read session metadata; D/llm-gateway-protocol § Gateway hint headers` |
| Resilience | Model interface (retries/watchdogs/fallback); Context (compaction); Control flow (resume/stall) | 10 retries w/ exponential backoff, no re-run after a completed block (partial output kept + "may be incomplete" notice, `-p` auto-continues ≤3×); four streaming watchdogs; fallback model chain; context-limit retry with reduced `max_tokens` then compaction; auto-compact "stops auto-compacting after a few attempts" on thrashing; interrupted tool call marked cut-off on resume (`CLAUDE_CODE_RESUME_INTERRUPTED_TURN`); SIGTERM runs `SessionEnd` hooks and exits 143; workflow agents restart on stall up to 5×; `/goal` auto-retries 3× then pauses; usage-limit auto-wait (`autoContinueAtUsageLimit`); MCP auto-reconnect | `D/errors § Automatic retries, § The response above may be incomplete; D/network-config § Streaming idle watchdogs; D/model-config § Fallback model chains; D/how-claude-code-works § When context fills up; D/troubleshooting § Auto-compaction stops with a thrashing error; D/sessions § What a resumed session restores; D/headless § Stop a run with SIGTERM; D/workflows § When an agent stalls and restarts; D/goal § Other errors retry or pause the goal` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Performance | Context (layering, deferral, trimming); Model interface (cache TTL); Output (parallel tools); Control flow (background) | Prefix-ordered requests and cache-safe updates via messages (system reminders); MCP schemas deferred; compaction as a forked cached request; `/rewind` truncates to a cached prefix; idle compaction before 1 h TTL expiry; read-only tools run concurrently; background subagents and auto-backgrounded commands; workflow fan-out holds siblings $\leq$5 s so they share the first agent's cache; WebFetch 15-min cache; `--bare` skips discovery; cache hit rate is alerted on internally | `D/prompt-caching § How the cache is organized`, `§ Compacting the conversation`, `§ Rewinding the conversation`, `§ Subagents and the cache`; `D/workflows § Prompt caching in a fan-out`; `SDK/agent-loop § Parallel tool execution`; `D/headless § Start faster with bare mode`; `B/lessons-from-building-claude-code-prompt-caching-is-everything § LESSONS LEARNED` |
| Evaluation | not documented for the harness itself | No harness test suite or agent evals are documented. Adjacent: `/insights` analyses up to 200 local sessions into a report; skills are evaluated by running prompts with the skill on/off (skill-creator plugin); the auto-mode classifier's first request "validates the Sonnet 5 default"; "Evals" listed as a workflow use case | `D/costs § Analyze your usage patterns`; `D/skills § Evaluating skills`; `D/permission-modes § How auto mode evaluates actions`; `B/a-harness-for-every-task-dynamic-workflows-in-claude-code § Use cases` |

### Design decisions (one line each: the choice this harness made within a primitive)

- Input: one entry point for everything that starts a turn (typed prompts, piped stdin, queued mid-turn messages delivered after the current tool batch, scheduled/cron prompts, cross-session and channel messages, background-subagent reports), all passing through `UserPromptSubmit`. `D/hooks § UserPromptSubmit`; `D/interactive-mode § When Claude Code sends what you queued`
- Context: a stable cached prefix (system prompt + tools, then project context) with every later update appended as a message or `<system-reminder>`; MCP schemas deferred behind `ToolSearch`; compaction on reaching the window via a forked request that re-injects CLAUDE.md, memory, five recent files and invoked skills. `D/prompt-caching § How the cache is organized`; `D/context-window § What survives compaction`
- Model interface: direct Anthropic Messages API with streaming, 10 exponential-backoff retries that never re-issue after a completed tool block, four idle watchdogs, a user-configured fallback model chain, and capability-rejection downgrade for gateways. `D/errors § Automatic retries`; `D/llm-gateway-protocol § Automatic retry and error forwarding`
- Output: a fixed built-in tool set plus MCP; read-only calls run concurrently, mutating calls sequentially; shell commands run under an OS sandbox (Seatbelt/bubblewrap) with a local domain-allowlist proxy; large results spill to files; file edits are checkpointed for `/rewind`. `SDK/agent-loop § Parallel tool execution`; `D/sandboxing § OS-level enforcement`; `D/checkpointing § Automatic tracking`
- Control flow: loop until a text-only response, with the decision to stop extendable by Stop hooks and `/goal`'s evaluator model; a six-step permission pipeline whose default gate is a second model (auto mode); delegation through an `Agent` tool (depth 3, 20 concurrent, background by default) and, at scale, a model-written JavaScript workflow the runtime executes. `SDK/agent-loop § Turns and messages`;

SDK/permissions § How permissions are evaluated; D/workflows § When to use a workflow

### Hard cases

- **WebFetch's inner model call:** a tool (output) that itself calls the model and decides what the main model sees (a summary, not the page). Resolved by rule 21: the tool call is output; the summarisation is a context decision (and a prompt-injection defence, D/security § Additional safeguards); the extra request is model interface. D/tools-reference § WebFetch tool behavior
- **Auto-mode classifier:** an approval gate the harness consumes (control flow) implemented as a separate model request with its own stripped-down context (D/permission-modes § How auto mode evaluates actions). Resolved: the decision is control flow; the request is model interface; the stripping of tool results is context assembly for that side call. Its "server-side classifier review" variant runs inside the API as part of the session's model requests, i.e. partly outside the harness. D/permission-modes § Server-side classifier review
- **Prompt/agent hooks and the /goal evaluator:** harness-sequenced model calls that decide whether to go again. Resolved by rule 27 (a program that sequences model calls is control flow). D/hooks § Prompt-based hooks; D/goal § How evaluation works
- **Dynamic workflows:** the model writes a JS program at run time whose agent() calls sequence model calls → control flow (rule 27); the Workflow tool call that launches it is output; the script's intermediate results live "in script variables" rather than any context. D/workflows § How a workflow runs
- **Compaction:** the trigger (window threshold) and the result (summary replaces history) are context; the summarisation is a model call (model interface); the thrashing stop after "a few attempts" is a loop decision (control flow/resilience). D/model-config § Default auto-compact thresholds; D/troubleshooting § Auto-compaction stops with a thrashing error
- **Context-limit retry with reduced max_tokens:** a retry of a failed request (model interface) that changes the request's parameters (context). Placed under model interface because the trigger is a rejected request and the content is unchanged; noted as a boundary case. D/errors § Automatic retries
- **Hook rewrites:** PreToolUse.updatedInput changes the action before execution (output executor side); PostToolUse.updatedToolOutput changes what the model is shown (context). Decomposed per rule 1. D/hooks § PreToolUse, § PostToolUse
- **Two kinds of model switch:** opusplan (plan vs execution) and subagent model fields are keyed to the work → control flow; --fallback-model (overloaded/unavailable) and safety-flag fallback → model interface (rule 22). D/model-config § Model aliases, § Fallback model chains, § Automatic model fallback
- **Subagent output scanning:** modifies a tool result before the model reads it → context (what the model is shown) and a safety concern; not output, because nothing is executed. D/sub-agents § Subagent output scanning
- **Checkpoints and session titles:** file snapshots are a user artifact (output) that persists; the title is produced by a cheap-model call keyed to the work (control flow) and stored for the operator/user (persistence). D/checkpointing § Automatic tracking; D/sessions § Name your sessions
- **Queued messages mid-turn:** capturing the message is input; deciding to hand it to the model "as soon as those tool calls finish, within the same turn" is a loop decision (control flow). D/interactive-mode § When Claude Code sends what you queued
- **Agent teams' mailbox:** messages arrive as turns for the recipient (input) but the delivery, idle notifications and task-claim locking are an outer harness's control flow; the classifier reviewing each SendMessage is a safety gate. Depth: lead → teammates (no nested teams); results return via idle notifications containing the final answer. D/agent-teams § Context and communication, § Limitations
- **Server-side classifier review and the server-side prompt-injection probe:** DOES NOT FIT the five primitives cleanly, because the safety decision is made inside the API during the model request rather than by the harness; the harness only consumes the verdict (denies when none arrives after ten responses). D/permission-modes § Server-side classifier review; D/glossary § Prompt injection
- **EndConversation:** a tool whose only effect is to stop the loop; by rule 21 the call is output and the stop is control flow; its un-removability is a safety choice. D/tools-reference § EndConversation tool behavior

**Summary row**

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| Prompts, stdin, queued messages, events, tool results, AskUserQuestion answers | Cached prefix + appended reminders; CLAUDE.md/memory; deferred MCP; compaction | Messages API streaming; 10 retries, watchdogs, fallback chain | Built-in + MCP tools; sandboxed Bash; concurrent read-only calls; checkpoints | Loop until no tool calls; six-step permission gate; Agent/workflow delegation |

## C.2 OpenAI Codex CLI

**Source:** https://github.com/openai/codex · **Version:** commit f0c5077c34ec7a76b4ad9134835925e6ae3d72ec (2026-10-15) · **Language:** Rust (workspace under `codex-rs/`) **Path conventions:** all paths are relative to `codex-rs/`. `core/` = crate `codex-core`; `session/` = `core/src/session/`; `tools/` under core = `core/src/tools/`; top-level crates (`codex-api`, `sandboxing`, `rollout`, `otel`, `prompts`, `protocol`, ...) are siblings of `core/`. Line ranges are `L<start>--L<end>` at the pinned commit. **What it is:** Codex is OpenAI's terminal/IDE coding agent. A `Session` (one thread) receives `Op` submissions over a channel, runs each user turn as a task that loops "sample → run tool calls → sample again" until the model stops calling tools, and talks to the OpenAI Responses API over HTTP SSE or a Responses WebSocket. It wraps execution in OS sandboxes (Seatbelt / seccomp+bwrap / Windows restricted token), gates risky actions with user or "Guardian" (model-reviewer) approvals, persists every thread as a JSONL rollout, and can spawn sub-agent threads that are full `Session`s of their own.

### Control flow

- **Submission loop (session-level dispatcher):** `submission_loop` `select!`s on shutdown, the `rx_sub` channel and mailbox updates, then matches each `Op` (TurnInput, Interrupt, ExecApproval, UserInputAnswer, Compact, Review, Shutdown...). `core/src/session/handlers.rs:L440--L727`
- **Turn admission modes:** `turn_input::handle` routes a TurnInput by mode: `StartOrSteer`, `StartIfIdle`, `ContinueIfIdle` (recovery requires an internal response item), `Steer { expected_turn_id }`. `core/src/session/turn_input.rs:L230--L296`
- **Task spawn replaces the active task:** `Session::spawn_task` first calls `abort_all_tasks(TurnAbortReason::Replaced)` then `start_task`, which registers the task on `active_turn`, seeds pending input from the mailbox, and sets a fresh `CancellationToken`. `core/src/tasks/mod.rs:L273--L340`
- **Abort / interrupt:** `abort_all_tasks` takes the active turn, cancels the task and clears pending approvals; on `Interrupted` it may immediately start a new turn for pending work. `core/src/tasks/mod.rs:L564--L592`
- **Outer task loop (RegularTask::run):** calls `run_turn` repeatedly until there is no pending input or a terminal error was recorded. `core/src/tasks/regular.rs:L105--L124`
- **Inner turn loop (run_turn):** `loop { drain pending input → capture StepContext → build sampling input → run_sampling_request → decide follow-up }`; `needs_follow_up = model_needs_follow_up || has_pending_input`; stops with `break` when `!needs_follow_up` after stop hooks run. `core/src/session/turn.rs:L453--L578`, `L604`, `L718--L821`
- **Stop condition from the model:** `try_run_sampling_request` sets `needs_follow_up = true` when any output item is a tool call, and on `ResponseEvent::Completed { end_turn: Some(false) }`. `core/src/stream_events_utils.rs:L351--L354`; `core/src/session/turn.rs:L3112--L3118`
- **Stop hooks can force another iteration:** `run_turn_stop_hooks` result `should_block` with a continuation prompt records a hook message, sets `stop_hook_active` and `continue`s the loop; `should_stop` breaks. `core/src/session/turn.rs:L718--L768`
- **Mid-turn roll-over on context limit:** `should_roll_over = auto_compaction_enabled && needs_follow_up && (new_context_window requested || token_limit_reached || background candidate ready)` triggers `run_auto_compact(..MidTurn..)` then `continue`. `core/src/session/turn.rs:L654--L716`

- **Steering (user message while a turn runs):** `steer_input` validates the active task is a Regular task (Review/Compact are non-steerable), then pushes the input onto `turn_state.pending_input` for the next loop iteration. `core/src/session/turn_input.rs:L799--L889`
- **Pending-input drain point:** `get_pending_input` splits off `turn_state.pending_input.items` and appends mailbox items only if the turn currently accepts mailbox delivery. `core/src/session/input_queue.rs:L400--L432`
- **Sampling retry loop:** `run_sampling_request` wraps `try_run_sampling_request` in a `loop` whose continuation is decided by `handle_response_stream_error` (retry budget `stream_max_retries`), and returns early with `needs_follow_up: true` on preempt. `core/src/session/turn.rs:L1707--L1835`
- **Approval gate consumed by harness (user):** `Session::request_command_approval` registers a `oneshot` under the approval id, emits `EventMsg::ExecApprovalRequest`, and awaits `rx_approve` (defaulting to `Abort` if dropped); `exec_approval` resolves it, and `Abort` calls `interrupt_task`. `core/src/session/mod.rs:L2923--L3012`; `core/src/session/handlers.rs:L177--L207`
- **Approval precedence:** hooks first, then Guardian (model reviewer) if enabled/strict, else user. `core/src/tools/approvals.rs:L479--L498`, `L541--L557`
- **Session-scoped approval cache:** `with_cached_approval` returns `ApprovedForSession` without asking when every cache key already holds that decision. `core/src/tools/sandboxing.rs:L72--L110`
- **Sub-agent spawn (effect of `spawn_agent` tool):** `thread_spawn::spawn` builds a `SessionSource::SubAgent(ThreadSpawn{parent_thread_id, depth, ...})` and calls `agent_control.spawn(SpawnRequest{ input: AgentInput::Message{ mode: TriggerTurn }, ... })`. `core/src/thread_spawn.rs:L58--L167`
- **Depth and fan-out limits:** `exceeds_thread_spawn_depth_limit(depth, max_depth)` (`depth > max_depth`) rejects a spawn with "Agent depth limit reached. Solve the task yourself."; `reserve_spawn_slot` returns `AgentLimitReached { max_threads }` when the registry is full. `core/src/agent/registry.rs:L85--L109`; `core/src/tools/handlers/multi_agents/spawn.rs:L74--L80`
- **Sub-agent results return:** `wait_agent` polls child status futures with `timeout_at(deadline, ...)` (default `DEFAULT_WAIT_TIMEOUT_MS`) and collects final statuses; `send_input` steers or starts a child turn through `start_or_steer_turn`. `core/src/tools/handlers/multi_agents/wait.rs:L95--L99`, `L181--L200`; `core/src/agent/control.rs:L137--L165`
- **Delegate (review / guardian) sub-session:** `run_codex_thread_one_shot` → `run_codex_thread_interactive` → `Session::spawn(SessionSpawnArgs{ thread_source: Subagent|GuardianReview, agent_control: Provided, ... })`; child events/ops are forwarded via `forward_session_io` tasks. `core/src/codex_delegate.rs:L105--L228`, `L231--L262`
- **Routing keyed to the work (review model):** the review task clones config, disables web search/collab, sets `base_instructions = REVIEW_PROMPT`, `approval_policy = Never`, and `model = config.review_model` before delegating. `core/src/tasks/review.rs:L105--L140`
- **Routing keyed to the work (guardian model):** the reviewer session resolves its own `review_model` and reasoning effort separately from the parent. `core/src/guardian/review_session.rs:L330--L372`
- **Code mode: model-written program sequences tools (output), but can call back into the harness:** nested calls from a cell go through `CodeModeCellDelegate::invoke_tool` → `DispatchMessage::InvokeTool` to the turn worker; the program sequences tools only, not model calls. `core/src/tools/code_mode/delegate.rs:L317--L360`
- **Session token budget (stop):** `record_rollout_budget_usage` returns `SessionBudgetExceeded` once weighted usage reaches `limit_tokens`. `core/src/agent/control/budget.rs:L12--L17`; `core/src/rollout_budget.rs:L48--L65`
- **Model-requested context reset (`new_context_window` tool):** the handler calls `session.request_new_context_window()`, which the run_turn roll-over check consumes via `take_new_context_window_request`. `core/src/tools/handlers/new_context_window.rs:L44--L56`; `core/src/session/turn.rs:L659--L663`

### Input

- **Op channel + public thread API:** `CodexThread::submit(op)` sends into `io.tx_sub`; `next_event()` reads `io.rx_event`. `core/src/codex_thread.rs:L258--L260`, `L773--L775`

- **User message submission op:** `Op::TurnInput { request, mode, reply }` is "Submit turn input using the requested routing behavior." `protocol/src/protocol.rs:L703--L710`
- **User turn ingestion:** `start_or_steer` builds `task_input` from merged additional context plus `pending_turn_input(...)` and calls `spawn_task(turn_context, task_input, RegularTask::new())`. `core/src/session/turn_input.rs:L428--L449`
- **Person's answer the MODEL consumes:** `request_user_input` tool awaits `session.request_user_input(...)` and serializes the `RequestUserInputResponse` back as the tool result; delivered by `Op::UserInputAnswer { id, response }`. `core/src/tools/handlers/request_user_input.rs:L78--L100`; `protocol/src/protocol.rs:L791--L797`
- **Tool results as input to the next request:** in-flight tool futures are drained in order and recorded via `record_annotated_conversation_items`. `core/src/session/turn.rs:L2565--L2592`
- **Inter-agent mail / mailbox:** `Op::InterAgentCommunication` and mailbox updates (`mailbox_updates`, `notify_mailbox`) feed pending input; mailbox delivery can preempt a response between items. `core/src/session/handlers.rs:L447--L479`, `L649--L656`; `core/src/session/turn.rs:L2913--L2932`
- **Hook-contributed input:** `drain_async_hook_results` records results of async hooks before/after the user prompt. `core/src/hook_runtime.rs:L782`; `core/src/session/turn.rs:L185`, `L591`

### Context

- **Base instructions come from the model catalog:** `render_model_instructions` reads `model_info.model_messages.instructions_template`; the bundled default is `BASE_INSTRUCTIONS = include_str!("../prompt.md")` in models-manager, overridable by `config.base_instructions`. `prompts/src/model_instructions.rs:L8--L17`; `models-manager/src/model_info.rs:L16`, `L46--L48`
- **Prompt assembly per sampling request:** `build_prompt` packs `input` (history), `tools = tool_router.model_visible_specs()`, `parallel_tool_calls: true`, `base_instructions`, `output_schema`; with incremental tools (WebSocket continuation) tools and instructions are sent empty. `core/src/session/turn.rs:L1666--L1694`
- **History normalization before send:** `for_prompt` → `normalize_history`: ensure every call has an output, drop orphan outputs, strip images/audio the model can't take. `core/src/context_manager/history.rs:L603--L617`, `L968--L982`
- **Tool-output truncation on record:** `record_item_with_metadata` applies `truncate_function_output_payload(output, policy, ...)` to Function/CustomToolCallOutput using the model's `truncation_policy` (live history only; rollout keeps full payload). `core/src/context_manager/history.rs:L517--L519`, `L568--L577`
- **World-state sections (diff-injected context):** `WorldStateSection` trait — typed sections (`agents_md`, `environment`, `permissions`, `tools`, `model`, ...) each render a diff vs. the previous snapshot; `WorldStateSnapshot::merge_patch_from` produces an RFC 7386 merge patch. `core/src/context/world_state/mod.rs:L302--L330`, `L381--L425`
- **Per-step world-state injection:** `record_step_world_state_if_changed` renders the step's world state, records only the changed fragments into history, and persists a `RolloutItem::WorldState` patch. `core/src/session/mod.rs:L3762--L3793`
- **AGENTS.md discovery:** walks from project root (marker `.git` by default) down to cwd, concatenating every `AGENTS.md`, bounded by `project_doc_max_bytes`; skipped for untrusted projects. `core/src/agents_md.rs:L1--L18`, `L58--L68`; refreshed per step in `capture_step_context_inner`. `core/src/session/mod.rs:L3915--L3924`
- **AGENTS.md as a world-state section:** `AgentsMdState` renders replacement/removal notices when the instructions change. `core/src/context/world_state/agents_md.rs:L11--L60`
- **Environment context:** `FileSystemContext { workspace_roots, permission_profile }` and `NetworkContext` are rendered into the model-visible environment section. `core/src/context/environment_context.rs:L10--L30`, `L213`
- **Developer instructions + extension contributors:** `build_initial_context_with_world_state` collects developer sections, plugin recommendations, and `context_contributors().contribute_thread_context/turn_context` fragments into developer/user slots. `core/src/session/mod.rs:L4473--L4560`
- **Skills injected on mention:** `build_skills_and_plugins` loads prompts for explicitly mentioned skills

(load_skill_prompts) and converts each fragment into a response item recorded into history. core/src/session/turn.rs:L1099, L1165--L1190, L1222--L1236; recorded at L409--L416

- **Memory read-path context:** MemoryContextFragment::{ReadInstructions (developer), ExtractionEvidence (user)}, capped below 10k tokens; memory root is $CODEX_HOME/memories. core/src/context/memory.rs:L1--L30; memories/read/src/lib.rs:L13
- **Tool catalog for the step:** built_tools builds a ToolRouter from MCP tools, connectors and tool-suggest candidates; capture_step_context_inner joins MCP refresh and tool build into a StepContext. core/src/session/turn.rs:L1884--L1973; core/src/session/mod.rs:L3959--L4030
- **Deferred tools (tool search):** ToolExposure::Deferred tools are omitted from the initial list and discoverable through the tool_search tool. tools/src/tool_executor.rs:L48--L62; core/src/tools/handlers/tool_search.rs:L68--L85
- **Compaction trigger math:** context_window_token_status computes token_limit_reached (auto-compact limit + fallback buffer, or effective_context_window_percent of the window) and turn_end_compaction_threshold_reached (model_post_turn_compact_threshold_percent). core/src/session/context_window.rs:L53--L130
- **Pre-turn / post-turn compaction:** run_pre_sampling_compact runs before the user input is recorded; post-turn compaction runs when foreground_due || candidate.is_some(). core/src/session/turn.rs:L192--L212, L779--L817
- **Compaction implementation selection:** run_auto_compact chooses token-budget reset (Feature::TokenBudget), remote ResponsesCompactionV2, or local summarization. core/src/session/turn.rs:L1507--L1570; core/src/compact_remote_v2.rs:L112--L140
- **Local summarization compaction:** run_compact_task_inner_impl sends history + SUMMARIZATION_PROMPT (prompts/templates/compact/prompt.md) to the model, then build_compacted_history keeps initial context + up to COMPACT_USER_MESSAGE_MAX_TOKENS = 20_000 tokens of recent user messages + the summary. core/src/compact.rs:L62--L64, L264--L330, L702--L760
- **Background (speculative) compaction:** async_compaction::maybe_start launches a parallel compaction once usage passes async_compaction_threshold_percent of the window, deduped by history version/model/hash. core/src/session/async_compaction.rs:L1--L9, L62--L95
- **Model-queryable context remaining (get_context_remaining tool):** reports base_window_tokens_remaining from context_window_token_status. core/src/tools/handlers/get_context_remaining.rs:L85--L92

### Model interface

- **Client lifetimes:** ModelClient lives for the session; ModelClientSession is created per turn, caches a Responses WebSocket and the x-codex-turn-state sticky-routing token. core/src/client.rs:L1--L26, L295--L303
- **Wire format:** ResponsesApiRequest { model, input, tools, tool_choice: "auto", parallel_tool_calls, reasoning, store: false, stream: true, include: ["reasoning.encrypted_content", ...], prompt_cache_key, text, ... } posted to /responses. core/src/client.rs:L1040--L1095; codex-api/src/endpoint/responses.rs:L110--L139
- **Transport selection + fallback:** ModelClientSession::stream tries stream_responses_websocket when the provider supports it and falls back to ResponsesHttpExecutor::stream_responses_http; try_switch_fallback_transport permanently disables WebSockets for the session. core/src/client.rs:L2363--L2435
- **SSE event parsing + idle timeout:** process_sse maps response.output_item.done, response.output_text.delta, response.completed|incomplete to ResponseEvents; each poll is timeout(idle_timeout, stream.next()) with default DEFAULT_STREAM_IDLE_TIMEOUT_MS = 300_000. codex-api/src/sse/responses.rs:L345--L470, L554--L558; model-provider-info/src/lib.rs:L66, L513--L516
- **Retry policy:** handle_response_stream_error: terminal errors (context exceeded, usage limit, invalid request…) return None delay; retryable ones use server Retry-After else exponential backoff with ±10% jitter; budget DEFAULT_STREAM_MAX_RETRIES = 5; unbounded connection retries (5s→60s doubling) behind Feature::UnboundedConnectionRetries; after exhausting the budget on WebSocket it switches

to HTTPS and resets the counter. core/src/responses_retry.rs:L23--L24, L57--L176; protocol/src/error.rs:L410--L440; codex-client/src/retry.rs:L43--L52; model-provider-info/src/lib.rs:L67, L506--L508

- **Content-filter retry injects guidance (context effect):** on ContentFilter the retry path records a ContentFilterGuidance user fragment before retrying. core/src/responses_retry.rs:L67--L82
- **Receiving the stream:** try_run_sampling_request opens client_session.stream(prompt, model_info, ...) and loops on stream.next() under preempt/cancellation. core/src/session/turn.rs:L2667--L2715
- **Token usage from the wire:** on Completed { token_usage, ... } the session records usage (record_token_usage_info) and emits a token-count event. core/src/session/turn.rs:L3069--L3118, L3295--L3301
- **Auth-expiry recovery:** PendingUnauthorizedRetry retries a request once after refreshing credentials on 401. core/src/client.rs:L1582--L1636, L1781--L1925
- **Prompt cache key:** prompt_cache_key is sent per request (override or thread-derived) so the server reuses cached prefixes. core/src/client.rs:L646--L668, L1060, L1090

### Output

- **Tool call recognition:** ToolRouter::build_tool_call turns FunctionCall, CustomToolCall, client-executed ToolSearchCall, etc. into a ToolCall. core/src/tools/router.rs:L257--L335
- **Dispatch:** handle_output_item_done logs the call, persists the item, and pushes tool_runtime.handle_tool_call(call, token) into the in-flight queue; non-tool items become turn items and are emitted to the client. core/src/stream_events_utils.rs:L315--L432
- **Concurrency of tool calls from one response:** ToolCallRuntime::handle_tool_call spawns each call on tokio; parallel-safe tools take a read lock on parallel_execution: RwLock<()>, others a write lock (serialized). core/src/tools/parallel.rs:L51, L141--L237; core/src/tools/router.rs:L246--L250
- **Streaming text to the person:** OutputTextDelta is parsed and emitted as assistant text deltas; the TUI appends them via on_agent_message_delta → handle_streaming_delta. core/src/session/turn.rs:L3120--L3135; tui/src/chatwidget/streaming.rs:L206--L208
- **Diff artifact for the user:** TurnDiffTracker accumulates file changes and emits EventMsg::TurnDiff { unified_diff } after each response. core/src/session/turn.rs:L441--L443, L3307--L3317
- **Exec tool (exec_command / write_stdin):** Unified Exec manages PTY processes with output caps (DEFAULT_MAX_OUTPUT_TOKENS = 10_000, UNIFIED_EXEC_OUTPUT_MAX_BYTES = 1 MiB) and yield-time bounds. core/src/unified_exec/mod.rs:L1--L25, L68--L76; core/src/tools/handlers/unified_exec/exec_command.rs:L109; core/src/tools/handlers/unified_exec/write_stdin.rs:L39
- **Exec timeout:** DEFAULT_EXEC_COMMAND_TIMEOUT_MS = 10_000; ExecExpiration::{Timeout, DefaultTimeout, Cancellation, TimeoutOrCancellation}. core/src/exec.rs:L63, L173--L223
- **Orchestrator (approval → sandbox → run → retry):** ExecApprovalRequirement::{Skip, Forbidden, NeedsApproval} decides whether to ask; on sandbox denial it retries without a sandbox only if policy allows (Never/OnRequest do not retry). core/src/tools/orchestrator.rs:L164--L222, L439--L470
- **Sandbox technology:** get_platform_sandbox → macOS MacosSeatbelt, Linux LinuxSeccomp (bwrap + seccomp + Landlock), Windows WindowsRestrictedToken/WindowsMxc. sandboxing/src/manager.rs:L50--L63; protocol/src/sandbox.rs:L29--L35; sandboxing/src/{seatbelt.rs,bwrap.rs,landlock.rs}; linux-sandbox/src/seccomp.rs
- **Remote executor:** codex-exec-server is a JSON-RPC server for spawning/controlling subprocesses via codex-utils-pty, used for non-local environments. exec-server/README.md (top), exec-server/src/lib.rs:L1--L12
- **Patch tool safety:** assess_patch_safety auto-approves patches confined to writable roots (still run in a sandbox), rejects under Never, otherwise asks. core/src/safety.rs:L67--L125, L146--L180
- **MCP tool calls:** handle_mcp_tool_call prepares the call, applies approval policy, and invokes the server via prepared_call.call_with_preparation. core/src/mcp_tool_call.rs:L131--L145, L473--L478

- **Code mode executor (`exec` tool):** the `exec` tool runs model-written JS in a V8 host process reached over gRPC; nested tool calls are brokered back to the harness. `code-mode-protocol/src/lib.rs:L54`; `core/src/code_mode_host.rs:L1--L4`; `core/src/tools/code_mode/execute_handler.rs:L176--L179`
- **Sub-agent tool surface:** `spawn_agent`, `send_input`, `wait_agent`, `resume_agent`, `close_agent` handlers. `core/src/tools/handlers/multi_agents/{spawn.rs,send_input.rs,wait.rs,resume_agent.rs,close_agent.rs}`
- **Review sub-agent as a tool-free delegate:** results come back as events through `forward_events`. `core/src/codex_delegate.rs:L200--L228`

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Input/Context (record of the conversation); Control flow (resume) | Every thread is a JSONL rollout at `~/.codex/sessions/.../rollout-<ts>-<id>.jsonl` written by `RolloutRecorder`; `should_persist_response_item` filters items; `reconstruct_rollout` replays from the latest compaction checkpoint on resume; SQLite mirrors metadata; approvals persist as execpolicy amendments; memories pipeline writes `$CODEX_HOME/memories` | `rollout/src/recorder.rs:L77--L88`; `rollout/src/policy.rs:L80--L100`; `core/src/session/rollout_reconstruction.rs:L208--L250`; `core/src/thread_manager.rs:L1319--L1332`; `state/src/lib.rs:L1--L5`; `core/src/session/handlers.rs:L184--L198`; `memories/README.md` (When it runs) |
| Safety & security | Output (sandbox, exec policy); Control flow (approvals, Guardian) | OS sandboxes per platform; execpolicy rules `Allow/Prompt/Forbidden` plus dangerous-command matcher; patch writable-root check; approval precedence hooks→Guardian→user; untrusted projects skip AGENTS.md; subagent depth/thread caps; tool-output metadata cleared before sending unless internal | `sandboxing/src/manager.rs:L50--L63`; `execpolicy/src/decision.rs:L9--L16`; `core/src/exec_policy.rs:L318--L400`; `core/src/safety.rs:L67--L125`; `core/src/tools/approvals.rs:L479--L498`; `core/src/agents_md.rs:L64--L66`; `core/src/agent/registry.rs:L85--L99`; `core/src/client.rs:L1072--L1076` |
| Observability | across primitives (what is recorded) | OTel spans per sampling (`codex.sampling`, `handle_responses` with `gen_ai.usage.*` fields), tool-decision and tool-result events, approval counters, per-call `call_trace`, analytics events, token-count events to the client | `core/src/session/turn.rs:L2653--L2658, L2715--L2729`; `otel/src/events/session_telemetry.rs:L637--L665, L1195--L1204, L1297--L1308`; `core/src/tools/sandboxing.rs:L100--L107`; `core/src/stream_events_utils.rs:L326--L331` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Resilience | Model interface (retries, fallback, timeouts); Context (compaction); Control flow (resume/recovery) | Stream retry budget with backoff and `Retry-After`; WS→HTTP fallback; 300 s idle timeout; auth-refresh retry; auto compaction pre/mid/-post turn and background; `Op::RecoverTurn/ContinueIfIdle` for recovery; delegate failure cleanup | `core/src/responses_retry.rs:L57--L176`; `core/src/client.rs:L2363--L2435`; `codex-api/src/sse/responses.rs:L554--L558`; `core/src/session/turn.rs:L654--L716`; `core/src/session/handlers.rs:L574--L590`; `core/src/codex_delegate.rs:L170--L179` |
| Performance | Model interface (streaming, WS prewarm, cache key); Output (parallel tools); Context (truncation, deferred tools, diff-based world state) | WebSocket connection reuse and prewarm; `prompt_cache_key`; parallel tool execution gate; tool-output truncation; deferred tools kept out of the initial tool list; world-state merge-patch injection; background compaction | `core/src/client.rs:L17--L26, L1060`; `core/src/tools/parallel.rs:L213--L217`; `core/src/context_manager/history.rs:L568--L577`; `tools/src/tool_executor.rs:L57--L62`; `core/src/context/world_state/mod.rs:L404--L425`; `core/src/session/async_compaction.rs:L62--L95` |
| Evaluation | — (test suites) | ~249 integration suites under `core/tests/suite/` driven by a wiremock Responses mock (`sse`, `ev_function_call`, `mount_sse_once_match`), plus unit tests colocated as `*_tests.rs` | `core/tests/suite/` (listing); `core/tests/common/responses.rs:L27--L31`, `L753`, `L971`, `L1133` |

### Design decisions (one line each: the choice this harness made within a primitive)

- Input: a single `Op` channel per thread; user text during a running turn is queued (steer) rather than starting a new turn, and the model's questions come back as a typed `UserInputAnswer` op.
- Context: history is the Responses item list itself; environment/instructions are typed "world-state sections" injected as diffs, tool outputs are token-truncated in live history only, and compaction is pluggable (local summary, remote v2, or budget reset).
- Model interface: Responses API only (`WireApi::Responses`), WebSocket-first with session-wide HTTP fallback, server-advised retry timing, `store: false` with encrypted reasoning replay.
- Output: every risky action goes through one orchestrator (approval → platform sandbox → run → sandbox-denied retry); tools run concurrently under a read/write gate; a V8 "code mode" lets the model batch tool calls in JS.
- Control flow: one task per turn, inner loop until the model emits no tool call (or `end_turn=false`), with compaction roll-over, stop hooks, and sub-agents that are full `Session`s coordinated by `AgentControl` with depth/thread caps.

### Hard cases

- **Guardian (auto-review of approvals):** a harness-consumed gate (control flow) whose decision is produced by a *model* in a separate `Session` (`ThreadSource::GuardianReview`). Rule: "an approval the harness consumes is control flow"; the reviewer itself is an inner harness (multi-agent rule). Placed as

control flow, with the reviewer's own loop as a nested harness. `core/src/guardian/mod.rs:L1--L3`; `core/src/codex_delegate.rs:L133--L137`

- **Content-filter retry that injects guidance:** retrying is model interface, but it also records a user fragment into history (context). Split per rule "assign by function". `core/src/responses_retry.rs:L67--L82`
- **Incremental tools over WebSocket:** `build_prompt` deliberately sends empty tools/instructions when the server already holds them (`previous_response_id`). What the model is shown is unchanged, but the request contents depend on transport state — placed as model interface with a context dependency. `core/src/session/turn.rs:L1675--L1689`; `core/src/client.rs:L2137--L2160`
- **Remote compaction v2:** the *server* summarizes (`CompactionImplementation::ResponsesCompactionV2`); the harness decides when (context) but the summarizer is the model/provider. Placed as context (the fitting step) with the call as model interface. `core/src/compact_remote_v2.rs:L112--L140`
- **Code mode:** the model-written program runs in V8 and sequences only tools (output per rule), yet nested calls re-enter the harness's approval/sandbox path, so each nested call is a normal output-side tool call. `core/src/tools/code_mode/delegate.rs:L317--L360`
- **`new_context_window / get_context_remaining` tools:** tools whose only effect is on context (reset request) or that expose context state to the model; the call is output, the effect is context/control flow per the "effect" rule. `core/src/tools/handlers/new_context_window.rs:L44--L56`
- **World-state diffing:** context mechanism that also writes a `RolloutItem::WorldState` for persistence; the model-facing part is context, the file write is persistence. `core/src/session/mod.rs:L3762--L3793`
- **Steer vs. mailbox vs. hooks as inputs:** three different channels (user steer queue, inter-agent mailbox, async hook results) all merge into `pending_input` before the next request; all are input, drained by control flow. `core/src/session/input_queue.rs:L400--L432`
- **DOES NOT FIT (partially): plan mode streaming parser.** `PlanModeStreamState` parses the assistant's streamed text into a "proposed plan" item and can suppress/redirect deltas — it is output rendering that also changes what is recorded as the last agent message (feeds control flow's stop decision). It straddles output and control flow by design. `core/src/session/turn.rs:L1992`, `L2709`, `L2846--L2858`

**Summary row**

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| Op channel; steer queue; mailbox; UserInputAnswer | Responses items + world-state diffs; truncation; 3 compaction modes | Responses API, WS→HTTP fallback, 5 retries w/ backoff | Sandboxed exec/-patch/MCP; parallel gate; V8 code mode | Task per turn; loop until no tool call; approvals; sub-agent Sessions |

## C.3 OpenCode

**Source:** https://github.com/anomalyco/opencode · **Version:** commit 654b79b27e79da735d6b07d766c8d7431a71d1b7 (2026-10-15); `@opencode/core` package version 2.0.29 · **Language:** TypeScript (Bun runtime, Effect 4 library) **Path conventions:** `core/` = `packages/core/src/`, `ai/` = `packages/ai/src/`, `schema/` = `packages/schema/src/`, `codemode/` = `packages/codemode/src/`, `server/` = `packages/server/src/`, `recorder/` = `packages/http-recorder/src/`. Line ranges are `Lstart--Lend` at the pinned commit. Note: the brief's expected layout (`packages/opencode/src/session/prompt.ts`, `processor.ts`, `llm.ts`, `system.ts`) no longer exists at this commit; the equivalents are `core/session/prompt.ts` (input preparation), `core/session/runner/llm.ts` (the loop), `core/session/runner/step.ts` (one model call plus its tools), `core/session/model-request.ts` (request assembly), `core/session/compaction.ts`, `core/session/system-prompt.ts`. **What it is:** OpenCode is an open-source coding agent harness by Anomaly (anomalyco) that runs as a local server with TUI, desktop, and web clients; a session is a durable, event-sourced record in SQLite that a per-session "runner" fiber drains step by step. The model is reached through the in-house `@opencode/ai` client, which compiles a canonical `LLMRequest` into a provider

protocol (Anthropic Messages, OpenAI Responses, Bedrock Converse, Gemini, ...) over HTTP/SSE or a stateful WebSocket, with a Vercel AI SDK bridge for third-party provider packages.

### Control flow

- **Entry: `Session.prompt` admits input then rings the doorbell:** reconciles the inbox row, prepares the prompt, admits it, and calls `execution.wake(sessionID)` unless `resume === false`. `core/session/session.ts:L145--L178`
- **Run coordinator (one execution fiber per session):** `loop` drains, then re-drains while a `pendingWake` doorbell was rung mid-drain; `wake` coalesces scopes ("input" subsumes "steer"); `interrupt` marks `stopping`, claims recorded wakes and interrupts the owner fiber. `core/session/run-coordinator.ts:L74--L85`, `L136--L145`, `L147--L167`
- **Execution service routes drains and handles `DrainResult`:** `Complete` ends, `Moved` and `Reloaded` re-enter `drain` with a continuation step. `core/session/execution.ts:L87--L110`; `DrainResult` enum `core/session/runner/index.ts:L23--L28`
- **The outer loop (`SessionRunner.drain`):** `while (true) { next = advanceToStep(); if not Ready return; continuing = runStep(ctx, step); step++ }`. `core/session/runner/llm.ts:L191--L198`
- **Step boundary (`advanceToStep`):** at each boundary the runner checks `lifecycle.isClosed()`, pulls the next inbox item under a per-session lock, handles `compaction` and `move` controls inline, and otherwise promotes pending steers/queued input before loading context; a promoted input resets `step = 1`. `core/session/runner/llm.ts:L74--L189`
- **Steer vs queue delivery:** `Promotable` doc: "steer" promotes steers only (mid-work), "input" also admits one queued item at an idle boundary. `core/session/inbox.ts:L43--L48`; `nextPromotable` `L407--L424`; `promote` `L496--L531`; a pending compaction is pulled ahead of earlier steers `L533--L550`; `Delivery = "steer" | "queue"` `schema/session-inbox.ts:L11`
- **Inner step policy (`runStep`):** before every attempt it runs auto-compaction (`compacted.status === "completed"` → new assistant id and `continue`), then prepares the request and dispatches on the attempt `Outcome`. `core/session/runner/llm.ts:L209--L330`
- **Stop condition:** `needsContinuation = toolChoice !== "none" && record.needsContinuation` (a step with executed tool calls continues; a text-only finish hands back); `finish === "length"` → `OutputLimit`. `core/session/runner/step.ts:L281--L283`
- **Step budget:** `stepLimitReached = agent.info.steps !== undefined && step >= agent.info.steps`; on the final step the runner appends `MAX_STEPS_PROMPT` as an assistant message and sets `toolChoice: "none"` while keeping tool definitions to preserve the cached prefix. `core/session/runner/llm.ts:L226--L247`; prompt text `core/session/runner/max-steps.ts:L1--L16`; `steps` field `schema/agent.ts:L33`; tool calls on that step fail with "Tools are disabled after the maximum agent steps" `core/session/runner/step.ts:L91--L101`
- **Output-limit continuation:** after `finish === "length"` the runner reads the last 3 user/assistant rows; if all three are assistant `length` finishes it fails with `output-limit`, else publishes the synthetic `CONTINUE_AFTER_OUTPUT_LIMIT` and goes again. `core/session/runner/llm.ts:L290--L319`; constant `L38--L39`
- **Interrupted-stream continuation:** `Continue` outcome waits the retry delay, publishes `CONTINUE_AFTER_INCOMPLETE_STREAM` and starts a fresh assistant message. `core/session/runner/llm.ts:L281--L289`; constant `L35--L36`
- **Retry wait is part of the loop:** `Retry` outcome → `retry.wait` (publishes `RetryScheduled`, sleeps) then the `while (true)` re-attempts with refreshed context. `core/session/runner/llm.ts:L275--L280`, `L216--L219`; `wait` `core/session/runner/retry.ts:L97--L113`
- **Overflow → compact → go again:** `Compacted` outcome sets `recoverOverflow = false` (one overflow compaction per step) and re-attempts. `core/session/runner/llm.ts:L265--L271`, `L320--L323`
- **Approval gate consumed by the harness:** `Permission.assert` evaluates rules (`deny` → `BlockedError`, `allow` → continue, `ask` → create request and await a `Deferred`). `core/permission.ts:L231--L263`; the reply endpoint resolves it, `"always"` persists a rule, a `reject` with no message fails every pending ask in the session. `core/permission.ts:L265--L325`

- **A decline ends the step:** declined tool calls are failed with "The user declined this tool call" and the attempt is interrupted (`Effect.interrupt`) rather than returned to the model. `core/session/runner/step.ts:L207--L215, L274--L277`
- **Interrupt with resume:** after interrupting, if the next promotable item is a steer or a control, the coordinator is woken with scope "steer" so queued next-turn prompts stay parked. `core/session/execution.ts:L153--L170`
- **Sub-agent delegation (outer-harness side):** the `subagent` tool walks `parentID` to compute depth and refuses beyond `experimental.subagent_depth` (default 1); it creates a child session with `parentID`, prompts it, and in foreground blocks on the child's job; background returns immediately. `core/tool/plugin/subagent.ts:L117--L133, L185--L198, L205--L218, L229--L240`
- **Child session is a harness of its own:** the job's `run` is `sessions.resume(childSessionID)` and the result is the text of the last completed assistant message. `core/session/subagent-job.ts:L45--L63`
- **How results return:** foreground → tool result text wrapped in `<subagent sessionID=... state="completed">`; background → a synthetic user message delivered to the parent when the job settles. `core/tool/plugin/subagent.ts:L252--L266`; `core/session/subagent-completion.ts:L21--L46`
- **Routing to a cheap model keyed to the work (title):** the title agent uses `model.small(primary.providerID)` plus a minimal-reasoning variant (`none|minimal|low`), forked so it never delays the step. `core/session/context.ts:L95--L119, L186`; fork `core/session/runner/llm.ts:L177--L180`
- **Compaction model keyed to the work:** the `compaction` agent may carry its own model; otherwise the session model is used. `core/session/compaction.ts:L213--L219`
- **Agent profiles (bundles):** `build` (default, all-allow + question), `general`/`explore` sub-agents with deny lists and a custom system prompt, hidden `compaction`/`title`/`summary` agents. `core/plugin/agent.ts:L80--L140`; default ruleset `schema/agent.ts:L46--L51`
- **Plan mode (bundle):** a `plan` agent denies `edit` outside `~/.opencode/plan/*` (control flow), rewrites blocked-edit errors (output), and re-injects a `<system-reminder>` per request (context). `core/plugin/plan.ts:L33--L42, L44--L53, L57--L60`
- **Crash/restart resume:** at boot, sessions whose execution claim survived are resumed with a synthetic "The server restarted..." notice; a durable per-turn counter stops after `DEFAULT_MAX_ATTEMPTS = 10`. `core/session/execution/restart.ts:L77--L99, L193--L231`; budget `L33`

### Input

- **HTTP handler → `session.prompt`:** `text`, `files`, `agents`, `skills`, `metadata`, `delivery`, `resume` are passed straight through. `server/handlers/session.ts:L307--L322`
- **Prompt preparation (`SessionPrompt.prepare`):** fires the plugin hook `session.prompt` (plugins may rewrite text/files/delivery), materializes attachments with concurrency 8, resolves `skills` to their text, and builds a `UserPayload`. `core/session/prompt.ts:L29--L91`
- **Attachments:** `data:` URLs or `file:` URIs; 20 MiB cap; `?start=&end=` line ranges for text; a directory becomes a sorted listing. `core/session/prompt.ts:L93--L133, L151--L201, L203`
- **Image normalization on input:** images are resized via `Image.Service` (resizer unavailable → passthrough). `core/session/prompt.ts:L135--L149`
- **Durable admission:** the inbox row is inserted into SQLite (`admit`, `reconcile`, `cancel`, `steer`, `queue`). `core/session/inbox.ts:L155, L169, L230--L256, L295`
- **Harness-originated input (synthetic):** `Session.synthetic` admits text the harness produced (restart notice, job completions) through the same inbox. `core/session/session.ts:L273--L314`
- **Answer the MODEL consumes (question tool):** the form answers are rendered as `User has answered your questions: ...` and returned as the tool result. `core/tool/plugin/question.ts:L38--L46, L62--L112`
- **Background shell completion event:** `notifyWhenDone` waits on the job and admits a synthetic message with the output. `core/tool/plugin/shell.ts:L157--L189`
- **Sub-agent completion event:** delivered as a synthetic message tagged `<subagent ... state=...>`. `core/session/subagent-completion.ts:L37--L44`

### Context

- **Selection (`SessionContext.select`):** fixes session, agent, tool snapshot (narrowed by merged agent+session permissions) and instruction sources, loaded in parallel. `core/session/context.ts:L121--L161`; permission narrowing `L131`
- **Instruction ordering for cache reuse:** "Ordered from most to least shared across sessions so the baseline stays a reusable prompt-cache prefix": Code Mode catalog, MCP, references, skills, AGENTS.md discovery, builtins (date/env), per-session entries. `core/session/context.ts:L147--L158`
- **Load (`SessionContext.load`):** resolves the model and reads history after the latest replayable compaction. `core/session/context.ts:L163--L179`
- **System prompt assembly:** `[agent.system ?? default prompt rendered with tool guidance, initial instructions]`; messages via `toLLMMessages`. `core/session/model-request.ts:L100--L120`
- **Default system prompt text:** 15 lines ("You are an AI agent running in OpenCode…"), with `${OPENCODE_TOOL_GUIDANCE}` filled per available tool. `core/session/runner/prompt/system.txt:L1--L15`; `core/session/system-prompt.ts:L9--L28`
- **Built-in instructions:** today's date and an `<env>` block (working dir, workspace root, git, platform, tmp). `core/instructions/builtins.ts:L24--L53`
- **AGENTS.md discovery:** global `AGENTS.md` plus ancestor-walk from the location to the project root, watched for changes. `core/config/plugin/instruction.ts:L36--L62`, `L76--L86`
- **Catalogs rendered into the baseline:** skills `<available_skills>` `core/skill/instructions.ts:L16--L33`; MCP `<mcp_instructions>` `core/mcp/instructions.ts:L17--L29`; Code Mode catalog `core/codemode/instructions.ts:L9--L30`
- **Instruction epochs:** changed instruction sources are frozen into a durable `InstructionsUpdated` event with rendered delta text, so later requests see "Today's date is now…" style updates as system messages. `core/session/instruction-state.ts:L38--L61`, `L80--L87`; rendered as `Message.system` `core/session/runner/to-llm-message.ts:L291--L292`
- **History read bounded by compaction:** rows from the latest completed compaction `seq` onward; native checkpoints only when the target model can replay them. `core/session/history.ts:L21--L26`, `L78--L108`, `L118--L135`
- **Projection of history into model messages:** synthetic/skill/shell messages become `user` role; a compaction becomes a `<conversation-checkpoint>` user message. `core/session/runner/to-llm-message.ts:L269--L327`, `L330--L334`
- **Tool definitions snapshot:** tools wholly denied by permission are dropped; `codemode: false` tools are advertised directly, the rest only inside the single `execute` tool; definitions sorted by name. `core/tool.ts:L225--L287`; JSON Schema generation `core/tool/runtime.ts:L21--L27`, `L174--L188`
- **Plugin hook that rewrites what the model sees:** `session.context` sees system, messages, tool descriptions/schemas and the default `maxTokens`; renamed tools are mapped back by object identity. `core/session/model-request.ts:L236--L274`
- **Media fitting:** unsupported modalities are replaced by an ERROR text; inline images are dropped from oldest once the total exceeds 25 MiB until under 15 MiB. `core/session/model-request.ts:L43--L46`, `L132--L147`, `L189--L216`; applied `L297`
- **Output token budget fitted to the window:** `outputLimit` = min(catalog limit, 256k), reduced to the room left after `measured + estimated×1.15`, floor 1,024. `core/session/model-request.ts:L47--L57`, `L88--L98`
- **Auto-compaction trigger:** `due` when `estimateContext >= ceiling`; ceiling = window - max(10 %, 16,000) or window - configured `buffer`; skipped right after a completed compaction or before a measured response. `core/session/compaction.ts:L205--L240`, `L242--L251`, `L911--L920`; defaults `auto: true, keep: 15_000` `L197`; settings `L41--L47`
- **Prompt size estimate = provider-measured + locally estimated:** anchors on the last assistant message with a usage count from the same provider, estimates everything after at ~4 chars/token. `core/session/compaction.ts:L880--L905`, `L924--L930`, `L950--L953`
- **Summary compaction:** splits history into `older` (summarized) and `recent` (kept verbatim within `keep` tokens, starting at a user boundary); sends older + summary prompt; one template nudge before failing. `core/session/compaction.ts:L272--L316`, `L708--L744`; template `L104--L135`

- **Tool outputs in the verbatim part are cut to 1,250 chars.** core/session/compaction.ts:L100, L813--L826
- **Overflow compaction (provider said too long):** captured before any output starts, then compaction aims at 70 %/50 %/35 % of the rejected request size. core/session/runner/step.ts:L113--L120, L155--L167; SHRINK_STEPS core/session/compaction.ts:L95, L429--L448
- **Manual /compact:** an inbox control item processed at the step boundary. core/session/session.ts:L246--L264; core/session/runner/llm.ts:L115--L163
- **Native (provider-side) compaction:** for models with compaction.type === "native" the provider returns one opaque item that replaces the window; the harness only appends the newest user messages. core/session/compaction.ts:L318--L332
- **Tool result truncation before it enters history:** 2,000 lines / 50 KiB, overflow written to a file with a [showing lines 1-N of M; full output saved to ...] marker. core/tool-output.ts:L13--L16, L67--L118; applied in the step core/session/runner/step.ts:L127
- **Tool input repair (what the model's call is seen as):** stringified JSON, numeric strings, placeholder nulls are repaired against the schema before execution. core/plugin/tool-input-repair.ts:L7--L36
- **Prompt-cache key / affinity:** promptCacheKey and x-session-affinity derive from the root session (children and forks share the parent's). core/session/model-request.ts:L278--L295; core/session/affinity.ts:L6--L8
- **Searchable memory for the model:** not found (no model-facing memory store; core/environment/memory.ts is an in-memory filesystem).

### Model interface

- **Streaming request:** the step calls llm.stream(request, options) and folds every LLMEvent through a publisher; it reads to the end "so the next request can reuse this response". core/session/runner/step.ts:L103--L138; client ai/route/client.ts:L602--L607, L618--L622
- **Request construction:** LLM.request({model, http.headers, promptCacheKey, system, messages, tools, toolChoice, generation, providerOptions}) with x-opencode-session-id/User-Agent headers. core/session/model-request.ts:L279--L302
- **Wire format:** per-provider protocol modules (anthropic-messages.ts, openai-responses.ts, bedrock-converse.ts, gemini.ts, ...) and framings (SSE) compile the canonical request and decode the stream. ai/protocols/ (directory), ai/route/client.ts:L423--L497
- **Route/header hooks:** session.model.request may change baseURL/headers; session.http.request/http.response wrap the raw HTTP exchange. core/session/model-request.ts:L305--L320, L337--L365
- **HTTP timeouts:** headers 300 s, inter-chunk 900 s, whole-request none by default. ai/route/transport/http.ts:L100--L146; defaults ai/schema/options.ts:L56, L62
- **Stateful WebSocket transport:** bound per session when model.transport === "websocket"; rotates after 55 min, 15 s connect, 30 min idle; after 5 consecutive stream failures the session stays on HTTP. core/session/model-request.ts:L368--L392; core/session/model-transport.ts:L22--L27, L186--L196, L308
- **Error classification:** isContextOverflowFailure and isRetryable (rate limit, provider internal, transport unless delivery rejected, incomplete stream, unknown; x-should-retry header overrides). ai/provider-error.ts:L72--L75, L81--L110
- **Retry policy:** exponential from 2 s capped at 10 s per gap, 10 retries, jittered; provider retry-after honored up to 15 min; at most 3 timeout retries; session.retry plugin hook may override the decision. core/session/runner/retry.ts:L31--L62, L64--L92
- **Retry vs continue vs recover-full:** before output started → Retry; after output started on an interrupted stream → Continue; WebSocket retry-full/rotate-and-retry-full → RecoverFull. core/session/runner/step.ts:L182--L204, L263--L271, L291--L295
- **Unknown finish reason is an error:** turned into InvalidProviderOutputError("incomplete-stream"). core/session/runner/step.ts:L171--L181
- **Model resolution:** session model must be in the available catalog, else ModelUnavailableError. core/session/runner/model.ts:L75--L91

- **Token usage and cost:** normalized `input/output/reasoning/cache.read/cache.write`, cost by context tier. `core/session/usage.ts:L11--L19, L22--L36`
- **Vercel AI SDK bridge:** `AISDK.Service.model(RuntimeInfo)` wraps a `LanguageModelV3` as an `@opencode/ai LanguageModel` for third-party provider packages. `core/aisdk.ts:L237--L252`
- **Backup model for a request that could not be served (title only):** if the small model's attempt fails, the primary model is tried. `core/session/title.ts:L129--L133`

### Output

- **Streaming render path:** provider events are published as durable `Text.Started/Delta/Ended`, `Reasoning.*`, `Tool.Called/Progress/Success/Failed`, `Step.Started/Streamed/Ended/Failed`; deltas batched every 100 ms. `core/session/runner/publish-llm-event.ts:L443--L470, L106, L194--L200`
- **Concurrent tool execution from one response:** each `tool-call` event forks a fiber immediately; all are awaited after the stream ends. `core/session/runner/step.ts:L121--L134, L151`
- **Tool executor:** `execute.before` hook → input decode/validate → `tool.execute` → `execute.after` hook → content normalization (image resize). `core/tool.ts:L103--L156, L263--L284`; validation `core/tool/runtime.ts:L29--L61, L88--L100`
- **Shell tool:** 2-minute default timeout, `AGENT=1 OPENCODE=1` env, jobs with foreground block or background; no OS sandbox found, gating is by permission after tree-sitter parsing of the command into per-command resources and touched directories. `core/tool/plugin/shell.ts:L24, L117--L155, L199--L279`; parser `core/shell/parse.ts:L7, L392`
- **Edit tool asks with a diff preview:** `permission.assert({action: "edit", metadata: {files: [preview]}})` before applying. `core/tool/plugin/edit.ts:L271--L283`
- **Read/external-directory gating:** `authorizeRead` asks `external_directory` for paths outside the location, then `read` per file. `core/file-access.ts:L125--L142, L144--L161`
- **MCP tools as output:** each discovered MCP tool is registered under a server namespace, permission-asserted, then `callTool`. `core/tool/mcp.ts:L41--L77`
- **Code Mode `execute` tool:** the model submits JavaScript; tool calls inside it are routed through the same `executeTool` (hooks + permissions); `fetch` only, no imports/fs/timers. `core/codemode/tool.ts: L62--L69, L71--L108`; sandbox = custom tree-walking interpreter over acorn AST with value limits and optional timeout. `codemode/interpreter/execute.ts:L1, L56--L68`; `codemode/interpreter/limits.ts:L6--L13`
- **File-change artifact for the user:** a git-backed snapshot is captured before and after each step and the changed paths are attached to `Step.Ended`. `core/session/runner/step.ts:L74--L76, L234--L260`; `core/snapshot.ts:L43--L68, L90--L105`
- **Question tool shows a form:** `forms.ask` with one field per question, "Type your own answer" custom option. `core/tool/plugin/question.ts:L62--L112, L119--L132`
- **Tool failure settlement:** interrupted, declined, provider-failed, and result-missing calls are failed with distinct errors so no call stays unsettled. `core/session/runner/step.ts:L62--L64, L207--L232`

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Input, Context, Control flow | SQLite via drizzle: `session_v2` (model, agent, permissions, token totals, execution claim), `session_message`, inbox; a durable, per-aggregate sequenced event log projected into those tables; execution claim written at turn start and released only by a terminal event (shutdown keeps it for resume); "always" permission replies saved per project; git snapshots for revert; tool-output spill files kept 7 days | `core/session/sql.ts:L22--L77`; `core/bus.ts:L21--L48`; `core/session/projector.ts:L1--L35`; `core/session/execution.ts:L76--L86`, `L111--L148`; `core/session/store.ts:L62--L74`; `core/permission.ts:L299--L305`; `core/session/revert.ts:L21--L40`; `core/tool-output.ts:L15` |
| Safety & security | Output (gates consumed by Control flow) | Rulesets `{action, resource, effect}` with last-match-wins and default `ask`; defaults ask on `*.env`; sub-agent profiles carry deny lists; shell commands parsed into per-command resources; external directories need approval; declines tunnel as defects so a "no" can never become model-visible tool output; compaction checkpoint tells the model it is "not new instructions"; `<system-reminder>` blocks declared harness-owned; sub-agent depth limit; Code Mode interpreter has no fs/imports; no OS-level sandbox found; secrets redacted only in test cassettes and scrubbed from the test env | `core/permission.ts:L87--L101`; `schema/agent.ts:L46--L51`; `core/plugin/agent.ts:L104--L117`; `core/shell/parse.ts:L18--L21`; `core/file-access.ts:L125--L142`; `core/permission.ts:L247--L253`, `core/session/model-request.ts:L400--L418`; `core/session/runner/to-llm-message.ts:L316--L321`; `core/session/runner/prompt/system.txt:L5`; `core/tool/plugin/subagent.ts:L129--L133`; `core/codemode/tool.ts:L63--L64`; `recorder/redaction/secrets.ts:L3--L14`; `packages/core/script/test.ts:L10--L22` |
| Observability | Model interface, Output | Durable event log doubles as the trace; per-step tokens/cost on `Step.Ended`; per-session token/cost columns and a stats aggregator; named Effect spans on every service function (`Effect.fn("SessionRunner.drain")`); WebSocket lifecycle counter metric; warning logs on overflow and transport fallback; event logger for catalog updates | `core/session/runner/step.ts:L246--L260`; `core/session/sql.ts:L45--L50`; `core/session/stats.ts:L17--L46`; `core/session/runner/llm.ts:L57`; `core/session/model-transport.ts:L28--L33`; `core/session/runner/step.ts:L161--L165`; `core/event-logger.ts:L7--L17` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Resilience | Model interface, Context, Control flow | Jittered exponential retry with retry-after and timeout caps; overflow-triggered compaction with shrinking targets; output-limit and interrupted-stream continuations; WebSocket→HTTP fallback after 5 failures; header/chunk timeouts; crash recovery from the execution claim with a 10-attempt budget; stale running compactions and tool calls settled on resume; pending permission asks rejected on shutdown | `core/session/runner/retry.ts:L42--L62`; `core/session/compaction.ts:L95`, `L429--L448`; `core/session/runner/llm.ts:L281--L319`; `core/session/model-transport.ts:L186--L196`; `ai/route/transport/http.ts:L100--L146`; `core/session/execution/restart.ts:L77--L99`; `core/session/runner/llm.ts:L332--L385`; `core/permission.ts:L133--L146` |
| Performance | Context, Model interface, Output | Baseline instructions ordered as a stable cache prefix; `promptCacheKey` shared across a session tree; tool definitions kept on the budget-exhausted step to keep the cached prefix; parallel context source loading; tool calls run concurrently as they stream; 100 ms delta batching; start snapshot captured while the request is in flight; Code Mode catalog and tool JSON-schema caches; tool output truncation and image byte budget | `core/session/context.ts:L147--L158`, `L142`; `core/session/model-request.ts:L295`; `core/session/runner/llm.ts:L243--L244`; `core/session/runner/step.ts:L121--L134`, `L74--L76`; `core/session/runner/publish-llm-event.ts:L106`; `core/tool.ts:L246--L254`; `core/tool/runtime.ts:L10--L19`; `core/tool-output.ts:L13--L14`; `core/session/model-request.ts:L43--L44` |
| Evaluation | — (all primitives) | 164 test files under `packages/core/test` (e.g. `session-runner-recorded.test.ts` replays provider traffic from `@opencode/http-recorder` cassettes; `session-compaction.test.ts`, `permission.test.ts`, `session-run-coordinator.test.ts`); 52 under `packages/ai/test`; a simulated provider package; test runner scrubs provider keys from the environment. Model-graded or task-success agent evals: not found | `packages/core/test/session-runner-recorded.test.ts:L1--L30`; `packages/simulation/src/backend/simulated-provider.ts`; `packages/core/script/test.ts:L10--L22` |

**Design decisions (one line each: the choice this harness made within a primitive)**

- Input: every input, including harness-generated notices, is a durable inbox row with a `steer` (interrupt the current step boundary) or `queue` (next turn) delivery, promoted only at step boundaries under a per-session lock.
- Context: history is event-sourced and re-projected into model messages on every call; compaction is a

summary message in that history (or an opaque provider checkpoint), triggered at ~90 % of the window using provider-measured counts plus a local estimate, with the newest 15k tokens kept verbatim.
- Model interface: an in-house canonical `LLMRequest` compiled per provider protocol, streamed over HTTP/SSE or a stateful WebSocket, with retry decisions split by whether output had already started.
- Output: tool calls are executed concurrently as they stream, every tool is permission-asserted inside its own code, most tools are reachable only through a single `execute` tool that runs model-written JavaScript in a custom interpreter.
- Control flow: one fiber per session drains a durable inbox; a step continues while tool calls were made, stops on a text-only finish, and is bounded by an optional per-agent step count; sub-agents are full child sessions (depth 1 by default) whose results come back as tool text or a synthetic message.

### Hard cases

- **Code Mode `execute` tool:** the model writes a program that sequences tool calls (`tools.ns["name"](input)`), which looked like control flow; rule "a program that sequences only tools is output" resolves it as output, and the interpreter cannot call the model (`core/codemode/tool.ts:L62--L69`). Its per-request catalog in the system prompt is context (`core/codemode/instructions.ts:L19--L30`).
- **`Permission.assert` lives inside each tool:** by file location it is output; by function it is an approval gate the harness consumes (the model never sees a plain decline, the step is interrupted), so it is control flow; the `"always"` rule save is persistence.
- **Question tool:** the form is output, the answers are input (the model consumes them), and a dismissal is treated like a permission decline (control flow) via the same defect tunnel (`core/tool/plugin/question.ts:L90--L97`).
- **Subagent tool:** the call is output; depth check, child creation, blocking and backgrounding are control flow; the child is a separate harness (`sessions.resume`); foreground results return as tool text, background results as a synthetic user message, which makes the same information input in one mode and output in the other.
- **Compaction service calls the model:** `SessionCompaction` decides when and what to summarize (context) but also issues `llm.stream` itself (`core/session/compaction.ts:L512`) and runs the retry policy (`L450--L459`). The brief says context "never calls the model"; resolved by assigning the trigger/split/budget to context, the summary request itself to model interface, and the `Compacted` → `re-attempt` loop to control flow. Partly DOES NOT FIT: native compaction (`L318--L332`) delegates context assembly to the provider, so the harness's context primitive holds only an opaque item it cannot inspect.
- **`MAX_STEPS_PROMPT` appended as an assistant message:** the step budget is control flow, but it is enforced by editing what the model sees (an assistant-role message plus `toolChoice: "none"`), so the mechanism straddles control flow and context (`core/session/runner/llm.ts:L240--L244`).
- **Instruction epochs:** changed AGENTS.md/date/env are frozen into durable `InstructionsUpdated` events rendered as system messages in the transcript; this is context, but it is implemented as persistence of history, and history is immutable, so stale instruction text stays in old events.
- **Plugin hooks (`session.context`, `session.model.request`, `http.request`, `tool.execute.before/after`, `permission.evaluate`, `session.retry`):** a single hook bus whose entries belong to five different primitives; decomposed by what each hook can change rather than where it is registered.
- **Durable event bus:** the same event log is persistence (projected into tables), observability (the trace), and the output channel (clients subscribe); the bus itself is out of scope as transport, its contents are assigned by what they record.
- **Title model fallback:** the "try the small model, then the primary" fallback (`core/session/title.ts:L129--L133`) is a backup model when a request could not be served (model interface), while choosing the small model in the first place is control flow.
- **WebSocket `RecoverFull`:** the transport reports it cannot replay an incremental exchange (model interface), but the decision to resend the full request is taken by the runner's outcome loop (control flow) (`core/session/runner/step.ts:L182--L188`, `core/session/runner/llm.ts:L324--L326`).
- **Synthetic messages:** "The server restarted…", "Your last response hit the output token limit…" are produced by control flow, admitted as input, and shown to the model as user-role messages; assigned as input with control-flow origin.

### Summary row

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| Durable inbox rows: user, synthetic, shell, skill; steer or queue | Event-sourced history re-projected; cached-prefix instructions; summary or native compaction | In-house LLMRequest compiled per protocol; SSE or WebSocket; split retry policy | Concurrent streamed tool fibers; permission-asserted tools; Code Mode interpreter | One fiber per session drains inbox; continue while tools called; child sessions |

## C.4 pi (pi-mono coding agent)

**Source:** https://github.com/badlogic/pi-mono · **Version:** commit 5ab7fe74c400097b386662205cd93900695b8a71 (2026-10-15) · **Language:** TypeScript (Node/Bun) **Path conventions:** `agent/` = `packages/agent/src/`; `ai/` = `packages/ai/src/`; `ca/` = `packages/coding-agent/src/`; `ex/` = `packages/coding-agent/examples/extensions/`; `ev/` = `packages/evals/`. All paths relative to the repo root. Package scope at this commit is `@earendil-works/*` (pi-agent-core, pi-ai, pi-coding-agent). **What it is:** pi is a terminal coding agent by Mario Zechner (badlogic) with a layered design: `pi-agent-core` holds a minimal agent loop, `pi-ai` holds provider adapters (Anthropic, OpenAI, Google, Bedrock, Mistral, ...) behind one `StreamFunction`, and `pi-coding-agent` wraps both in an `AgentSession` with JSONL session trees, compaction, retries, built-in tools, and an extension system; interactive TUI, print, JSON, and RPC modes sit on top. The model is reached over HTTPS/SSE (or WebSocket for some providers) through provider SDKs or raw fetch, with `maxRetries: 0` so pi owns retry policy.

### Control flow

- **Inner/outer loop (`runLoop`):** inner `while (hasMoreToolCalls || pendingMessages.length > 0)` issues one model call per iteration; outer `while (true)` re-enters only if follow-up messages arrived or `finishTurn` asked for one more context-only turn, else breaks and emits `agent_end`. `agent/agent-loop.ts:L163--L321`
- **Stop on error/abort:** an assistant `stopReason` of `"error"` or `"aborted"` is a hard exit: `finishTurn` is called, `turn_end`/`agent_end` emitted, loop returns. `agent/agent-loop.ts:L245--L256`
- **Go-again decision:** `hasMoreToolCalls = !executedToolBatch.terminate`; if the response had no tool calls and no steering/follow-up is queued, the loop ends. `agent/agent-loop.ts:L258--L278`, `L295--L318`
- **`finishTurn` decision hook:** returns `{action:"end"}` (stop now) or `{action:"continue"}` (force one more request); undefined keeps normal scheduling. `agent/agent-loop.ts:L286--L298`, `agent/types.ts:L146--L158`
- **Early termination from tools:** a batch ends the run only when every finalized tool result sets `terminate: true` (`shouldTerminateToolBatch`); a blocked `beforeToolCall` may set it too. `agent/agent-loop.ts:L690--L692`, `L745--L755`, `agent/types.ts:L66--L74`
- **Steering vs. follow-up queues:** `Agent.steer()` injects after the current turn's tools finish; `followUp()` waits until the agent would otherwise stop; drain modes `"one-at-a-time"` (default) or `"all"`. `agent/agent.ts:L143--L174`, `L247--L248`, `L299--L306`, `L496--L503`; polled in the loop at `agent/agent-loop.ts:L176`, `L204--L206`, `L295`, `L302--L308`
- **One run at a time:** `prompt()` throws if a run is active; `continue()` resumes from a user/toolResult tail, else drains steering/follow-up queues. `agent/agent.ts:L373--L381`, `L384--L411`, `L507--L530`
- **Abort:** `Agent.abort()` fires the run's `AbortController`; `AgentSession.abort()` also cancels retry backoff, compaction, branch summary, then waits for idle. `agent/agent.ts:L341--L343`; `ca/core/agent-session.ts:L2464--L2474`
- **Session post-run loop:** `_startRun` calls `agent.prompt(messages)` then loops `_handlePostAgentRun()` → `agent.continue()` while retry/overflow-recovery/queued messages/extension `agent_before_settle` ask for more. `ca/core/agent-session.ts:L1837--L1882`, `L1943--L1981`, `L1983--L2005`

- **Auto-retry budget (session level):** `_prepareRetry` increments `_retryAttempt`, stops past `maxRetries`, sleeps `retryDelayMs()` (abortable), omits the failed assistant message from projection, and returns true so the loop re-issues the request. Defaults `enabled: true, maxRetries: 3, baseDelayMs: 2000`. `ca/core/agent-session.ts:L3792--L3832`; `ca/core/settings-defaults.ts:L18--L26`
- **Overflow recovery is one-shot:** `_checkCompaction` case 1 compacts and retries the turn once (`_overflowRecoveryAttempted` guard); case 2 (completed response) compacts without retry; case 3 threshold compacts without retry. `ca/core/agent-session.ts:L2977--L3115`
- **Virtual-model routing keyed to the work:** `prepareRequest` resolves a virtual model to a physical one per request with `reason: "user" | "continuation" | "retry"`, storing router state as a session entry; a router (extension-registered) picks the model. `ca/core/agent-session.ts:L815--L843`; `ca/core/model-runtime.ts:L1003--L1034`; `ca/core/virtual-models.ts:L1--L11, L43--L50`
- **Extension boundary hooks that can force another turn:** `turn_end` and `agent_before_settle` handlers return `continue: true` and may commit boundary drafts; the session validates `canContinue` first. `ca/core/agent-session.ts:L847--L897, L1983--L2005`
- **`tool_call` approval gate (extension):** `beforeToolCall` → `runner.emitToolCall`; first handler returning `{block:true}` wins; the `permission-gate` example asks `ctx.ui.select` and blocks when no UI. `ca/core/agent-session.ts:L659--L682`; `ca/core/extensions/runner.ts:L1249--L1267`; `ex/permission-gate.ts:L13--L33`
- **Project trust gate:** `resolveProjectTrusted` checks a persisted trust store, an extension `project_trust` handler, a `defaultProjectTrust` setting ("ask" default), then prompts; untrusted cwd means project `.pi/` extensions/skills/prompts/SYSTEM.md are not loaded. `ca/core/project-trust.ts:L47--L97`; `ca/core/trust-manager.ts:L30--L39, L45--L59, L67--L97`
- **Sub-agents (example extension, not core):** the `subagent` tool spawns a separate `pi --mode json -p --no-session` process per agent, in `single`, `parallel`, or `chain` modes, and returns the child's final output as the tool result (depth 1; results return as tool-result text). `ex/subagent/index.ts:L298, L342--L348, L470--L475`; `ex/subagent/README.md`
- **Nested tool calls from tools:** `ctx.executeTool()` runs through `runToolCall` with the session's hooks; ids are `<parent>/<n>` and never enter the transcript (sequences only tools → output by rule, but concurrency is serialized by a queue). `ca/core/nested-tool-calls.ts:L161--L200`; `agent/agent-loop.ts:L811--L819`; `ca/core/extensions/types.ts:L384--L395`

### Input

- **TUI submit:** the editor's `onSubmit` trims text, handles slash commands locally, then `session.prompt(text)`; while streaming it calls `prompt(text, {streamingBehavior:"steer"})`; Alt+Enter queues `followUp`. `ca/modes/interactive/interactive-mode.ts:L3200--L3202, L3364--L3385, L4508`
- **Print/JSON mode:** initial message plus `messages[]` are sent via `session.prompt()` sequentially. `ca/modes/print-mode.ts:L131--L137`
- **RPC mode:** JSONL commands `prompt`, `steer`, `follow_up`, `abort`, `clear_queue` map onto session methods with `source: "rpc"`. `ca/modes/rpc/rpc-mode.ts:L394--L433`
- **`@file` arguments:** CLI `@path` args become text content and image attachments before the first prompt. `ca/cli/file-processor.ts:L24--L25`
- **`AgentSession.prompt()` capture path:** extension-command check → `input` extension handlers (transform/handled) → `/skill:` and prompt-template expansion → queue if streaming, else auth check and `_startRun`. `ca/core/agent-session.ts:L2058--L2141, L2007--L2025`; `ca/core/extensions/runner.ts:L1525--L1564`; `ca/core/extensions/types.ts:L1171--L1187`
- **Queue capture:** `_queueSteer`/`_queueFollowUp` build a `user` message (text + images) and hand it to `agent.steer`/`followUp`. `ca/core/agent-session.ts:L2241--L2293`
- **Prompt normalization in core:** a string prompt becomes `{role:"user", content:[text, ...images]}`. `agent/agent.ts:L413--L430`
- **Tool results as input:** each executed call becomes a `toolResult` message (`content`, `details`, `isError`, `durationMs`) pushed into context. `agent/agent-loop.ts:L930--L944, L274--L277`
- **Streamed tool output:** `onUpdate` partial results flow out as `tool_execution_update` events (bash emits snapshots of its `OutputAccumulator`). `agent/agent-loop.ts:L779--L788, L821--L857`; `ca/core/`

tools/bash.ts:L276--L294

- **User !bash results:** executeBash runs a user command outside the model loop and records a bashExecution message that convertToLlm renders as a user message (unless !! excluded). ca/core/agent-session.ts:L3871--L3903; ca/core/messages.ts:L152--L161
- **Extension-injected messages:** before_agent_start handlers may return a custom message appended after the user prompt; pi.sendUserMessage starts a run. ca/core/agent-session.ts:L1926--L1936; ca/core/extensions/runner.ts:L1427--L1477
- **Image inputs:** prompt images are resized per the model's inputLimits before being attached. ca/core/agent-session.ts:L2027--L2040

### Context

- **System prompt sections:** buildSystemPromptSections produces ordered sections preamble, tools, rules, docs, addendum, project_context, skills, cwd, plus custom sections, each wrapped in a same-named XML tag so later patches can replace one section. ca/core/system-prompt.ts:L128--L193
- **Default preamble/rules text:** "You are an expert coding assistant operating inside pi..."; rules merge per-tool guidelines, extension guidelines, and two fixed rules. ca/core/system-prompt.ts:L155--L169, L88--L125
- **Prompt diffing into the transcript:** diffSystemPromptSections yields a SystemMessage.sections patch only when changed; _preparePromptAndToolLoadout appends it before the next request. ca/core/system-prompt.ts:L217--L229; ca/core/agent-session.ts:L1743--L1755, L899--L933
- **Project instructions:** AGENTS.override.md/AGENTS.md/CLAUDE.md from ~/.pi/agent and every ancestor of cwd (worktree shadowing handled), rendered as <project_instructions path=...>. ca/core/resource-loader.ts:L184--L203, L232--L274; ca/core/system-prompt.ts:L79--L86
- **SYSTEM.md / APPEND_SYSTEM.md overrides:** project .pi/SYSTEM.md replaces the preamble; APPEND_SYSTEM.md becomes addendum. ca/core/resource-loader.ts:L1214--L1233; ca/core/agent-session.ts:L1714--L1730
- **Skills in prompt:** only name/description/location of each SKILL.md is listed under <available_skills>; the model loads the file with read. ca/core/skills.ts:L358--L388
- **Tool declarations as transcript deltas:** declareToolChanges computes toolsAdded/toolsRemoved on a system message before each request so replay yields exactly context.tools. agent/agent-loop.ts:L333--L363
- **Tool loadout/exposure:** tools have exposure direct | model-only | codemode | deferred | hidden; tool_search (BM25) activates deferred tools for the next request; hidden declarations are stripped from every request. ca/core/extensions/types.ts:L516; ca/extensions/tool-search/index.ts:L14, ca/extensions/tool-search/tool.ts:L192--L206; ca/core/agent-session.ts:L1772--L1790
- **Canonical context = session projection:** prepareRequest replaces context.messages with sessionManager.buildSessionProjection().messages, which walks the tree path, applies the newest compaction (firstKeptEntryId) and context_edit replacements. ca/core/agent-session.ts:L788--L811; ca/core/session-manager.ts:L476--L512, L543--L573
- **convertToLlm:** maps bashExecution, custom, branchSummary, compactionSummary roles to user messages with prefixes and passes system/user/assistant/toolResult through. ca/core/messages.ts:L148--L196
- **Extension context / context_with_system hooks:** run inside transformContext before every call; context sees non-system messages (clone), context_with_system owns the full list. ca/core/sdk.ts:L436--L440; ca/core/extensions/runner.ts:L1305--L1366; agent/agent-loop.ts:L388--L397
- **Forced prompt projection:** a before_agent_start systemPrompt override collapses system messages into one head for the request only; the transcript keeps structured sections. ca/core/agent-session.ts:L1792--L1807
- **Compaction trigger:** shouldCompact fires when contextTokens > contextWindow - reserveTokens (defaults reserveTokens 16384, keepRecentTokens 20000); checked before the next assistant response and after the run. ca/core/compaction/compaction.ts:L120--L130, L267--L270; ca/core/agent-session.ts:L768--L786, L3079--L3113
- **Cut point:** walks back from newest accumulating estimated tokens until keepRecentTokens, then picks

the nearest valid cut (turn boundary; can split a turn and summarize its prefix). ca/core/compaction/compaction.ts:L446--L480, L872--L905
- **Summary generation:** a separate model call with a fixed structured prompt (## Goal, ## Constraints & Preferences, ## Progress ...), update prompt when a previous summary exists, cacheRetention: "none", wrapped in retryAssistantCall. ca/core/compaction/compaction.ts:L507--L520, L619--L639, L682--L693
- **Compaction persisted as a tree entry:** appendCompaction stores summary, firstKeptEntryId, tokensBefore, usage, and the current system message. ca/core/session-manager.ts:L1261--L1287
- **Token estimation:** provider usage.totalTokens when present, else chars / 3.5 and 4800 chars per image. ai/utils/estimate.ts:L15--L19, L38--L40
- **Tool-output trimming:** read/bash truncate to 2000 lines or 50 KB (head for read, tail for bash) with full output spilled to a temp file. ca/core/tools/truncate.ts:L11--L12; ca/core/tools/bash.ts:L255; ca/core/tools/read.ts:L96; ca/core/bash-executor.ts:L56--L104, L130--L145
- **Image policy in context:** blockImages setting strips images in convertToLlm; tool-result images are resized per model limits after tool_result hooks. ca/core/sdk.ts:L296--L310; ca/core/agent-session.ts:L705--L711
- **Branch summaries:** navigating the tree back can append a branch_summary entry rendered as a user message. ca/core/session-manager.ts:L1600--L1615; ca/core/messages.ts:L170--L175

### Model interface

- **StreamFunction contract:** (model, normalizedTranscript, options) => AssistantMessageEventStream; errors must terminate the stream with stopReason "error"|"aborted". ai/types.ts:L379--L394
- **Loop → stream boundary:** streamAssistantResponse resolves the API key via getApiKey (expiring tokens), calls streamFunction, and folds start/*_delta/done/error events into a partial AssistantMessage in context. agent/agent-loop.ts:L399--L407, L414--L458
- **Session stream function:** modelRuntime.streamSimple → lazyStream (async setup behind a sync stream) → prepareRequest (auth/headers) → provider streamSimple. ca/core/sdk.ts:L420--L431; ca/core/model-runtime.ts:L723--L749; ai/api/lazy.ts:L46--L62; ai/models.ts:L907--L914
- **Anthropic adapter:** builds a @anthropic-ai/sdk client (API key or OAuth bearer with Claude Code identity headers), calls client.beta.messages.create(params, {maxRetries: 0, signal, timeout}), decodes SSE by hand. ai/api/anthropic-messages.ts:L603--L645, L1018--L1060, L475--L500, L534
- **Provider-level retry:** retryProviderRequest mirrors SDK policy (408/409/429/5xx or x-should-retry), honors retry-after(-ms) up to maxRetryDelayMs (default 60 s, else fail fast so the outer policy sees it), otherwise min(0.5·2^n, 8)s with 25 % jitter, abortable. ai/utils/provider-retry.ts:L25--L37, L53--L69, L107--L128; ai/api/anthropic-messages.ts:L658--L666
- **Assistant-level retry classifier:** isRetryableAssistantError matches overloaded/429/5xx/network text but excludes quota/billing patterns; retryAssistantCall does bounded exponential backoff baseDelayMs·2^(n-1) capped at 60 s. ai/utils/retry.ts:L7--L28, L30--L60, L121--L137, L196--L246, L257--L262
- **Overflow detection:** regex table over provider error texts (prompt is too long, request_too_large, ...) with non-overflow exclusions (rate limit), used to divert to compaction instead of retry. ai/utils/overflow.ts:L37--L63, L76--L80; ca/core/agent-session.ts:L3739--L3743
- **Strict-tools fallback within the adapter:** a 400 "compiled grammar too large"/"Too many strict tools" rebuilds params without preferred strict schemas and retries once. ai/api/anthropic-messages.ts:L667--L691
- **Stream termination:** abort → stopReason "aborted", exceptions → "error" with errorMessage; done carries the final message with usage and cost. ai/api/anthropic-messages.ts:L897--L933; ai/utils/event-stream.ts:L26--L58
- **Prompt caching:** cache_control on the last user/system block and last tool declaration; retention short/long/none; sessionId forwarded for cache-aware backends. ai/api/anthropic-messages.ts:L1529--L1555, L1659; ai/types.ts:L239--L249

- **Thinking mapping:** pi `ThinkingLevel` maps to Anthropic effort or `budget_tokens` (adaptive vs. budget thinking), interleaved-thinking beta header. `ai/api/anthropic-messages.ts:L944--L1012`, `L1296--L1305`, `L195`
- **Wire-level hooks:** `onPayload` (extension `before_provider_request` may replace the body), `onResponse` (`after_provider_response`), `onProviderStreamEvent`. `ai/api/anthropic-messages.ts:L653--L656`, `L693`, `L700`; `ca/core/sdk.ts:L382--L409`; `ai/types.ts:L174--L182`, `L218--L223`
- **Transport/options:** `transport: "sse"|"websocket"|...`, `timeoutMs`, `maxRetries`, `maxRetryDelayMs`, custom `fetch`, custom `headers`. `ai/types.ts:L144`, `L157--L210`, `L235--L238`
- **Credential handling:** OAuth tokens auto-refresh under a file lock with a timeout; `auth.json` written with mode `0o600` in a `0o700` dir. `ai/auth/resolve.ts:L119--L142`; `ca/core/auth-storage.ts:L37`, `L71`
- **Direct-route virtual model outside the loop:** `streamSimple` on a virtual model routes with `reason: "direct"` and withholds the caller's credentials when the provider differs. `ca/core/model-runtime.ts:L725--L743`

### Output

- **Tool execution pipeline:** `prepareToolCall` (lookup, `prepareArguments`, schema validation, `beforeToolCall`) → `executePreparedToolCall` (`tool.execute(id, args, signal, onUpdate)` in try/catch, duration measured) → `finalizeExecutedToolCall` (`afterToolCall` may replace content/details/isError). `agent/agent-loop.ts:L708--L777`, `L821--L857`, `L859--L910`
- **Parallel vs. sequential:** default `"parallel"`: preflight sequentially, run allowed tools concurrently with `Promise.all`, emit results in source order; `"sequential"` if configured or any called tool has `executionMode: "sequential"`. `agent/agent-loop.ts:L508--L523`, `L586--L660`; `agent/types.ts:L308--L317`
- **Truncated-output guard:** if the response `stopReason` is `"length"`, every tool call is failed with an error result instead of executed. `agent/agent-loop.ts:L263--L270`, `L478--L503`
- **Built-in tool set:** `read, bash, powershell, edit, write, grep, find, ls` (default four: read, bash, edit, write). `ca/core/tools/index.ts:L95--L105`; `ca/core/system-prompt.ts:L64`
- **bash executor:** `spawn(shell, args, {cwd, detached, env: process.env + bin dir})`, stdout/stderr streamed, optional per-call timeout, abort kills the whole process group with `SIGKILL` (`taskkill /T` on Windows). `ca/core/tools/bash.ts:L97--L166`; `ca/utils/shell.ts:L138--L150`, `L187--L215`
- **edit tool:** exact unique `oldText`→`newText` replacements against the original file, BOM/LF normalization, fuzzy-match fallback with diff details; file mutations serialized per real path. `ca/core/tools/edit.ts:L152`, `L190--L194`; `ca/core/tools/edit-diff.ts:L28--L37`; `ca/core/tools/file-mutation-queue.ts:L32--L61`
- **read tool:** text with `offset`/`limit` and head truncation; images returned as image blocks (resized) or omitted for non-vision models. `ca/core/tools/read.ts:L96`, `L159--L177`, `L80--L83`
- **grep/find:** spawn ripgrep (`rg`, downloaded if missing). `ca/core/tools/grep.ts:L119--L121`, `L168`
- **MCP tools:** the built-in `mcp` extension registers each connected server's tools via `pi.registerTool` (hidden exposure when only reachable from codemode) and forwards calls with `client.callTool`. `ca/extensions/mcp/index.ts:L387--L460`; `ca/extensions/mcp/tools.ts:L337--L339`
- **codemode tool:** the model writes JavaScript that calls other tools; it runs in a QuickJS WASM sandbox (256 MiB heap, 2 workers max, 10k-token output cap, optional `timeout_ms`), with no Node/fs/network; nested calls use `ctx.executeTool`. `ca/extensions/codemode/tool.ts:L1--L10`, `L118--L120`; `ca/extensions/codemode/execute.ts:L50--L59`, `L254`, `L510--L522`
- **Rendering streamed fragments:** the TUI updates a streaming assistant component on each `message_update` and creates tool components as tool calls appear; JSON mode writes each event as one JSON line; print mode prints final assistant text. `ca/modes/interactive/interactive-mode.ts:L3524--L3536`; `ca/modes/print-mode.ts:L108--L112`, `L139--L158`
- **Extension UI as output:** `ctx.ui.select/confirm/input/editor` render dialogs (also used by approval gates). `ca/core/extensions/types.ts:L149--L157`, `L240`

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Context (session transcript) | Append-only JSONL under `~/.pi/agent/sessions/--<cwd>--/<ts>_<id>.jsonl`, v3 entries with `id`/`parentId` tree; file created on first user/assistant message; branches via `branch()`; model/thinking changes and usage persisted as entries; `context_edit` entries persist recovery omissions | `ca/core/session-manager.ts:L41, L589--L594, L1057--L1083, L1166--L1214, L1244, L1360, L1579--L1584; ca/core/agent-session.ts: L1147--L1168, L1241--L1257` |
| Safety & security | Control flow (gates), Output (sandbox), Model interface (secrets) | Project trust gate before loading repo-controlled extensions/skills; `tool_call` block hook (example permission gate); OS sandbox only via example extension (`@anthropic-ai/sandbox-runtime`: sandbox-exec/bubblewrap); codemode scripts in QuickJS with no fs/network; auth file 0600; bash inherits full `process.env` (no secret scrubbing found); nothing found for prompt-injection defenses beyond trust gating | `ca/core/project-trust.ts:L47--L97; ca/core/trust-manager.ts:L30--L39; ex/permission-gate.ts:L13--L33; ex/sandbox/index.ts:L1--L10; ca/extensions/codemode/tool.ts:L118; ca/core/auth-storage.ts:L37; ca/utils/shell.ts:L138--L150` |
| Observability | (observability) | Lifecycle events (`agent_start/turn_*/message_*/tool_execution_*`) to listeners/extensions; `usage` (input/output/cache/cost) on every assistant message and `usage` session entries; `getContextUsage`/`getSessionStats`; crash log; redacted bug-report bundle; `TelemetryContext` span interface passed through options but no `startSpan` call found in `ai/` | `agent/agent.ts:L565--L612; ai/types.ts:L451--L470, L583; ca/core/session-manager.ts: L1244; ca/core/agent-session.ts:L4267--L4290; ca/core/crash-log.ts:L124--L144; ca/core/bug-report.ts:L27, L51; packages/telemetry/src/index.ts:L1--L22; ai/types.ts:L159--L160` |
| Resilience | Model interface (retries), Context (compaction), Control flow (recovery loop) | Two retry tiers (provider HTTP retry with retry-after cap; assistant-level exponential backoff, 3 attempts); overflow → compact-and-retry once; `length` stop fails tool calls; abortable sleeps; resume from JSONL on restart; compaction itself retried | `ai/utils/provider-retry.ts:L107--L128; ai/utils/retry.ts: L196--L246; ca/core/agent-session.ts:L3040--L3077, L3792--L3832; agent/agent-loop.ts: L263--L270; ca/core/compaction/compaction.ts:L619--L639` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Performance | Model interface (cache), Output (parallel tools), Context (trimming) | Prompt `cache_control` + `sessionId`; `CacheWarmer` re-sends a 1-token request at 90 % of cache TTL when expected savings $\geq$ \$0.05; parallel tool execution; streaming; 50 KB/2000-line tool output caps; lazy provider module loading | `ai/api/anthropic-messages.ts:L1529--L1555`; `ca/core/cache-warmer.ts:L15--L32`, `L284--L296`; `agent/agent-loop.ts:L586--L660`; `ca/core/tools/truncate.ts:L11--L12`; `ai/api/lazy.ts:L74--L80` |
| Evaluation | (evaluation) | 651 `*.test.ts` files (vitest) across packages (coding-agent 207, ai 180, agent 5), incl. agent-session retry/compaction/branching suites; `packages/evals` runs the real agent via `vitest-evals` harness with Docker variants and LLM-graded docs evals (`autoevals`) | `packages/coding-agent/test/` (listing); `ev/package.json:L1--L23`; `ev/evals/smoke.eval.ts:L5--L18`; `ev/src/harness.ts:L1--L50` |

### Design decisions (one line each: the choice this harness made within a primitive)

- Input: three capture surfaces (TUI editor, print/JSON CLI, JSONL RPC) all funnel into `AgentSession.prompt()`, with mid-run input split into "steer" (after this turn) vs. "follow-up" (after the agent would stop), one message at a time by default.
- Context: the transcript is a persisted tree; every request is a projection of the active branch (compaction entry + kept tail + context edits), the system prompt is a set of named XML sections diffed into system-message patches, and tool declarations are transcript deltas, so replaying the file reproduces the exact request.
- Model interface: one `StreamFunction` contract across ~15 provider adapters with SDK retries disabled (`maxRetries: 0`) and pi's own two-tier retry (HTTP-level with retry-after cap, assistant-level with error-text classification), overflow routed to compaction rather than retry.
- Output: tools are plain `{name, parameters (TypeBox), execute}` objects; parallel execution by default with per-file mutation serialization; no sandbox in core (OS sandboxing and permission prompts are extensions); a QuickJS-sandboxed `codemode` tool lets the model script tool calls.
- Control flow: a two-level loop in `pi-agent-core` (tool calls/steering inner, follow-ups outer) plus a session-level outer loop for retries/overflow recovery/extension continuations; sub-agents are not in core but an example spawns child `pi` processes; per-request model routing via virtual models.

### Hard cases

- **`before_provider_request` payload hook:** lives in the adapter (`onPayload`) at the wire boundary, but an extension can rewrite the request body, i.e. decide what the model is shown. By the function rule it is Context when it changes content; the transport callback itself stays Model interface. `ai/api/anthropic-messages.ts:L653--L656`; `ca/core/sdk.ts:L382--L386`
- **Virtual-model routing:** `prepareRequest` chooses the physical model per request by reason (user/-continuation/retry) — keyed to the work, so Control flow — even though it is implemented inside the model-runtime layer; the `reason: "retry"` branch after a failed request overlaps Model interface's "backup model" (resolved: the router decides based on work state, so Control flow). `ca/core/agent-session.ts:L815--L843`; `ca/core/model-runtime.ts:L1003--L1034`
- **Strict-tools 400 fallback:** an in-adapter request rebuild on a specific 400 (drops strict schemas). It alters the request contents, but the trigger is "request could not be served" — assigned Model interface by the unavailability rule. `ai/api/anthropic-messages.ts:L667--L691`
- **Compaction summary call:** a second model call issued by the Context primitive (summary generation).

Rule: Context never calls the model; here it does, through the same `StreamFunction`. Resolved as Context mechanism using the Model interface as a service; noted as a borderline.

- **Nested tool calls / codemode:** a script the model writes sequences only tools (Output by rule), yet it also reaches `models.*` (classifiers, image models) from inside the sandbox — a model call issued from inside a tool. `ca/extensions/codemode/tool.ts:L1--L10`; `ca/extensions/codemode/execute.ts:L50`. Partially DOES NOT FIT: a tool that calls models is neither pure Output nor Control flow in the given scheme.
- **`agent_before_settle` / `turn_end` boundaries:** extensions may commit "boundary drafts" (entries appended to the session: Context) and return `continue` (Control flow) in one hook; decomposed per rule 28. `ca/core/agent-session.ts:L847--L897`, `L1983--L2005`
- **`pi-durable` package:** a second, experimental harness in the same repo (durable task graph, child tasks, commit-before-show) sharing `pi-ai`; not deconstructed here beyond noting it exists. `packages/durable/README.md:L1--L6`, `L581`
- **Telemetry package:** span interface exists and `telemetryContext` is passed through options, but no span is started in `ai/`, `agent/`, or `ca/` at this commit — recorded as "interface present, emission not found". `packages/telemetry/src/index.ts:L1--L22`; `ai/api/simple-options.ts:L48`

### Summary row

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| TUI/CLI/RPC prompts, steer/follow-up queues, tool results | Tree-projected JSONL transcript, sectioned prompt, threshold compaction | One StreamFunction, SSE, two-tier retry, overflow classifier | Parallel TypeBox tools, bash/edit/read, QuickJS codemode | Inner tool loop, outer follow-up loop, retry/overflow re-entry, routing |

## C.5 Hermes Agent

**Source:** https://github.com/NousResearch/hermes-agent · **Version:** commit 7bf3766f3db72d4643d1a78d18e51c1d9ddf1d6d (2026-10-15) · **Language:** Python (TypeScript for the TUI/desktop front-ends, out of scope) **Path conventions:** paths are relative to the repo root (`hermes-agent/`). `L` = line. Clone read as untrusted data; nothing executed. **What it is:** A personal agent by Nous Research that runs one agent core (`AIAgent`) behind a prompt_toolkit CLI/TUI, ~25 messaging-platform gateway adapters (Telegram, Discord, Slack, ...), an OpenAI-compatible HTTP server, an ACP adapter, a cron scheduler and a desktop app. The model is reached through an OpenAI-SDK chat-completions client by default, with alternative wire transports for Anthropic Messages, OpenAI Responses ("codex") and Bedrock Converse selected by `api_mode`; providers are plugins and the model slug is config.

### Control flow

- **Turn entry + lease:** `AIAgent.run_conversation` builds a task/turn id and admits a durable turn lease before forwarding to the loop (one turn per session at a time). `agent/turn_facade.py:L22`, `agent/turn_facade.py:L83`
- **Main loop:** `_run_conversation_turn` runs `while (api_call_count < max_iterations and iteration_budget.remaining > 0) or _budget_grace_call`; each iteration = begin → prepare → assemble → preflight gate → API retry loop → normalize → `run_tool_round` if tool calls else `finish_text_response`; post-loop `finalize_turn`. `agent/conversation_loop.py:L1656--L1706`
- **Phase dispatch:** each phase is a sibling function returning a verdict with `action` ∈ {`fallthrough`, `continue`, `break`, `return`}; `_run_phase` copies verdict fields back into `_LoopState`. `agent/conversation_loop.py:L1389--L1502`
- **Inner retry loop:** `_run_api_retry_loop` loops `while retry_count < max_retries` over guard → build → call → check, with `handle_api_interrupt` / `handle_api_error` branches. `agent/conversation_loop.py:L1505--L1534`

- **Step budget:** `IterationBudget` is a thread-safe counter with `consume()`/`refund()`; `max_iterations` defaults to `sys.maxsize` (unlimited, shared with subagents); a tool round whose only tool was `execute_code` refunds its iteration; a reasoning-only stall that triggers a model fallback sets `_budget_grace_call` for one extra call. `agent/iteration_budget.py:L25--L56`, `run_agent.py:L241`, `agent/turn_tool_round.py:L199--L200`, `agent/turn_truncation.py:L630--L635`
- **Stop gates (go again vs. hand back):** on a text answer, `apply_stop_gates` runs verify-on-stop → `pre_verify` plugin hook → kanban stop guard; a nudge appends a user-role row and continues the loop, holding the answer as a budget-exhaustion fallback. `agent/turn_stop_gates.py:L105--L140`, `agent/turn_stop_gates.py:L35--L50`
- **Tool-call guardrail halt:** a `ToolCallGuardrailController` decision can end the turn (`guardrail_halt`) after a tool round; identical-call stall threshold is 3. `agent/turn_tool_round.py:L179--L193`, `agent/tool_guardrails.py:L30`, `agent/tool_guardrails.py:L307`
- **Approval gate (harness consumes the answer):** `check_all_command_guards` runs floor blocks → yolo/`off` bypass (process `--yolo`, session `/yolo`, `approvals.mode`) → permanent allowlist → unattended-context policy → `detect_dangerous_command` → `_human_decision`; answers are `once/session/always/deny/timeout`; in gateway ask mode the tool returns `status="pending_approval"` instead of blocking. `tools/approval.py:L1151--L1200`, `tools/approval.py:L506--L510`, `tools/approval.py:L796--L802`, `tools/approval_prompt.py:L37`, `tools/terminal_tool.py:L1019--L1028`
- **Interrupt / redirect crossing a response:** if a redirect arrived while the request was in flight the response is discarded and a restart is armed; `_apply_active_turn_redirect` then appends a checkpoint + correction row so role alternation holds. `agent/turn_api_call.py:L148--L167`, `agent/conversation_loop.py:L322--L340`
- **Gateway steer:** a message arriving while a turn runs is injected via `agent.steer()` or queued; after a tool batch, pending steer is folded onto tool results once the per-turn budget is enforced. `gateway/run_inbound.py:L649--L664`, `agent/tool_executor.py:L1209--L1220`
- **Delegation (sub-agents):** `delegate_task` refuses when `_delegate_depth >= delegation.max_spawn_depth` (default 1 = flat); children are full `AIAgent` instances built with `child_depth = parent + 1` and role orchestrator/leaf derived from depth. `tools/delegate_tool.py:L520--L527`, `tools/delegate_tool_config.py:L151--L159`, `tools/delegate_tool.py:L199--L234`
- **Child run, parallelism and result return:** `_run_single_child` runs a pre-built child on a worker thread, heartbeats the parent, and returns `{status ∈ completed|interrupted|failed, exit_reason}`; a batch runs on `DaemonThreadPoolExecutor(max_workers=batch.max_children)` polled with short waits; `background=True` returns a dispatch handle. `tools/delegate_tool.py:L342--L406`, `tools/delegate_tool_dispatch.py:L118--L145`, `tools/delegate_tool_dispatch.py:L470--L478`
- **Work-keyed model routing (auxiliary):** titling, compression summaries, vision etc. resolve via `_resolve_auto_route` (main model → per-task fallback chain → discovery chain); titles come from one small-model call off the critical path. `agent/auxiliary_client.py:L4659--L4684`, `agent/title_generator.py:L1--L5`
- **Mixture-of-Agents turn:** `/moa` gathers reference-model context before each iteration while the normal loop keeps tool calling. `agent/moa_loop.py:L1--L5`
- **Cron turns:** the scheduler runs `agent.run_conversation` on a worker thread under an inactivity watchdog (default 600 s) on a fresh `AIAgent(platform="cron")`. `cron/scheduler.py:L1961--L1963`, `cron/scheduler.py:L2496--L2525`
- **Codex app-server handoff:** when `api_mode == "codex_app_server"` the whole turn is handed to an external codex subprocess; on failure a fallback is activated and the generic loop retries the same user turn. `agent/conversation_loop.py:L1650--L1651`, `agent/conversation_loop.py:L1537--L1552`

### Input

- **CLI REPL:** a worker thread drains `_pending_input` (queue) and dispatches each input; `_looks_like_slash_command` routes to `process_command`, else `self.chat(...)`; `/skill-name` is rewritten into a user message. `hermes_cli/cli_tui_runtime_mixin.py:L31--L47`, `hermes_cli/cli_tui_runtime_mixin.py:L139`, `cli.py:L710`, `cli.py:L1300`
- **CLI chat turn:** `chat()` runs the agent on a daemon thread; input typed meanwhile goes to `_`

interrupt_queue and is re-queued as the next turn; the call is run_conversation(user_message, conversation_history[:-1], stream_callback, task_id, persist_user_message). hermes_cli/cli_chat_turn_mixin.py:L49--L60, hermes_cli/cli_chat_turn_mixin.py:L115--L118, hermes_cli/cli_chat_turn_mixin.py:L326--L330

- **Gateway event shape:** every adapter normalizes to MessageEvent(text, message_type, user_id, ...); adapters subclass BasePlatformAdapter; in-tree adapters (signal, webhook, bluebubbles, weixin, yuanbao, whatsapp_cloud, api_server) plus plugin adapters (telegram, discord, slack, matrix, teams, email, irc, ...). gateway/platforms/event.py:L45--L51, gateway/platforms/base.py:L1876, gateway/platforms/ (listing), plugins/platforms/ (listing)
- **Gateway inbound pipeline:** _handle_message = admit → pending-reply intercepts → authority admission → running-session handling (steer/interrupt/preempt) → idle commands → _handle_message_with_agent. gateway/run_inbound.py:L1196--L1300
- **Authorization and bot-loop budget at admission:** unauthorized users get a pairing code (DM) or are ignored before any session state is touched; bot-authored messages are capped by a sliding window per conversation. gateway/run_inbound.py:L201--L225, gateway/bot_loop_guard.py:L1--L5
- **Gateway → agent call:** the turn runner calls agent.run_conversation(api_message, **kwargs) with persist_user_message, display kind and the raw platform message id. gateway/run_turn_runner.py:L934--L936
- **HTTP and webhook input:** OpenAI-compatible aiohttp server (/v1/chat/completions, /v1/responses, /v1/runs, ...); webhook payloads are treated as untrusted and get no file/system execution toolset. gateway/platforms/api_server.py:L1--L5, toolsets.py:L41
- **ACP input:** the ACP agent calls agent.run_conversation(user_message=..., conversation_history=state.history). acp_adapter/server.py:L816--L819
- **Model asks the user (clarify):** clarify is an inline agent tool whose UI is a platform callback; the gateway intercepts the reply and its result returns to the model. tools/clarify_tool.py:L1--L6, tools/clarify_tool.py:L111, gateway/run_inbound.py:L339
- **Tool results captured:** every executed tool's result is committed as a role: tool row (_commit_tool_result); invalid names get an error tool row so each call has a matching result. agent/tool_executor.py:L1066, agent/turn_tool_round.py:L116--L121
- **Multimodal:** native image parts attached to the user turn stay visible to vision_analyze for the turn. agent/conversation_loop.py:L1749

### Context

- **User message staging:** messages = list(conversation_history) + staged user dict appended; current_turn_user_idx recorded. agent/turn_context.py:L1104--L1115
- **System prompt as three cache tiers:** stable (SOUL.md from HERMES_HOME or default identity, guidance, pinned skills, coding brief) → context (caller system_message, project context files, workspace) → volatile (skills index, memory, plugin sections, profile line, timestamp, runtime environment). Never re-rendered mid-session. agent/system_prompt.py:L730--L793, agent/system_prompt.py:L796--L805, agent/system_prompt.py:L544--L550, agent/prompt_builder.py:L1652--L1700
- **Project context files:** .hermes.md → AGENTS.md chain → CLAUDE.md → .cursorrules, loaded from cwd at startup only; cap 20 000 chars floor, dynamic 6 % of the window up to 500 000, overridable by context_file_max_chars. agent/prompt_builder.py:L1783--L1788, agent/system_prompt.py:L709--L722, agent/prompt_builder.py:L1156--L1171
- **Memory files:** MEMORY.md (2 200 chars) and USER.md (1 375 chars) are rendered into the volatile tier; the memory tool edits them. agent/system_prompt.py:L516--L541, tools/memory_tool_store.py:L186, tools/memory_tool_store.py:L618
- **External memory providers:** one plugin provider at a time; build_system_prompt() block plus per-turn prefetch_all(query) text injected into the request. agent/memory_manager.py:L1--L4, agent/memory_manager.py:L501--L511, agent/turn_context.py:L903--L931
- **Skills index + progressive disclosure:** names/descriptions in the prompt; skill_view loads a SKILL.md on demand as a tool result. agent/system_prompt.py:L301--L314, tools/skills_tool.py:L2--L5

- **Byte-stable prompt restored from DB:** on resume the stored `system_prompt` row is reused if it matches the runtime; rebuilt only after compression (`invalidate_system_prompt`). `agent/conversation_loop.py:L749--L772`, `agent/system_prompt.py:L808--L826`, `hermes_state_common.py:L377--L380`
- **Tool definitions:** `get_tool_definitions(enabled/disabled toolsets)` at init; `_HERMES_CORE_TOOLS` is the always-present core. `agent/agent_init.py:L1118`, `model_tools.py:L215`, `toolsets.py:L11--L39`
- **Tool Search deferral:** MCP/plugin and selected core tools are replaced in the model-visible array by `tool_search / tool_describe / tool_call` bridge tools, catalog rebuilt per assembly. `tools/tool_search.py:L1--L7`
- **Request assembly order:** `build_api_messages` → MoA context → canonicalization; "ORDER IS LOAD-BEARING" for cache breakpoints; system row prepended. `agent/turn_request_assembly.py:L106--L130`, `agent/turn_context.py:L1344`
- **Compaction trigger:** `ContextCompressor(threshold_percent=0.50, protect_first_n=3, protect_last_n=20)`; `should_compress_info` fires when `tokens >= threshold_tokens` unless in cooldown or the ineffective breaker; per-model threshold overrides. `agent/context_compressor.py:L2811--L2832`, `agent/context_compressor.py:L3006--L3015`, `agent/context_compressor.py:L2659--L2671`
- **Compaction algorithm:** prune old tool-result bodies (no LLM) → pick boundaries on the pruned copy → auxiliary-model summary → clean orphaned tool pairs. `agent/context_compressor.py:L5528--L5575`, `agent/context_compressor.py:L3285--L3300`
- **Where compaction runs:** turn-start compaction; a pre-API preflight gate comparing the assembled request with the threshold (insufficient progress blocks further passes); 413 / context-length provider errors compress and restart. `agent/turn_context_compaction.py:L129`, `agent/turn_preflight_gate.py:L20--L120`, `agent/turn_overflow.py:L1--L6`, `agent/turn_overflow.py:L463`
- **Gateway session hygiene:** a second compaction layer at 0.85 of the window between turns. `gateway/run_turn_hygiene.py:L94--L102`
- **Tool-output truncation:** 40 % head / 60 % tail with one marker; limits 50 000 bytes, 2 000 lines, 2 000 chars/line; `read_file` pages by char budget and reports `next_offset`. `tools/tool_output_truncate.py:L1--L12`, `tools/tool_output_limits.py:L13--L15`, `tools/file_tools.py:L96--L113`
- **Spill-to-file instead of truncating:** results over the per-tool threshold are written to `$HERMES_HOME/cache/spillover/{id}.txt` and replaced by preview + path; `enforce_turn_budget` caps the batch. `tools/tool_result_storage.py:L1--L6`, `tools/tool_result_storage.py:L293--L300`, `tools/tool_result_storage.py:L359`
- **Subdirectory hints:** AGENTS.md/CLAUDE.md found while navigating are appended to the tool result, never to the system prompt. `agent/subdirectory_hints.py:L1--L5`
- **Untrusted-result wrapping:** web/browser/MCP results ≥ 32 chars are wrapped in `<untrusted_tool_result>` with delimiter neutralization. `agent/tool_dispatch_helpers.py:L484--L495`, `agent/tool_dispatch_helpers.py:L572--L595`
- **History source on resume:** CLI loads `get_messages_as_conversation(session_id, repair_alternation=True)`; gateway loads `load_transcript` then hygiene. `hermes_cli/cli_agent_setup_mixin.py:L593--L596`, `hermes_state_messages.py:L1537--L1545`, `gateway/run_turn_prepare.py:L693--L700`
- **Pluggable context engine:** `context.engine` selects a plugin that owns `should_compress/compress` and may expose tools. `agent/context_engine.py:L1--L7`

### Model interface

- **Transport registry:** `anthropic`, `codex`, `chat_completions`, `bedrock` transports behind `ProviderTransport` (convert_messages/convert_tools/build_kwargs/normalize_response/validate_response). `agent/transports/__init__.py:L18--L31`, `agent/transports/base.py:L12--L51`
- **Client construction:** `_build_client` picks the Anthropic SDK, the MoA facade, boto3 Bedrock, or the OpenAI SDK client (shared httpx) by `api_mode`; GPT-5.x on a direct OpenAI URL auto-upgrades `chat_completions` → `codex_responses`. `agent/agent_init.py:L1029--L1042`, `agent/agent_init.py:L990--L1026`, `agent/agent_init.py:L482--L499`

- **Request kwargs:** `_build_api_kwargs_for_mode` branches anthropic / bedrock / codex / chat-completions builders. `agent/chat_completion_helpers.py:L1555--L1572`
- **Per-attempt request build:** re-apply reasoning echo for the provider, re-render prompt-cache decoration, strip images for rejecting models, LLM request middleware (trace kept), `pre_api_request` hook, optional `HERMES_DUMP_REQUESTS` dump. `agent/turn_api_request.py:L93--L205`, `agent/turn_api_request.py:L152--L153`
- **Prompt-cache decoration:** `build_prompt_cache_plan` places Anthropic `cache_control` markers; OpenAI-style history is converted to Anthropic content blocks. `agent/prompt_caching.py:L298--L308`, `agent/anthropic_message_convert.py:L1--L4`
- **Call issue:** `perform_api_call` chooses streaming (`_should_stream`: preferred even with no consumer, off for ACP/external-process providers) and runs through `run_llm_execution_middleware`. `agent/turn_api_call.py:L67--L167`, `agent/turn_api_call.py:L46--L64`
- **Streaming receive:** `_StreamingCall` issues `chat.completions.create(**stream_kwargs)` with `stream_options.include_usage`; the Anthropic path uses `messages.stream`; a cross-turn stale-stream breaker runs first. `agent/chat_completion_helpers.py:L3993--L4006`, `agent/chat_completion_helpers.py:L2734`, `agent/chat_completion_helpers.py:L2957--L2966`, `agent/chat_completion_helpers.py:L3412`
- **Non-stream call on a worker thread:** per-request client so interrupts close only that connection; a stale-call detector kills and raises. `agent/chat_completion_helpers.py:L1292--L1305`
- **Retry policy:** `api_max_retries` default 3; wait = provider `Retry-After` (capped) else `jittered_backoff(retry, base 2.0, max 60)` = min(base·2^(n-1), max) + jitter; adaptive policy for 429. `agent/agent_init.py:L1441--L1446`, `agent/turn_recovery.py:L1401--L1430`, `agent/retry_utils.py:L121--L130`, `agent/turn_api_error.py:L406--L417`
- **Error classification:** `classify_api_error` → structured `ClassifiedError` (retryable / should_compress / rotate credential / fallback); plugins may override via `transform_api_error_classification`. `agent/error_classifier.py:L974--L980`, `hermes_cli/plugins.py:L126--L130`
- **Backup model when unservable:** `try_activate_fallback` walks `_fallback_chain`, swapping client/model/provider in place, on classified failover or after retries exhaust; a persistent credential pool gives same-provider failover. `agent/chat_completion_helpers.py:L2086--L2121`, `agent/turn_api_error.py:L341`, `agent/turn_api_error.py:L380`, `agent/credential_pool.py:L1`
- **Post-exhaustion auto-recovery:** when retries and the chain are spent on a transient outage, the turn parks with a countdown and re-enters the retry loop (`auto_recovery_cycles` default 5). `agent/turn_recovery_autorecover.py:L1--L7`
- **Response validation + normalization:** `validate_response_shape` via the transport; `normalize_model_response` yields an assistant message with string content. `agent/turn_recovery.py:L1524--L1530`, `agent/turn_response_intake.py:L120--L128`
- **Degenerate-response ladders:** empty/thinking-only → partial-stream recovery → nudge → prefill continuation ×2 → budgeted retries → fallback provider; `finish_reason=length` → continuation nudges (up to 4), truncated tool-call retries with max_tokens boosts, repetition abort. `agent/turn_empty_response.py:L1--L5`, `agent/turn_truncation.py:L1--L6`, `agent/repetition_guard.py:L141`
- **Usage receipt:** `record_response_usage` folds `response.usage` into the compressor, session counters and state.db. `agent/turn_usage.py:L73--L79`

### Output

- **Tool set:** core tools (terminal, process_manage, read/write/patch/search files, web, vision, image, skills, browser_, *todo, memory, session_search, clarify, execute_code, delegate_task, cronjob, kanban_*, computer_use). `toolsets.py:L11--L39`
- **Dispatch precedence:** inline agent-level tools (`INLINE_TOOL_EXECUTORS` table) → `delegate_task` → context-engine tools → memory-provider tools → registry via `handle_function_call`. `agent/tool_executor.py:L1669--L1733`, `agent/inline_tool_executors.py:L253--L313`
- **Batch planning:** >1 calls are split into parallel/sequential segments; the parallel-safe set is read-only tools plus path-non-overlapping file ops. `run_agent.py:L1299--L1325`, `agent/tool_dispatch_helpers.py:L212--L257`, `agent/tool_dispatch_helpers.py:L33--L44`

- **Concurrent execution:** `_ConcurrentBatch` runs on `DaemonThreadPoolExecutor(max_workers)` with a batch deadline; results appended in call order. `agent/tool_executor.py:L1573--L1620`, `agent/tool_executor.py:L1500--L1501`
- **Registry call path:** `handle_function_call` → aliases → bridge unwrap → request middleware → `_pre_dispatch_guards` (`pre_tool_call` hook, fail-closed on timeout) → `registry.dispatch` → `post_tool_call` → `transform_tool_result`. `model_tools.py:L873--L967`, `model_tools.py:L823--L846`, `hermes_cli/plugins_dispatch.py:L51`
- **Terminal tool:** plan → acquire env → bounded pre-exec guard → approval guards → foreground or background spawn; description tells the model output is auto-truncated to a file. `tools/terminal_tool.py:L1360--L1475`, `tools/terminal_tool.py:L160--L165`
- **Execution backends (sandboxes):** local, docker, singularity/apptainer, modal, daytona, vercel_sandbox, ssh, plus plugin envs; built by `_ENV_BUILDERS`. `tools/terminal_tool_backends.py:L54`, `tools/terminal_tool_backends.py:L239--L241`
- **Programmatic tool calling:** `execute_code` runs a model-written Python script in a persistent per-conversation kernel that calls Hermes tools over RPC; only stdout returns. `tools/code_execution_tool.py:L1--L10`, `tools/code_kernel.py:L1--L5`
- **Browser:** agent-browser CLI over local headless Chromium, cloud backends, a user CDP endpoint or Camofox; accessibility-tree snapshots with `@eN` refs. `tools/browser_tool.py:L1--L8`
- **Persist-before-execute:** the assistant tool-call row is flushed to SQLite before any tool runs; a failed append ends the turn. `agent/turn_tool_round.py:L131--L153`
- **Rendering and delivery:** single-writer stream ownership with delta/reasoning fan-out; open stream boxes flushed before tools; gateway `delivery.py` routes responses to targets and `response_filters.py` decides whether a turn is delivered to the chat. `agent/stream_delivery.py:L1--L3`, `agent/turn_tool_round.py:L160--L164`, `gateway/delivery.py:L1--L2`, `gateway/response_filters.py:L1--L4`
- **Sub-agent spawn tool:** `delegate_task` (output) whose effect is the control-flow delegation above. `tools/delegate_tool.py:L488`

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Output (persist-before-execute), Context (prompt/history restore), Input (session resume) | SQLite `state.db` in WAL mode: `sessions`, `messages` (role/content/tool_calls/reasoning/...), `system_prompts`, `session_model_usage`, FTS5 search; `_flush_messages_to_session_db` dedups by marker; memory in `MEMORY.md/USER.md`; spillover files; trajectories | `hermes_state.py:L177`, `hermes_state.py:L460--L466`, `hermes_state_common.py:L451--L470`, `agent/session_persistence.py:L467--L511`, `agent/turn_finalizer.py:L608`, `agent/turn_finalizer.py:L635` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Safety & security | Output (sandbox, approvals, path denies), Context (injection scans, untrusted wrapping), Control flow (gate decision) | OS is the only boundary; hardline + dangerous command regexes with human gate; write-denied credential paths (`.ssh`, `.env`, ...); SSRF-safe URL policy; secret redaction of logs/tool output; threat-pattern scan of context files, skills and tool results; `UnscopedSecretError` secret scoping | `SECURITY.md:L58--L65`, `tools/approval_detection.py:L107`, `tools/approval_detection.py:L228`, `agent/file_safety.py:L176--L196`, `agent/file_safety.py:L210`, `tools/url_safety.py:L1--L9`, `agent/redact.py:L1--L3`, `agent/prompt_builder.py:L83--L110`, `tools/skills_guard.py:L240--L253`, `agent/tool_dispatch_helpers.py:L544--L562` |
| Observability | across (recorded at Model interface and Output) | per-call usage folded into state.db; `pre/post_api_request`, `pre/post_tool_call`, `pre/post_auxiliary_call` plugin hooks; gateway monitoring events exported over OTLP (opt-in); `HERMES_DUMP_REQUESTS`; middleware traces on requests | `agent/turn_usage.py:L73--L79`, `hermes_state_common.py:L484`, `hermes_cli/plugins.py:L106--L135`, `agent/monitoring/events.py:L16--L66`, `agent/monitoring/otlp_exporter.py:L1--L5`, `agent/turn_api_request.py:L152--L153` |
| Resilience | Model interface (retries, fallback, auto-recovery), Context (compaction on overflow), Output (bounded guards) | 3 retries with Retry-After/jittered backoff; fallback chain and credential pool; post-exhaustion ladder; 413/context-length → compress+restart; stale-stream breaker; turn liveness watchdog; cron inactivity watchdog; resume after restart from persisted tool-call rows | `agent/turn_recovery.py:L1401--L1430`, `agent/chat_completion_helpers.py:L2086--L2121`, `agent/turn_recovery_autorecover.py:L1--L7`, `agent/turn_overflow.py:L1--L6`, `agent/chat_completion_helpers.py:L4005`, `agent/turn_liveness.py:L90`, `cron/scheduler.py:L1961--L1963`, `agent/turn_tool_round.py:L131--L153` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Performance | Context (cache tiers, truncation, spill), Output (parallel tools), Model interface (streaming, cache_control) | system prompt byte-stable in stable→context→volatile tiers; compression is the only sanctioned cache break; head/tail truncation and spill files; parallel-safe tool segments; streaming by default; Anthropic cache markers; tool-definition cache | `agent/system_prompt.py:L730--L793`, `agent/system_prompt.py:L808--L826`, `tools/tool_output_truncate.py:L1--L12`, `agent/tool_dispatch_helpers.py:L212--L257`, `agent/turn_api_call.py:L46--L64`, `agent/prompt_caching.py:L298--L308`, `model_tools.py:L215` |
| Evaluation | across | 6 392 pytest files run per-file-isolated by `scripts/run_tests.sh`; markers for `integration/live` provider canaries; `evals/` harnesses (compaction recall eval, provider_fallback, tool_search, token_accounting, ...); `batch_runner.py` runs the agent over JSONL datasets; `mini_swe_runner.py` runs SWE tasks in the execution environments | `scripts/run_tests.sh:L1--L15`, `pyproject.toml:L691--L700`, `evals/compaction/README.md:L1--L12`, `batch_runner.py:L1--L6`, `mini_swe_runner.py:L1--L6`, `tests/evals/` (listing) |

### Design decisions (one line each: the choice this harness made within a primitive)

- Input: one normalized `MessageEvent/run_conversation(user_message, conversation_history, task_id)` seam shared by CLI, ~25 gateway adapters, HTTP, ACP and cron; mid-turn messages become steer/redirect rather than new turns.
- Context: a three-tier byte-stable system prompt (identity → project files → volatile memory/skills/time) rebuilt only after compression; compaction at 50 % of the window by prune-then-summarize with a separate auxiliary model; oversized tool output is spilled to files, not dropped.
- Model interface: OpenAI chat-completions wire by default with pluggable transports (Anthropic, Responses, Bedrock) and provider plugins; streaming always on; 3 retries with backoff, then a configurable fallback chain, then a bounded auto-recovery wait.
- Output: a fixed core tool set sent on every call plus a Tool Search bridge for the long tail; tool batches split into parallel-safe segments; terminal execution behind pluggable sandboxes (local/docker/modal/...) and a regex-plus-human approval gate; tool-call rows persisted before execution.
- Control flow: synchronous phase-verdict loop with an unlimited default iteration budget, verify-on-stop nudges that re-enter the loop, flat delegation by default (`max_spawn_depth=1`) with threaded children, and task-keyed auxiliary model routing.

### Hard cases

- **Tool Search bridge (`tool_search/tool_describe/tool_call`):** a tool call the model makes (output) whose only job is to decide which tool schemas the model sees (context); resolved by "assign by function": the catalog read is context, the unwrapped `tool_call` re-dispatch is output. `tools/tool_search.py:L1--L7`, `model_tools.py:L701--L709`
- **`execute_code` programmatic tool calling:** a program the model writes at run time that sequences only tools → output (rule 27); but the loop refunds the iteration it used, which is control flow. `tools/code_execution_tool.py:L1--L10`, `agent/turn_tool_round.py:L199--L200`
- **Subdirectory hints and untrusted wrapping inside the tool result path:** code in the output executor that changes what the model is shown; assigned to context by rule 20. `agent/subdirectory_hints.py:L1--L5`, `agent/tool_dispatch_helpers.py:L572--L595`

- **Stop-gate nudges:** the harness appends a synthetic user row and goes again; the decision is control flow, the row's text is context, and it is not input (no person wrote it). `agent/turn_stop_gates.py:L105--L140`
- **Approval inside the terminal tool:** the gate runs inside the tool executor and blocks the worker thread (CLI) or returns `pending_approval` and ends the call (gateway); the person's answer is consumed by the harness → control flow by rule 26, even though the code lives in `tools/`. `tools/terminal_tool.py:L1013--L1047`
- **Reasoning-only stall → model fallback:** a model swap triggered by unusable output rather than an HTTP failure; placed in model interface because the request "could not be served" with content, not keyed to the work. `agent/turn_truncation.py:L630--L635`
- **Prompt-cache decoration:** `cache_control` markers are wire-format (model interface) but the tiered ordering of the system prompt exists only to keep them effective (context design); split across both. `agent/prompt_caching.py:L298--L308`, `agent/system_prompt.py:L730--L793`
- **Gateway session hygiene (0.85 compaction):** compaction performed by the gateway outside the agent turn; still context (fits the window) but hosted in a different process layer. `gateway/run_turn_hygiene.py:L94--L102`
- **Codex app-server mode — DOES NOT FIT cleanly:** with `api_mode == "codex_app_server"` the whole turn (loop, tool execution, model calls) is handed to an external OpenAI Codex subprocess and Hermes only projects its events into rows and arms a fallback on failure; the five primitives collapse into one opaque external harness that is not a sub-agent (no delegation, no outer loop). `agent/conversation_loop.py:L1650--L1651`, `agent/conversation_loop.py:L1537--L1552`
- **ACP / external-process "providers":** `acp://` base URLs make the "model" another agent process; streaming is disabled and the model interface boundary is another harness. `agent/turn_api_call.py:L46--L56`
- **Mixture-of-Agents:** several model calls per iteration issued by the harness (control flow) whose outputs are appended to the user message (context). `agent/moa_loop.py:L1--L5`

**Summary row**

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| One MessageEvent seam: CLI, 25 gateways, HTTP, ACP, cron, steer | Tiered byte-stable prompt, memory files, 50 % prune-then-summarize compaction | Pluggable transports, streaming, 3 retries, fallback chain, auto-recovery | Core tools + bridge, parallel segments, sandboxed terminal, approval gate | Phase-verdict loop, unlimited budget, stop-gate nudges, flat threaded delegation |

## C.6 LangGraph (prebuilt ReAct agent on the Pregel runtime)

**Source:** https://github.com/langchain-ai/langgraph · **Version:** commit 5eb60aef6f3837a1fb632a8abc6cf8e8da81785b (2026-10-14; packages `langgraph` 1.2.15, `langgraph-prebuilt` 1.1.1, `langgraph-checkpoint`; the local clone holds this single squashed commit, so no file history is available) · **Language:** Python **Path conventions:** `LG/` = `libs/langgraph/langgraph/`; `PB/` = `libs/prebuilt/langgraph/prebuilt/`; `CP/` = `libs/checkpoint/langgraph/`; `T/` = `libs/*/tests/`. "User-written" marks a decision the framework leaves to the person building the graph. **What it is:** LangGraph (LangChain Inc.) is a framework for writing agent control flow as a graph of nodes over a shared, checkpointed state; the Pregel runtime executes the graph in bulk-synchronous supersteps. The canonical harness deconstructed here is `create_react_agent` (`PB/chat_agent_executor.py`): an `agent` node that calls a LangChain chat model and a `tools` node (`ToolNode`) that runs the model's tool calls, cycling until the model stops calling tools. The model is reached through a `langchain_core` `BaseChatModel` (`.invoke`/`.ainvoke`), which lives outside this repository; LangGraph itself contains no provider client.

### Control flow

- **ReAct loop shape (user-written graph, built by the prebuilt):** `agent` and `tools` nodes are added and the `tools` node's edge returns to the entrypoint, so the loop is an edge, not a `while`. `PB/chat_agent_executor.py:L867--L872`, `L985--L990`
- **Go again / stop decision:** `should_continue` reads the last message; no `tool_calls` → `END` (or `post_model_hook` / `generate_structured_response`), otherwise route to `tools`. `PB/chat_agent_executor.py:L831--L859`; attached as a conditional edge `L964--L968`
- **v2 fan-out, one task per tool call:** `should_continue` returns one `Send("tools", ToolCallWithContext(...))` per tool call, so each call becomes its own Pregel task. `PB/chat_agent_executor.py:L849--L859`; `Send` type `LG/types.py:L738--L826`
- **`return_direct` hand-back:** tools flagged `return_direct` end the loop after they run; `route_tool_responses` returns `END` when the newest ToolMessages came from such a tool. `PB/chat_agent_executor.py:L597`, `L970--L990`
- **post_model_hook router (user-written hook, framework routing):** after the hook, pending tool calls without ToolMessages are re-sent to `tools`; a trailing ToolMessage routes back to the entrypoint; else END/structured. `PB/chat_agent_executor.py:L919--L962`
- **Step budget inside the agent (`remaining_steps`):** a managed value computed as `scratchpad.stop - scratchpad.step`. `LG/managed/is_last_step.py:L18--L24`; `stop = step + recursion_limit + 1` `LG/pregel/_loop.py:L1858--L1859`; `_are_more_steps_needed` returns True when `< 2` steps remain and the response has tool calls `PB/chat_agent_executor.py:L620--L634`; the model's response is then replaced by an AIMessage "Sorry, need more steps to process this request." `L684--L694`
- **Recursion limit (runtime step budget):** `tick()` sets `status = "out_of_steps"` when `step > stop`. `LG/pregel/_loop.py:L640--L643`; `stream()` then raises `GraphRecursionError` `LG/pregel/main.py:L3093--L3102`; error class `LG/errors.py:L68--L88`; default `DEFAULT_RECURSION_LIMIT = int(getenv("LANGGRAPH_DEFAULT_RECURSION_LIMIT", "10007"))` `LG/_internal/_config.py:L32`
- **Pregel superstep loop:** `while loop.tick(): ... runner.tick(...) ... loop.after_tick()`; channel updates from step N are visible only in step N+1. `LG/pregel/main.py:L3050--L3079`
- **Natural stop:** the loop ends when `prepare_next_tasks` yields no tasks (`status = "done"`). `LG/pregel/_loop.py:L686--L689`
- **Which node runs next (the scheduler):** `prepare_next_tasks` first consumes pending `Send` packets from the reserved `TASKS` topic (PUSH tasks), then PULL tasks for nodes whose trigger channels were updated. `LG/pregel/_algo.py:L391--L512` (PUSH `L441--L465`, PULL `L489--L511`); a node triggers when a trigger channel's version exceeds the version it last saw `L1259--L1276`
- **Apply writes between steps:** `apply_writes` sorts tasks by path for determinism, records `versions_seen`, updates channels and bumps versions. `LG/pregel/_algo.py:L254--L258`, `L293--L344`
- **Conditional edges as routing-by-channel:** `add_conditional_edges` stores a `BranchSpec` `LG/graph/state.py:L982--L1030`; at run time `_route` reads fresh state, calls the user's path function, maps results through `path_map`, and writes to `branch:to:<node>` channels `LG/graph/_branch.py:L146--L167`, `L192--L225`; `attach_branch` installs the branch as a writer on the source node `LG/graph/state.py:L1577--L1624`; plain edges write the same way `L1551--L1575`; each node is triggered by its own `branch:to:` ephemeral channel `L1525--L1532`
- **`Command(goto=...)` from a node or tool:** `_control_branch` turns `goto` into TASKS (`Send`) or `branch:to:` writes; `Command.PARENT` raises `ParentCommand`. `LG/graph/state.py:L1749--L1775`; `ParentCommand` is bubbled to the parent graph by `run_with_retry` `LG/pregel/_retry.py:L618--L631`; the exception class `LG/errors.py:L130--L134`
- **Send packets are tasks of the next (or same) superstep:** `prepare_push_task_send` turns `TASKS[idx]` into a task for the named node `LG/pregel/_algo.py:L937--L996`; `accept_push` schedules pushes emitted mid-step `LG/pregel/_loop.py:L584--L621`; `TASKS` is a non-accumulating `Topic(Send)` `LG/pregel/main.py:L814--L819`
- **Static breakpoints (`interrupt_before`/`interrupt_after`):** `should_interrupt` returns tasks whose node is listed, if any channel changed since the last interrupt `LG/pregel/_algo.py:L156--L186`; checked before running tasks `LG/pregel/_loop.py:L696--L701` and after the step is checkpointed `L787--L792`; the prebuilt passes the user's lists straight to `compile` `PB/chat_agent_executor.py:L995--L1002`

- **Dynamic pause (`interrupt()`):** first call in a task raises `GraphInterrupt` carrying an `Interrupt` whose id hashes the task namespace `LG/types.py:L1038--L1046`; on re-run it returns the stored resume value instead `L1019--L1036`; `GraphInterrupt` `LG/errors.py:L103--L108`
- **Pause bookkeeping:** the runner saves the interrupt as an `INTERRUPT` pending write `LG/pregel/_runner.py:L586--L592`; at the root graph the exception is swallowed, state is emitted and the run returns `LG/pregel/_loop.py:L1491--L1530`
- **Resume (harness-consumed answer):** `Command(resume=...)` maps to a `RESUME` write `LG/pregel/_io.py:L56--L78`; `_first` requires a checkpointer and demands interrupt ids when several interrupts are pending `LG/pregel/_loop.py:L992--L1009`; the task scratchpad looks up its resume value `LG/pregel/_algo.py:L1279--L1344`; resume is inferred from `None` input or a `Command` `LG/pregel/_loop.py:L915--L928`
- **Approval gate is user-written:** the framework ships only the `HumanInterrupt`/`HumanResponse` shapes (`accept`/`ignore`/`response`/`edit`); no code in the harness consumes them. `PB/interrupt.py:L51--L84`, `L87--L105`
- **Model routing keyed to the work (user-written):** `model` may be a callable `(state, runtime) -> BaseChatModel` resolved on every `agent` step. `PB/chat_agent_executor.py:L563--L564`, `L599--L618`, `L674--L677`
- **Second model call for structured output:** after the loop ends, `generate_structured_response` runs as its own node. `PB/chat_agent_executor.py:L744--L767`, wiring `L899--L910`
- **Node concurrency within a superstep:** all ready tasks are submitted to the executor and awaited with `FIRST_COMPLETED`; the first non-interrupt failure cancels the rest. `LG/pregel/_runner.py:L259--L290`, `L618--L636`, `L652--L699`
- **Cooperative drain (operator stop):** `RunControl.request_drain()` is checked at the superstep boundary; the loop sets `status="draining"` and `stream()` raises `GraphDrained` after checkpointing. `LG/runtime.py:L79--L104`; `LG/pregel/_loop.py:L706--L708`; `LG/pregel/main.py:L3103--L3106`; `LG/errors.py:L55--L65`
- **Per-node error-handler nodes (user-written handler, framework routing):** `compile` assigns default/explicit handler nodes `LG/graph/state.py:L1290--L1344`; the runner reroutes a failed task to its handler instead of panicking `LG/pregel/_runner.py:L298--L324`
- **Sub-agents / subgraphs:** a compiled agent is used as a node of another graph via `name` `PB/chat_agent_executor.py:L466--L468`; nesting is detected by `CONFIG_KEY_TASK_ID` in config `LG/pregel/_loop.py:L351`; a child's `Command(graph=PARENT, goto=[Send...])` is bubbled and merged by `ToolNode` so a tool can hand work to the parent `PB/tool_node.py:L903--L927`. Results return as channel writes to the parent task; depth is unbounded (namespace `parent|child:task_id`).
- **`tools_condition` helper:** standalone routing function (`"tools"` or `"__end__"`) for user-built loops. `PB/tool_node.py:L1616--L1619`
- **Hook nodes (user-written):** `pre_model_hook` runs before every `agent` step, `post_model_hook` after it; both are ordinary nodes the user supplies. `PB/chat_agent_executor.py:L876--L881`, `L891--L896`

### Input

- **Person's message enters as graph input:** `map_input` turns the input dict into writes on the input channels. `LG/pregel/_io.py:L81--L97`; applied as a null-task write set on the first tick `LG/pregel/_loop.py:L1086--L1121`; the hidden `START` node forwards it into state `LG/graph/state.py:L1508--L1514`
- **Coercion of raw messages:** `add_messages` converts dicts/tuples to `BaseMessage`s and assigns UUIDs to id-less messages. `LG/graph/message.py:L190--L211`; ids are fixed before writes are stored `LG/pregel/_loop.py:L1087--L1089`, `L461--L463`
- **Tool results captured as `ToolMessage`:** raw tool returns are normalized (`Command`, `ToolMessage`, or list) `PB/tool_node.py:L1466--L1487`; non-string outputs are JSON-dumped `L317--L344`; failures become `ToolMessage(status="error")` `L1022--L1029`
- **Human answer the model consumes:** when a tool calls `interrupt()`, the resume value is returned to the tool code on re-run (`L1023--L1036` of `LG/types.py`) and becomes input only if the tool puts it in its `ToolMessage` (user-written). Shape: `PB/interrupt.py:L87--L105`

- **State edits from outside (`Command(update=...)`):** mapped to null-task channel writes. `LG/pregel/_io.py:L76--L78`
- **Run-scoped `context` (operator-supplied, not model-visible unless user code shows it):** built into `Runtime` and passed to nodes and tools. `LG/pregel/main.py:L2963--L2973`; `ToolRuntime.context` `PB/tool_node.py:L814--L825`

### Context

- **Prompt assembly (user chooses form):** `_get_prompt_runnable` prepends a `SystemMessage` (from `str`/`SystemMessage`) to `state["messages"]`, or runs a user callable/Runnable over the whole state. `PB/chat_agent_executor.py:L137--L170`
- **What the model is shown each call:** `state["messages"]`, or `llm_input_messages` produced by `pre_model_hook` (which does not alter stored history). `PB/chat_agent_executor.py:L636--L658`; contract documented `L396--L424`
- **Tool descriptions:** `bind_tools(tool_classes + llm_builtin_tools)` unless the user already bound them; `_should_bind_tools` cross-checks names (OpenAI- and Anthropic-style schemas). `PB/chat_agent_executor.py:L582--L589`, `L173--L217`
- **Conversation as reducer-merged state:** `AgentState.messages` uses `add_messages`, which upserts by id and applies `RemoveMessage`. `PB/chat_agent_executor.py:L57--L62`; `LG/graph/message.py:L218--L237`
- **Window fitting (left to the user):** the only primitives are `RemoveMessage(id=REMOVE_ALL_MESSAGES)` to replace history `LG/graph/message.py:L212--L216` and the `pre_model_hook` slot `PB/chat_agent_executor.py:L396--L424`; trimming/summarization code: not found in this repository (grep for `trim_messages`/`summariz` hits only docstrings).
- **History validation before the call:** every `AIMessage.tool_calls` id must have a `ToolMessage`, else `INVALID_CHAT_HISTORY`. `PB/chat_agent_executor.py:L243--L271`
- **Node input is read from channels:** `_proc_input` collects the node's subscribed channels plus managed values, then applies the schema mapper. `LG/pregel/_algo.py:L1347--L1391`; conditional edges read a fresh copy including the node's own writes `L189--L225`; node reads/writes go through `CONFIG_KEY_READ`/`CONFIG_KEY_SEND` `LG/pregel/_read.py:L74--L91`, `LG/pregel/_write.py:L106--L126`
- **Long-term memory for the model (cross-thread store):** `BaseStore.get/put/search` `CP/store/base/__init__.py:L708`, `L756`, `L784`, `L866`; injected into tools via `InjectedStore` `PB/tool_node.py:L1441--L1449`; retrieval and what to show the model are user-written.
- **Hidden state injected into tools (not shown to the model):** `InjectedState`/`ToolRuntime` args are filled from graph state at call time and any model-supplied values for those names are stripped. `PB/tool_node.py:L1400--L1439`, `L1455--L1463`; `ToolRuntime` fields `L1756--L1764`
- **State snapshot travels with each tool task:** `ToolCallWithContext` carries `state` inside the `Send`; `_extract_state` hydrates it (or reads channels). `PB/tool_node.py:L306--L314`, `L1315--L1347`
- **Structured-output prompt:** an optional system prompt is prepended for the final structured call. `PB/chat_agent_executor.py:L756--L758`
- **Agent name stamped on responses:** `response.name = name` (used in multi-agent transcripts). `PB/chat_agent_executor.py:L681--L682`
- **Injection of `config`/`store`/`writer`/`runtime` into node functions by signature:** `LG/_internal/_runnable.py:L168--L219`, `L392--L419`

### Model interface

- **The call:** `static_model.invoke(model_input, config)` (async `ainvoke`); the model is a `langchain_core` `BaseChatModel`, outside this repo. `PB/chat_agent_executor.py:L672--L679`, `L699--L707`; string ids resolved with `init_chat_model` `L569--L580`
- **Structured output call:** `model.with_structured_output(schema).invoke(messages, config)`. `PB/chat_agent_executor.py:L761--L766`
- **Receiving a token stream:** `StreamMessagesHandler` is a LangChain callback: `on_chat_model_start` records namespace/metadata, `on_llm_new_token` forwards each `ChatGenerationChunk`, `on_llm_end`

emits the final message once. LG/pregel/_messages.py:L139--L158, L160--L173, L175--L188, L106--L113; installed when "messages" is in stream_mode LG/pregel/main.py:L2902--L2918; modes LG/types.py:L131--L133

- **Retries of a failed request (per node, user-configured):** RetryPolicy(initial_interval=0.5, backoff_factor=2.0, max_interval=128.0, max_attempts=3, jitter=True, retry_on=default_retry_on) LG/types.py:L427--L446; default predicate retries ConnectionError and HTTP 5xx, not ValueError/TypeError/OSError etc. LG/_internal/_retry.py:L1--L29; loop with exponential backoff and jitter LG/pregel/_retry.py:L641--L682; policy comes from the node or the graph L579; create_react_agent sets none and Pregel defaults to () LG/pregel/main.py:L786, L832--L834
- **Backup model when the request cannot be served:** not found. The dynamic-model callable is keyed to state/context, not to failure (PB/chat_agent_executor.py:L599--L606).
- **Timeouts per attempt:** TimeoutPolicy(run_timeout, idle_timeout, refresh_on) LG/types.py:L461--L523; async watchdog tasks race the node LG/pregel/_retry.py:L422--L470; NodeTimeoutError is deliberately retryable LG/errors.py:L191--L207; sync nodes cannot use timeouts LG/pregel/_retry.py:L580--L583
- **Sync/async mismatch guard:** an async model callable invoked synchronously raises. PB/chat_agent_executor.py:L664--L670

### Output

- **Tool executor (ToolNode):** parses the last AIMessage.tool_calls (or a single ToolCallWithContext) PB/tool_node.py:L1258--L1300; runs all calls concurrently with a thread pool executor.map L829--L832 or asyncio.gather L863--L866; results combined into {"messages": [...]} L870--L895
- **One call's execution:** tool.invoke(call_args, config) after injection PB/tool_node.py:L960--L966; unknown tool name → error ToolMessage listing valid tools L1302--L1313, template L116--L118
- **Error-to-message policy (user-configurable):** default handler only converts argument-validation errors (ToolInvocationError) and re-raises everything else PB/tool_node.py:L391--L399; handle_tool_errors may be bool/str/callable/exception types L1002--L1029; templates L119--L129
- **Interceptor around each call (user-written):** wrap_tool_call(request, execute) can retry, cache, rewrite or short-circuit a call. PB/tool_node.py:L1061--L1088
- **Tool returning Command:** validated and passed through so the tool's EFFECT (state update / goto) reaches the scheduler. PB/tool_node.py:L1473--L1474, L897--L928
- **Interrupt raised inside a tool is never caught:** GraphBubbleUp passes through the error handling. PB/tool_node.py:L986--L996
- **Rendering to the person (stream modes):** values after each step LG/pregel/_loop.py:L751--L759; updates/tasks per task L1571--L1621; _emit filters by subscribed modes L1535--L1569; final value of invoke is the last values chunk plus collected interrupts LG/pregel/main.py:L4000--L4043
- **Custom and tool-progress streams:** stream_writer for "custom" LG/pregel/main.py:L2932--L2951; "tools" mode installs StreamToolCallHandler (on_tool_start/on_tool_end) L2921--L2929, LG/pregel/_tools.py:L35, L228--L258; tools push partial output with ToolRuntime.emit_output_delta PB/tool_node.py:L1766--L1784
- **Sandboxing of tool execution:** not found; tools run in-process in the caller's executor (PB/tool_node.py:L829--L832).
- **Argument validation without execution:** ValidationNode checks tool calls against schemas and returns error ToolMessages for the model to fix. PB/tool_validator.py:L47

### Production concerns

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Persistence | Control flow (loop state) and Context (conversation state) | `Checkpoint` = channel values + versions + versions_seen per thread; `BaseCheckpointSaver.put` after every superstep and `put_writes` per task; `durability` `sync`/`async`/`exit` (default `async`); `thread_id` required; backends `InMemorySaver`, `SqliteSaver`, `PostgresSaver`; resume loads latest checkpoint or a given `checkpoint_id` (time travel); cross-thread `BaseStore` | `CP/checkpoint/base/__init__.py:L93--L124, L301--L342; LG/pregel/_loop.py:L785--L786, L498--L528, L1261--L1263, L1479--L1489; LG/types.py:L98--L104; LG/pregel/main.py:L2682--L2686, L2695--L2696, L3078--L3079; LG/pregel/_loop.py:L1783--L1813; CP/checkpoint/memory/__init__.py:L34; libs/checkpoint-sqlite/langgraph/checkpoint/sqlite/__init__.py:L49; libs/checkpoint-postgres/langgraph/checkpoint/postgres/__init__.py:L45` |
| Safety & security | Output (tool args), Control flow (pauses), Persistence layer | No sandbox (not found). Approval = user-written `interrupt()` before/inside a tool. Model cannot forge `InjectedState`/`InjectedStore` args (stripped). Resume values failing an interrupt's `response_schema` are never fed back to the model. Checkpoint deserializer warns it must not see untrusted data; `LANGGRAPH_STRICT_MSGPACK` / msgpack allowlist; `EncryptedSerializer` wraps payloads; `TAG_HIDDEN` keeps nodes out of streams/breakpoints | `PB/tool_node.py:L1455--L1463, L997--L1001; CP/checkpoint/serde/jsonplus.py:L104--L135; CP/checkpoint/base/__init__.py:L742--L771; CP/checkpoint/serde/encrypted.py:L8--L35; LG/pregel/_algo.py:L175--L182` |
| Observability | Across (traces via LangChain callbacks) | `on_chain_start/end/error` per run with `ls_integration: langgraph` metadata; per-task metadata `langgraph_step/node/triggers/path/checkpoint_ns`; `checkpoints/tasks/debug` stream modes; retry attempts logged; `get_state/get_state_history`; `ExecutionInfo` (thread/run/task ids, attempt). Token accounting: not found in LangGraph (left to `langchain_core`/LangSmith). | `LG/pregel/main.py:L2865--L2890, L3108--L3112; LG/pregel/_algo.py:L653--L659; LG/pregel/_loop.py:L665--L684, L711; LG/pregel/_retry.py:L676--L680; LG/pregel/main.py:L1464, L1514; LG/runtime.py:L27--L52` |

| Concern | Where it lives (primitive) | Mechanism | Evidence |
|---|---|---|---|
| Resilience | Model interface (retry/timeout), Control flow (resume, error handlers), Output (tool errors) | `RetryPolicy` with backoff; `TimeoutPolicy` with idle/run watchdogs; `step_timeout` on a whole superstep; on resume, finished tasks' pending writes are re-applied so only unfinished tasks re-run; failed tasks routed to error-handler nodes; first failure cancels sibling tasks; `GraphDrained` on SIGTERM-style drain; tool errors returned to the model as messages. Compaction on overflow: not in framework (user `pre_model_hook`). | `LG/pregel/_retry.py:L641--L682, L422--L470; LG/pregel/main.py:L781, L3060; LG/pregel/_loop.py:L805--L821, L691--L694; LG/pregel/_runner.py:L298--L324, L652--L699; LG/pregel/_loop.py:L706--L708; PB/tool_node.py:L1022--L1029` |
| Performance | Output (parallel tools), Control flow (parallel nodes, caching), Model interface (streaming) | Tool calls run concurrently (thread pool / gather); nodes in a superstep run concurrently; node-level `CachePolicy(key_func, ttl)` with `BaseCache` lookup before execution; per-step input cache; `trigger_to_nodes` avoids scanning all nodes; token streaming; `durability="async"/"exit"` defers checkpoint writes | `PB/tool_node.py:L829--L832, L863--L866; LG/pregel/_runner.py:L259--L277; LG/types.py:L529--L538; LG/pregel/_loop.py:L1701--L1714; LG/pregel/main.py:L3056--L3057; LG/pregel/_algo.py:L667--L686, L1357--L1359, L474--L481; CP/cache/base/__init__.py:L15` |
| Evaluation | Across | Unit/integration test suites per package (prebuilt: `test_react_agent.py` 46 tests, `test_tool_node.py` 42; runtime: `test_pregel.py` 172, `test_retry.py` 105, `test_time_travel.py` 50, `test_interruption.py` 6); a `checkpoint-conformance` package for savers. Agent-quality evals: not found in this repository. | `libs/prebuilt/tests/test_react_agent.py; libs/prebuilt/tests/test_tool_node.py; libs/langgraph/tests/test_pregel.py; libs/langgraph/tests/test_retry.py; libs/checkpoint-conformance/tests/test_validate_memory.py` |

### Design decisions (one line each: the choice this harness made within a primitive)

- Input: everything (user message, tool result, resume value, external `Command(update)`) enters as a write to a named channel of a typed state, merged by a user-chosen reducer (`add_messages` upserts by message id).
- Context: the model sees exactly `state["messages"]` (optionally prefixed by one system message); fitting to the window is delegated entirely to a user-written `pre_model_hook`, with `REMOVE_ALL_MESSAGES` as the only built-in rewrite primitive.
- Model interface: no client of its own; a LangChain `BaseChatModel` is invoked once per `agent` node, with retries/timeouts expressed as per-node policies that are off by default.
- Output: `ToolNode` runs every tool call of a response concurrently in-process, converts results and (configurable) errors into `ToolMessage`s, and lets a tool return a `Command` to reach the scheduler.
- Control flow: the loop is a user-written graph executed as Pregel supersteps; "go again" is a conditional edge, "stop" is an empty task set or `recursion_limit`, pauses are exceptions persisted as checkpoint

writes and resumed by re-running the node.

### Hard cases

- **Conditional edge `should_continue`:** user code, but it decides go-again vs. stop and is executed by the scheduler as a writer on the node; placed in control flow by function (rule: assign by function). Note the whole graph is user-authored control flow; the prebuilt merely authors it for you.
- **`remaining_steps` managed value:** it is injected into the node's *state* (so it looks like context), but no model ever sees it; `call_model` uses it to replace the model's answer with a canned message. Placed in control flow (step budget). Its side effect — a synthetic AIMessage written into the conversation — is a context effect of a control-flow decision.
- **`Send("tools", ToolCallWithContext(state=...))`:** a scheduling packet (control flow) that also carries a state snapshot (context) to the tool task. Split: the `Send` is control flow; the snapshot and `_extract_state` are context.
- **Tool returning `Command(goto=...)` or `Command(graph=PARENT)`:** the tool call is output; its effect is routing, so `_combine_tool_outputs` and `_control_branch` are control flow (rule: a tool's EFFECT is assigned to what it changes).
- **`interrupt()` inside a tool and `HumanResponse`:** the pause/resume machinery is control flow, but the harness never consumes an approval; the framework hands the raw resume value to user code in the tool, which decides whether to execute. So the approval gate is user-written, and the value is input only if the tool shows it to the model.
- **Dynamic model callable:** a model change keyed to state/context (planning vs execution) is control flow by the rules, but the selection logic is user-written; the framework only re-resolves it before each call. No failure-keyed fallback exists, so "backup model" is absent from the model-interface primitive.
- **Checkpointer as conversation memory:** the same mechanism that makes the loop resumable (persistence of control-flow state) is what makes a `thread_id` a chat history (persistence of context). Recorded once under persistence, spanning both primitives.
- **Pregel channel/version scheduler:** it is a general dataflow scheduler (nodes fire when subscribed channels change), not a loop with explicit go-again/hand-back/stop. It fits control flow, but the "shape of the loop" is emergent from edges and versions rather than coded; the only explicit stop conditions are "no tasks" and the step budget.
- **`DEFAULT_RECURSION_LIMIT = "10007"`:** the value found at this commit (`LG/_internal/_config.py:L32`) differs from the widely documented default of 25; the clone has a single squashed commit, so the change cannot be traced. Recorded as found, not interpreted.
- **Run `context` / `Runtime`:** operator-supplied static data that is neither a person's message nor a tool result; it is closest to configuration (out of scope) but is delivered on the request path to tools, so it is noted under Input without a strong claim.
- **Error-handler nodes and `wrap_tool_call`:** both are user-written code that the framework schedules; they are resilience mechanisms whose primitive depends on what the user's code does (a retry inside `wrap_tool_call` is output-side; an error handler that routes is control flow).

### Summary row

| Input | Context | Model interface | Output | Control flow |
|---|---|---|---|---|
| Messages and tool results become reducer-merged channel writes | System prompt plus state messages; fitting left to user hook | One LangChain chat-model invoke per step; per-node retries optional | ToolNode runs tool calls concurrently, errors become ToolMessages | User-written graph run as Pregel supersteps; interrupts checkpointed |

## The wider set, at summary level

## C.7 SWE-agent

**Source:** https://github.com/SWE-agent/SWE-agent · **Version:** commit 3ea751c087f32b16e039a2233dd6eefecef325d5 (2026-07-16) · **Language:** Python **Path conventions:** `ag/` = `sweagent/agent/`; `rn/` = `sweagent/run/`; `tl/` = `sweagent/tools/`; `ev/` = `sweagent/environment/` . All paths relative to the repo root. **What it is:** SWE-agent is the Princeton/Stanford research harness that turns a language model into a software-engineering agent for fixing GitHub issues and SWE-bench instances, one task per run. A `DefaultAgent` repeats a thought-action-observation loop against a bash session in a SWE-ReX Docker deployment, with optional `RetryAgent` attempts judged by a reviewer model. The model is reached synchronously through `litellm.completion` behind an `AbstractModel.query` interface that can also be a human, a trajectory replay, or a test stub.

### Control flow

- **Step loop:** `DefaultAgent.run` sets up the env and repeats `while not step_output.done: step()`; `step()` calls `forward_with_handling(self.messages)`, appends the action and observation to history, updates `info`, saves the trajectory. `ag/agents.py:L1265--L1294`, `L1235--L1263`
- **Stop rule:** `done=True` when the action is `exit`, when the observation carries the `«SWE_AGENT_SUBMISSION»` marker and `/root/model.patch` is read as the submission, or when any exit-class error ends with `attempt_autosubmission_after_error` (`exit_cost`, `exit_context`, `exit_format`, `exit_api`, timeouts). `ag/agents.py:L949--L956`, `L870--L905`, `L823--L868`, `L1154--L1218`; `tl/tools.py:L372--L376`
- **Budgets:** `per_instance_cost_limit`, `total_cost_limit` and `per_instance_call_limit` are checked in `_update_stats` after every model call and raise cost/call-limit errors; `total_execution_timeout` (1800 s) is checked at the start of `forward`; `max_consecutive_execution_timeouts` (3) ends the run. `ag/models.py:L73--L78`, `L632--L670`; `ag/agents.py:L1017--L1019`, `L967--L975`; `tl/tools.py:L139--L151`
- **Requery sub-loop:** `forward_with_handling` requeries the model up to `max_requeries` (3) on format errors, blocklisted actions, bash syntax errors and content-policy violations, feeding an error template back instead of running anything. `ag/agents.py:L1107--L1152`, `L158--L161`
- **Attempts and multi-sampling:** `RetryAgent.run` hard-resets the env and starts a new `DefaultAgent` while `ScoreRetryLoop`/`ChooserRetryLoop.retry()` allows (max_attempts, cost_limit, accept_score), using reviewer/chooser model calls to pick the best submission; `AskColleagues`/`BinaryTrajectoryComparison` samplers issue several model calls per step and keep one completion; `RunBatch` runs instances in a `ThreadPoolExecutor` with `num_workers`. `ag/agents.py:L390--L440`, `L321--L326`, `L1031--L1041`; `ag/reviewer.py:L524--L546`, `L617--L645`; `ag/action_sampler.py:L76--L94`, `L266--L304`; `rn/run_batch.py:L268--L289`, `L87--L88`

### Input

- **Problem statement sources:** `TextProblemStatement`, `FileProblemStatement`, `GithubIssue` (fetched over the GitHub API with `GITHUB_TOKEN`) and `SWEBenchMultimodalProblemStatement` (issue images downloaded and base64-embedded) are captured once at setup. `ag/problem_statement.py:L68--L126`, `L128--L155`, `L157--L216`
- **Tool results re-enter:** `add_step_to_history` appends the assistant action (with `tool_calls`, `thinking_blocks`) then a templated observation as a `user` message, or as a `tool` message carrying the single `tool_call_id` under function calling. `ag/agents.py:L714--L746`, `L675--L713`
- **Environment state:** after every action `get_state` runs the bundles' state commands and reads `/root/state.json`; the dict is captured on the step and templated into the observation. `tl/tools.py:L317--L348`; `ag/agents.py:L992--L993`
- **Human and interactive modes:** `HumanModel._query` reads the action from stdin with `input()` (multi-line commands supported); `RunShell` runs a `ShellAgent` interactively; no mid-run user message path exists for the batch agent. `ag/models.py:L384--L423`; `rn/run_shell.py:L79--L97`

### Context

- **System and instance prompts:** `add_system_message_to_history` renders the Jinja `system_template`; demonstrations are added next; `add_instance_template_to_history` renders `instance_template` plus optional `strategy_template` with the problem statement and the initial state. `ag/agents.py:L600--L615`, `L617--L656`, `L748--L760`, `L65--L90`
- **Tool descriptions:** `command_docs` is generated from the bundles' command definitions for the templates; under function calling the JSON tool schemas are passed as `tools` on every request. `tl/tools.py:L122--L126`, `L194--L197`; `ag/models.py:L708--L709`
- **History processors:** the `messages` property filters history to this agent and chains processors; `LastNObservations` replaces older observations with "Old environment output: (n lines omitted)", `ClosedWindowHistoryProcessor` and `RemoveRegex` edit content, `ImageParsingHistoryProcessor` turns image markup into image parts. `ag/agents.py:L539--L551`; `ag/history_processors.py:L85--L177`, `L215--L259`, `L305--L338`, `L340--L361`
- **Window fitting:** observations longer than `max_observation_length` (100,000 chars) are clipped with `<response clipped>`; input tokens are counted with `litellm.utils.token_counter` before the call and `ContextWindowExceededError` ends the run with autosubmission; no compaction or summary exists. `ag/agents.py:L728--L746`, `L69--L79`, `L1176--L1179`; `ag/models.py:L690--L703`
- **Prompt caching:** `CacheControlHistoryProcessor` marks the last `last_n_messages` user/tool messages with `cache_control` and clears older marks; `_history_to_messages` copies the marks onto the wire messages. `ag/history_processors.py:L261--L302`; `ag/models.py:L840--L873`

### Model interface

- **Client:** `LiteLLMModel._single_query` calls `litellm.completion` with model name, temperature, top_p, api_version, a rotated API key from `choose_api_key`, config `fallbacks`, `tools` when function calling, `max_tokens` for Anthropic and a `swe-agent/<version>` User-Agent. `ag/models.py:L679--L742`, `L155--L190`
- **Retries:** `query` wraps the call in tenacity `Retrying` with `stop_after_attempt(retries)` (default 20) and `wait_random_exponential(min_wait=10, max_wait=120)`, never retrying context-window, cost-limit, auth, not-found or content-policy errors. `ag/models.py:L794--L838`, `L55--L63`
- **Response shape and cost:** the response becomes `{"message", "tool_calls", "thinking_blocks"}`; cost comes from `litellm.cost_calculator.completion_cost` and failing cost lookup is an error unless limits are 0. `ag/models.py:L743--L782`
- **No streaming; pacing:** completion is synchronous; `_sleep` enforces a global minimum `delay` between calls; `AbstractModel.query` is also implemented by `HumanModel`, `ReplayModel` and test models. `ag/models.py:L722--L733`, `L672--L677`, `L311--L326`, `L875--L903`

### Output

- **Tool set and parsing:** bash plus tool bundles installed into the container; `parse_function` (default `FunctionCallingParser`, which requires exactly one tool call; also thought-action, XML, JSON and code-block parsers) turns the completion into `(thought, action)` where the action is one shell command string. `tl/tools.py:L75--L155`, `L252--L316`, `L378--L380`; `tl/parsing.py:L371--L455`
- **Executor:** `handle_action` runs the command in the persistent bash session via `SWEEnv.communicate`, which sends a SWE-ReX `BashAction` with `execution_timeout` (30 s); on timeout the session is interrupted and a cancelled-command template becomes the observation. `ag/agents.py:L958--L991`; `ev/swe_env.py:L197--L233`, `L192--L194`
- **Sandbox:** `EnvironmentConfig.deployment` defaults to `DockerDeploymentConfig(image="python:3.11")`; `_init_deployment` starts the deployment and creates one bash session; the repo is copied and reset per attempt. `ev/swe_env.py:L24--L50`, `L176--L190`, `L128--L148`
- **Blocklist and single action:** `should_block_action` blocks interactive commands (prefix blocklist, standalone list, `block_unless_regex`) before execution; multiple tool calls in one response are a format error, so there is no parallel execution. `tl/tools.py:L353--L369`, `L34--L72`; `tl/parsing.py:L441--L455`

- **Result presentation:** the submission is the patch read from `/root/model.patch`; `save_predictions` writes a `.pred` JSON; `SaveApplyPatchHook` saves and optionally applies the patch locally and `OpenPRHook` opens a pull request. `ag/agents.py:L870--L905`; `rn/common.py:L370--L379`; `rn/hooks/apply_patch.py:L36--L50`; `rn/hooks/open_pr.py:L127--L148`

### Production concerns

- **Persistence:** a `.traj` JSON (history, trajectory steps with `query`, info, replay config) is rewritten after every step; `RunReplay` replays a trajectory through `ReplayModel`; `RunBatch.should_skip` skips instances that already have a trajectory. `ag/agents.py:L779--L788`, `L762--L777`; `rn/run_replay.py:L66--L95`; `ag/models.py:L464--L500`; `rn/run_batch.py:L376--L409`
- **Safety & security:** Docker isolation through SWE-ReX (output); command blocklist and requery on blocked actions (output/control flow); API keys read from config/env with key rotation and `GITHUB_TOKEN` from env; no approval gate was found. `ev/swe_env.py:L24--L30`; `tl/tools.py:L353--L369`; `ag/models.py:L155--L190`; `ag/problem_statement.py:L149--L151`
- **Observability:** per-instance trace/debug/info log files; `InstanceStats/GlobalStats` token and cost accounting logged after every call; agent hooks (`on_model_query`, `on_action_executed`, `on_step_done`); a trajectory inspector web viewer. `rn/run_single.py:L139--L147`; `ag/models.py:L273--L309`, `L632--L651`; `ag/hooks/abstract.py:L10--L54`; `rn/inspector_cli.py:L1--L30`
- **Resilience:** tenacity retries on the model call; autosubmission on every error class, including extracting the last `diff` from state when the runtime is dead; session interrupt after command timeouts; batch loop isolates per-instance exceptions and only `_BreakLoop` on total-cost or keyboard interrupt. `ag/models.py:L809--L838`; `ag/agents.py:L823--L868`, `L967--L985`; `rn/run_batch.py:L309--L332`
- **Performance:** cache-control marks for Anthropic prompt caching; observation elision and 100k-char clipping; multi-worker batch threads with a random startup delay; minimum delay between model calls. `ag/history_processors.py:L261--L302`; `ag/agents.py:L728--L746`; `rn/run_batch.py:L268--L298`; `ag/models.py:L672--L677`
- **Evaluation:** pytest suite covering agent, models, parsing, history processors and runs; `SweBenchEvaluate` hook merges predictions and submits them to `sb-cli` during batch runs. `tests/test_agent.py:L1--L40`; `rn/hooks/swe_bench_evaluate.py:L19--L45`, `L74--L92`

### Hard to place

- **HumanModel:** a person types the action where the model's completion would be; by function it is Input (a person's message) sitting behind the `AbstractModel.query` model interface. The rules resolve it as input, but the code does not separate the two. `ag/models.py:L344--L444`
- **Autosubmission:** on error the harness itself runs `git add -A && git diff --cached > /root/model.patch` and reads the file; a tool-like effect (output) initiated by a control-flow decision, not by the model. `ag/agents.py:L851--L868`
- **State commands:** tool bundles run shell commands after every action solely to compute the state dict shown in the next observation; execution machinery (output) whose function is context assembly, resolved by the assign-by-function rule as context. `tl/tools.py:L337--L348`
- **Reviewer and samplers:** `Reviewer.review`, `Chooser.choose` and the action samplers issue extra model calls whose outputs are scores or picks consumed by the harness, not shown to the agent; model interface calls sequenced by control flow. `ag/reviewer.py:L416--L453`, `L329--L373`; `ag/action_sampler.py:L266--L304`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Single problem statement (text, file, GitHub issue, images) plus templated bash observations; no mid-run messages |

| Primitive | Design decision |
|---|---|
| Context | Jinja system/instance templates, demos, history processors eliding old observations, 100k-char clipping, cache marks; no compaction |
| Model interface | Synchronous litellm.completion, tenacity 20 random-exponential retries, litellm fallbacks, no streaming |
| Output | Exactly one shell command per step in a SWE-ReX Docker bash session; blocklist; patch from /root/model.patch |
| Control flow | Step until submit or exit; cost, call and timeout budgets; requery on format errors; RetryAgent attempts with reviewer |

## C.8 OpenHands

**Source:** https://github.com/All-Hands-AI/OpenHands · **Version:** commit aa593c72103862d4f1851988ecacd30f514368dc (2026-10-16) · **Language:** TypeScript (Agent Canvas UI) over Python (software-agent-sdk) **Path conventions:** `ui/` = `src/` in OpenHands/OpenHands at the commit above. The agent loop, model client and tools no longer live in that repo; they live in https://github.com/OpenHands/software-agent-sdk, read at commit 5fa4dea24c52fe3ac53fa4e3e67a1376b72189e2 (2026-10-15): `sdk/` = `openhands-sdk/openhands/sdk/`, `tools/` = `openhands-tools/openhands/tools/`, `srv/` = `openhands-agent-server/openhands/agent_server/`, `sdkt/` = `tests/` in that repo. **What it is:** OpenHands (All Hands AI / OpenHands org) is at this commit "Agent Canvas", a self-hosted React control center that starts conversations on an Agent Server over REST and WebSocket and can run the OpenHands agent or third-party ACP agents (Claude Code, Codex, Gemini CLI, Pi, OpenCode). The agent itself is the SDK's `Agent` driven by `LocalConversation`, which the Agent Server hosts locally, in Docker, on VMs or in OpenHands Cloud. The model is reached through `litellm` (chat completions or Responses API) with tenacity retries and profile-based fallbacks.

### Control flow

- **Run loop:** `LocalConversation._run` loops `while True` under the state lock: break on PAUSED/STUCK; on FINISHED break unless a Stop hook denies and injects feedback (then continue); stuck check; clear WAITING_FOR_CONFIRMATION; `agent.step()`; then break on WAITING_FOR_CONFIRMATION, on budget exceeded, or when `iteration >= max_iteration_per_run` (default 500). `sdk/conversation/impl/local_conversation.py:L1933--L2063`, `L221`
- **Stop rule inside a step:** a text-only response sets FINISHED ("awaits user input"); a FinishTool action sets FINISHED via `_ActionBatch.finalize` unless an iterative-refinement follow-up is injected; `send_message` after FINISHED/STUCK resets to IDLE so the next `run()` continues. `sdk/agent/response_dispatch.py:L259--L281`; `sdk/agent/agent.py:L381--L412`; `local_conversation.py:L1845--L1854`
- **Budgets and stuck detection:** `max_budget_per_run` compares accumulated cost across all LLMs (agent, condenser) and emits `MaxBudgetReached`; `StuckDetector` nudges once on an action-error streak, then marks STUCK on repeated action/observation, error, monologue, alternating or context-window-error patterns. `local_conversation.py:L727--L775`, `L2031--L2037`; `sdk/conversation/stuck_detector.py:L104--L155`
- **Approval gate:** `_requires_user_confirmation` runs the security analyzer on the batch (Finish/Think never require it), records a `SecurityAnalysisEvent`, and `confirmation_policy.should_confirm(risk)` (AlwaysConfirm, NeverConfirm, ConfirmRisky with HIGH threshold) sets WAITING_FOR_CONFIRMATION; the next `run()` executes the pending unmatched actions as implicit confirmation; Canvas posts accept/reject through `EventService.respondToConfirmation`; PreToolUse hooks block individual actions. `sdk/agent/agent.py:L1131--L1200`, `L707--L722`; `sdk/security/confirmation_policy.py:L43--L60`; `ui/hooks/mutation/use-respond-to-confirmation.ts:L12--L32`; `sdk/hooks/conversation_hooks.py:L127--L177`, `L355--L388`
- **Delegation and routing:** `DelegateExecutor` spawns one `LocalConversation` per sub-agent (persisted under the parent, inheriting MCP config, confirmation policy and `max_iteration_per_run`), runs delegated

tasks in parallel threads and returns each agent's final response as the observation; Canvas's `launch_child_conversation` client tool starts an independent conversation (local worktree or cloud) whose result arrives later as a message; `classify_and_switch_llm` asks a classifier model and switches the LLM profile keyed to the task; `/plan` routes a message to a planner conversation. `tools/delegate/impl.py:L210--L245`, `L287--L365`, `L112--L131`; `ui/api/launch-child-conversation-client-tool.ts:L14--L20`, `L61--L108`; `sdk/tool/builtins/classify_and_switch_llm.py:L1--L10`, `L147--L195`; `ui/hooks/chat/use-plan-mode-interceptor.ts:L16--L29`, `L60--L85`

### Input

- **Canvas chat input:** `useSendMessage` turns the chat payload into `{role:"user", content:[text, image]}` and `AgentServerConversationService.sendMessage` posts it with `ConversationClient.sendEvent(..., {run: true})` to the Agent Server (per-conversation host and session key on cloud); events come back over a WebSocket with reconnect. `ui/hooks/use-send-message.ts:L13--L70`; `ui/api/conversation-service/agent-server-conversation-service.api.ts:L475--L510`; `ui/hooks/use-websocket.ts:L29--L64`, `L118--L145`
- **SDK message capture:** `send_message` wraps text into `MessageEvent(source="user")`, resets terminal states to IDLE, and attaches knowledge-skill content triggered by the message as `extended_content`; the loop deliberately does not break on FINISHED so a message sent concurrently is picked up on the next iteration. `local_conversation.py:L1823--L1886`, `L2019--L2030`
- **Tool results re-enter:** `_execute_action_event` runs one tool and returns `ObservationEvent` or, on `ValueError`, an `AgentErrorEvent` for self-correction; `_ActionBatch.emit` emits results in order after the batch. `sdk/agent/agent.py:L1499--L1559`, `L352--L380`
- **Harness-generated user messages:** stuck nudges, Stop-hook feedback, content-filter and malformed-tool-call corrections are injected as `role="user"` messages from `source="environment"` or `"user"`. `local_conversation.py:L759--L767`, `L1972--L1984`; `sdk/agent/agent.py:L798--L825`

### Context

- **System prompt:** `static_system_message` builds from a typed section registry preset (default or planning), or an inline `system_prompt` plus tool guidance, or a custom Jinja template; `AgentContext` contributes repo skills (`AGENTS.md`), knowledge skills, `system_message_suffix`, memory and secrets descriptions. `sdk/agent/base.py:L357--L393`, `L174--L204`, `L451--L500`; `sdk/context/agent_context.py:L56--L110`, `L265--L297`
- **Message assembly:** `prepare_llm_messages` takes the incrementally maintained `state.view`, lets the condenser return either a `View` or a `Condensation` event (which ends the step without a model call), then converts events to messages; images for non-vision models are swapped for references. `sdk/agent/utils.py:L581--L634`; `sdk/agent/agent.py:L742--L785`
- **Compaction:** `LLMSummarizingCondenser` condenses when token count exceeds the effective max (HARD), when events exceed `max_size` 240 (SOFT) or on request, keeps the first `keep_first` 2 events, and replaces forgotten events with an LLM-written summary. `sdk/context/condenser/llm_summarizing_condenser.py:L64--L70`, `L136--L204`, `L224--L276`
- **Overflow recovery:** `LLMContextWindowExceedError` or a malformed-history rejection emits `CondensationRequest` and returns to the loop, which condenses on the next step. `sdk/agent/agent.py:L826--L876`
- **Caching and truncation:** prompt-cache markers with `prompt_cache_retention`, retried without caching when the provider minimum is not met; terminal output is middle-truncated at `MAX_CMD_OUTPUT_SIZE` 30,000 chars with the full output saved to a file. `sdk/llm/llm.py:L602--L610`, `L1837--L1855`; `tools/terminal/definition.py:L192--L199`; `tools/terminal/constants.py:L18`; `sdk/utils/truncate.py:L50--L62`

### Model interface

- **Client:** `LLM.completion` resolves runtime metadata, formats messages and tools in `_prepare_completion_params`, and calls `litellm` (chat completions, or the Responses API via `responses`); non-native models get prompt-mocked tool calls through `NonNativeToolCallingMixin`. `sdk/llm/llm.py:L1752--L1816`, `L1415--L1445`, `L2011--L2040`; `sdk/llm/mixins/non_native_fc.py:L27--L80`
- **Retries:** tenacity `stop_after_attempt(num_retries=5)` with `wait_exponential(multiplier=8, min=8, max=64)` on `LLM_RETRY_EXCEPTIONS` (connection, 502, 429, 503, timeout, 500, no response) and transient HTTP errors, skipping quota exhaustion; a 401 refreshes the key and retries once. `sdk/llm/llm.py:L180--L188`, `L379--L382`, `L1141--L1160`, `L1856--L1867`; `sdk/llm/utils/retry_mixin.py:L77--L116`
- **Fallback:** after retries fail on a transient error, `FallbackStrategy.try_fallback` tries alternate LLM profiles in order (nested fallback disabled, metrics merged into the primary), and budget errors are never bypassed. `sdk/llm/fallback_strategy.py:L45--L131`; `sdk/llm/llm.py:L1014--L1038`
- **Streaming:** `stream` flag with an `on_token` callback and `stream_idle_timeout`; streaming is silently downgraded when no callback is wired; the agent opens a `StreamContext` per step. `sdk/llm/llm.py:L505--L512`, `L393--L400`, `L1789--L1800`; `sdk/agent/agent.py:L694--L705`

### Output

- **Tool set:** `openhands-tools` (terminal, file_editor, grep, glob, browser_use, task_tracker, delegate, planning_file_editor, apply_patch) plus MCP and skill tools; each `ToolDefinition` pairs an action schema with a `ToolExecutor` and declares resources for locking. `tools/terminal/definition.py:L185--L199`; `sdk/tool/tool.py:L283--L328`, `L347--L400`, `L529--L545`
- **Dispatch:** `_handle_tool_calls` converts each tool call into an `ActionEvent` (thought and reasoning attached to the first), then `_ActionBatch.prepare` truncates the batch at the first FinishAction, partitions hook-blocked actions and executes the rest. `sdk/agent/response_dispatch.py:L154--L201`; `sdk/agent/agent.py:L226--L306`, `L628--L656`
- **Parallelism:** `ParallelToolExecutor` runs a batch sequentially by default (`tool_concurrency_limit` 1) or in a `ThreadPoolExecutor` with per-resource locks when raised. `sdk/agent/parallel_executor.py:L51--L126`; `sdk/agent/base.py:L314--L322`; `sdk/agent/agent.py:L473--L475`
- **Sandbox:** tools run in the Agent Server's workspace, which can be the local machine or an ephemeral Docker/Kubernetes workspace; Canvas picks the backend (local, Docker, VM, cloud). `software-agent-sdk/README.md:L37--L38`; `README.md:L35--L41`; `srv/docker_runtime/__init__.py:L1`; `srv/docker_runtime/provisioning.py:L1--L20`
- **Presentation:** events are persisted then published to subscribers (visualizer prepended for terminal rendering); Canvas renders them from the WebSocket with per-tool visualizers and a confirmation prompt. `local_conversation.py:L432--L470`; `ui/components/features/chat/tool-visualizers/index.ts:L1--L20`

### Production concerns

- **Persistence:** `ConversationState` keeps events in an `EventLog` on a `FileStore` (one JSON file per event under a lock with a length marker) and autosaves `base_state.json`; a conversation resumes from `persistence_dir` and can be forked (context plus control flow state). `sdk/conversation/event_store.py:L34--L60`, `L188--L244`; `sdk/conversation/state.py:L233--L234`, `L429--L432`; `local_conversation.py:L353--L374`, `L409--L412`, `L804--L830`
- **Safety & security:** security analyzers (model-reported `security_risk`, defense-in-depth policy rails, Grayswan, ToolShield) feed confirmation policies (control flow); PreToolUse hooks block actions; `SecretRegistry` masks secrets in emitted events; sub-agent scope narrows tools; the Agent Server requires a session API key. `sdk/security/llm_analyzer.py:L10--L28`; `sdk/security/analyzer.py:L29--L60`; `sdk/security/defense_in_depth/policy_rails.py:L1--L40`; `sdk/subagent/scope.py:L38--L60`; `sdk/conversation/secret_registry.py:L24--L75`; `srv/credential_binding.py:L1--L30`
- **Observability:** `Telemetry.on_response` records latency, cost and tokens into `Metrics`, optionally logs each LLM call as JSON and annotates a span; Laminar `observe` spans wrap steps and tool executions;

Canvas reports usage in a metrics modal. sdk/llm/utils/telemetry.py:L190--L242, L401--L470; sdk/observability/laminar.py:L155--L234, L235--L260; sdk/agent/agent.py:L1527--L1537

- **Resilience:** retries and fallback on the model call; condensation on overflow; stuck nudge; iteration and budget caps; event-log resume; ConversationRunError carries the persistence dir; Canvas WebSocket reconnects. local_conversation.py:L2064--L2080; sdk/agent/agent.py:L826--L876; ui/hooks/use-websocket.ts:L118--L145
- **Performance:** the view is maintained incrementally to avoid O(n) rebuilds per step; prompt caching; parallel tool execution; streaming; runtime metadata cached with a 1 h TTL. sdk/agent/utils.py:L592--L597; sdk/llm/llm.py:L1780--L1786; sdk/agent/parallel_executor.py:L110--L126
- **Evaluation:** SDK unit tests under tests/sdk (agent, conversation, llm, context, hooks) and behavior integration tests run by run_infer.py; Canvas has vitest suites and Playwright mock-LLM and live end-to-end configs. sdkt/integration/run_infer.py:L125--L321; sdkt/sdk/agent/test_agent_context_window_condensation.py:L1--L30; playwright.mock-llm.config.ts:L88--L100

### Hard to place

- **The repo is a shell:** at this commit OpenHands/OpenHands holds no loop, model client or tools; AGENTS.md assigns those to software-agent-sdk, and most of Canvas is UI framework and transport (out of scope), yet Canvas keeps control-flow pieces (plan-mode routing, child-conversation launch, confirmation answers, router-at-start). The harness is split across three repos and the primitives cross repo boundaries. AGENTS.md:L30--L37; ui/api/conversation-service/agent-server-conversation-service.api.ts:L456--L470
- **ACP agents:** ACPAgent drives a third-party agent (Claude Code, Codex, ...) through its own step; OpenHands' loop, persistence and confirmation then wrap another complete harness, which the multi-agent rule treats as an inner harness, although here it is a peer at depth 0 rather than a delegate. sdk/agent/base.py:L1077--L1080; sdk/agent/acp_agent.py:L1718--L1730, L3952--L3960
- **Model-predicted risk:** add_security_risk_prediction=True adds a security_risk field to every tool schema, so the approval gate (control flow) is fed by the model's own output rather than by the harness; by rule it is output whose effect is control flow. sdk/agent/agent.py:L790--L795; sdk/security/llm_analyzer.py:L20--L28
- **Condenser as loop step:** when the condenser returns a Condensation, the step emits it and returns without a model call, so a context operation spends one loop iteration; the stop-hook path likewise fabricates a user message to keep the loop going. sdk/agent/agent.py:L748--L756; local_conversation.py:L1963--L1991

### Summary

| Primitive | Design decision |
|---|---|
| Input | Canvas posts user events to an Agent Server; SDK wraps them as MessageEvents; tool results are ObservationEvents |
| Context | Section-registry system prompt with AGENTS.md skills; event view condensed by LLM summary on token or size limit |
| Model interface | litellm chat or Responses API; 5 exponential retries; profile fallbacks; optional token streaming |
| Output | Typed ToolDefinitions with executors in a local or Docker workspace; batch truncated at Finish; optional parallelism |
| Control flow | Status-machine loop with iteration and budget caps, stuck detector, risk-based confirmation, hooks, threaded delegation |

## C.9 Aider

**Source:** https://github.com/Aider-AI/aider · **Version:** commit 5dc9490bb35f9729ef2c95d00a19ccd30c26339c (2026-05-22) · **Language:** Python **Path conventions:** a/ = aider/; c/ = aider/coders/; b/ = benchmark/

; t/ = tests/basic/. All paths relative to the repo root. **What it is:** Aider is a terminal pair-programming tool by Paul Gauthier (Aider-AI) that edits files in a local git repository and commits each change. The model is reached through LiteLLM (litellm.completion) over HTTPS with streaming. There is no native tool calling on the main path: the model writes edits in a text "edit format" (SEARCH/REPLACE blocks, unified diffs, whole files) that the harness parses and applies, and a bounded "reflection" loop feeds parse, lint, and test failures back as the next user message.

### Control flow

- **Outer REPL loop (Coder.run):** while True reads a line, calls run_one, and returns on EOF; main wraps it in a loop that catches SwitchCoder and rebuilds the coder when the user changes mode or model. c/base_coder.py:L876--L892; a/main.py:L1159--L1178
- **Reflection loop and stop rule (run_one):** while message: sends one request; the loop goes again only when reflected_message was set, and stops with a warning once num_reflections >= max_reflections (class default 3). c/base_coder.py:L924--L944, L101
- **Go-again triggers:** after a reply, send_message sets reflected_message on newly mentioned files, on an edit-format ValueError from apply_updates, on lint errors (after confirm_ask), and on test errors (after confirm_ask), each returning early so run_one re-sends. c/base_coder.py:L1561--L1566, L1585--L1623, L2296--L2336
- **Approval gates the harness consumes:** confirm_ask gates adding mentioned files, creating or editing files not yet in chat, running model-suggested shell commands (explicit yes required), adding command output, and attempting lint/test fixes; "don't ask again" and group "all/skip" answers are remembered per question. a/io.py:L807--L860; c/base_coder.py:L2191--L2242, L2450--L2485
- **Architect delegation (depth 1):** ArchitectCoder.reply_completed optionally asks "Edit the files?", builds a second Coder on editor_model with the editor edit format and no repo map, runs the architect's reply as that coder's message, then folds cost and commit hashes back into the parent. c/architect_coder.py:L11--L48; a/models.py:L625--L645
- **Model routing keyed to the work:** the weak_model writes commit messages and history summaries and the editor_model performs the architect's edit step; both come from model settings, not from failures. a/models.py:L603--L623; a/repo.py:L342--L364; a/history.py:L7--L13

### Input

- **Terminal input:** get_input shows files in chat and the mode in a prompt_toolkit prompt; preproc_user_input dispatches /commands, scans the text for file mentions, and offers to scrape URLs into the message. c/base_coder.py:L898--L922, L964--L984; a/io.py:L523--L545
- **Scripted input:** --message and --message-file call coder.run(with_message=...), which sends one message and returns the reply text. a/main.py:L1126--L1140; c/base_coder.py:L876--L881
- **File-watcher input ("AI comments"):** FileWatcher.process_changes turns AI! or AI? comments found in changed files into a request, adding the files to chat and including tree-sitter context around each comment. a/watch.py:L181--L283
- **Results re-enter as text:** approved shell-command output, /run and /test output, lint and test errors, and the edit-format failure report are either appended as user/assistant message pairs or become the next reflected_message. c/base_coder.py:L1609--L1623; a/commands.py:L1013--L1060; c/editblock_coder.py:L79--L124

### Context

- **System prompt and chunk order:** fmt_system_prompt fills the edit-format prompt with fence choice, platform and shell info, reply language, and lazy/overeager reminders; format_chat_chunks assembles ordered chunks: system, examples, read-only files, repo map, summarized history, chat files, current messages, and a reminder placed as system or appended to the last user message if it fits. c/base_coder.py:L1174--L1224, L1226--L1331; c/chat_chunks.py:L16--L26

- **Repo map:** `RepoMap` extracts definitions and references with tree-sitter (cached by mtime in a `diskcache` SQLite directory), ranks files with NetworkX PageRank personalized toward chat files and mentioned identifiers, and binary-searches the tag count so the rendered tree fits `map_tokens` (default 1024, times 8 when no files are in chat). `a/repomap.py:L103--L168`, `L217--L265`, `L470--L529`, `L629--L709`; `c/base_coder.py:L709--L748`
- **History compaction:** `ChatSummary.summarize_real` keeps a tail under half the budget, summarizes the head with the weak model, and recurses to depth 3; `summarize_start` runs it in a background thread whenever `done_messages` exceeds `max_chat_history_tokens`, and `format_chat_chunks` joins the thread before use. `a/history.py:L33--L130`; `c/base_coder.py:L1002--L1046`
- **Window check:** `check_tokens` compares the estimated prompt to `max_input_tokens` and asks before sending; a `ContextWindowExceededError` sets `exhausted` and prints a report instead of retrying. `c/base_coder.py:L1396--L1417`, `L1536--L1546`
- **Prompt caching:** with `--cache-prompts`, `add_cache_control_headers` marks the system/examples, repo-map, and chat-files chunks `cache_control: ephemeral`, and `warm_cache` re-sends the cacheable prefix with `max_tokens=1` every ~5 minutes to keep it warm. `c/chat_chunks.py:L28--L56`; `c/base_coder.py:L1333--L1394`

### Model interface

- **Client:** `Model.send_completion` builds the request (temperature, an optional single forced `tools` function, `extra_params`, ollama `num_ctx`, a 600 s timeout) and calls `litellm.completion`; litellm is imported lazily with `drop_params = True`. `a/models.py:L985--L1037`, `L28`; `a/llm.py:L21--L45`
- **Streaming:** `show_send_output_stream` iterates chunks, accumulates content, reasoning, and function-call deltas, and raises `FinishReasonLength` when `finish_reason == "length"`. `c/base_coder.py:L1900--L1975`
- **Retries and prefill continuation:** `send_message` loops with exponential backoff from 0.125 s until `RETRY_TIMEOUT = 60` for litellm exceptions whose table entry says `retry=True`; on `FinishReasonLength` it appends the partial reply as an assistant prefill and re-sends when the model supports it. `c/base_coder.py:L1449--L1513`; `a/exceptions.py:L13--L57`; `a/models.py:L26`
- **Summarizer and commit-message fallback:** `ChatSummary` and `get_commit_message` try `[weak_model, main_model]` in order through `simple_send_with_retries`, which has its own backoff loop. `a/history.py:L98--L130`; `a/models.py:L1039--L1084`; `a/repo.py:L342--L364`

### Output

- **Edit formats as the tool set:** each `Coder` subclass pairs a prompt with a parser and applier: `EditBlockCoder.get_edits` parses SEARCH/REPLACE blocks and fenced shell blocks, `UnifiedDiffCoder` parses hunks, `WholeFileCoder` whole files, `PatchCoder` the apply-patch format. `c/editblock_coder.py:L15--L41`, `L439--L470`; `c/udiff_coder.py:L49--L94`; `c/__init__.py:L1--L34`
- **Fuzzy application and failure report:** `replace_most_similar_chunk` tries exact, whitespace-insensitive, and `...` elision matches; failed blocks produce a `ValueError` with the block, "did you mean" lines, and instructions to resend only the failed blocks. `c/editblock_coder.py:L157--L189`, `L79--L124`
- **Apply pipeline:** `apply_updates` runs parse, dry run, `prepare_to_edit` (per-file permission and dirty commit), and apply; edited files are auto-committed with a generated commit message and, if enabled, linted and tested. `c/base_coder.py:L2269--L2336`, `L2375--L2395`, `L1681--L1696`; `a/repo.py:L131--L140`
- **Shell execution:** model-suggested commands run after approval through `run_cmd` (pexpect interactive spawn on a POSIX TTY, otherwise `subprocess.Popen`) in the repo root with no sandbox; output is added to the chat only after a second confirmation. `c/base_coder.py:L2434--L2485`; `a/run_cmd.py:L11--L24`, `L42--L62`, `L89--L121`
- **Rendering:** streamed markdown is drawn live through `MarkdownStream`, reasoning tags are reformatted, and `/diff` shows the auto-commit diff. `c/base_coder.py:L1977--L1992`; `a/io.py:L1014--L1042`

### Production concerns

- **Persistence:** chat history is appended to `.aider.chat.history.md` and reloaded with `--restore-chat-history` (context); git commits are the durable artifact and `/undo` reverts aider's own commits (output); repo-map tags live in `.aider.tags.cache.v4` (context). `a/io.py:L1117--L1137`; `c/base_coder.py:L519--L523`; `a/commands.py:L553--L575`; `a/repomap.py:L35--L43`
- **Safety & security:** no sandbox; safety is the `confirm_ask` gates (control flow), gitignore and aiderignore checks before editing, and `--dry-run` (output). `c/base_coder.py:L2191--L2242`, `L2450--L2460`; `a/repo.py:L523--L566`
- **Observability:** per-message token and cost report from `completion.usage` including cache read/write tokens; `--llm-history-file` logs full prompts and responses; opt-in analytics send `message_send` and `exit` events with redacted model names to PostHog or Mixpanel. `c/base_coder.py:L1994--L2126`; `a/io.py:L754--L766`; `a/analytics.py:L213--L253`
- **Resilience:** backoff retries on transient provider errors (model interface), prefill continuation on output-length cutoff (model interface), the `exhausted` report on context overflow (context), and Ctrl-C handling that records the interruption in the transcript. `c/base_coder.py:L1456--L1513`, `L1575--L1583`, `L1628--L1680`
- **Performance:** streaming, prompt-cache headers plus keepalive pings, background summarization thread, repo-map memoization keyed by chat files and mentions, and lazy litellm import. `c/base_coder.py:L1340--L1394`; `a/repomap.py:L576--L627`; `a/llm.py:L21--L45`
- **Evaluation:** `benchmark/benchmark.py` runs Exercism and SWE-bench tasks in Docker per model and edit format; unit tests cover the coder loop, edit-block parsing, and the repo map. `b/benchmark.py:L679--L700`; `t/test_editblock.py:L1--L30`; `t/test_repomap.py:L1--L30`; `t/test_coder.py:L1--L30`

### Hard to place

- **Edit formats:** one class per format carries both the prompt (what the model is shown) and the parser/applier (what is done with the reply); by function the prompt text is context and `get_edits`/`apply_edits` are output, so the rules split a single component. `c/editblock_coder.py:L15--L41`; `c/editblock_prompts.py:L8--L28`
- **File mentions in the reply:** a model mention of a file name has the effect of a tool call (add file to context) without any tool; the rules resolve it as a context effect behind a control-flow approval gate, but the capture is text scanning, not a tool-call format. `c/base_coder.py:L1714--L1782`, `L1561--L1566`
- **Cache warming:** a background thread issues real model requests (`max_tokens=1`) whose response is discarded; it is a model-interface call made purely as a performance measure for context, with no input, output, or control-flow role. `c/base_coder.py:L1340--L1394`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Terminal REPL, scripted message, or AI! file comments; results re-enter as plain text message pairs |
| Context | Ordered chunks: system, examples, PageRank repo map, files, weak-model summarized history; optional cache headers |
| Model interface | LiteLLM completion with streaming, exponential backoff to 60 s, assistant-prefill continuation on length cutoff |
| Output | Text edit formats parsed and fuzzily applied to files, auto-commit, lint/test, approved unsandboxed shell commands |
| Control flow | Reflection loop up to 3 re-sends on edit/lint/test failure; confirm gates; architect-to-editor two-coder handoff |

## C.10 Cline

**Source:** https://github.com/cline/cline · **Version:** commit 23791b9323aadd91d7dbaea8a8fa262cd2f7ad0b (2026-10-15) · **Language:** TypeScript (Bun/Node) **Path conventions:** `ag/` = `sdk/packages/agents/src/`; `co/` = `sdk/packages/core/src/`; `ll/` = `sdk/packages/llms/src/`; `sh/` = `sdk/packages/shared/src/`; `cli/` = `apps/cli/src/`. All paths relative to the repo root. **What it is:** Cline is an open-source coding agent (Cline Bot Inc.) shipped as a VS Code extension, a CLI, and an SDK. At this commit the harness is a layered monorepo: `@cline/agents` holds the stateless loop (`AgentRuntime`), `@cline/core` adds sessions, persistence, compaction, hooks, the tool set, and sub-agents, and `@cline/llms` reaches providers through the Vercel AI SDK (`streamText`) behind a gateway registry. Tool calls use the providers' native tool-calling wire format.

### Control flow

- **Iteration loop and stop rule:** `execute` loops `while iteration < maxIterations`, issues one model request per iteration, executes the reply's tool calls, and returns `completed` when a turn has no tool calls or a `completesRun` tool (such as `submit_and_exit`) succeeded; exceeding `maxIterations` throws and the run fails. `ag/agent-runtime.ts:L836--L838`, `L949--L972`, `L1005--L1027`, `L2383--L2403`
- **Completion reminders:** when a required completion tool exists and a turn has no tool calls, a reminder user message is appended and the loop continues instead of stopping. `ag/agent-runtime.ts:L744--L786`, `L955--L963`
- **Budgets and safety stops:** `MistakeTracker` counts consecutive mistakes (default 6) and either injects guidance or stops; `LoopDetectionTracker` (CLI default: soft at 3, hard at 5 identical calls) appends a recovery notice or escalates to the mistake limit; any hook returning `{stop}` raises `ControlledStopError`. `co/runtime/safety/mistake-tracker.ts:L80--L164`; `co/runtime/safety/loop-detection.ts:L66--L124`; `co/runtime/orchestration/session-runtime-orchestrator.ts:L471--L510`, `L1465--L1498`; `ag/agent-runtime.ts:L2777--L2788`; `cli/runtime/defaults.ts:L6--L9`
- **Approval gate:** `prepareToolExecution` resolves a per-tool policy and, when `autoApprove === false`, awaits `requestToolApproval`; the CLI answers with a terminal `[y/N]` prompt or a file-based desktop IPC polled for up to 5 minutes, and a rejection becomes an error tool result. `ag/agent-runtime.ts:L2477--L2541`; `cli/utils/approval.ts:L68--L113`; `co/runtime/tools/tool-approval.ts:L29--L102`
- **Sub-agents (depth 1 per spawn, inner harness is a `SessionRuntime`):** the `spawn_agent` tool builds a delegated `SessionRuntime` with its own tools, `maxIterations`, and parent id, awaits `run(task)`, and returns text, iteration count, and usage as the tool result; `AgentTeam` composes worker/reviewer teams. `co/extensions/tools/team/spawn-agent-tool.ts:L117--L204`; `co/extensions/tools/team/delegated-agent.ts:L149--L158`; `co/extensions/tools/team/multi-agent.ts:L239--L245`, `L535--L587`
- **Plan/Act modes as a bundle:** plan mode swaps the system prompt (context), installs a `beforeTool` command guard that rejects file-editing shell commands (output filter), and exposes `switch_to_act_mode` so the model can change mode (control flow). `sh/prompt/cline.ts:L22--L58`; `co/runtime/orchestration/runtime-builder.ts:L546--L566`; `co/extensions/tools/command-guard.ts:L1--L23`

### Input

- **Host entry:** `LocalRuntimeHost.runTurn` takes a prompt with optional images and file paths, records `session.input_sent`, and either runs the turn or enqueues it (`queue` or `steer` delivery) when a run is active. `co/runtime/host/local-runtime-host.ts:L1142--L1224`; `co/runtime/orchestration/user-input-builder.ts:L10--L37`
- **Mid-turn steering:** `steer` prompts are consumed at every request boundary by `consumePendingUserMessage`, which appends them as a user message before the request; `queue` prompts drain after the turn finishes. `ag/agent-runtime.ts:L1687--L1695`, `L2302--L2321`; `co/runtime/host/local-runtime-host.ts:L861--L868`; `co/runtime/turn-queue/pending-prompt-service.ts:L54--L120`

- **Tool results re-enter:** each executed call becomes a `tool` message holding a `tool-result` part (`isError` on failure or rejection) pushed onto `state.messages`. `ag/agent-runtime.ts:L2628--L2647`, `L974--L982`
- **Model-consumed answers:** the `ask_question` tool returns the person's chosen option as a tool result (input by rule), distinct from the harness-consumed approval. `co/extensions/tools/definitions.ts:L818--L837`; `cli/utils/approval.ts:L119--L125`

### Context

- **System prompt:** `buildClineSystemPrompt` fills the ACT or YOLO template (platform, date, IDE, cwd, workspace metadata, rules) and appends mode instructions; the orchestrator merges extension rules and passes the result to the runtime. `sh/prompt/cline.ts:L152--L213`; `sh/prompt/system/act.ts:L1--L36`; `co/runtime/orchestration/session-runtime-orchestrator.ts:L139--L152`, `L990--L1027`
- **Request assembly:** `prepareModelRequest` clones the transcript, lists tool definitions, appends pending steer input, runs `prepareTurn` (compaction) and then `beforeModel` hooks that may replace messages, tools, or options; the orchestrator's `beforeModel` runs message builders that convert to the provider's message format. `ag/agent-runtime.ts:L1656--L1736`; `co/runtime/orchestration/session-runtime-orchestrator.ts:L1152--L1192`
- **Hook context injection:** `beforeRun`, `beforeTool`, and `afterTool` hooks may return `appendContext`, which is buffered and flushed as one user message after the tool results. `ag/agent-runtime.ts:L1099--L1133`, `L2457--L2466`, `L2606--L2614`
- **Compaction trigger and strategies:** `createContextCompactionPrepareTurn` estimates request tokens, scales the budget by observed provider under-counting (capped at 4x), compacts when the estimate reaches 90% of the effective input limit, and targets 70%; `agentic` compaction calls a summarizer model over messages before a cut index that preserves 20k recent tokens, `basic` compaction drops and truncates by projection policy. `co/extensions/context/compaction.ts:L267--L420`; `co/extensions/context/compaction-shared.ts:L15--L37`; `co/extensions/context/agentic-compaction.ts:L116--L175`; `co/extensions/context/budget-projection/project.ts:L479--L560`
- **Oversized tool results:** tools marked `resultPolicy: "cache-oversized"` have results longer than `maxCachedResultChars` stored in a tool-result cache and replaced by a preview in the transcript. `co/runtime/orchestration/session-runtime-orchestrator.ts:L1160--L1180`, `L1001--L1005`

### Model interface

- **Gateway and providers:** `DefaultGateway.stream` resolves provider and model from a registry, validates model-side tools, derives `maxTokens`, and calls the provider's `stream`; built-in providers wrap AI SDK modules (Anthropic via `@ai-sdk/anthropic`). `ll/providers/gateway.ts:L273--L408`; `ll/providers/vendors/anthropic.ts:L15--L70`
- **Streaming and tool-call wire format:** `streamText` runs with `maxRetries: 5`, `experimental_repairToolCall`, and native tools; the runtime consumes `text-delta`, `reasoning-delta`, `tool-call-delta`, `usage`, and `finish` events and classifies a finish error (`classifyProviderError`, `isRetryableProviderError`). `ll/providers/ai-sdk.ts:L1086--L1097`, `L2436--L2446`; `ag/agent-runtime.ts:L1808--L1830`, `L1877--L1895`, `L1988--L2008`
- **Turn-level retries:** the runtime re-issues a turn that failed with a retryable provider error up to 3 times with 1 s to 15 s exponential backoff, only when nothing was streamed; an AI SDK middleware retries empty streams up to 3 times. `ag/agent-runtime.ts:L72--L91`, `L1202--L1322`; `ll/providers/middleware/retry-empty-response.ts:L97--L101`, `L273--L340`
- **Prompt caching:** Anthropic-compatible routing attaches `cache_control` according to per-provider prompt-cache routes. `ll/providers/routing/anthropic-compatible.ts:L51--L91`, `L199--L227`

### Output

- **Tool set:** default tools are `read_files`, `search_codebase`, `run_commands`, `fetch_web_content`, `editor` (or `apply_patch`), `skills`, `ask_question`, and optionally `submit_and_exit`; MCP server tools, plugin

tools, and team tools are appended by the runtime builder. co/extensions/tools/definitions.ts:L915--L981; co/runtime/orchestration/runtime-builder.ts:L568--L598

- **Executor and concurrency:** executeToolCalls prepares every call, then runs adjacent parallel-mode calls under Promise.all and sequential ones one at a time; each tool carries timeoutMs (default 30 s), retryable, and maxRetries (default 3). ag/agent-runtime.ts:L2343--L2381; sh/tools/create.ts:L108--L138
- **Shell execution:** run_commands runs its commands concurrently through createShellExecutor, which spawns the default shell (detached on POSIX) with a rolling output collector capped at maxOutputChars and a per-command timeout; there is no OS sandbox for commands, while plugins run in a SubprocessSandbox or WorkerSandbox. co/extensions/tools/definitions.ts:L180--L263; co/extensions/tools/executors/bash.ts:L717--L760, L1133--L1182; co/runtime/tools/subprocess-sandbox.ts:L126--L160; co/runtime/tools/worker-sandbox.ts:L91--L130
- **Result presentation:** tool-started, tool-updated (streaming command output), tool-finished, and assistant-text-delta events are emitted to hosts (CLI TUI, VS Code webview) through the session event bridge. ag/agent-runtime.ts:L2543--L2556, L2578--L2590, L2692--L2760
- **Checkpoints:** hook-based checkpoints snapshot the working tree with a stash-style commit plus an untracked-files parent, kept under private git refs per session for restore from the UI. co/hooks/checkpoint-hooks.ts:L249--L260, L349--L360, L437--L450, L497--L540

### Production concerns

- **Persistence:** session rows live in a SQLite sessions.db (provider, model, cwd, status, messages_path); messages are persisted through persistSessionMessages on runtime events and reloaded from messages_path on resume (context); compaction state is saved to a sidecar. co/services/storage/sqlite-session-store.ts:L25--L90; co/runtime/host/local-runtime-host.ts:L352--L366, L754--L770, L1620--L1627; co/runtime/host/history.ts:L418--L447
- **Safety & security:** per-tool approval policies with an approval callback (control flow), the plan-mode command blacklist and hook skip/stop controls (output filters), provider-executed tool calls skipped locally, and command timeouts. ag/agent-runtime.ts:L2420--L2500; co/extensions/tools/command-guard.ts:L1--L23; cli/runtime/tool-policies.ts:L30--L86
- **Observability:** CORE_TELEMETRY_EVENTS (session, task.tokens, task.tool_used, task.provider_api_error, task.mistake_limit_reached) through a TelemetryService with OpenTelemetry and Langfuse adapters; the runtime logs request diagnostics and emits status-notice events for retries and compaction. co/services/telemetry/core-events.ts:L43--L82; co/services/telemetry/TelemetryService.ts:L11--L40; ll/services/langfuse-telemetry.ts:L1--L40; ag/agent-runtime.ts:L1721--L1733
- **Resilience:** provider-error turn retry, one-shot overflow recovery (compact and retry), max-tokens recovery by compaction then up to 3 concision nudges, a single continuation after an unknown finish, AI SDK request retries honoring retry-after, and abort propagation. ag/agent-runtime.ts:L1163--L1200, L1363--L1448, L1481--L1596; ll/providers/ai-sdk.ts:L1086--L1097
- **Performance:** streamed deltas, parallel tool groups, prompt-cache routing, oversized-result caching, a char-based token estimator corrected by provider feedback, and idle-timeout sandboxes. ag/agent-runtime.ts:L2362--L2380; co/extensions/context/compaction.ts:L335--L372; co/runtime/tools/subprocess-sandbox.ts:L148--L160
- **Evaluation:** 290 Bun unit-test files across agents, core, and llms (the runtime alone has a 4,920-line test), provider VCR fixtures, CLI end-to-end tests, and evals/ with smoke tests and cline-bench run under Harbor in Docker or Daytona. ag/agent-runtime.test.ts:L1--L40; ll/tests/provider-vcr/README.md:L1--L31; cli/cli.e2e.test.ts:L1--L40; evals/e2e/run-cline-bench.ts:L1--L40

### Hard to place

- **Provider-executed tools:** Anthropic web_search runs inside the provider; the harness sees it only as tool-call-delta events with execution metadata, records it as model tool activity, and skips local execution; the tool runs on the model side, so the output primitive is partly outside the harness, and the

rules do not cover it. ag/agent-runtime.ts:L1877--L1895, L2422--L2432; ll/providers/vendors/anthropic.ts:L29--L47

- **Agentic compaction:** a context component (prepareTurn) itself issues a model request to the summarizer; by function the trigger and cut are context and the summarizer call is model interface, but the call is sequenced from inside context assembly, not by the loop. co/extensions/context/agentic-compaction.ts:L116--L175; co/extensions/context/compaction-shared.ts:L678--L720
- **Mistake and loop trackers:** one component both appends guidance as a user message (context) and decides to abort the run (control flow); the rules split it by effect. co/runtime/orchestration/session-runtime-orchestrator.ts:L1465--L1498; co/runtime/safety/mistake-tracker.ts:L128--L160
- **Subprocess hook files:** user hook scripts are bundles that can cancel a run (control flow), modify context, and observe events (observability); each decomposes under the rules. co/hooks/subprocess.ts:L133--L180; co/hooks/hook-file-hooks.ts:L760--L770

### Summary

| Primitive | Design decision |
|---|---|
| Input | Host runTurn with text, images, files; steer queue consumed per request; tool results as tool messages |
| Context | Templated system prompt plus rules, cloned transcript, hook context, estimator-driven compaction at 90% via summarizer |
| Model interface | Gateway over Vercel AI SDK streamText, native tools, 5 SDK retries plus 3 turn retries with backoff |
| Output | Native tool calls in parallel groups with timeouts; shell spawn, editor, MCP; git-ref checkpoints |
| Control flow | Loop until no tool calls or completion tool; maxIterations, mistake and loop trackers, approval gates, sub-agents |

## C.11 Gemini CLI

**Source:** https://github.com/google-gemini/gemini-cli · **Version:** commit b25669d84e1a404e30c176d779e732e25dafcec0 (2026-10-16) · **Language:** TypeScript (Node) **Path conventions:** core/ = packages/core/src/; cli/ = packages/cli/src/. All paths relative to the repo root. **What it is:** Gemini CLI is Google's open-source terminal coding agent. The core package holds the agent loop (GeminiClient), chat history (GeminiChat), the tool scheduler, policy engine, hooks, sandboxing and sub-agents; the cli package is an Ink/React terminal UI that drives the loop and runs tool calls. The model is reached through the @google/genai SDK (API key, Vertex AI, or a gateway) or through the Code Assist OAuth backend, always as a streaming generateContentStream call.

### Control flow

- **Turn loop and budgets:** sendMessageStream runs one model call through processTurn and recurses with boundedTurns - 1 for continuations; recursion is capped by MAX_TURNS = 100 and the session by maxSessionTurns, and a pre-flight overflow check or a MaxSessionTurns event ends the run. core/core/client.ts:L79, L644--L663, L727--L744, L940--L1000
- **Tool re-entry loop lives outside the client:** LegacyAgentProtocol collects ToolCallRequest events, calls scheduler.schedule, and re-sends the completed calls' responseParts as the next request, stopping when a response has no tool calls, on a STOP_EXECUTION tool error, or on a fatal tool error; the interactive CLI does the same in handleCompletedTools -> submitQuery(..., {isContinuation: true}). core/agent/legacy-agent-session.ts:L199--L215, L260--L343; cli/ui/hooks/useGeminiStream.ts:L2109--L2130, L2314--L2326
- **Stop rule:** when a turn ends with no pending tool calls, checkNextSpeaker asks a utility model whether the model should speak next and, if so, the harness sends "Please continue."; LoopDetectionService

watches repeated tool calls (5), repeated content chunks (10) and an LLM check after 30 turns, injecting a corrective message on the first detection and aborting on the second. core/core/client.ts:L777--L795, L840--L857, L905--L937, L1302--L1328; core/utils/nextSpeakerChecker.ts:L14--L43; core/services/loopDetectionService.ts:L29--L65, L185--L215, L260--L300

- **Approval gates and hooks:** the scheduler's _processToolCall runs the BeforeTool hook, then checkPolicy -> PolicyEngine.check (TOML rules plus approval mode DEFAULT-/AUTO_EDIT/YOLO/PLAN) yielding ALLOW/DENY/ASK_USER, escalates taint-risk invocations to ASK_USER, and resolves ASK_USER through resolveConfirmation on the message bus, cancelling the whole batch on a user Cancel; BeforeAgent hooks can stop or block a run, and AfterAgent hooks can stop it or force one more turn. core/scheduler/scheduler.ts:L640--L770; core/scheduler/policy.ts:L53--L75; core/policy/policy-engine.ts:L301--L304, L350--L357; core/core/client.ts:L956--L988, L1004--L1058; core/hooks/types.ts:L43--L54
- **Model routing and sub-agents:** per prompt a ModelRouterService composite (Fallback, Override, ApprovalMode, classifier strategies, Default) picks the model, which then sticks for the sequence; AgentTool runs a LocalAgentExecutor whose runInternal loops executeTurn until complete_task, maxTurns or a maxTimeMinutes deadline, and returns the result as a tool result (built-in agents: codebase investigator, CLI help, generalist). core/core/client.ts:L797--L826; core/routing/modelRouterService.ts:L39--L75; core/agents/agent-tool.ts:L217--L243; core/agents/local-executor.ts:L582--L598, L685--L718; core/agents/registry.ts:L286--L288

### Input

- **User message:** the CLI's submitQuery hands the typed parts to geminiClient.sendMessageStream; @path references are expanded by handleAtCommand through the read_many_files tool before submission. cli/ui/hooks/useGeminiStream.ts:L1910--L1916; cli/ui/hooks/atCommandProcessor.ts:L690--L699
- **Tool results:** completed calls' functionResponse parts become the next user content; a queued user steering hint is prepended to that content; if the model was switched by a quota error the responses are only recorded to history. cli/ui/hooks/useGeminiStream.ts:L2280--L2326; core/agent/legacy-agent-session.ts:L278--L308, L341--L342
- **Answers the model asked for:** AskUserTool receives the user's answers from the confirmation payload and returns them as JSON llmContent. core/tools/ask-user.ts:L171--L179, L187--L248
- **IDE context:** open files and cursor state from the IDE companion are appended as a user part when no tool call is pending. core/core/client.ts:L459--L470, L755--L771

### Context

- **System prompt:** PromptProvider.getCoreSystemPrompt composes snippet sections (mandates, workflows, sandbox, git, user memory, skills, plan-mode tool list) chosen by model generation, or loads a GEMINI_SYSTEM_MD override file with substitutions. core/prompts/promptProvider.ts:L47--L135; core/prompts/snippets.ts:L136--L336
- **Project and user memory:** GEMINI.md files are read from the global ~/.gemini dir, upward from the cwd to the project root, and from extensions, then concatenated into the user-memory section. core/tools/memoryTool.ts:L11; core/utils/memoryDiscovery.ts:L209, L295, L317, L342, L463--L471
- **History shape:** GeminiChat keeps a comprehensive history and derives a curated one (invalid model turns dropped) that is what gets sent. core/core/geminiChat.ts:L220--L255, L1270--L1278
- **Compaction:** before each call tryCompressChat runs when the last prompt token count reaches 50% of the model's limit; it truncates to budget, splits keeping the last 30%, summarizes the rest with a utility model into a <state_snapshot>, and falls back to truncation only after an inflated result; an optional ContextManager pipeline (masking, distillation, rolling summary processors) replaces this when enabled. core/context/chatCompressionService.ts:L45, L269, L467--L560, L595--L614; core/core/client.ts:L670--L725, L1226--L1283; core/context/contextManager.ts:L90--L122
- **Tool output masking:** tryMaskToolOutputs replaces bulky older tool outputs once protected recent output exceeds 50k tokens, and the request is token-estimated against the window before sending.

core/context/toolOutputMaskingService.ts:L25--L28; core/core/client.ts:L727--L744, L1288--L1297

### Model interface

- **Client:** createContentGenerator builds a GoogleGenAI SDK client (API key, Vertex, gateway, proxy agent) or the Code Assist OAuth generator, wrapped in ModelMappingContentGenerator and LoggingContentGenerator, optionally RecordingContentGenerator. core/core/contentGenerator.ts:L210--L239, L293--L310, L385--L427
- **Streaming and retries:** generateContentStream is wrapped in retryWithBackoff (default 10 attempts, 5s to 30s backoff on 429/5xx); mid-stream invalid-content or network errors are retried up to 4 attempts with exponential backoff and a RETRY event tells the UI to discard partial output. core/core/geminiChat.ts:L105--L109, L664--L700, L739--L794, L1180--L1230; core/utils/retry.ts:L20, L42--L46, L205
- **Fallback model:** a persistent 429 calls handleFallback, which selects the next available model from the policy chain and consults a user handler whose intent (retry_always, retry_once, stop, ...) decides whether to switch. core/fallback/handler.ts:L26--L80, L157--L196
- **Tool-call format:** native Gemini functionCall parts; Turn.run turns each into a ToolCallRequest event and yields Finished with the finish reason and usage. core/core/turn.ts:L270--L300, L380--L414, L508--L521

### Output

- **Tool set:** core tools registered in Config (ls, read_file, grep/ripgrep, glob, edit, write_file, shell, web_fetch, web_search, memory, ask_user, activate_skill, agent tools) plus MCP tools discovered into the ToolRegistry. core/config/config.ts:L4014--L4060; core/tools/tool-registry.ts:L287--L302, L370, L663; core/tools/mcp-tool.ts:L501
- **Executor and parallelism:** the scheduler dequeues contiguous parallelizable calls as one batch, validates them concurrently, and executes all scheduled calls with Promise.all, streaming live output into state. core/scheduler/scheduler.ts:L519--L566, L777--L821
- **Shell execution:** commands run under node-pty with a 16MB buffer, an inactivity timeout, and optional backgrounding. core/services/shellExecutionService.ts:L50, L454--L470; core/tools/shell.ts:L554--L660
- **Sandbox:** createSandboxManager picks macOS sandbox-exec (seatbelt profiles), Linux bubblewrap with seccomp, or a Windows manager; the CLI can also run itself inside a docker/podman container. core/services/sandboxManagerFactory.ts:L22--L36; core/sandbox/macos/MacOsSandboxManager.ts:L178; core/sandbox/linux/LinuxSandboxManager.ts:L303--L318; cli/utils/sandbox.ts:L121, L402--L403
- **Result presentation and truncation:** shell output above 40,000 chars is saved to the project temp dir and replaced by a truncated excerpt; stream events are rendered incrementally by processGeminiStreamEvents. core/scheduler/tool-executor.ts:L202--L232; core/config/config.ts:L483; cli/ui/hooks/useGeminiStream.ts:L1917--L1920

### Production concerns

- **Persistence:** context/input record: ChatRecordingService writes each session as a JSONL file under the project temp chats/ dir (sub-agents nested under the parent id) for resume. core/services/chatRecordingService.ts:L779--L823
- **Safety & security:** control flow and output: TOML policy engine with approval modes and non-interactive deny default, taint-risk escalation to ASK_USER, OS sandboxes, hook shell commands with a 60s timeout. core/policy/policy-engine.ts:L301--L304; core/scheduler/scheduler.ts:L687--L696; core/hooks/hookRunner.ts:L252--L262
- **Observability:** OpenTelemetry logs, metrics and traces exported over OTLP gRPC/HTTP plus Clearcut; logApiResponse records duration, status and token usage including cached tokens. core/telemetry/sdk.ts:L15--L21; core/telemetry/loggers.ts:L307--L330; core/telemetry/types.ts:L691

- **Resilience:** model interface and context: connection and mid-stream retries, model fallback on persistent 429, pre-flight overflow detection and compaction, loop recovery. `core/core/geminiChat.ts:L739--L794`; `core/core/client.ts:L727--L744`, `L1302--L1328`
- **Performance:** output and context: parallel tool batches, streaming, compaction at 50% of the window, tool-output masking, model routing to cheaper models by classifier. `core/scheduler/scheduler.ts:L519--L566`; `core/routing/strategies/classifierStrategy.ts:L31--L32`
- **Evaluation:** `evals/` behavioral evals (38 `.eval.ts`) check what the model chooses to do; `integration-tests/` (48 `.test.ts`) replay recorded API responses through `FakeContentGenerator`. `evals/README.md:L1--L20`; `core/core/contentGenerator.ts:L216--L228`

### Hard to place

- **Model calls that steer the loop:** `checkNextSpeaker`, the LLM loop check and the classifier router are model calls whose only effect is a go-again or routing decision; by the routing rule they are control flow even though each is itself a request to the model interface. `core/utils/nextSpeakerChecker.ts:L43--L80`; `core/services/loopDetectionService.ts:L260--L300`; `core/routing/strategies/classifierStrategy.ts:L132`, `L171`
- **Hooks as bundles:** BeforeAgent `additionalContext` is context, BeforeTool arg modification is output, AfterAgent stop/continue is control flow, and BeforeModel/AfterModel hooks edit the request inside the chat layer; the hook runner spawning a shell process fits no primitive. `core/core/client.ts:L978--L988`; `core/scheduler/scheduler.ts:L670--L674`; `core/core/geminiChat.ts:L1173--L1176`; `core/hooks/hookRunner.ts:L7`
- **Turn loop split across packages:** the client only ever issues one model call per `processTurn`; the model/tool alternation is owned by the caller (CLI hook or `LegacyAgentProtocol`), so the "loop" is control flow assembled from two files. `core/core/client.ts:L644--L663`; `core/agent/legacy-agent-session.ts:L260--L343`

### Summary

| Primitive | Design decision |
|---|---|
| Input | CLI/ACP parts with @file expansion; tool results and AskUser answers re-enter as functionResponse parts |
| Context | Snippet-composed system prompt, hierarchical GEMINI.md memory, curated history, 50% threshold snapshot compression, tool masking |
| Model interface | @google/genai or Code Assist streaming, 10-attempt backoff, mid-stream retries, policy-chain model fallback |
| Output | Registry of core and MCP tools, parallel batches, node-pty shell, seatbelt/bwrap/container sandboxes, 40k truncation |
| Control flow | Caller-driven tool loop, 100-turn cap, next-speaker and loop checks, policy/confirmation gate, hooks, classifier routing, sub-agents |

## C.12 Goose (Block)

**Source:** https://github.com/block/goose · **Version:** commit 5eca486214c86945ac6de2603ee27f51083822fb (2026-10-16) · **Language:** Rust **Path conventions:** `g/` = `crates/goose/src/`; `sm/` = `crates/goose/src/agents/state_machine/`; `ga/` = `crates/goose-agent/src/`; `gp/` = `crates/goose-providers/src/`; `gpt/` = `crates/goose-provider-types/src/`; `cli/` = `crates/goose-cli/src/`. All paths relative to the repo root. **What it is:** Goose is an open-source general-purpose agent started by Block and now attributed to the Agentic AI Foundation, with a CLI, a desktop app and a server sharing one Rust core. Each user message runs an ordered state machine of "operations" over a persisted SQLite conversation, with a single inference step at the end of the order. The model is reached through a `Provider` trait with many HTTP implementations (Anthropic, OpenAI, Google, Bedrock, Ollama, ...) and some ACP/CLI-wrapper providers that run their own tool loop.

### Control flow

- **State machine loop:** `build_state_machine` orders operations (slash command, tool approval, skills, recipe, tool execution, remaining tool calls, subagent, steer, bang shell, compaction, tool-pair compaction, doctor, project, retry, empty response, stop hook, exit on error, max turns) and appends the `Inference` step; `StateMachine::step` runs the first applicable operation and returns `Continuing`, or `Yielded` when none applies, and `run` repeats load-step-apply until a yield or cancel. `g/agents/state_machine_builder.rs:L43--L136`; `ga/machine.rs:L121--L190`, `L211--L226`
- **Go-again rule:** inference applies only when the projected conversation ends with a provider turn (`should_infer`) and there is no trailing error; after tool results are appended the next step infers again, and a plain assistant reply leaves nothing applicable, so the turn ends. `ga/inference.rs:L277--L286`, `L377--L403`
- **Budgets:** `MaxTurnsOperation` ends the turn with a fixed message once assistant turns since kickoff reach `max_turns` (session config, `GOOSE_MAX_TURNS`, default 1000) and advertises a `<turn-budget>` part after half is used; `ExitOnErrorOperation` yields on a trailing error. `sm/ops_maxturns.rs:L14--L67`; `g/agents/agent.rs:L52`, `L565--L569`; `sm/ops_exit_on_error.rs:L21--L32`
- **Approval gate:** `ToolApprovalOperation` (skipped in `Chat` mode) runs tool inspectors, marks denied requests non-executable and emits `ActionRequired` confirmation messages; `stream_state_machine_turn` waits for all confirmation answers, then re-runs the machine; modes are `Auto`, `Approve`, `SmartApprove`, `Chat`. `sm/ops_tool_approval.rs:L43--L148`; `g/agents/agent.rs:L793--L837`; `gpt/goose_mode.rs:L22--L32`
- **Stop hook, steering, sub-agents:** `StopHookOperation` lets a plugin deny ending the turn up to `stop_hook_block_cap` times; `SteerOperation` drains queued user messages between model and tool turns; `SubagentOperation` with `ForegroundSubagentRunner` runs a delegated `SubAgent` session (own recipe, max_turns and provider) through `stream_subagent_turn` and delivers its outcome back as a message. `sm/ops_stop_hook.rs:L80--L119`; `sm/ops_steer.rs:L60--L97`; `g/agents/subagent_handler.rs:L35--L82`, `L102--L130`; `g/agents/platform_extensions/summon.rs:L396--L421`, `L460--L470`

### Input

- **User message:** `Agent::reply` -> `reply_impl` -> `reply_with_state_machine_inner`, which persists the message with `session_manager.add_message` before the machine runs; the CLI calls `agent.reply` per prompt. `g/agents/agent.rs:L953--L1035`, `L637--L660`; `cli/session/mod.rs:L1333--L1340`
- **Mid-turn messages:** `Agent::steer` pushes into a per-session queue that `SteerOperation` consumes; a message carrying an `ElicitationResponse` completes a pending MCP elicitation instead of starting a turn. `g/agents/agent.rs:L293--L302`, `L996--L1031`
- **Tool results:** executed, declined and unparseable requests are recorded into one user message of tool responses, emitted and persisted as an `AppendMessage` effect. `sm/ops_toolcalling.rs:L1044--L1100`; `sm/session.rs:L52--L56`

### Context

- **System prompt:** `PromptManager::build_system_prompt` renders `system.md` (extension list and instructions, mode, date) or a session `system_prompt_override`, then appends hints and per-operation `prompt_parts`; inference tools are gathered from every operation. `g/agents/prompt_manager.rs:L114--L160`, `L201--L213`; `g/prompts/system.md:L1--L30`; `sm/inference_preparation.rs:L40--L90`; `ga/machine.rs:L143--L155`
- **Project files:** `.goosehints` and `AGENTS.md` (configurable `CONTEXT_FILE_NAMES`) are loaded from the global config dir and the working directory. `g/hints/load_hints.rs:L10--L21`, `L232--L252`
- **Per-turn context event:** a `<turn-context>` user-role message with working dir, context limit and operation `moim_parts` (such as the turn budget) is appended to the request when it changed. `g/agents/moim.rs:L92--L122`; `sm/ops_maxturns.rs:L20--L28`
- **Compaction:** `CompactionOperation` fires when tokens/context_limit exceeds the threshold (default 0.8, lowered so compaction starts by 225k tokens) or reactively after a `ContextLengthExceeded` error (at most 2), summarizes with the provider via `compact_messages` and replaces the conversation; `ToolPairCompactionOperation` summarizes old tool pairs past a cutoff; providers that manage their own

context skip both. sm/ops_compaction.rs:L24, L192--L294; g/context_mgmt/mod.rs:L25--L42, L87; crates/goose-context-management/src/lib.rs:L32; sm/ops_tool_pair_compaction.rs:L65--L83

- **Request projection and caching:** before sending, only agent-visible messages are kept, the conversation is repaired and consecutive messages merged; the Anthropic format sets cache_control: ephemeral on the request. ga/inference.rs:L434--L439; g/agents/reply_parts.rs:L293--L306; gpt/formats/anthropic.rs:L126--L134

### Model interface

- **Provider trait:** stream is the one required method, complete collects the stream, and retry_config is overridable; concrete providers are built from ProviderDef::from_env. gpt/base.rs:L520--L565; g/providers/base.rs:L31--L68
- **HTTP client and streaming:** the Anthropic provider POSTs v1/messages through with_retry, decodes SSE lines into a message stream, and retries once with thinking history stripped if the API rejects replayed thinking blocks; default timeouts are 600s per request and 30s connect. gp/anthropic.rs:L184--L262, L434--L449; gp/api_client.rs:L19--L20
- **Retry policy:** RetryConfig defaults to 3 retries with 1s initial delay, x2 backoff, 30s cap, retrying rate-limit, server and network errors and honoring a server retry_delay. gpt/retry.rs:L14--L17, L107--L157
- **Stream consumption:** InferenceRunner::infer calls provider.stream, records usage per chunk, tags assistant chunks with provider/model metadata, drops duplicate tool requests, and retries an empty response up to 3 times; provider errors become error messages in the conversation; no backup model was found. ga/inference.rs:L77, L443--L520; g/agents/reply_parts.rs:L282--L352

### Output

- **Tool set:** every tool is an MCP extension reached through an ExtensionLease: builtin platform extensions (developer shell/editor, summon with delegate, code execution, skills) and external stdio/HTTP MCP servers. sm/ops_toolcalling.rs:L841--L848; g/agents/extension_manager/builtin.rs:L12--L26; g/agents/platform_extensions/developer/shell.rs:L229--L245
- **Executor and parallelism:** ToolExecutionOperation::run dispatches each executable request (pre-tool hooks, lease.call, post-tool hooks) and merges the result streams with select_all, so calls from one response run concurrently and finish in any order. sm/ops_toolcalling.rs:L397--L470, L910--L944, L1010--L1100
- **Sandbox:** a session may carry a Docker container id; builtin extensions are then started with docker exec inside it, otherwise tools run as host processes. g/agents/container.rs:L3--L12; g/agents/extension_manager/builtin.rs:L19--L26; sm/ops_toolcalling.rs:L443
- **Result size:** shell output is capped at 2000 lines / 50,000 bytes with the full output saved to a file, and any tool text over 200,000 chars (GOOSE_MAX_TOOL_RESPONSE_SIZE) is written to a temp file and replaced by a reference. g/agents/platform_extensions/developer/shell.rs:L159--L164; g/agents/large_response_handler.rs:L5--L30
- **Presentation:** events flow through an mpsc channel to the caller's stream; the CLI renders them through a streaming markdown buffer with MCP spinners. g/agents/agent.rs:L862--L920; cli/session/mod.rs:L1342--L1344

### Production concerns

- **Persistence:** context record: sessions and messages live in a SQLite sessions.db (WAL) under the data dir; every machine effect is applied through SessionManager::apply_effects and the machine reloads the session each step, so a turn can be resumed after a crash. g/session/session_manager.rs:L34, L1066--L1080; sm/session.rs:L33--L79; g/agents/agent.rs:L712
- **Safety & security:** control flow and output: Goose modes, a persisted tool permission store (tool_permissions.json), an LLM permission judge for read-only classification, a prompt-injection scanner (threshold 0.8) that forces approval, and shell hooks with a 30s default timeout. g/

permission/permission_store.rs:L45--L90; g/permission/permission_judge.rs:L145--L158; g/security/scanner.rs:L126--L130; g/security/security_inspector.rs:L29--L53; g/hooks/mod.rs:L43, L1031--L1040

- **Observability:** tracing spans carry `gen_ai.*` attributes (usage, tool arguments, finish reasons), exported via OTLP, a Langfuse layer, and PostHog product telemetry. g/agents/gen_ai_telemetry.rs:L52--L70; g/otel/otlp.rs:L75--L143; g/tracing/langfuse_layer.rs:L155--L157; g/posthog.rs:L17--L50
- **Resilience:** model interface, context and control flow: provider retries, reactive compaction on context overflow, empty-response retries, recipe `RetryOperation` with success checks and an on-failure command, cancellation tokens checked between steps. sm/ops_retry.rs:L227--L266; sm/ops_compaction.rs:L203--L218; ga/machine.rs:L159--L171
- **Performance:** output and context: concurrent tool streams, streamed provider responses, prompt-cache headers, tool-pair summarization, session naming done by a background task. g/agents/agent.rs:L662--L682; gpt/formats/anthropic.rs:L126--L134
- **Evaluation:** `evals/harbor` runs terminal-bench-2 style benchmarks across models and builds; `goose-self-test.yaml` is a self-testing recipe; the state machine has its own test module. evals/harbor/README.md:L1--L14; goose-self-test.yaml:L1--L12; sm/mod.rs:L36--L37

### Hard to place

- **Toolshim:** for models without native tool calling, the text reply is sent to a separate interpreter model that extracts tool calls, and the system prompt is rewritten to ask for JSON; this is a second model call that produces the wire format, so it sits in the model interface with a context side effect, not a routing decision. g/providers/toolshim.rs:L1--L12; g/agents/reply_parts.rs:L248--L257
- **Permission judge:** an LLM classifies pending tool calls as read-only to auto-approve them; the decision is a harness-consumed approval (control flow) made by a model call. g/permission/permission_judge.rs:L145--L158
- **Operations as bundles:** every control-flow operation also contributes `prompt_parts`, `moim_parts` and `inference_tools` to the request, so one object mixes control flow, context and the tool set and must be decomposed per method. ga/machine.rs:L143--L155; ga/operation.rs:L89--L139
- **Providers that run their own loop:** `ProviderDef::RUNS_OWN_TOOL_LOOP` hands the session's MCP servers to an external agent (ACP/CLI wrappers), and `manages_own_context` disables compaction; output and control flow then live outside this harness, so it degrades to input plus a thin model interface. g/providers/base.rs:L40--L42; g/agents/state_machine_builder.rs:L66, L89--L96

### Summary

| Primitive | Design decision |
|---|---|
| Input | User message persisted first; steer queue and elicitation answers; tool results batched into one message |
| Context | Templated system prompt plus hints, per-turn context event, 0.8 threshold summarization, tool-pair summaries, prompt caching |
| Model interface | Provider trait with required stream; reqwest SSE; 3 retries exponential; empty-response retries; no backup model |
| Output | All tools are MCP extensions; concurrent select_all execution; optional Docker container; 50k/200k output spill |
| Control flow | Ordered operation pipeline with one inference step; 1000-turn cap; approval wait and rerun; stop hook; foreground sub-agents |

## C.13 OpenAI Agents SDK (Python)

**Source:** https://github.com/openai/openai-agents-python · **Version:** commit 17d994bd3045dae71acda676a4ef4762ed8ecd94 (2026-10-15) · **Language:** Python **Path conventions:** a/

= src/agents/; ri/ = src/agents/run_internal/; m/ = src/agents/models/. All paths relative to the repo root. **What it is:** The Agents SDK is OpenAI's library for building agents as `Agent` objects (instructions, tools, handoffs, guardrails) run by a `Runner` that loops model call, tool execution and handoff until a final output. The canonical shipped agent is an `Agent` with `@function_tool` tools run via `Runner.run`/`run_streamed`. The model is reached through the OpenAI Python SDK, by default the Responses API over HTTPS (optionally a Responses websocket), with Chat Completions, LiteLLM and any-llm providers selectable by model-name prefix.

### Control flow

- **Run loop:** `AgentRunner._run_impl` runs `while True`, calls `run_single_turn` once per iteration, then branches on `turn_result.next_step`: `NextStepRunAgain` continues, `NextStepFinalOutput` runs output guardrails and returns, `NextStepHandoff` swaps `current_agent` and loops, `NextStepInterruption` returns a result holding pending approvals. `a/run.py:L1032--L1062`, `a/run.py:L1331--L1372`, `a/run.py:L1543--L1552`; `ri/run_steps.py:L164--L190`
- **Stop rule and turn budget:** a response with no tool calls or handoffs is the final output; after tools ran, `check_for_final_output_from_tools` applies `Agent.tool_use_behavior` (`"run_llm_again"` default, `"stop_on_first_tool"`, a StopAtTools name list, or a callable); `current_turn += 1` and `MaxTurnsExceeded` once `current_turn > max_turns` (`DEFAULT_MAX_TURNS = 10`). `ri/turn_resolution.py:L776--L809`; `a/agent.py:L396--L418`; `a/run.py:L1597--L1606`; `a/run_config.py:L45`; `a/exceptions.py:L354--L363`
- **Guardrails the harness consumes:** input guardrails run on turn 0 only (sequential ones before the model call, parallel ones gathered with the model task), output guardrails run on the final output; a tripwire raises `InputGuardrailTripwireTriggered` / `OutputGuardrailTripwireTriggered` and ends the run. `a/run.py:L1046--L1062`, `a/run.py:L1833--L1845`; `ri/guardrails.py:L116--L170`, `ri/guardrails.py:L172--L225`; `a/guardrail.py:L72--L100`
- **Approval gate:** `function_needs_approval` evaluates `FunctionTool.needs_approval` before execution; unapproved calls become `ToolApprovalItem`s inside `NextStepInterruption`, the caller calls `RunState.approve/reject` and resumes by passing the `RunState` back as `input` to `Runner.run`. `ri/tool_execution.py:L1306--L1315`; `ri/run_steps.py:L179--L190`; `a/run_state.py:L1380--L1410`; `a/run.py:L279--L300`
- **Handoffs and sub-agents:** a `Handoff` is a function tool whose invocation returns the target `Agent`; `execute_handoffs` calls `on_invoke_handoff`, applies `input_filter` or `nest_handoff_history`, and returns `NextStepHandoff`, so the same loop continues with the new agent (one hop per turn, same run, same session); `Agent.as_tool` instead runs a nested `Runner.run` with its own `max_turns` and returns its output as a tool result. `a/handoffs/__init__.py:L126--L182`, `a/handoffs/__init__.py:L320--L350`; `ri/turn_resolution.py:L544--L600`, `ri/turn_resolution.py:L605--L690`; `a/agent.py:L606--L640`

### Input

- **Caller input:** `Runner.run(starting_agent, input, ...)` accepts a string, a list of Responses input items, or a `RunState` to resume; a string is normalized to one user message by `input_to_new_input_list`. `a/run.py:L279--L300`; `a/items.py:L813--L826`
- **Tool results re-enter:** each executed call becomes a `ToolCallOutputItem` (`function_call_output` keyed by `call_id`) in `generated_items`; `_prepare_turn_input_items` converts caller input plus generated items into the next turn's input list. `a/items.py:L431`, `a/items.py:L845--L860`; `ri/run_loop.py:L349--L358`
- **Session input:** `prepare_input_with_session` loads `session.get_items()` history and concatenates the new input (or lets `session_input_callback` merge them); with a server-managed conversation only the new items are prepared. `ri/session_persistence.py:L424--L510`; `a/run.py:L716--L745`
- **Streamed input events:** `run_streamed` puts each raw model event on an event queue as `RawResponsesStreamEvent` (receiving the stream is model interface; the queue is the harness's input side of rendering). `ri/run_loop.py:L2621--L2652`; `a/stream_events.py:L11--L60`

### Context

- **Instructions:** `Agent.get_system_prompt` returns the static `instructions` string or calls `instructions(context, agent)` (sync or async); `get_prompt` resolves a server-side prompt object; both are gathered at the start of `run_single_turn`. `a/agent.py:L1129--L1170`; `ri/run_loop.py:L2838--L2841`
- **Tool and handoff descriptions:** `get_all_tools` merges MCP server tools with function tools filtered by `is_enabled`; handoffs are listed as function tools; name collisions resolved per `tool_name_collision_policy`. `a/agent.py:L286--L316`; `ri/run_loop.py:L2836--L2850`
- **Per-call assembly and filter:** the request is instructions plus caller input plus all run items so far; `call_model_input_filter` may rewrite instructions and input, then items are de-duplicated before `get_model`. `ri/run_loop.py:L2946--L2956`; `ri/turn_preparation.py:L51--L95`
- **Session history and server-held conversations:** the `Session` protocol (`get_items`/`add_items`/`pop_item`) is implemented by `SQLiteSession`, SQLAlchemy/Redis/Mongo/Dapr/encrypted extensions, and `OpenAIConversationsSession`; with `conversation_id` or `previous_response_id`, `OpenAIServerConversationTracker.prepare_input` sends only new deltas and records the last `response_id`. `a/memory/session.py:L53--L95`; `a/memory/sqlite_session.py:L18--L60`; `a/memory/openai_conversations_session.py:L29--L60`; `ri/oai_conversation.py:L124--L140`, `ri/oai_conversation.py:L350--L393`
- **Window fitting:** `OpenAIResponsesCompactionSession` calls `responses.compact` after a write when at least 10 candidate items exist (default trigger); `ToolOutputTrimmer` replaces tool outputs over 500 chars from older turns with a preview; `model_settings.truncation="auto"` delegates truncation to the server. `a/memory/openai_responses_compaction_session.py:L68--L71`, `a/memory/openai_responses_compaction_session.py:L92--L170`; `ri/session_persistence.py:L713--L735`; `a/extensions/tool_output_trimmer.py:L88--L115`; `a/model_settings.py:L123`

### Model interface

- **Client:** the `Model` ABC exposes `get_response` and `stream_response`; `OpenAIResponsesModel` builds kwargs and calls `client.responses.create(**kwargs)` (or a Responses websocket), `OpenAIChatCompletionsModel` calls `chat.completions.create`; `MultiProvider` maps the model-name prefix (`openai/` or none, `litellm/`, `any-llm/`) to a `ModelProvider`. `m/interface.py:L37--L136`; `m/openai_responses.py:L513`, `m/openai_responses.py:L590--L640`, `m/openai_responses.py:L860--L900`; `m/openai_chatcompletions.py:L55`, `m/openai_chatcompletions.py:L742`; `m/multi_provider.py:L62--L75`, `m/multi_provider.py:L155--L200`; `ri/turn_preparation.py:L134--L145`
- **Retries and timeout:** `get_response_with_retry` wraps the call with `model_settings.retry` (`ModelRetrySettings` max_retries, backoff), provider advice from `get_openai_retry_advice` (honours Retry-After), exponential backoff with jitter, and `model_settings.timeout` raising `ModelTimeoutError`; a rewind hook restores session and tracker state before replay. `ri/run_loop.py:L3024--L3047`; `ri/model_retry.py:L175--L205`, `ri/model_retry.py:L389--L420`, `ri/model_retry.py:L574`; `a/retry.py:L210--L231`; `m/_openai_retry.py:L41--L112`; `a/model_settings.py:L188`, `a/model_settings.py:L215`
- **Streaming:** `stream_response_with_retry` consumes the provider's event stream and yields raw `TResponseStreamEvent`s, with usage added to `context_wrapper.usage` at the end. `ri/run_loop.py:L2621--L2652`; `m/openai_responses.py:L683--L720`; `ri/run_loop.py:L3065`
- **Tool-call format:** `Converter.convert_tools` emits Responses `{"type": "function", "strict": ...}` entries for function tools and handoffs, plus hosted tool configs; output items are Responses `function_call` items. `m/openai_responses.py:L2115--L2251`, `m/openai_responses.py:L2360--L2370`

### Output

- **Tool set:** `FunctionTool` (name, description, JSON schema, `on_invoke_tool`, `strict_json_schema`, `needs_approval`, `timeout`, tool guardrails) built by `@function_tool`; hosted tools (`WebSearchTool`, `FileSearchTool`, `CodeInterpreterTool`, `HostedMCPTool`, `ImageGenerationTool`) run inside the provider; `ComputerTool`, `ShellTool`, `ApplyPatchTool`, `LocalShellTool` run locally. `a/tool.py:L455--L530`, `a/tool.py:L2621`, `a/tool.py:L815--L1300`

- **Response processing:** `process_model_response` sorts output items into function runs, handoffs, computer actions, shell and apply-patch calls, custom tools and MCP approval requests, then `execute_tools_and_side_effects` runs the plan. `ri/turn_resolution.py:L3133--L3175`, `ri/turn_resolution.py:L941--L990`
- **Executor and parallelism:** `_FunctionToolBatchExecutor` creates one `asyncio.Task` per function call, bounded by `tool_execution.max_function_tool_concurrency`, isolating sibling failures and applying per-tool timeouts. `ri/tool_execution.py:L1548--L1600`, `ri/tool_execution.py:L1648--L1670`, `ri/tool_execution.py:L2341--L2371`; `a/run_config.py:L143--L163`
- **Sandbox:** `SandboxAgent` plus `RunConfig(sandbox=...)` run shell, file and patch tools inside a `DockerSandboxSession` or a `UnixLocalSandboxSession` (asyncio subprocess with PTY). `a/sandbox/sandbox_agent.py:L31--L60`; `a/sandbox/sandboxes/docker.py:L287`; `a/sandbox/sandboxes/unix_local.py:L261`
- **Result presentation:** `RunResult.final_output` and `RunResultStreaming.stream_events()` (`RawResponsesStreamEvent`, `RunItemStreamEvent`, `AgentUpdatedStreamEvent`); shell outputs are truncated to `max_output_length` before being shown to the model. `a/result.py:L547`, `a/result.py:L660`, `a/result.py:L1005`; `a/stream_events.py:L11--L60`; `ri/tool_execution.py:L1008--L1047`

### Production concerns

- **Persistence:** context: session history written by `save_result_to_session` to any `Session` store (SQLite, SQLAlchemy, Redis, MongoDB, Dapr, encrypted wrapper, OpenAI Conversations); control flow: `RunState.to_json/from_json` serialize an interrupted run for resume after approvals. `ri/session_persistence.py:L778--L800`; `a/extensions/memory/sqlalchemy_session.py`; `a/run_state.py:L842`, `a/run_state.py:L1899`, `a/run_state.py:L2423`
- **Safety & security:** control flow: `needs_approval` gate and input/output guardrails; output: tool input/output guardrails with `allow/reject_content/raise_exception` behaviours, Docker sandbox for shell and patches, mount-credential redaction in errors; no prompt-injection fencing of tool output found. `a/tool_guardrails.py:L60--L118`; `a/guardrail.py:L72--L100`; `a/sandbox/_mount_security.py:L487--L511`
- **Observability:** tracing spans (`agent_span`, `generation_span`, `function_span`, `handoff_span`, `guardrail_span`) batched by `BatchTraceProcessor` and posted by `BackendSpanExporter` to `api.openai.com/v1/traces/ingest` with 3 retries; `Usage` sums requests and input/output tokens per run; `RunHooks`/`AgentHooks` expose lifecycle callbacks. `a/tracing/create.py:L89--L123`, `a/tracing/create.py:L155--L235`; `a/tracing/processors.py:L45--L90`, `a/tracing/processors.py:L698--L771`; `a/usage.py:L111--L141`; `a/lifecycle.py:L13--L118`
- **Resilience:** model interface: retry policy, timeout, request rewind; output: `ToolTimeoutError` per tool; control flow: `MaxTurnsExceeded` handler via run error handlers, resumable `RunState`, session item rewind on failed writes. `ri/model_retry.py:L574`; `a/exceptions.py:L387`, `a/exceptions.py:L417`; `a/run.py:L1628`; `ri/session_persistence.py:L1622`
- **Performance:** streaming, concurrent tool tasks, parallel guardrails gathered with the model call, `PromptCacheKeyResolver` deriving `prompt_cache_key` from session or group id, `prompt_cache_retention`, `ToolOutputTrimmer`. `ri/prompt_cache_key.py:L17--L90`; `a/model_settings.py:L150`; `a/run.py:L1841`
- **Evaluation:** 172 pytest files under `tests/` with a `ScriptedModel` double that scripts model turns; no agent-eval harness (datasets, graders) found in the SDK. `src/agents/testing/model.py:L249`; `pyproject.toml:L256--L259`

### Hard to place

- **Server-managed conversation:** with `conversation_id/previous_response_id` the provider keeps the transcript; the harness's context primitive only sends deltas and the provider does assembly, so part of Context lives outside the harness. `ri/oai_conversation.py:L518--L630`
- **Hosted tools:** `WebSearchTool`, `CodeInterpreterTool`, `HostedMCPTool` execute inside the provider during the model call; the harness declares them (context) and reads results from the response (model

interface), with no local Output step. a/tool.py:L853, a/tool.py:L1158, a/tool.py:L1189
- **Server-side compaction:** OpenAIResponsesCompactionSession fits the window (context) by calling responses.compact, an extra provider request issued from a session write rather than from the loop. a/memory/openai_responses_compaction_session.py:L92--L170
- **tool_use_behavior callable:** lives on Agent and receives tool results (output) but decides whether to go again (control flow); assigned by function. ri/turn_resolution.py:L776--L809
- **Sandbox memory generation:** SandboxMemoryGenerationManager turns rollouts into memory for the model (context) by issuing its own model calls (control flow) outside the main loop. a/sandbox/memory/manager.py:L43

### Summary

| Primitive | Design decision |
|---|---|
| Input | String or Responses items plus Session history; tool outputs re-enter as run items; RunState resumes |
| Context | Instructions callable, tools and handoffs as functions, full item list; server compaction and trimmer |
| Model interface | OpenAI Responses API via SDK; prefix-routed providers; configurable retry, timeout, streaming |
| Output | Function tools as concurrent asyncio tasks; hosted tools; Docker or local sandbox for shell and patches |
| Control flow | NextStep loop with max_turns=10; guardrail tripwires; approval interruptions; handoffs swap agent in place |

## C.14 Google Agent Development Kit (Python)

**Source:** https://github.com/google/adk-python · **Version:** commit f78149a97e109e9c947ade06e3a9ab8a45385e0e (2026-10-15) · **Language:** Python **Path conventions:** adk/ = src/google/adk/; fl/ = src/google/adk/flows/llm_flows/. All paths relative to the repo root. **What it is:** ADK is Google's framework for building agents as a tree of BaseAgents (an LlmAgent with tools and sub-agents, plus SequentialAgent, ParallelAgent, LoopAgent workflow agents) executed by a Runner over a Session of Events held by a session service. The canonical agent is an LlmAgent with FunctionTools run by Runner.run_async. The model is reached through BaseLlm adapters: Gemini over the google-genai client (Gemini API or Vertex AI), with Claude (AnthropicLlm) and LiteLLM providers resolved by model name.

### Control flow

- **Runner loop:** Runner.run_async gets or creates the session, appends the user event, picks agent_to_run (_find_agent_to_run: the agent whose pending function call the message answers, else the last-replying transferable LlmAgent, else root), then iterates active_agent.run_async(ctx) inside _exec_with_plugin, appending every non-partial event to the session, and runs post-invocation compaction. adk/runners.py:L1124--L1136, adk/runners.py:L1190--L1215, adk/runners.py:L1336--L1370, adk/runners.py:L1815--L1846, adk/runners.py:L1531--L1557
- **Flow loop and stop rule:** BaseLlmFlow.run_async runs while True: _run_one_step_async (request processors, one model call, response processors, function calls) and breaks when the step's last event is_final_response() (no function calls or responses, not partial, or a long-running tool id) or the context is aborted or end_invocation is set. fl/base_llm_flow.py:L332--L353, fl/base_llm_flow.py:L377--L450; adk/events/event.py:L286--L309
- **Budgets:** increment_llm_call_count runs before every model call and raises LlmCallsLimitExceededError once calls exceed RunConfig.max_llm_calls (default read from ADK_MAX_LLM_CALLS); LoopAgent.max_iterations bounds workflow loops, and an escalate action exits them. fl/core/_model_call.py:L227; adk/agents/invocation_context.py:L63--L96, adk/

agents/invocation_context.py:L596--L609; adk/agents/run_config.py:L41--L45, adk/agents/run_config.py:L264--L282; adk/agents/loop_agent.py:L92--L130

- **Gates the harness consumes:** plugin then agent before_model_callback/before_tool_callback may return a replacement and short-circuit (first non-None wins); a tool whose check_require_confirmation is true emits an adk_request_confirmation long-running call that pauses the invocation until a ToolConfirmation function response arrives, which the confirmation request processor replays. fl/core/_finalizer.py:L150--L195; adk/plugins/plugin_manager.py:L277--L325; adk/tools/function_tool.py:L404--L430; fl/tools/_confirmation.py:L259--L300; adk/agents/invocation_context.py:L745--L770
- **Sub-agents:** AutoFlow adds a transfer_to_agent tool and instructions listing peers and parent; the tool sets actions.transfer_to_agent and the postprocessor runs the target's run_async in the same invocation and session; AgentTool instead spawns a nested Runner with in-memory services and returns its last text as a tool result; SequentialAgent, ParallelAgent (one TaskGroup task and branch per child) and LoopAgent sequence children without a model. fl/extensions/_agent_transfer.py:L44--L98; fl/core/_function_call_postprocessor.py:L72--L140; adk/tools/agent_tool.py:L224--L300; adk/agents/parallel_agent.py:L134--L136, adk/agents/parallel_agent.py:L176--L200; adk/agents/sequential_agent.py:L96--L130

### Input

- **User message:** run_async(user_id, session_id, new_message: types.Content, state_delta, run_config); the message becomes an Event(author='user') appended through session_service.append_event, with input blobs optionally saved as artifacts. adk/runners.py:L1124--L1136, adk/runners.py:L705--L735, adk/runners.py:L1667--L1700
- **Tool results re-enter:** _build_response_event wraps each tool result as a function-response Event; parallel results are merged into one event by merge_parallel_function_response_events; the event is yielded, appended to the session, and rebuilt into contents by the next step's content processor. fl/tools/_caller.py:L321--L360; fl/tools/_batch_executor.py:L61--L118, fl/tools/_batch_executor.py:L146--L170
- **Answers the model consumes:** request_input and get_user_choice are long-running tools that pause the invocation; the user's function response arrives as the next new_message and _find_agent_to_run routes it to the agent that asked. adk/tools/_request_input_tool.py:L27--L56; adk/tools/get_user_choice_tool.py:L23; adk/runners.py:L1815--L1846
- **Resume input:** for a resumable app, run_async with an invocation_id and no message resumes the paused invocation from stored events. adk/runners.py:L1254--L1300; adk/apps/_configs.py:L29--L48

### Context

- **Ordered request processors:** SingleFlow assembles one LlmRequest (model, contents, GenerateContentConfig, tools_dict) by running basic, auth, request_confirmation, instructions, identity, compaction, interactions, contents, context_cache, planning, code_execution and output_schema processors in that order. fl/single_flow.py:L42--L80; fl/base_llm_flow.py:L451--L485; adk/models/llm_request.py:L52--L95
- **Instructions:** _build_instructions appends the root global_instruction, the agent's static_instruction, and instruction (string or InstructionProvider, with {state_key} injection from session state) as system instruction, fencing dynamic text between begin/end markers. fl/prompt/_instructions.py:L69--L160; adk/agents/llm_agent.py:L327--L354, adk/agents/llm_agent.py:L824--L860
- **Conversation:** _ContentLlmRequestProcessor rebuilds contents from session events filtered by branch and visibility (include_contents='default') or only the current turn ('none'), rearranging function responses and dropping orphans; _maybe_append_user_content guarantees a trailing user turn. fl/context/_contents.py:L116--L200, fl/context/_contents.py:L473--L640; adk/models/base_llm.py:L234--L275

- **Memory:** `BaseMemoryService.search_memory` (in-memory, Vertex AI Memory Bank, Vertex RAG) is reached by `PreloadMemoryTool.process_llm_request`, which injects past conversations into the instructions, or by the `load_memory` function tool. `adk/memory/base_memory_service.py:L44--L141`; `adk/tools/preload_memory_tool.py:L33--L75`; `adk/tools/load_memory_tool.py:L38--L60`
- **Window fitting:** `EventsCompactionConfig` (sliding-window `compaction_interval`/`overlap_size` after an invocation, `token_threshold` at request time) summarizes events with `LlmEventSummarizer` into a compaction event that `_process_compaction_events` substitutes for raw events; `ContextFilterPlugin` keeps the last N invocations; `ContextCacheConfig` enables Gemini context caching. `adk/apps/_configs.py:L50--L85`; `adk/apps/compaction.py:L675--L800`; `adk/apps/llm_event_summarizer.py:L29--L160`; `fl/context/_compaction.py:L33--L100`; `adk/plugins/context_filter_plugin.py:L101--L120`; `fl/context/_cache.py:L36--L80`

### Model interface

- **Client:** `BaseLlm.generate_content_async(llm_request, stream)` is an async generator of `LlmResponse`; `Gemini` calls `genai.Client().aio.models.generate_content` or `generate_content_stream` (or the Interactions API); `LLMRegistry.resolve` picks the adapter class by model-name regex (Gemini, `AnthropicLlm` for `claude-*`, `LiteLlm` for other providers). `adk/models/base_llm.py:L132--L160`; `adk/models/google_llm.py:L99`, `adk/models/google_llm.py:L228--L383`; `adk/models/registry.py:L62--L84`, `adk/models/registry.py:L147--L200`; `adk/models/anthropic_llm.py:L1099`; `adk/models/lite_llm.py:L3679`
- **Streaming:** `call_llm_async` passes `stream=True` when `RunConfig.streaming_mode` is `SSE` and yields partial `LlmResponse`s as partial events, running `after_model_callback` on the final one. `fl/core/_model_call.py:L170--L310`; `adk/agents/run_config.py:L145`
- **Retries and fallback:** `Gemini.retry_options` (`types.HttpRetryOptions`) is handed to the SDK's `HttpOptions`; `FallbackModel` tries each listed model in turn on retriable status codes (429 and friends) unless a response was already yielded; `ReflectAndRetryModelPlugin` feeds model errors back as guidance. `adk/models/google_llm.py:L190--L205`, `adk/models/google_llm.py:L427--L462`; `adk/models/_fallback_model.py:L207--L260`, `adk/models/_fallback_model.py:L352--L410`; `adk/plugins/_reflect_retry_model_plugin.py:L36--L70`
- **Tool-call format:** each `BaseTool.process_llm_request` registers a `FunctionDeclaration` on `GenerateContentConfig.tools` and itself in `tools_dict`; responses carry `types.FunctionCall` parts that the postprocessor dispatches by name. `adk/tools/base_tool.py:L144--L200`; `adk/tools/function_tool.py:L118--L150`

### Output

- **Tool set:** `BaseTool` subclasses: `FunctionTool` (declaration generated from a Python callable, argument validation, optional confirmation), `LongRunningFunctionTool`, `AgentTool`, MCP, OpenAPI and toolsets, plus Google-hosted search, code execution and data tools. `adk/tools/base_tool.py:L57--L200`; `adk/tools/function_tool.py:L76--L150`, `adk/tools/function_tool.py:L370--L453`
- **Executor and parallelism:** `handle_function_call_list_async` prepares one `asyncio.Task` per function call and executes them concurrently, wrapping `tool.run_async(args, tool_context)` with plugin and agent before/after tool callbacks and an `execute_tool` span, then merges the responses into one event. `fl/tools/_batch_executor.py:L215--L250`, `fl/tools/_batch_executor.py:L384--L404`; `fl/tools/_caller.py:L471--L640`, `fl/tools/_caller.py:L663--L671`
- **Code execution:** an agent's `code_executor` runs model-written code via `BuiltInCodeExecutor` (inside Gemini), `UnsafeLocalCodeExecutor` (`exec` in process), `ContainerCodeExecutor` (Docker image), or GKE, Vertex and Agent Engine sandboxes; request and response processors extract code blocks and feed results back. `adk/code_executors/built_in_code_executor.py:L31`; `adk/code_executors/unsafe_local_code_executor.py:L120`; `adk/code_executors/container_code_executor.py:L126--L240`; `fl/extensions/_code_execution.py:L146--L190`
- **Result presentation:** every step is yielded to the caller as an `Event` (an `LlmResponse` with author, `actions`, `partial`), and `is_final_response()` marks the user-facing answer; `output_key` writes the

final text into session state. adk/events/event.py:L87--L160, adk/events/event.py:L286--L309; adk/agents/llm_agent.py:L1093--L1120
- **Tool effects as actions:** EventActions carries state_delta, artifact_delta, transfer_to_agent, escalate, requested_tool_confirmations and end_of_agent, which the session service and loops apply. adk/events/event_actions.py:L78--L204

### Production concerns

- **Persistence:** context: Session (state plus events) through BaseSessionService.append_event in memory, SQLite (aiosqlite), SQLAlchemy DatabaseSessionService, or Vertex AI sessions; artifacts via BaseArtifactService (file, GCS, in-memory); memory via memory services. adk/sessions/base_session_service.py:L166--L235; adk/sessions/sqlite_session_service.py:L184--L212; adk/sessions/database_session_service.py:L310--L340; adk/artifacts/base_artifact_service.py
- **Safety & security:** control flow: tool confirmation gate and before-callback vetoes; context: _fencing.py fences tool descriptions and quotes untrusted or other-agent text to blunt prompt injection, _restricted_pickle limits what session state can deserialize; output: container and remote code sandboxes; auth preprocessor and credential service for tool credentials. fl/context/_fencing.py:L98--L140, fl/context/_fencing.py:L206--L223; adk/sessions/_restricted_pickle.py; fl/single_flow.py:L51--L54
- **Observability:** OpenTelemetry spans call_llm (request, response, token usage attributes) and execute_tool with gen_ai semantic conventions; exporters configured from OTEL_EXPORTER_OTLP_* or Google Cloud; LoggingPlugin, DebugLoggingPlugin and BigQueryAgentAnalyticsPlugin; TokenUsage accounting. adk/telemetry/tracing.py:L251--L369, adk/telemetry/tracing.py:L623--L717; adk/telemetry/setup.py:L45--L114; adk/telemetry/_token_usage.py:L96; adk/plugins/logging_plugin.py
- **Resilience:** model interface: SDK HTTP retries and FallbackModel; output: ReflectAndRetryToolPlugin returns failures to the model up to max_retries, on_tool_error callbacks; control flow: on_model_error callbacks, resumable invocations (ResumabilityConfig), rewind_async to roll back state and artifacts, an abort signal checked each step. adk/plugins/reflect_retry_tool_plugin.py:L47--L131; adk/apps/_configs.py:L29--L48; adk/runners.py:L1394--L1432; adk/agents/invocation_context.py:L439--L460
- **Performance:** SSE streaming, concurrent tool tasks, ParallelAgent branches, Gemini context caching (ContextCacheConfig, GeminiContextCacheManager), event compaction and ContextFilterPlugin trimming. adk/agents/context_cache_config.py:L27--L84; adk/models/gemini_context_cache_manager.py:L84; adk/agents/parallel_agent.py:L134--L136
- **Evaluation:** AgentEvaluator.evaluate runs eval sets through LocalEvalService with trajectory, final-response match, LLM-as-judge, rubric, hallucination and safety metrics; 682 test files under tests/. adk/evaluation/agent_evaluator.py:L118--L130, adk/evaluation/agent_evaluator.py:L302--L390; adk/evaluation/local_eval_service.py:L131; pyproject.toml:L470--L473

### Hard to place

- **transfer_to_agent:** a tool call (output) whose effect is control flow, but the target runs inside the same invocation, session and event stream, so the "inner agent" is not a separate harness; transfer depth is unbounded via peer and parent hops. fl/core/_function_call_postprocessor.py:L72--L140
- **Agent routing by history:** _find_agent_to_run decides control flow by reading the session's events (context) rather than from the loop's own state. adk/runners.py:L1815--L1846
- **Plugins and callbacks are bundles:** a before_model_callback returning content both skips the model call (control flow) and stands in for the response (model interface); a before_tool_callback can replace a tool result (output); each resolved by function. adk/plugins/base_plugin.py:L41--L105; fl/core/_finalizer.py:L150--L195
- **Hosted execution:** BuiltInCodeExecutor and Google-hosted tools run inside Gemini; the harness declares them (context) and never executes (output). adk/code_executors/built_in_code_executor.py:L31
- **output_key into state:** model text is written to session state (persistence) and read back by {key}

instruction injection, so an output effect feeds the next context. adk/agents/llm_agent.py:L1093--L1120; fl/prompt/_instructions.py:L85--L95

### Summary

| Primitive | Design decision |
|---|---|
| Input | types.Content user message appended as an Event; tool results and HITL answers re-enter as function-response events |
| Context | Ordered request processors rebuild LlmRequest from session events each step; instructions, memory tool, compaction, caching |
| Model interface | BaseLlm adapters resolved by model name; Gemini via google-genai; SSE streaming; SDK retries; FallbackModel |
| Output | BaseTool subclasses run concurrently as asyncio tasks with callbacks; code executors from in-process to Docker |
| Control flow | Runner picks agent, flow loops until final response; max_llm_calls; confirmation pauses; transfers run sub-agents in place |

## C.15 AutoGen (agentchat AssistantAgent and its runtime)

**Source:** https://github.com/microsoft/autogen · **Version:** commit 027ecf0a379bcc1d09956d46d12d44a3ad9cee14 (2026-04-06) · **Language:** Python **Path conventions:** ac/ = python/packages/autogen-agentchat/src/autogen_agentchat/; core/ = python/packages/autogen-core/src/autogen_core/; ext/ = python/packages/autogen-ext/src/autogen_ext/; t/ = python/packages/autogen-agentchat/tests/. All paths relative to the repo root. **What it is:** AutoGen is Microsoft's layered agent framework: autogen-core is an actor-style message runtime, autogen-agentchat builds chat agents and teams on it, and autogen-ext supplies model clients, tools and code executors. AssistantAgent is the canonical tool-using agent; it reaches the model through a ChatCompletionClient (for example OpenAIChatCompletionClient over the openai SDK) and runs tools through a Workbench.

### Control flow

- **Per-turn tool loop:** on_messages_stream adds the incoming messages to the model context, queries memory, calls the model once, then _process_model_result loops for loop_iteration in range(max_tool_iterations): a text response returns immediately; tool calls are executed, their results added to context, and another model call is made unless this is the last iteration. ac/agents/_assistant_agent.py:L901--L1013, ac/agents/_assistant_agent.py:L1149--L1270
- **Stop rule and budget:** max_tool_iterations (default 1) bounds the loop; after it the agent either reflects (one more model call without tools) or emits a ToolCallSummaryMessage built from a format string; reflect_on_tool_use defaults to true only when a structured output type is set. ac/agents/_assistant_agent.py:L738--L739, ac/agents/_assistant_agent.py:L842--L853, ac/agents/_assistant_agent.py:L1302--L1325
- **Handoff as early exit:** if any executed call was a handoff tool, _check_and_handle_handoff returns a HandoffMessage carrying the tool-call context and the loop returns; multi-agent delegation then happens in the team. ac/agents/_assistant_agent.py:L1245--L1251, ac/agents/_assistant_agent.py:L1328--L1408
- **Team sequencing:** BaseGroupChat.run_stream starts a SingleThreadedAgentRuntime, publishes GroupChatStart, and reads an output queue until a GroupChatTermination; the manager's handle_agent_response applies the TerminationCondition and max_turns, then select_speaker (round-robin: next index modulo participants) and publishes GroupChatRequestPublish to the chosen ChatAgentContainer. ac/teams/_group_chat/_base_group_chat.py:L489--L558, ac/teams/_group_chat/_base_group_chat_manager.py:L135--L238, ac/teams/_group_chat/_round_robin_group_chat.py:L72--L82

- **Core runtime loop:** `SingleThreadedAgentRuntime._process_next` pops one envelope from an asyncio queue, runs intervention handlers (`on_send`/`on_publish`, which may return `DropMessage`), and dispatches to the agent's handler. `core/_single_threaded_agent_runtime.py:L332--L385`, `core/_single_threaded_agent_runtime.py:L671--L720`

### Input

- **Task entry:** `BaseChatAgent.run` wraps a string task as `TextMessage(source="user")`, accepts a message or message list, and calls `on_messages`; in a team the container buffers messages from other agents and passes them to `on_messages_stream` when asked to speak. `ac/agents/_base_chat_agent.py:L111--L153`, `ac/teams/_group_chat/_chat_agent_container.py:L86--L144`
- **Tool results re-enter the context:** each executed call yields a `FunctionExecutionResult` (content, `is_error`) and the batch is added as one `FunctionExecutionResultMessage`; `HandoffMessage.context` carries the prior tool calls into the receiving agent. `ac/agents/_assistant_agent.py:L1236--L1240`, `ac/agents/_assistant_agent.py:L1014--L1026`
- **Human input:** `UserProxyAgent` calls `input_func` (default `cancellable_input` over `input()`) and returns the text as a `TextMessage`, emitting `UserInputRequestedEvent` first; the model consumes it as the next turn. `ac/agents/_user_proxy_agent.py:L22--L27`, `ac/agents/_user_proxy_agent.py:L165--L170`, `ac/agents/_user_proxy_agent.py:L186--L200`

### Context

- **System prompt:** a single `SystemMessage` from the `system_message` argument (default tells the model to use tools and reply TERMINATE); it is prepended to the context's messages at call time. `ac/agents/_assistant_agent.py:L735--L737`, `ac/agents/_assistant_agent.py:L770`, `ac/agents/_assistant_agent.py:L1085--L1086`
- **History as `ChatCompletionContext`:** the agent keeps an `UnboundedChatCompletionContext` by default; `get_messages` returns what the model sees, and `on_reset` clears it. `ac/agents/_assistant_agent.py:L840`, `core/model_context/_chat_completion_context.py:L10--L60`, `ac/agents/_assistant_agent.py:L1626--L1628`
- **Window fitting variants:** `BufferedChatCompletionContext` keeps the last `buffer_size` messages (dropping a leading function-result), `TokenLimitedChatCompletionContext` pops middle messages until `remaining_tokens` or `token_limit` is satisfied, `HeadAndTailChatCompletionContext` keeps the first and last N; no summarization is built in. `core/model_context/_buffered_chat_completion_context.py:L34--L41`, `core/model_context/_token_limited_chat_completion_context.py:L57--L75`, `core/model_context/_head_and_tail_chat_completion_context.py:L41--L60`
- **Memory injection:** each `Memory.update_context` may rewrite the model context before the call, and the agent yields a `MemoryQueryEvent` with the retrieved items. `ac/agents/_assistant_agent.py:L1028--L1055`, `core/memory/_base_memory.py:L60--L94`
- **Client compatibility pass:** `_get_compatible_context` strips images when `model_info["vision"]` is false. `ac/agents/_assistant_agent.py:L1642--L1648`, `ac/utils/_utils.py:L36`

### Model interface

- **`ChatCompletionClient` protocol:** `create`/`create_stream` take messages, tool schemas, `json_output` and a `CancellationToken`, and expose `count_tokens`, `remaining_tokens` and `model_info`. `core/models/_model_client.py:L209--L295`
- **OpenAI client:** `BaseOpenAIChatCompletionClient.create` validates create args, awaits `chat.completions.create` linked to the cancellation token, records `RequestUsage`, and maps `tool_calls` to `FunctionCall`s with the content kept as `thought`; retries are delegated to the SDK via `max_retries`. `ext/models/openai/_openai_client.py:L663--L815`, `ext/models/openai/_openai_client.py:L131--L134`, `ext/models/openai/_openai_client.py:L1200`
- **Streaming:** `create_stream` iterates chunks, accumulates `content_deltas` and per-index tool-call fragments, and ends with a `CreateResult`; the agent re-emits text chunks as

ModelClientStreamingChunkEvent. ext/models/openai/_openai_client.py:L816--L1033, ac/agents/_assistant_agent.py:L1093--L1107
- **Fallback:** no backup-model routing found in the agent or client; ChatCompletionCache wraps a client to serve repeated requests from a cache store. ext/models/cache/_chat_completion_cache.py:L29--L70

### Output

- **Tool set and workbench:** callables become FunctionTools, and tools are wrapped in a StaticWorkbench (or the user passes workbenches); handoffs are separate tools; names must be unique. ac/agents/_assistant_agent.py:L774--L805, ac/agents/_assistant_agent.py:L827--L838, core/tools/_function_tool.py:L105--L125
- **Executor and concurrency:** all tool calls in one response run concurrently with asyncio.gather; _execute_tool_call parses arguments, tries handoff tools, then workbench.call_tool_stream, returning an error result for bad JSON or an unknown tool. ac/agents/_assistant_agent.py:L1197--L1215, ac/agents/_assistant_agent.py:L1536--L1624, core/tools/_static_workbench.py:L94--L124
- **Sandbox (code tools):** DockerCommandLineCodeExecutor writes each block to a work dir and runs timeout <s> <cmd> <file> inside a container (default 60 s, auto-remove); LocalCommandLineCodeExecutor runs subprocesses on the host with the same timeout. ext/code_executors/docker/_docker_code_executor.py:L85--L222, ext/code_executors/docker/_docker_code_executor.py:L327--L376, ext/code_executors/local/__init__.py:L45--L173
- **Result presentation:** ToolCallRequestEvent and ToolCallExecutionEvent are yielded as inner events, the final Response carries a TextMessage, StructuredMessage, ToolCallSummaryMessage or HandoffMessage, and Console prints the stream. ac/messages.py:L445--L500, ac/agents/_assistant_agent.py:L1498--L1535, ac/ui/_console.py:L82--L103

### Production concerns

- **Persistence:** save_state/load_state serialize the model context (context primitive) as AssistantAgentState; teams and the runtime save state the same way; nothing is written to disk by default. ac/agents/_assistant_agent.py:L1630--L1640, core/model_context/_chat_completion_context.py:L73
- **Safety & security:** code runs in Docker when that executor is chosen; runtime intervention handlers can drop or rewrite messages; CancellationToken cancels in-flight calls; no approval gate on tool calls or prompt-injection defense found. ext/code_executors/docker/_docker_code_executor.py:L85--L120, core/_single_threaded_agent_runtime.py:L691--L711
- **Observability:** the OpenAI client logs LLMCallEvent/LLMStreamStartEvent with prompt and usage tokens to the event logger, tools log ToolCallEvent, agentchat emits typed events on autogen_agentchat.events, and _telemetry adds OpenTelemetry tracing around runtime messages. ext/models/openai/_openai_client.py:L716--L717, core/logging.py:L10--L160, ac/__init__.py:L8--L11, core/_telemetry/_tracing.py
- **Resilience:** HTTP retries are the SDK's max_retries; code execution has a timeout; a TimeoutTermination and TokenUsageTermination exist for teams; no overflow compaction, only the trimming contexts. ext/models/openai/_openai_client.py:L1200, ac/conditions/_terminations.py:L229--L354
- **Performance:** concurrent tool execution, streaming chunks, ChatCompletionCache, and token-limited contexts. ac/agents/_assistant_agent.py:L1200, ext/models/cache/_chat_completion_cache.py:L29, core/model_context/_token_limited_chat_completion_context.py:L19
- **Evaluation:** pytest suites per package cover the assistant agent, group chats, termination conditions and executors with replay model clients. t/test_assistant_agent.py, t/test_group_chat.py, t/test_termination_condition.py

### Hard to place

- **Reflection call:** `_reflect_on_tool_use_flow` is a second model call whose request is assembled from the same context but with no tools; its decision to run is control flow, its assembly context, its call model interface, and it lives in one method. `ac/agents/_assistant_agent.py:L1409--L1497`
- **Handoff tool:** by rule the tool call is output and the delegation is control flow of the team; here the agent itself returns a `HandoffMessage` and the team's `Swarm` manager routes on it, so the delegation decision is split across two harness layers. `ac/agents/_assistant_agent.py:L1328--L1408`, `ac/teams/_group_chat/_swarm_group_chat.py`
- **Runtime intervention handlers:** `on_send`/`on_publish` can drop or rewrite any message between agents; they act on the transport, not the model request, so they sit between input and control flow rather than within one primitive. `core/_single_threaded_agent_runtime.py:L691--L711`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Chat messages from user or peers; tool results batched as one function-result message |
| Context | One system message plus a pluggable ChatCompletionContext (unbounded, buffered, token-limited); memory hooks |
| Model interface | ChatCompletionClient protocol; OpenAI SDK client with usage logging and streaming; SDK retries |
| Output | Workbench tools run concurrently via asyncio.gather; Docker or local code executors |
| Control flow | Bounded max_tool_iterations loop, then reflect or summarize; teams via actor runtime, termination conditions |

## C.16 CrewAI (one Agent executing a Task with tools)

**Source:** https://github.com/crewAIInc/crewAI · **Version:** commit 6115839576d68d2e83b9f7f31e6c55d471e1fc64 (2026-10-15) · **Language:** Python **Path conventions:** `c/` = `lib/crewai/src/crewai/`; `t/` = `lib/crewai/tests/`. All paths relative to the repo root. **What it is:** CrewAI (crewAI Inc.) is a multi-agent orchestration framework in which a `Crew` runs `Task`s through role-playing `Agent`s. Each task execution hands a prompt to an `AgentExecutor`, a state-machine `Flow` that loops a ReAct-style LLM call and tool execution until a final answer. The model is reached through the `LLM` class, which routes to native provider clients (OpenAI, Anthropic, Azure, Gemini, Bedrock) or falls back to `litellm.completion`.

### Control flow

- **Crew task sequencing:** `kickoff` prepares inputs then dispatches to `_run_sequential_process` or `_run_hierarchical_process` (which first creates a manager agent); both call `_execute_tasks`, a `for` loop over tasks that runs each synchronously via `task.execute_sync` or asynchronously via a future, feeding prior outputs forward. `c/crew.py:L1053--L1069`, `c/crew.py:L1522--L1529`, `c/crew.py:L1595--L1635`
- **Agent step loop as a Flow:** `AgentExecutor` is a `Flow`; `check_max_iterations` routes to `continue_reasoning` (text ReAct) or `continue_reasoning_native` (API tool calls), tool routers return to `initialized`, and `finalize` ends when `current_answer` is an `AgentFinish`. `c/agents/agent_executor.py:L2404--L2422`, `c/agents/agent_executor.py:L1828--L1909`, `c/agents/agent_executor.py:L2544--L2624`
- **Iteration budget:** `max_iter` (default 25) is checked by `has_reached_max_iterations`; on exhaustion `ensure_force_final_answer` appends a "give your final answer" instruction and makes one more LLM call. `c/agents/agent_builder/base_agent.py:L285--L287`, `c/utilities/agent_utils.`

py:L364--L375, c/agents/agent_executor.py:L1600--L1621, c/utilities/agent_utils.py:L377--L432

- **Task-level retries, timeouts and guardrails:** Agent.execute_task re-enters itself on exception up to max_retry_limit, optionally under a thread-pool max_execution_time; Task._execute_core then runs guardrails up to guardrail_max_retries times, re-executing the agent when one fails. c/agent/core.py:L862--L908, c/agent/core.py:L1029--L1068, c/task.py:L893--L910, c/task.py:L1327--L1395
- **Human-in-the-loop gate:** when human_input is set, invoke calls _handle_human_feedback, whose sync provider loops input() -> append feedback message -> re-run the flow until an empty reply. c/task.py:L233, c/agents/agent_executor.py:L3079--L3098, c/core/providers/human_input.py:L255--L265, c/core/providers/human_input.py:L319--L364

### Input

- **Crew inputs:** kickoff(inputs) normalizes a dict, runs before_kickoff callbacks, interpolates it into task and agent templates, and optionally runs crew-level planning. c/crews/utils.py:L243--L379
- **Tool results re-enter as text:** in text mode the observation is appended to the action text and pushed as an assistant message; in native mode each result becomes a role: "tool" message; tool errors are injected as "Observation: Error executing tool". c/agents/agent_executor.py:L1845--L1871, c/agents/agent_executor.py:L2014--L2046
- **Human feedback:** the feedback string is formatted with the human_feedback slice and appended to the executor's messages before re-running. c/core/providers/human_input.py:L230--L235, c/agents/agent_executor.py:L3537--L3549

### Context

- **System and user prompt assembly:** Prompts.task_execution concatenates i18n slices (role_playing, tools or no_tools, a skill block, a date block) into a system prompt and a task slice into the user prompt, substituting {role}, {goal}, {backstory}. c/utilities/prompts.py:L93--L142, c/utilities/prompts.py:L211--L260, c/translations/en.json:L8--L19
- **Task prompt enrichment:** _prepare_task_execution takes task.prompt(), adds an output schema and the prior tasks' context, then appends up to five recalled memories under a "Memories from past conversations" header; knowledge retrieval and training data are added next. c/agent/core.py:L678--L705, c/agent/core.py:L757--L823, c/agent/core.py:L957--L993
- **Message list per call:** _setup_messages builds system + history + user (+ trailing) messages with cache breakpoints; the executor's state.messages grows with every assistant and tool message within the task. c/agents/agent_executor.py:L510--L543, c/agents/agent_executor.py:L3274--L3283
- **Window fitting by summarization:** before each call _proactively_summarize_context asks the LLM whether the request exceeds 85% of the window and, if so, chunks and LLM-summarizes the history in place; a provider context-length error routes to recover_from_context_length, which summarizes when respect_context_window is set and otherwise exits. c/utilities/agent_utils.py:L632--L663, c/llms/base_llm.py:L427--L450, c/utilities/agent_utils.py:L942--L981, c/utilities/agent_utils.py:L1019--L1100
- **Cross-task memory store:** Memory is a pluggable store (LanceDB default) written after each task via remember_many and read by recall. c/memory/unified_memory.py:L76--L92, c/memory/unified_memory.py:L237--L248, c/agent/core.py:L1968

### Model interface

- **LLM.call:** emits a start event, runs before-call hooks, builds completion params, then dispatches to streaming or non-streaming handlers that call litellm.completion; native providers are substituted by _get_native_provider for OpenAI, Anthropic, Azure, Gemini and others. c/llm.py:L1846--L1960, c/llm.py:L1238--L1304, c/llm.py:L640--L692
- **Retries:** BaseLLM wraps every subclass call in run_with_rate_limit_retry: 3 attempts on throttling errors with exponential backoff (1 s to 8 s, 20% jitter), honoring a provider Retry-After. c/llms/base_

llm.py:L185--L206, c/llms/retry.py:L35--L38, c/llms/retry.py:L100--L143
- **Streaming:** _handle_streaming_response iterates litellm.completion(stream=True) chunks, accumulating text and tool-call arguments and emitting chunk events. c/llm.py:L815--L858, c/llm.py:L1146--L1194
- **Action format:** with native tool calling the response's tool calls are returned as objects; otherwise the text is parsed for Action:/Action Input: or Final Answer: by parser.parse, raising OutputParserError on malformed output; an unsupported-tools error downgrades to text mode. c/agents/agent_executor.py:L1726--L1747, c/agents/parser.py:L62--L126, c/agents/agent_executor.py:L1784--L1788

### Output

- **Tool execution (text mode):** execute_tool_and_check_finality drives ToolUsage._use, which rejects a repeated identical call, consults the cache, invokes the structured tool, and periodically re-appends the tool-format instructions to the result. c/agents/agent_executor.py:L1845--L1860, c/tools/tool_usage.py:L503--L521, c/tools/tool_usage.py:L558--L569, c/tools/tool_usage.py:L622--L631, c/tools/tool_usage.py:L760--L777
- **Tool execution (native mode):** execute_native_tool runs the pending calls in a ThreadPoolExecutor when _should_parallelize_native_tool_calls allows, else sequentially so a result_as_answer tool can short-circuit to AgentFinish. c/agents/agent_executor.py:L1966--L1985, c/agents/agent_executor.py:L2080--L2111, c/agents/agent_executor.py:L2014--L2046
- **Tool set and schema:** tools are BaseTools parsed to CrewStructuredTool; delegation and ask-question tools are added for other agents; native mode converts them to OpenAI function schemas. c/utilities/agent_utils.py:L173--L236, c/agent/core.py:L1355, c/utilities/agent_utils.py:L238--L306
- **No sandbox:** tools run in-process in the executor thread; a per-tool max_usage_count and tool_failure_policy are the only limits found. c/tools/tool_usage.py:L684--L690, c/agents/agent_builder/base_agent.py:L288--L290
- **Result presentation:** finalize logs the final answer through _show_logs; Task._execute_core wraps it in a TaskOutput (raw, pydantic, json) and fires task callbacks; the crew returns a CrewOutput. c/agents/agent_executor.py:L2622--L2624, c/task.py:L853--L890, c/task.py:L925--L937

### Production concerns

- **Persistence:** the long-term memory store (LanceDB or Qdrant) and task execution logs outlast the run; checkpoint config lets kickoff resume a restored crew. c/memory/unified_memory.py:L237--L248, c/crew.py:L1492--L1520, c/crew.py:L400--L406, c/crew.py:L1013
- **Safety & security:** guardrails (callable or LLM-judged) validate task output; hooks can block an LLM or tool call or request human approval; no execution sandbox found. c/task.py:L393--L406, c/hooks/llm_hooks.py:L137--L150, c/hooks/tool_hooks.py:L107--L121
- **Observability:** every step emits typed events on crewai_event_bus (task, LLM call, tool usage, memory retrieval); TokenCalcHandler and calculate_usage_metrics aggregate token counts; tracing and output_log_file are crew options. c/events/event_bus.py:L95--L110, c/events/event_bus.py:L572, c/agent/core.py:L1299, c/crew.py:L2273, c/crew.py:L343, c/crew.py:L412
- **Resilience:** rate-limit retries at the client, agent re-execution to max_retry_limit, execution timeout, parser-error recovery that re-prompts, context-length recovery by summarization. c/llms/retry.py:L124--L143, c/agent/core.py:L886--L888, c/agents/agent_executor.py:L2980--L3017
- **Performance:** tool result cache, streaming, parallel native tool calls, proactive summarization, max_rpm throttling via RPMController, prompt cache breakpoints. c/tools/tool_usage.py:L558--L569, c/agent/core.py:L1253--L1254, c/crew.py:L649, c/agents/agent_executor.py:L521--L525
- **Evaluation:** a large pytest suite with recorded cassettes covers crews, agents, tasks, LLMs and memory; utilities/evaluators holds agent and task evaluators. t/test_crew.py:L1--L40, t/cassettes, c/utilities/evaluators

### Hard to place

- **`AgentExecutor` as a Flow:** the per-task loop is itself expressed in CrewAI's `Flow` state machine (routers and listeners), so the loop body mixes control flow (`check_max_iterations`, `route_by_answer_type`) with context (`_setup_messages`) and output (`execute_tool_action`) in one class; the rules split it by function but the file does not. `c/agents/agent_executor.py:L187`, `c/agents/agent_executor.py:L2410--L2422`
- **Planning as an in-loop sub-harness:** when `planning_enabled`, `generate_plan` makes its own model calls, creates todos, and the executor runs each todo as a nested reasoning loop with replanning; a program that sequences model calls, so control flow, though it lives inside the executor. `c/agents/agent_executor.py:L554--L598`, `c/agents/agent_executor.py:L1312--L1379`, `c/agents/agent_executor.py:L2752--L2803`
- **Proactive summarization triggered by the model client:** `is_context_window_overflowing_threshold` lives on `BaseLLM` yet decides what the model is shown; by function it is context. `c/llms/base_llm.py:L427--L450`, `c/utilities/agent_utils.py:L645`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Crew inputs interpolated into templates; tool results and human feedback appended as messages |
| Context | Role/goal/backstory system prompt, task plus memory and knowledge; LLM summarization near window limit |
| Model interface | LLM class over native provider clients or litellm; rate-limit retries; streaming; native or ReAct text |
| Output | In-process BaseTool execution with cache and repeat check; parallel native calls; no sandbox |
| Control flow | Flow state machine per task to max_iter, task sequencing by crew, guardrail retries, human gate |

## C.17 smolagents (CodeAgent and ToolCallingAgent)

**Source:** https://github.com/huggingface/smolagents · **Version:** commit 96f33faaf028479119ec8d34507b47694cf14e34 (2026-10-06) · **Language:** Python **Path conventions:** `s/` = `src/smolagents/`; `t/` = `tests/`; `ex/` = `examples/`. All paths relative to the repo root. **What it is:** smolagents is Hugging Face's small agent library built around one ReAct loop (`MultiStepAgent`) with two concrete agents: `CodeAgent`, whose model writes a Python snippet per step that a sandboxed interpreter runs, and `ToolCallingAgent`, whose model emits JSON tool calls. The model is reached through a `Model` adapter (OpenAI-compatible, LiteLLM, HF Inference, Bedrock, local transformers/vLLM/MLX) that normalizes messages and returns a `ChatMessage`.

### Control flow

- **Step loop:** `_run_stream` runs `while not returned_final_answer and self.step_number <= max_steps`, building one `ActionStep` per iteration and calling `_step_stream`, which makes exactly one model call and executes its action. `s/agents.py:L540--L611`
- **Stop rule:** the loop ends when a step yields an `ActionOutput` with `is_final_answer` (the `final_answer` tool was called); optional `final_answer_checks` run first and an `AgentError` from them keeps the loop going. `s/agents.py:L577--L592`, `s/agents.py:L613--L618`
- **Step budget and forced answer:** `max_steps` (default 20) caps iterations; on exhaustion `_handle_max_steps_reached` makes one extra model call (`provide_final_answer`) and records an `AgentMaxStepsError` step. `s/agents.py:L294--L303`, `s/agents.py:L606--L608`, `s/agents.py:L625--L637`

- **Error policy inside the loop:** `AgentGenerationError` (model call failed) is re-raised and ends the run; any other `AgentError` (parse, tool, execution) is stored on the step and the loop continues; `interrupt()` sets a switch checked at each iteration. `s/agents.py:L594--L604`, `s/agents.py:L546--L547`, `s/agents.py:L754--L756`
- **Planning cadence and managed agents:** `planning_interval` schedules an extra planning model call at step 1 and every N steps; `managed_agents` are sub-agents exposed to the model as tools, each running its own `run()` loop (depth 1 per level) and returning a templated report string. `s/agents.py:L548--L566`, `s/agents.py:L369--L387`, `s/agents.py:L868--L890`

### Input

- **Task entry:** `run(task, images, additional_args)` stores the task, appends a sentence listing `additional_args` keys, resets memory unless `reset=False`, and appends a `TaskStep`; the CLI's `run_smolagent` is a thin wrapper over `agent.run(prompt)`. `s/agents.py:L470--L491`, `s/cli.py:L219--L259`
- **Tool results re-enter as observations:** `ToolCallingAgent.process_tool_calls` joins each tool's stringified result into `memory_step.observations`; `CodeAgent` stores "Execution logs" plus the truncated last expression value. `s/agents.py:L1430--L1442`, `s/agents.py:L1753--L1754`
- **Errors re-enter as tool responses:** `ActionStep.to_messages` renders a stored error as a `TOOL_RESPONSE` message beginning "Error:" with a retry instruction. `s/memory.py:L138--L150`
- **Human input:** no mid-run prompt in the loop; the only channel is the optional `UserInputTool`, whose result the model consumes via `input()`. `s/default_tools.py:L93--L103`

### Context

- **System prompt assembly:** `initialize_system_prompt` renders a Jinja template with the tool and managed-agent descriptions, authorized imports, custom `instructions`, and code-block tags; the template is `code_agent.yaml` (or `toolcalling_agent.yaml`). `s/agents.py:L1620--L1636`, `s/agents.py:L1265--L1274`, `s/prompts/code_agent.yaml:L1--L174`
- **History is the full memory:** `write_memory_to_messages` emits the system prompt plus every step's `to_messages` (assistant output, "Calling tools:" message, observation images, observation text, error); no window fitting, compaction or summarization exists. `s/agents.py:L758--L770`, `s/memory.py:L92--L151`
- **Role normalization and merging:** `get_clean_message_list` maps `TOOL_CALL` to assistant and `TOOL_RESPONSE` to user, merges consecutive same-role messages, and can flatten to text for models without multimodal input. `s/models.py:L282--L285`, `s/models.py:L332--L399`
- **Size limits by truncation only:** observations are head/tail truncated at 20,000 chars and executor print logs at 50,000 chars. `s/utils.py:L254--L267`, `s/local_python_executor.py:L57`, `s/local_python_executor.py:L1643--L1646`
- **Planning and summary variants:** planning calls wrap the memory in pre/post templates that include `remaining_steps`; `summary_mode` drops the system prompt and model outputs when a managed agent reports its work. `s/agents.py:L696--L713`, `s/memory.py:L92--L94`, `s/memory.py:L200--L206`

### Model interface

- **`Model` adapter:** `_prepare_completion_kwargs` builds the wire request (normalized messages, stop sequences, `response_format`, tool JSON schemas with `tool_choice="required"`); subclasses implement `generate`/`generate_stream` over their SDK, for example `OpenAIModel.generate` calling `client.chat.completions.create` and returning a `ChatMessage` with `TokenUsage`. `s/models.py:L502--L551`, `s/models.py:L1761--L1796`
- **Retries and rate limiting:** `ApiModel` wraps every call in a `Retrying` helper (3 attempts, exponential wait with jitter) whose predicate matches only rate-limit errors, and throttles with a `RateLimiter`; `LiteLLMRouterModel` delegates fallback routing to a litellm Router. `s/models.py:L39`, `s/models.py:L1174--L1182`, `s/models.py:L1194--L1202`, `s/models.py:L1363`

- **Streaming:** `generate_stream` yields `ChatMessageStreamDeltas` with `include_usage`; the agent aggregates them with `agglomerate_stream_deltas` and re-renders a rich `Live` panel per delta. `s/models.py:L1707--L1730`, `s/agents.py:L1660--L1672`
- **Action format:** `ToolCallingAgent` uses native tool calls and falls back to `parse_tool_calls` on text; `CodeAgent` extracts a code blob between `<code>`/`</code>` tags (or a JSON `{"code":...}` structured output) and appends the closing tag to history to nudge early stopping. `s/models.py:L583--L595`, `s/agents.py:L1326--L1333`, `s/agents.py:L1686--L1708`

### Output

- **Tool execution (`ToolCallingAgent`):** `execute_tool_call` checks the name against tools plus managed agents, substitutes state variables, validates arguments, and calls the tool with `sanitize_inputs_outputs=True`; multiple calls in one response run concurrently in a `ThreadPoolExecutor`. `s/agents.py:L1453--L1502`, `s/agents.py:L1419--L1432`
- **Code execution (`CodeAgent`):** the parsed snippet becomes a synthetic `python_interpreter` tool call and runs through `self.python_executor`; `executor_type` selects `local` or a remote sandbox. `s/agents.py:L1710--L1737`, `s/agents.py:L1598--L1618`
- **Local sandbox:** `LocalPythonExecutor` is an AST interpreter (`evaluate_python_code`) with an authorized-import allowlist, `DANGEROUS_MODULES`/`DANGEROUS_FUNCTIONS` denylists, a 10,000,000 operation cap, a 1,000,000 while-iteration cap and a 30 s timeout; `final_answer` is patched to raise `FinalAnswerException`. `s/local_python_executor.py:L57--L60`, `s/local_python_executor.py:L130--L154`, `s/local_python_executor.py:L1444--L1448`, `s/local_python_executor.py:L1583--L1668`
- **Remote sandboxes:** `E2BExecutor`, `DockerExecutor` (a Jupyter kernel-gateway container reached over WebSocket), `ModalExecutor` and `BlaxelExecutor` run the snippet elsewhere and return the final answer as a prefixed payload, with pickle deserialization refused unless `allow_pickle=True`. `s/remote_executors.py:L53--L96`, `s/remote_executors.py:L307--L332`, `s/remote_executors.py:L551--L677`
- **Result presentation:** the rich `AgentLogger` prints the code, execution logs and `Out:` value per step; final values pass through `handle_agent_output_types` (text, image, audio wrappers). `s/agents.py:L1738--L1762`, `s/monitoring.py:L130--L230`, `s/tools.py:L231--L249`

### Production concerns

- **Persistence:** memory lives only in the process (`AgentMemory.steps`); `save`/`to_dict`/`push_to_hub` serialize the agent definition (tools, prompts, model config) but not run state, and `RunResult` returns steps and token usage to the caller. `s/memory.py:L214--L235`, `s/agents.py:L892--L1010`, `s/agents.py:L196--L265`
- **Safety & security:** code runs in the AST interpreter with import allowlist and denylists or in a remote sandbox; API keys are excluded from `to_dict`; tool inputs/outputs are type-sanitized; no approval gate or prompt-injection defense found. `s/local_python_executor.py:L130--L154`, `s/models.py:L621--L625`, `s/tools.py:L244--L248`
- **Observability:** `Monitor.update_metrics` is registered as a step callback and sums input/output tokens and step duration; each step keeps `model_input_messages`, `model_output_message`, `token_usage` and `timing`; OpenTelemetry tracing is external via the `openinference` instrumentor. `s/monitoring.py:L81--L118`, `s/agents.py:L416--L435`, `t/test_telemetry.py:L26--L35`
- **Resilience:** rate-limit retries at the model client, a per-snippet execution timeout, error steps fed back to the model, and a forced final answer at `max_steps`; no resume after crash found. `s/models.py:L1174--L1182`, `s/local_python_executor.py:L1664--L1667`, `s/agents.py:L625--L637`
- **Performance:** parallel tool calls via threads, token streaming, stop sequences to cut generation, head/tail truncation of observations, and a client-side `RateLimiter`. `s/agents.py:L1419--L1432`, `s/agents.py:L1651--L1655`, `s/utils.py:L257--L267`, `s/models.py:L1173`
- **Evaluation:** a pytest suite covers agents, executors, models and memory, and `examples/smolagents_benchmark` scores agents on benchmark tasks. `t/test_agents.py:L1--L40`, `t/test_local_python_executor.py:L1--L40`, `ex/smolagents_benchmark/run.py:L1--L40`

### Hard to place

- **`final_answer` tool:** it is a tool the model calls (output) whose execution raises `FinalAnswerException` inside the executor, and the loop reads `is_final_answer` to stop; the rules split it cleanly (output executes, control flow consumes the flag) but the stop signal is carried by an exception thrown inside the sandbox. `s/local_python_executor.py:L1630--L1636`, `s/agents.py:L577--L592`
- **Shared `state` dict:** tool results that are images or audio are stored under a name in `self.state` and the model is told only "Stored 'image.png' in memory"; the same dict is pushed into the executor namespace, so it is a record kept for the model (context) that lives in the execution environment (output). `s/agents.py:L1393--L1400`, `s/agents.py:L490--L492`, `s/agents.py:L1444--L1451`
- **`final_answer_checks`:** user-supplied validators the harness runs on the model's answer and that can force another iteration; a harness-consumed gate, so control flow, though they execute arbitrary code like a tool. `s/agents.py:L613--L618`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Task string plus optional images and named args; tool results and errors re-enter as text observations |
| Context | Jinja system prompt plus full step memory replayed each call; truncation only, no compaction |
| Model interface | Pluggable Model adapters over provider SDKs; rate-limit retries; native tool calls or code blobs |
| Output | JSON tool calls run in threads, or Python snippets in an AST interpreter or remote sandbox |
| Control flow | ReAct while-loop to max_steps, stops on final_answer; optional periodic planning; managed agents as tools |

## C.18 Cursor Agent

**Source:** https://cursor.com/docs (official docs; source not published) · **Version:** documentation as of 2026-10-16 · **Language:** not documented **Path conventions:** `D` = https://cursor.com/docs (https://docs.cursor.com now returns a 308 redirect to `D`); every citation is `D/page § Section heading` and was read on 2026-10-16. Where a page states what happens but not how, I write "not documented". **What it is:** Cursor's Agent is the agentic coding harness inside the Cursor editor (sidepane), the Agents Window, the `agent` CLI and Cloud Agents, made by Anysphere (Cursor). The docs say an agent "is built on three components": instructions (system prompt and rules), tools and the model you choose, with instructions and tools tuned per model (`D/agent/overview § How Agent works`). Requests go from the client to Cursor's backend and on to third-party providers (OpenAI, Anthropic, Google) or to inference providers hosting Cursor's own models (`D/enterprise/privacy-and-data-governance § 1. LLM requests`).

### Control flow

- **Core loop and budgets:** the agent plans, calls tools, observes results and repeats; "There is no limit on the number of tool calls Agent can make during a task"; the stop rule (a response without tool calls) is not documented; no turn or token budget flag exists in the CLI reference, so spend is capped only by account usage limits and spending caps. `D/agent/overview § How Agent works`, `§ Tools`; `D/cli/reference/parameters § Global options`; `D/models-and-pricing § Top-ups and spending caps`
- **Stop hook as loop extender:** the `stop` hook receives `status` (completed, aborted, error) and `loop_count` and may return `followup_message` to auto-submit the next user turn; `loop_limit` caps this at 5 by default (`null` = no limit). `D/hooks § Reference § Hook events § stop`, `§ Configuration § Per-Script Configuration Options`

- **Approval gates the harness consumes:** Run Modes govern shell, MCP and Fetch calls: Auto-review (allowlisted calls run, other shell commands run in the sandbox, the rest go to an LLM classifier on the Cursor backend, Gemini 3.5 Flash Lite with Haiku fallback), Allowlist (no classifier), Run Everything (no prompts); "When the classifier blocks a call, Cursor can try another approach", and if the agent still wants it the user gets an approval prompt; hook decisions merge as deny over ask over allow and exit code 2 blocks; Plan Mode writes nothing until the user clicks to build after clarifying questions, research, plan creation and review. `D/agent/security/run-modes § Pick a mode`, `§ How Auto-review works`; `D/hooks § Configuration`, `§ Hook Types § Command-Based Hooks`; `D/agent/plan-mode § How it works`
- **Sub-agent delegation and model keyed to work:** the agent delegates via the Task tool; each subagent "operates in its own context window" and returns a final message plus an agent ID for resumption; foreground blocks, background returns immediately; several Task calls in one message run in parallel; a subagent launched by a subagent "can't launch further ones" (depth 2); the Explore subagent "runs in its own context window with a faster model", custom subagents set `model: inherit` or a specific ID, and Cursor Router (Auto with Balance/Intelligence) runs a classifier on each request by task type and complexity to pick the model. `D/subagents § How subagents work § Foreground vs background`, `§ Using subagents § Parallel execution`, `§ FAQ § Can subagents launch other subagents?`, `§ Custom subagents § Model configuration`; `D/agent/tools/search § Explore subagent`; `D/cursor-router § How it works`
- **Outer harness (Projects):** a coordinator agent on Cloud Agents "doesn't write code itself"; it plans and delegates to agents "running as many in parallel as the work needs" and starts a local agent when something must run on the user's machine; `/goal` gives a long-lived objective and `/loop` reruns a prompt on an interval. `D/agent/projects § How a Project works`, `§ Cloud by default, local when needed`; `D/agent/overview § Goals with /goal`; `D/skills § Built-in Cursor skills`

### Input

- **Prompt capture:** the sidepane chat input (with `@` mentions, image and voice input), `agent "prompt"` in the CLI, and `agent -p` for scripts with `CURSOR_API_KEY`. `D/agent/prompting § @ mentions`, `§ Image input`; `D/cli/overview § Interactive mode`, `§ Non-interactive mode`; `D/cli/headless § Setup`
- **Mid-turn messages:** Enter queues a message until the task finishes; "Send now" or Enter twice delivers it "at the agent's next tool call instead of cutting off work mid-action"; in the CLI, Enter steers at a safe boundary and a second Enter interrupts the turn. `D/agent/overview § Queued messages`, `§ Steer a running agent`
- **Model-consumed answers:** the Ask questions tool asks clarifying questions and "the agent continues reading files, making edits, or running commands" while waiting; answers are incorporated when they arrive. `D/agent/overview § Tools § Ask questions`
- **Tool results and events re-enter the conversation:** tool results appear in the Conversation segment of the context window; for Cloud Agents, subscription events (Slack replies, PR comments, CI results) "land as follow-ups in the same conversation", and bursts wake the agent once. `D/agent/prompting § Context usage`; `D/cloud-agent/capabilities § Subscriptions`

### Context

- **Rules:** Project Rules (`.cursor/rules/*.mdc`), User Rules, Team Rules and `AGENTS.md`; "rule contents are included at the start of the model context"; `alwaysApply`, `globs` (auto-attach when a matching file is in context), `description` (agent decides, via the Fetch Rules tool) or @-mention only; files referenced from a rule are not inlined; precedence is Team, then Project, then User Rules; nested `AGENTS.md` combine with parents. `D/rules § How rules work`, `§ Project rules § Rule anatomy`, `§ Referencing files from a rule`, `§ Team Rules § Format and how Team Rules are applied`, `§ AGENTS.md § Nested AGENTS.md support`; `D/agent/overview § Tools § Fetch Rules`
- **Skills:** discovered from `.cursor/skills/`, `.agents/skills/` and compatibility folders; "The agent is presented with available skills and decides when they are relevant based on context" and resources load on demand; `paths` scopes a skill to files; a skill used as a Custom Mode "stays in context on every turn". `D/skills § How skills work`, `§ Skill directories`, `§ Scoping a skill to specific`

files; D/agent/prompting § Custom Modes

- **Codebase retrieval:** Instant Grep "builds and queries its index on your machine" and Cursor "does not store embeddings of your codebase for search"; the agent gathers files through search and read tools, and "When Agent opens a match, that file content can still be included in the model request". D/agent/tools/search § Instant Grep, § Privacy and security
- **Summarization and window accounting:** "When the window gets close to full, Cursor compresses older parts of the conversation into a summary"; the context ring breaks the prompt into System prompt, Tools, Rules, Skills, MCP, Subagents, Summarized conversation and Conversation; the CLI /summarize (alias /compress) frees space on demand; preCompact hook can show a message but cannot block compaction. D/agent/prompting § Context usage; D/cli/using § Working with Agent § Selecting context; D/hooks § Reference § Hook events § preCompact
- **Memory:** no native memories feature is documented; the nearest is Projects' "Shared context", a set of files synced across the cloud and local machines where agents store research, artifacts, codebase knowledge and preferences; side chats use the parent thread as hidden reference context; postToolUse and sessionStart hooks can inject additional_context. D/agent/projects § Shared context; D/agent/overview § Side chats; D/hooks § Reference § Hook events § postToolUse, § sessionStart

### Model interface

- **Models and reach:** Claude, GPT, Gemini, Grok, Composer and others are selectable; "Models are hosted by the model provider, a trusted partner, or Cursor"; the client sends prompts and code context to Cursor and on to the provider; most models run under zero-data-retention agreements; BYOK API keys and custom base URLs are supported (not on Plus). D/models-and-pricing § FAQ § Where are models hosted?, § Plans; D/enterprise/privacy-and-data-governance § 1. LLM requests, § Models with data retention
- **Routing and fallback:** Auto has Cost, Balance and Intelligence modes served by Cursor Router; on Enterprise the router routes to an allowed model when one is blocked; "If a Fable request triggers a security guardrail, Cursor automatically routes it to Claude Opus"; a subagent's configured model falls back to a compatible one when blocked by an admin or plan. D/models-and-pricing § Auto modes; D/cursor-router § How it works; D/enterprise/privacy-and-data-governance § Models with data retention; D/subagents § Custom subagents § Model configuration § When the configured model won't be used
- **Context window limits:** Grok 4.7 defaults to 256k with long context to 500k at 2x above 256k; Claude 4.5+ models up to 1M at the same rate; GPT-5.x up to 1M with 2x input pricing; Max Mode (legacy plans) extends the window at API rate plus 20%. D/models-and-pricing § Included usage § Cursor models, § Third-party models § Model pricing, § Legacy request-based pricing § Max Mode
- **Streaming:** --output-format stream-json with --stream-partial-output streams text deltas; events are system (init with model), assistant, tool_call (started/completed) and result with duration_ms. D/cli/headless § Example scripts § Real-time progress tracking

### Output

- **Tool set:** Search files and folders, Web, Fetch Rules, Read files (including images for vision models), Edit files (applied automatically), Run shell commands, Browser, Image generation (saved to assets/), Ask questions; plus the Task tool for subagents. D/agent/overview § Tools; D/subagents § Using subagents § Parallel execution
- **Terminal execution:** "Cursor runs shell commands directly in your terminal" using the default terminal profile; CURSOR_AGENT is set in the shell; sudo passwords in the CLI go to sudo over a secure IPC channel and "the AI model never sees it". D/agent/tools/terminal § Terminal, § Troubleshooting § Disable heavy prompts for Cursor sessions; D/cli/overview § Sudo password prompting
- **Sandboxing:** macOS uses Seatbelt through sandbox-exec; Linux uses Landlock and seccomp (kernel 6.2+, Bubblewrap as a reported fallback backend); network is "Blocked by default, then opened by your network mode" (sandbox.json Only, sandbox.json + Defaults, Allow All); .git/config, .git/hooks, .vscode and .cursorignore are protected; a command that fails on a sandbox restriction may be

rerun outside the sandbox under classifier review. D/agent/security/run-modes § Sandboxing § How sandboxing works on your platform, § Network access, § Sandbox configuration

- **MCP:** servers configured in .cursor/mcp.json or ~/.cursor/mcp.json over stdio, SSE or Streamable HTTP; "Cursor asks for approval before using MCP tools by default"; responses appear in chat with expandable arguments and results; prompts, resources and elicitation are supported; Cursor "isolates server failures to prevent one server from affecting others". D/mcp § Installing MCP servers § Configuration locations, § What is MCP? § How it works, § Using MCP in chat § Tool approval, § Tool response, § FAQ § What happens if an MCP server crashes or times out?
- **Result presentation and rollback:** edits save to disk immediately and are reviewed in chat or with Ctrl+R in the CLI; checkpoints snapshot modified files before significant changes and are "stored locally and separate from Git"; print mode emits text, json (.result) or stream-json. D/agent/security § First-party tool calls; D/cli/using § Working with Agent § Review; D/agent/overview § Checkpoints; D/cli/headless § How it works § File modification in scripts

### Production concerns

- **Persistence:** chat transcripts are searchable from a local index and resumable (agent ls, agent resume, --continue); checkpoints live locally; Projects' shared context syncs across machines (context); Cloud Agent conversation state is kept indefinitely by default and VM snapshots are deleted after 30 days of inactivity. D/agent/overview § Conversation search; D/cli/overview § Sessions; D/agent/projects § Shared context; D/cloud-agent/security § What data is stored, where, and for how long
- **Safety & security:** "By default, sensitive actions require your manual approval" (output and control flow); terminal commands need approval, config-file edits need approval, .cursorignore hides files, "All MCP connections need your approval", agents "cannot make arbitrary network requests with default settings"; Auto-review "is not a security boundary"; CLI permissions Shell(), Read(), Write(), WebFetch(), Mcp() with "Deny rules take precedence over allow rules"; Cloud Agents keep Runtime Secrets out of the transcript and the model, sign commits with an HSM-backed Ed25519 key and never merge without review. D/agent/security § First-party tool calls, § Third-party tool calls, § Network requests; D/agent/security/run-modes § How Auto-review works § Auto-review is not a security boundary; D/cli/reference/permissions § Permission types, § Pattern matching; D/cloud-agent/security § Autonomy and prompt injection
- **Observability:** the context ring shows token use by category; hooks observe every event (afterShellExecution, afterAgentThought, sessionEnd) as spawned processes exchanging JSON over stdio; MCP Logs in the Output panel; Cloud Agent runs are logged, with audit logs streamable to a SIEM, webhook or S3. D/agent/prompting § Context usage; D/hooks § Hook categories; D/mcp § FAQ § How do I debug MCP server issues?; D/cloud-agent/security § Auditability
- **Resilience:** compaction when the window nears capacity; a failed subagent "returns an error status" and the parent can retry or resume it by agent ID; hooks fail open by default (failClosed: false) and a classifier-blocked call lets the agent "try another approach"; retry policy for failed model requests is not documented. D/agent/prompting § Context usage; D/subagents § FAQ § What happens if a subagent fails?; D/hooks § Configuration § Per-Script Configuration Options; D/agent/security/run-modes § How Auto-review works
- **Performance:** parallel subagents and the Explore subagent on a faster model; cache-read token pricing implies prompt caching but the mechanism is not documented; streaming deltas in the CLI; Cursor Router picks "the most cost-effective model"; "Requests are never downgraded in quality or speed". D/subagents § Using subagents § Parallel execution; D/models-and-pricing § Third-party models § Model pricing, § Top-ups and spending caps § What happens when I reach my limit?; D/cursor-router § How it works
- **Evaluation:** not documented (no eval or test-suite material on any page read).

### Hard to place

- **Auto-review classifier:** a second model call that decides whether a tool call runs. By the rules it is control flow (an approval gate the harness consumes), but it also makes its own read-only tool calls (`ReadFile`, `Grep`, `Glob`, `ListDir`) on the machine, which is output of a nested mini-harness; placed under control flow with that note. `D/agent/security/run-modes § How Auto-review works`
- **Fetch Rules tool:** a tool call the model makes (output) whose only effect is to decide what the model is shown next (context); assigned by function to context, consistent with the rule that code inside a tool deciding what the model sees is context. `D/agent/overview § Tools § Fetch Rules`
- **Cloud Agents versus local Run Modes:** "Run Modes apply to local agents"; Cloud Agents auto-run commands on their own VM and never ask, so the approval gate is absent and replaced by egress controls and human PR review. Same harness family, different control flow per deployment. `D/agent/security/run-modes § Changelog § Cloud Agents do not use Run Modes`; `D/cloud-agent/security § Autonomy and prompt injection`
- **Sandbox rerun on failure:** the agent may rerun a sandbox-blocked command outside the sandbox with classifier review, mixing an output concern (sandbox) with a control-flow gate; resolved by treating the rerun as a new tool call under the gate. `D/agent/security/run-modes § Sandboxing`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Chat or CLI prompt, queued or steered messages delivered at next tool call, Ask-questions tool answers mid-work |
| Context | Rules at prompt start, skills on demand, local grep search, compaction near capacity, no native memory |
| Model interface | Cursor backend to many providers, Auto routing by classifier, 256k to 1M windows, guardrail reroute |
| Output | Nine first-party tools plus MCP, Seatbelt or Landlock sandbox, network blocked by default, local checkpoints |
| Control flow | Unbounded tool loop, Run Mode gate with LLM classifier, two-level subagents, stop hook loops, coordinator Projects |

## C.19 GitHub Copilot cloud agent and agent mode

**Source:** https://docs.github.com/en/copilot (official docs; source not published) · **Version:** documentation as of 2026-10-16 · **Language:** not documented **Path conventions:** `G` = https://docs.github.com/en/copilot ; every citation is `G/page § Section heading` and was read on 2026-10-16. The docs now call the coding agent "Copilot cloud agent"; the legacy path `G/concepts/agents/coding-agent/about-coding-agent` serves the page titled "GitHub Copilot on GitHub.com" and is cited as `G/concepts/copilot-surfaces/copilot-on-github`. Agent mode (the IDE loop) is documented only at summary level on docs.github.com; its VS Code internals live on code.visualstudio.com and are out of scope here, so several mechanisms read "not documented". **What it is:** Copilot cloud agent is GitHub's hosted coding agent: given an issue, a prompt or a `@copilot` mention it works "in its own ephemeral development environment, powered by GitHub Actions", pushes commits to a `copilot/` branch and opens one pull request per task. Agent mode is the in-editor loop in VS Code, Visual Studio, JetBrains, Xcode and Eclipse that edits files and runs commands "iterating until the task is done". Models are reached through GitHub's Copilot service (hosts such as `api.*.githubcopilot.com`), with provider models from OpenAI, Anthropic, Google, Microsoft, Moonshot AI and xAI. `G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § About customizing Copilot cloud agent's development environment`; `G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Introduction`; `G/reference/ai-models/supported-models § Supported AI models in Copilot`

### Control flow

- **Cloud agent session loop:** a session researches the repository, plans, implements, runs tests and linters, then pushes; the inner model-tool loop is not documented; each session "has a maximum execution time of 59 minutes" that "cannot be extended or bypassed", with a shorter `timeout-minutes` allowed in `copilot-setup-steps.yml`. `G/concepts/copilot-surfaces/copilot-on-github § Chat and agentic experiences § Delegating work`, `§ Limitations and compatibility § Agentic work`
- **Stop rule:** a session ends when the work is pushed and (if asked) a pull request is opened, or when a person clicks Stop session, which "ends the GitHub Actions run and preserves any commits already pushed"; sessions "do not create pull requests automatically" unless the prompt asks. `G/how-tos/copilot-on-github/use-copilot-agents/manage-and-track-agents § Stop a session`; `G/how-tos/copilot-on-github/use-copilot-agents/research-plan-iterate § Research, plan, and iterate on code changes with Copilot cloud agent`
- **Approval gates the harness consumes (cloud):** no tool-level approvals during a session (MCP tools run "autonomously"); gates sit at the edges: only write-access users can trigger, GitHub Actions "workflows are not triggered until Copilot cloud agent's code is reviewed" and a write-access user clicks Approve and run workflows, and the requester cannot approve the PR; `preToolUse` hooks can "approve or deny tool executions". `G/how-tos/copilot-on-github/customize-copilot/configure-mcp-servers § Introduction`; `G/concepts/agents/cloud-agent/risks-and-mitigations § Copilot cloud agent can push code changes to your repository`; `G/concepts/agents/hooks § Types of hooks`
- **Approval gates (agent mode):** terminal commands "require your confirmation by default" unless the administrator or editor settings allow them to run automatically; enterprise managed settings control "whether users can bypass permission checks"; iteration or request caps are not documented on docs.github.com. `G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Introduction`; `G/concepts/copilot-surfaces/copilot-in-ides § Using Copilot in other IDEs § Entry points in JetBrains IDEs`
- **Sub-agents and automations:** custom agents (`.github/agents/NAME.md`) are instantiated when assigned to a task; agent mode in VS Code and JetBrains can hand a self-contained subtask to a subagent (depth and result return not documented); automations start a session on a schedule (hourly, daily, weekly) or on issue-created, PR-opened or PR-synchronized events with a chosen tool set. `G/concepts/agents/cloud-agent/about-custom-agents § About custom agents`; `G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Extending agent mode with tools`; `G/concepts/agents/cloud-agent/about-automations § Triggers`, `§ Tools and actions`

### Input

- **Entry points:** assigning an issue, the agents tab or panel, Copilot Chat (`/task` carries over the chat context), `@copilot` in a PR comment, seeding a repository, a failing Actions run, GitHub Mobile, IDEs, REST API, GitHub CLI, the GitHub MCP Server and integrations; an eyes reaction confirms the session started. `G/how-tos/use-copilot-agents/cloud-agent/create-a-pr § Starting GitHub Copilot sessions`; `G/how-tos/use-copilot-agents/cloud-agent/use-cloud-agent-on-github § Starting from Copilot Chat`, `§ Continuing work on a pull request § Tracking and continuing a session`
- **Mid-session steering:** a follow-up typed below the session log is applied "after it finishes its current tool call"; each steering message consumes AI credits; in agent mode a follow-up prompt redirects the agent before it finishes. `G/how-tos/copilot-on-github/use-copilot-agents/manage-and-track-agents § Steer an agent session`; `G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Steering an agent while it works`
- **Review comments re-enter:** `@copilot` mentions on any pull request, or batched review comments via Add to batch, start a new session on that PR's branch; comments from users without write access "are never shown to the agent"; hidden characters such as HTML comments are filtered before input reaches the agent; images dropped into an issue or prompt are processed. `G/how-tos/use-copilot-agents/cloud-agent/use-cloud-agent-on-github § Continuing work on a pull request § Asking Copilot`

to make changes, § Delegating review comments; G/concepts/agents/cloud-agent/risks-and-mitigations § AI prompts can be vulnerable to injection; G/how-tos/copilot-on-github/use-copilot-agents/research-plan-iterate § Provide visual context

- **Tool results re-enter:** not documented beyond session logs showing "tool calls" and "commands" with their results feeding the next step. G/how-tos/copilot-on-github/use-copilot-agents/manage-and-track-agents § Review session logs

### Context

- **Custom instructions:** .github/copilot-instructions.md (repository-wide), .github/instructions/*.instructions.md with applyTo globs (cloud agent and code review only on GitHub.com), nested AGENTS.md where "the nearest AGENTS.md file in the directory tree will take precedence", or a root CLAUDE.md/GEMINI.md, plus organization instructions; "Instructions are automatically added to requests that you submit to Copilot" and "all sets of relevant instructions are provided to Copilot"; excludeAgent: cloud-agent keeps a file from the cloud agent. G/how-tos/configure-custom-instructions/add-repository-instructions § Creating repository-wide custom instructions, § Creating path-specific custom instructions, § Custom instructions in use; G/concepts/prompting/response-customization § Precedence of custom instructions
- **Custom agents and skills:** an agent profile supplies "Custom instructions that define the agent's behavior and expertise" and a tools allowlist; agent skills are "folders of instructions, scripts, and resources" in .github/skills, .claude/skills or .agents/skills that Copilot loads when relevant (loading mechanism not documented). G/concepts/agents/cloud-agent/about-custom-agents § Agent profile format; G/concepts/agents/about-agent-skills § About agent skills
- **Memory:** Copilot Memory keeps repository-level facts (with code citations, created only from write-access users' activity) and user-level preferences; when Copilot creates context for an agent session it retrieves relevant memories, checks citations against the current branch and "Only validated facts are used"; unused entries are deleted after 28 days; used by cloud agent, code review, CLI and agentic autofix. G/concepts/agents/copilot-memory § How Copilot Memory stores, retains, and uses information § Repository-level facts, § Retention and validation, § Feature availability
- **Session context and codebase access:** chat and the cloud agent share session context, which is "separate from Copilot Memory"; the GitHub MCP server gives read-only access to the current repository by default; the agent reads the checked-out repository in its Actions environment; in agent mode VS Code adds the active file and selection and #codebase references the workspace. G/concepts/copilot-surfaces/copilot-on-github § Chat and agentic experiences § Session context, § Limitations and compatibility § Agentic work
- **Summarization and window handling:** not documented; the only window fact is that a 1M-token context "is available in Visual Studio Code and Copilot CLI only" and larger settings raise credit consumption. G/reference/ai-models/supported-models § Models with extended capabilities

### Model interface

- **Models:** the cloud agent offers Auto plus Claude (Opus 5, 5.5, Sonnet 5.5, Haiku 4.5, 5.5), Gemini (3.7, 3.8 Flash, 4 Argon), GPT (5.4 mini, 5.6 Luna/Sol/Terra, 6 Astra/Luna/Sol, 6.1 Sol), Grok (4.5 to 4.7) and MAI-Code-1.1-Flash; a reasoning-level dropdown appears for models that support it. G/how-tos/use-copilot-agents/cloud-agent/changing-the-ai-model § Supported models, § Reasoning level
- **How reached:** through GitHub's Copilot API hosts (api.*.githubcopilot.com, uploads.github.com must be reachable from the runner); wire format and streaming are not documented; Anthropic retains Fable 5/5.1 prompts for safety classifiers while other Claude models run under zero data retention. G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § Upgrading to larger GitHub-hosted GitHub Actions runners; G/reference/ai-models/supported-models § Supported AI models in Copilot
- **Auto selection and fallback:** "Where a model picker is not available, Auto will be used automatically"; Auto combines a system that "tracks real-time system health and availability" with one that "evaluates task complexity", excludes models outside the plan or blocked by policy, and gives a 10% discount;

GPT-5.3-Codex is the LTS model "used in the event no other models are available". G/how-tos/use-copilot-agents/cloud-agent/changing-the-ai-model § Introduction; G/concepts/models/auto-model-selection § Auto with task optimization, § Policies and availability, § Discount for using auto model selection; G/reference/ai-models/supported-models § Supported AI models in Auto model selection

- **Limits and billing:** usage is metered in AI credits (1 credit = $0.01) where "Each token is priced based on the model used"; Business includes 1,900 and Enterprise 3,900 credits per user per month; retry policy is not documented. G/concepts/billing/usage-based-billing-for-organizations-and-enterprises § What are GitHub AI Credits?, § How do AI credits work?

### Output

- **Tool set:** the cloud agent has a Bash tool (named in the firewall docs), file-editing and repository tools, the built-in GitHub MCP server (read-only token scoped to the current repository) and Playwright MCP server, plus repository-configured MCP servers; the exact built-in tool list is not documented. G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-firewall § Limitations; G/how-tos/copilot-on-github/customize-copilot/configure-mcp-servers § Introduction, § Customizing the built-in GitHub MCP server
- **Execution environment:** an ephemeral GitHub Actions job on a GitHub-hosted Ubuntu runner (larger, Windows or self-hosted ephemeral runners optional); `.github/workflows/copilot-setup-steps.yml` with a single `copilot-setup-steps` job preinstalls dependencies, honoring only `steps`, `permissions`, `runs-on`, `services`, `snapshot` and `timeout-minutes` (max 59); if a setup step fails "Copilot will skip the remaining setup steps and begin working". G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § Customizing Copilot's development environment with Copilot setup steps, § Using self-hosted GitHub Actions runners
- **Sandboxing and network:** an integrated firewall limits internet access by default to a recommended allowlist (package registries, container registries, certificate authorities) plus org and repo custom rules; it "only operates within the GitHub Actions appliance environment", covers only processes started through the Bash tool (not MCP servers or setup steps), and a blocked request adds a warning to the PR naming the address and command. G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-firewall § Overview, § Limitations, § Understanding the recommended firewall allowlist
- **MCP:** JSON in repository settings with `type` (`local`, `stdio`, `http`, `sse`) and a required `tools` allowlist (`*` for all); only tools, not resources or prompts; no OAuth remote servers; secrets as `COPILOT_MCP_*` variables; MCP tools "will be able to use the tools provided by the server autonomously" with no approval. G/how-tos/copilot-on-github/customize-copilot/configure-mcp-servers § Writing a JSON configuration for MCP servers, § Introduction, § Adding an MCP configuration to your repository
- **Result presentation:** commits pushed to a `copilot/` branch via "simple push operations" only (no direct `git push`), a draft pull request (or Diff view in the session before one is requested), screenshots and logs in session logs; Copilot "cannot mark its PRs ready for review, approve them, or merge them"; in agent mode edits stream into the editor and can be undone. G/concepts/agents/cloud-agent/risks-and-mitigations § Copilot cloud agent can push code changes to your repository; G/how-tos/copilot-on-github/use-copilot-agents/research-plan-iterate § Iterate on code changes; G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Steering an agent while it works

### Production concerns

- **Persistence:** session logs and shared sessions are kept and queryable (observability); Copilot Memory outlasts sessions with a 28-day unused-deletion timer (context); the Actions environment is ephemeral and only pushed commits survive a stop (output). G/how-tos/copilot-on-github/

use-copilot-agents/manage-and-track-agents § Query past sessions, § Stop a session; G/concepts/agents/copilot-memory § Retention and validation

- **Safety & security:** firewall and read-only GitHub token (output); write-access-only triggers, hidden-character filtering, `copilot/` branch confinement, workflow approval, requester cannot approve, secret scanning of Copilot's code, rulesets and branch protections still apply (control flow and output); `preToolUse` hooks can deny; MCP tools run autonomously so allowlisting read-only tools is recommended; `.github/hooks/*.json` run as shell commands. G/concepts/agents/cloud-agent/risks-and-mitigations § Copilot cloud agent can push code changes to your repository, § AI prompts can be vulnerable to injection, § Unvalidated code can introduce vulnerabilities; G/concepts/agents/hooks § Types of hooks; G/how-tos/copilot-on-github/customize-copilot/configure-mcp-servers § Writing a JSON configuration for MCP servers
- **Observability:** session logs show "Reasoning", "Tool calls" and "Commands", with token usage and session length on the overview; every commit links back to its session log; setup-step and Start MCP Servers output appears in the logs; JetBrains supports OpenTelemetry agent monitoring. G/how-tos/copilot-on-github/use-copilot-agents/manage-and-track-agents § Monitor agent activity, § Review session logs, § Trace commits to session logs; G/concepts/copilot-surfaces/copilot-in-ides § Using Copilot in other IDEs § Entry points in JetBrains IDEs
- **Resilience:** 59-minute hard timeout; setup failures are skipped rather than fatal; Auto routes on live service health; hooks have `timeoutSec` default 30 and run synchronously; model-request retries and compaction on overflow are not documented. G/concepts/copilot-surfaces/copilot-on-github § Limitations and compatibility § Agentic work; G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § Customizing Copilot's development environment with Copilot setup steps; G/concepts/models/auto-model-selection § Auto with task optimization; G/concepts/agents/hooks § Hook configuration format
- **Performance:** preinstalled dependencies via setup steps avoid trial-and-error installs; larger runners for CPU, memory and disk; cached tokens priced separately (caching mechanism not documented); keep hooks "under 5 seconds when possible". G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § Preinstalling tools or dependencies in Copilot's environment, § Upgrading to larger GitHub-hosted GitHub Actions runners; G/concepts/billing/usage-based-billing-for-organizations-and-enterprises § How do AI credits work?; G/concepts/agents/hooks § Performance considerations
- **Evaluation:** not documented beyond thumbs up or down feedback on Copilot's pull requests and comments. G/how-tos/use-copilot-agents/cloud-agent/review-copilot-prs § Give feedback on Copilot's work

### Hard to place

- **GitHub Actions as both executor and loop boundary:** the Actions job is the output executor (sandbox, runner) but its `timeout-minutes` and the Stop session button end the loop, so one artifact carries an output mechanism and a control-flow budget; placed in both. G/how-tos/copilot-on-github/customize-copilot/customize-cloud-agent/customize-the-agent-environment § Customizing Copilot's development environment with Copilot setup steps; G/how-tos/copilot-on-github/use-copilot-agents/manage-and-track-agents § Stop a session
- **Workflow approval after the session:** the Approve and run workflows gate is a human decision the harness consumes, but it fires after the agent has stopped, gating CI on the PR rather than the agent's loop; it is a control-flow gate of GitHub's PR pipeline, not of the agent harness. G/how-tos/use-copilot-agents/cloud-agent/review-copilot-prs § Manage GitHub Actions workflow runs
- **Copilot Memory validation:** a record kept for the model (context) that is re-validated against the current branch before use, which is a context-assembly step performed by a model-driven check; the docs do not say whether a model call is involved, so the mechanism is not documented. G/concepts/agents/copilot-memory § How Copilot Memory stores, retains, and uses information § Repository-level facts
- **Agent mode internals:** docs.github.com describes agent mode only as iterating with command confir-

mation; its loop, tool list and request caps are documented by the editor vendors, so the harness cannot be fully deconstructed from this source. `G/how-tos/copilot-in-your-ide/use-copilot-agents/use-agent-mode § Introduction`

### Summary

| Primitive | Design decision |
|---|---|
| Input | Issues, @copilot mentions, chat handoff, API and automations; steering applied after the current tool call; write-access only |
| Context | Layered instruction files, custom agents, skills and validated 28-day memory; summarization not documented |
| Model interface | GitHub-hosted Copilot API, Auto routing on health and task complexity, LTS fallback, AI-credit metering |
| Output | Ephemeral Actions runner with firewall, Bash plus read-only GitHub MCP, pushes to copilot/ branch, draft PR |
| Control flow | Background session with 59-minute cap, no in-session approvals, edge gates on triggers and workflows, automations |

**Table 2:** Design decisions by primitive, and the choices seen across the harnesses deconstructed in §5.

| Primitive | Decision | Choices seen |
|---|---|---|
| Input | source | terminal; editor; chat gateways; HTTP; issue tracker; scheduled trigger; a typed state channel |
| | mid-turn messages | delivered at the next step boundary (steer); after the turn (queue); as an interrupt; not at all |
| | the person's answers | consumed by the harness as approvals; returned to the model as tool results |
| Context | instructions | a system prompt alone; plus project files (`CLAUDE.md`, `AGENTS.md`); plus memory files; a sectioned prompt diffed into patches |
| | long-term memory | none; files preloaded into the instructions; a store searched by the agent through tools; summaries carried forward |
| | window fitting | drop old turns; summarize (prune, then summarize); compaction at a threshold with a verbatim tail; provider-native compaction; left to user code |
| | layout for caching | a stable prefix with changes appended; instruction changes frozen as dated events; byte-stable tiers |
| Model interface | client | provider SDK; in-house request compiled per protocol; one stream function over adapters; a library's chat-model call |
| | transport | HTTP with server-sent events; WebSocket, falling back to HTTP |
| | retries and backup | none; three to ten retries; retry split by whether output started; a fallback chain of models |
| | action format | native tool calls; calls parsed from text; a code block to execute |
| Output | tool set | one shell; a fixed built-in set; built-in plus MCP; plain tool objects; a tool node |
| | action format | structured calls; model-written code run in an interpreter (V8, QuickJS, in-process) |
| | execution environment | the host; an OS sandbox with a domain allowlist; a container; a remote sandbox |
| | parallelism | sequential; read-only calls concurrent; all calls concurrent as they stream |
| Control flow | loop shape | until no tool call; one task per turn; a user-written graph run as supersteps; a phase-verdict loop |
| | stopping rules | step budget; token or cost budget; unlimited; recursion limit; stop-gate nudges |
| | approval | a rule pipeline ending in a prompt; a classifier model; a reviewer model; each tool asserting its own permission |
| | delegation | an agent tool to a fixed depth; child sessions; sub-agents as full sessions; flat threaded delegation |

**Table 3:** Six production concerns as questions asked of each primitive. Each cell names what the concern changes in that primitive; "—" means we found nothing essential there in the harnesses examined.

| | **Persistence** *What outlasts the session?* | **Safety & security** *What could do harm, and who could make it?* | **Observability** *What did it do?* | **Resilience** *What if something breaks?* | **Performance** *What is slow or costly?* | **Evaluation** *Does it still work?* |
|---|---|---|---|---|---|---|
| Input | an unfinished session offered back | approvals asked of a person | refusals recorded | input ending; interrupts | — | yes to every approval in a test run |
| Context | memories; resumed transcripts | untrusted content; poisoned memory; roles and authority | — | compaction when too long | caching the unchanged prefix; trimming results | the instructions under test |
| Model interface | — | — | tokens, latency, cost per call | retries; a backup model | streaming | the model under test |
| Output | — | sandboxes; never-lists; secrets hidden; pinned images | what each action did | time limits on actions | parallel actions; smaller models for side tasks | the tools under test |
| Control flow | the task in progress; a diary of every message | runaway loops; the record kept out of reach | each decision | resume after a crash | — | the loop under test |

**Table 4:** The five primitives, the step of the agent cycle each serves, and the CoALA component it corresponds to.

| Primitive | Mechanism | In the agent cycle | CoALA |
|---|---|---|---|
| Input | capture requests and results | perceive | observations |
| Context | assemble what the model is shown | working and long-term memory | working memory; retrieval |
| Model interface | request a response | decide | reasoning (by the model) |
| Output | show words, carry out actions | act | grounding actions; learning (writes) |
| Control flow | go again, hand back, or stop | the cycle | decision cycle |

**Table 5:** Six harnesses by primitive. Each cell is the design decision that harness made within the primitive; the evidence for each is in Appendix C.

| | Claude Code | Codex CLI | opencode | pi | Hermes Agent | LangGraph |
|---|---|---|---|---|---|---|
| Input | prompts, stdin, queued mid-turn messages, scheduled prompts, sub-agent reports, tool results, answers to an ask-user tool | one operation channel per thread; steering queued mid-turn; mailbox; typed answers to the model's questions; tool results | durable inbox rows (person, shell, skill, synthetic), delivered as steer or queue at step boundaries | terminal, print and RPC modes into one prompt call; steer and follow-up queues | one message event from CLI, 25 messaging gateways, HTTP, ACP, cron; mid-turn steering | writes to a typed state channel, merged by a reducer: messages, tool results, resume values |
| Context | cached prefix (system, tools, CLAUDE.md) plus appended reminders; deferred MCP schemas; compaction re-injects memory and recent files | the Responses item list; world-state sections as diffs; project files walked to cwd; three compaction modes, with background compaction | event-sourced history re-projected per call; cached instructions as dated events; compaction at ∼90%, 15k tail, or provider-native | projection of a session tree; sectioned prompt diffed into patches; compaction past a reserve, 20k tail | three-tier byte-stable prompt; memory and project files; compaction at 50% by prune then summarize; tool-search bridge | the state's messages plus one system message; window-fitting left to a user hook |
| Model interface | Messages API, streaming; 10 retries; idle watchdogs; user-configured fallback chain | Responses API only; WebSocket, falling back to HTTP; 5 retries with server-advised timing; no server-side storage | in-house request compiled per protocol; SSE or WebSocket; retry split by whether output started | one stream function over ∼15 adapters; SDK retries off; two-tier retry; overflow routed to compaction | chat-completions client with pluggable transports; streaming; 3 retries, fallback chain, auto-recovery | one LangChain chat-model call per step; no client, streaming or fallback of its own |
| Output | built-in tools plus MCP; read-only calls concurrent; OS sandbox with domain allowlist; results spill to files; edit checkpoints | one orchestrator: approve, sandbox, run, retry on denial; OS sandboxes on three platforms; read–write gate for parallel tools; V8 code mode | tools run concurrently as they stream; each asserts its own permission; most reached through a code-mode interpreter | plain tool objects, parallel by default; no core sandbox; QuickJS code mode | core tools plus bridge; parallel-safe segments; sandboxed terminal (local, Docker, Modal, . . . ); approval gate | tool node runs calls concurrently in-process; errors become messages; tools may return routing commands |
| Control flow | loop until no tool calls; six-step permission gate, with a classifier model by default; agent tool to depth 3; model-written workflows | task per turn; loop until no tool call; hooks, Guardian reviewer, then person; sub-agents as full sessions with depth caps | one fibre per session; continue while tools called; per-agent step budget; child sessions, depth 1 | inner tool loop, outer follow-up loop, session loop for retries; virtual-model routing | phase-verdict loop; unlimited budget; stop-gate nudges; flat threaded delegation; task-keyed routing | a user-written graph run as supersteps; stop on empty task set or recursion limit; interrupts checkpointed |